

Professional Level – Options Module

Advanced Financial Management

March/June 2017 – Sample Questions



Time allowed: 3 hours 15 minutes

This question paper is divided into two sections:

Section A – This ONE question is compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Formulae and tables are on pages 9–13.

Do NOT open this question paper until instructed by the supervisor.

This question paper must not be removed from the examination hall.

P4 Paper

Think Ahead

ACCA

The Association of
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The question paper begins on page 3.

Section A – This ONE question is compulsory and MUST be attempted

- 1 The eight-member board of executive directors (BoD) of Chrysos Co, a large private, unlisted company, is considering the company's long-term business and financial future. The BoD is considering whether or not to undertake a restructuring programme. This will be followed a few years later by undertaking a reverse takeover to obtain a listing on the stock exchange in order to raise new finance. However, a few members of the BoD have raised doubts about the restructuring programme and the reverse takeover, not least the impact upon the company's stakeholders. Some directors are of the opinion that an initial public offering (IPO) would be a better option when obtaining a listing compared to a reverse takeover.

Chrysos Co was formed about 15 years ago by a team of five senior equity holders who are part of the BoD and own 40% of the equity share capital in total; 30 other equity holders own a further 40% of the equity share capital but are not part of the BoD; and a consortium of venture capital organisations (VCOs) own the remaining 20% of the equity share capital and have three representatives on the BoD. The VCOs have also lent Chrysos Co substantial debt finance in the form of unsecured bonds due to be redeemed in 10 years' time. In addition to the BoD, Chrysos Co also has a non-executive supervisory board consisting of members of Chrysos Co's key stakeholder groups. Details of the supervisory board are given below.

Chrysos Co has two business units: a mining and shipping business unit, and a machinery parts manufacturing business unit. The mining and shipping business unit accounts for around 80% of Chrysos Co's business in terms of sales revenue, non-current and current assets, and payables. However, it is estimated that this business unit accounts for around 75% of the company's operating costs. The smaller machinery parts manufacturing business unit accounts for the remaining 20% of sales revenue, non-current and current assets, and payables; and around 25% of the company's operating costs.

The following figures have been extracted from Chrysos Co's most recent financial statements:

Profit before depreciation, interest and tax for the year to 28 February 2017

	\$m
Sales revenue	16,800
Operating costs	(10,080)
Profit before depreciation, interest and tax	6,720

Financial position as at 28 February 2017

	\$m
Non-current assets	
Land and buildings	7,500
Equipment	5,400
Current assets	
Inventory	1,800
Receivables	900
Total assets	15,600
Equity	
Share capital (\$1 par value per share)	1,800
Reserves	5,400
Non-current liabilities	
4.50% unsecured bonds 2026 (from the VCOs)	4,800
Other debt	1,050
Current liabilities	
Payables	750
Bank overdraft	1,800
Total equity and liabilities	15,600

Corporate restructuring programme

The purpose of the restructuring programme is to simplify the company's gearing structure and to obtain extra funding to expand the mining and shipping business in the future. At present, Chrysos Co is having difficulty obtaining additional funding without having to pay high interest rates.

Machinery parts manufacturing business unit

The smaller machinery parts manufacturing business unit will be unbundled either by having its assets sold to a local supplier for \$3,102 million after its share of payables have been paid; or

The smaller machinery parts manufacturing business unit will be unbundled through a management buy-out by four managers. In this case, it is estimated that its after-tax net cash flows will increase by 8% in the first year only and then stay fixed at this level for the foreseeable future. The cost of capital related to the smaller business unit is estimated to be 10%. The management buy-out team will pay Chrysos Co 70% of the estimated market value of the smaller machinery parts manufacturing business unit.

Mining and shipping business unit

Following the unbundling of the smaller machinery parts manufacturing business unit, Chrysos Co will focus solely on the mining and shipping business unit, prior to undertaking the reverse takeover some years into the future.

As part of the restructuring programme, the existing unsecured bonds lent by the VCOs will be cancelled and replaced by an additional 600 million \$1 shares for the VCOs. The VCOs will pay \$400 million for these shares. The bank overdraft will be converted into a 15-year loan on which Chrysos Co will pay a fixed annual interest of 4.50%. The other debt under non-current liabilities will be repaid. In addition to this, Chrysos Co will invest \$1,200 million into equipment for its mining and shipping business unit and this will result in its profits and cash flows growing by 4% per year in perpetuity.

Additional financial information

Chrysos Co aims to maintain a long-term capital structure of 20% debt and 80% equity in market value terms. Chrysos Co's finance director has assessed that the 4.50% annual interest it will pay on its bank loan is a reasonable estimate of its long-term cost of debt, based on the long-term capital structure above.

Although Chrysos Co does not know what its cost of capital is for the mining and shipping business unit, its finance director has determined that the current ungeared cost of equity of Sidero Co, a large quoted mining and shipping company, is 12.46%. Chrysos Co's finance director wants to use Sidero Co's ungeared cost of equity to calculate its cost of capital for the mining and shipping business unit.

The annual corporation tax rate on profits applicable to all companies is 18% and it can be assumed that tax is payable in the year incurred. All the non-current assets are eligible for tax allowable depreciation of 12% annually on the book values. The annual reinvestment needed to keep operations at their current levels is equivalent to the tax allowable depreciation.

Details of the supervisory board

The non-executive supervisory board provides an extra layer of governance over the BoD. It consists of representatives from the company's internal stakeholder groups including the finance providers, employees and the company's management. It ensures that the actions taken by the BoD are for the benefit of all the stakeholder groups and to the company as a whole. Any issues raised in board meetings are resolved through negotiation until an agreed position is reached.

Required:

- (a) Explain what a reverse takeover involves and discuss the relative advantages and disadvantages to a company, such as Chrysos Co, of obtaining a listing through a reverse takeover as opposed to an initial public offering (IPO). (9 marks)
- (b) Prepare a report for the board of directors of Chrysos Co which includes:
- (i) An extract of the financial position and an estimate of Chrysos Co's value to the equity holders, after undertaking the restructuring programme. (18 marks)
 - (ii) An explanation of the approach taken and assumptions made in estimating Chrysos Co's value to the equity holders, after undertaking the restructuring programme. (5 marks)
 - (iii) A discussion of the impact of the restructuring programme on Chrysos Co and on the venture capital organisations. (10 marks)
- Professional marks will be awarded in part (b) for the format, structure and presentation of the report. (4 marks)
- (c) Discuss why the attention Chrysos Co pays to its stakeholders represented on the supervisory board may change once it has obtained a listing. (4 marks)

(50 marks)

Section B – TWO questions ONLY to be attempted

- 2 Bournelorth Co is an IT company which was established by three friends ten years ago. It was listed on a local stock exchange for smaller companies nine months ago.

Bournelorth Co originally provided support to businesses in the financial services sector. It has been able to expand into other sectors over time due to the excellent services it has provided and the high quality staff whom its founders recruited. The founders have been happy with the level of profits which the IT services have generated. Over time they have increasingly left the supervision of the IT services in the hands of experienced managers and focused on developing diagnostic applications (apps). The founders have worked fairly independently of each other on development work. Each has a small team of staff and all three want their teams to work in an informal environment which they believe enhances creativity.

Two apps which Bournelorth Co developed were very successful and generated significant profits. The founders wanted the company to invest much more in developing diagnostic apps. Previously they had preferred to use internal funding, because they were worried that external finance providers would want a lot of information about how Bournelorth Co is performing. However, the amount of finance required meant that funding had to be obtained from external sources and they decided to seek a listing, as two of Bournelorth Co's principal competitors had recently been successfully listed.

25% of Bournelorth Co's equity shares were made available on the stock exchange for external investors, which was the minimum allowed by the rules of the exchange. The founders have continued to own the remaining 75% of Bournelorth Co's equity share capital. Although the listing was fully subscribed, the price which new investors paid was lower than the directors had originally hoped.

The board now consists of the three founders, who are the executive directors, and two independent non-executive directors, who were appointed when the company was listed. The non-executive directors have expressed concerns about the lack of frequency of formal board meetings and the limited time spent by the executive directors overseeing the company's activities, compared with the time they spend leading development work. The non-executive directors would also like Bournelorth Co's external auditors to carry out a thorough review of its risk management and control systems.

The funds obtained from the listing have helped Bournelorth Co expand its development activities. Bournelorth Co's competitors have recently launched some very successful diagnostic apps and its executive directors are now afraid that Bournelorth Co will fall behind its competitors unless there is further investment in development. However, they disagree about how this investment should be funded. One executive director believes that Bournelorth Co should consider selling off its IT support and consultancy services business. The second executive director favours a rights issue and the third executive director would prefer to seek debt finance. At present Bournelorth Co has low gearing and the director who is in favour of debt finance believes that there is too much uncertainty associated with obtaining further equity finance, as investors do not always act rationally.

Required:

- (a) Discuss the factors which will determine whether the sources of finance suggested by the executive directors are used to finance further investment in diagnostic applications (apps). (8 marks)
- (b) (i) Identify the risks associated with investing in the development of apps and describe the controls which Bournelorth Co should have over its investment in development. (6 marks)
- (ii) Discuss the issues which determine the information Bournelorth Co communicates to external finance providers. (3 marks)
- (c) (i) Explain the insights which behavioural finance provides about investor behaviour. (3 marks)
- (ii) Assess how behavioural factors may affect the share price of Bournelorth Co. (5 marks)

(25 marks)

- 3** Buryecs Co is an international transport operator based in the Eurozone which has been invited to take over a rail operating franchise in Wirtonia, where the local currency is the dollar (\$). Previously this franchise was run by a local operator in Wirtonia but its performance was unsatisfactory and the government in Wirtonia withdrew the franchise.

Buryecs Co will pay \$5,000 million for the rail franchise immediately. The government has stated that Buryecs Co should make an annual income from the franchise of \$600 million in each of the next three years. At the end of the three years the government in Wirtonia has offered to buy the franchise back for \$7,500 million if no other operator can be found to take over the franchise.

Today's spot exchange rate between the Euro and Wirtonia \$ is €0.1430 = \$1. The predicted inflation rates are as follows:

Year	1	2	3
Eurozone	6%	4%	3%
Wirtonia	3%	8%	11%

Buryecs Co's finance director (FD) has contacted its bankers with a view to arranging a currency swap, since he believes that this will be the best way to manage financial risks associated with the franchise. The swap would be for the initial fee paid for the franchise, with a swap of principal immediately and in three years' time, both these swaps being at today's spot rate. Buryecs Co's bank would charge an annual fee of 0.5% in € for arranging the swap. Buryecs Co would take 60% of any benefit of the swap before deducting bank fees, but would then have to pay 60% of the bank fees.

Relevant borrowing rates are:

	Buryecs Co	Counterparty
Eurozone	4.0%	5.8%
Wirtonia	Wirtonia bank rate + 0.6%	Wirtonia bank rate + 0.4%

In order to provide Buryecs Co's board with an alternative hedging method to consider, the FD has obtained the following information about over-the-counter options in Wirtonia \$ from the company's bank.

The exercise price quotation is in Wirtonia \$ per €1, premium is % of amount hedged, translated at today's spot rate.

Exercise price	Call options	Put options
7.75	2.8%	1.6%
7.25	1.8%	2.7%

Assume a discount rate of 14%.

Required:

- (a) **Discuss the advantages and drawbacks of using the currency swap to manage financial risks associated with the franchise in Wirtonia.** (6 marks)
- (b) (i) **Calculate the annual percentage interest saving which Buryecs Co could make from using a currency swap, compared with borrowing directly in Wirtonia, demonstrating how the currency swap will work.** (4 marks)
 - (ii) **Evaluate, using net present value, the financial acceptability of Buryecs Co operating the rail franchise under the terms suggested by the government of Wirtonia and calculate the gain or loss in € from using the swap arrangement.** (8 marks)
- (c) **Calculate the results of hedging the receipt of \$7,500 million using the currency options and discuss whether currency options would be a better method of hedging this receipt than a currency swap.** (7 marks)

(25 marks)

- 4 Toltuck Co is a listed company in the building industry which specialises in the construction of large commercial and residential developments. Toltuck Co had been profitable for many years, but has just incurred major losses on the last two developments which it has completed in its home country of Arumland. These developments were an out-of-town retail centre and a major residential development. Toltuck Co's directors have blamed the poor results primarily on the recent recession in Arumland, although demand for the residential development also appears to have been adversely affected by it being located in an area which has suffered serious flooding over the last two years.

As a result of returns from these two major developments being much lower than expected, Toltuck Co has had to finance current work-in-progress by a significantly greater amount of debt finance, giving it higher gearing than most other construction companies operating in Arumland. Toltuck Co's directors have recently been alarmed by a major credit agency's decision to downgrade Toltuck Co's credit rating from AA to BBB. The directors are very concerned about the impact this will have on the valuation of Toltuck Co's bonds and the future cost of debt.

The following information can be used to assess the consequences of the change in Toltuck Co's credit rating.

Toltuck Co has issued an 8% bond, which has a face or nominal value of \$100 and a premium of 2% on redemption in three years' time. The coupon on the bond is payable on an annual basis.

The government of Arumland has three bonds in issue. They all have a face or nominal value of \$100 and are all redeemable at par. Taxation can be ignored on government bonds. They are of the same risk class and the coupon on each is payable on an annual basis. Details of the bonds are as follows:

Bond	Redeemable	Coupon	Current market value
			\$
1	1 year	9%	104
2	2 years	7%	102
3	3 years	6%	98

Credit spreads, published by the credit agency, are as follows (shown in basis points):

Rating	1 year	2 years	3 years
AA	18	31	45
BBB	54	69	86

Toltuck Co's shareholder base can be divided broadly into two groups. The majority of shareholders are comfortable with investing in a company where dividends in some years will be high, but there will be low or no dividends in other years because of the cash demands facing the business. However, a minority of shareholders would like Toltuck Co to achieve at least a minimum dividend each year and are concerned about the company undertaking investments which they regard as very speculative. Shareholders from both groups have expressed some concerns to the board about the impact of the fall in credit rating on their investment.

Required:

- (a) Calculate the valuation and yield to maturity of Toltuck Co's \$100 bond under its old and new credit ratings. (10 marks)
- (b) Discuss the factors which may have affected the credit rating of Toltuck Co published by the credit agency. (8 marks)
- (c) Discuss the impact of the fall in Toltuck Co's credit rating on its ability to raise financial capital and on its shareholders' return. (7 marks)

(25 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i(E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_0 = \frac{D_0(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = b r_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \quad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model

$$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$$

Where:

$$d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$$

$$d_2 = d_1 - s\sqrt{t}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Answers

- 1 (a) A reverse takeover enables a private, unlisted company, like Chrysos Co, to gain a listing on the stock exchange without needing to go through the process of an initial public offering (IPO). The private company merges with a listed 'shell' company. The private company initially purchases equity shares in the listed company and takes control of its board of directors. The listed company then issues new equity shares and these are exchanged for equity shares in the unlisted company, thereby the original private company's equity shares gain a listing on the stock exchange. Often the name of the listed company is also changed to that of the original unlisted company.

Advantages relative to an IPO

1. An IPO can take a long time, typically between one and two years, because it involves preparing a prospectus and creating an interest among potential investors. The equity shares need to be valued and the issue process needs to be administered. Since with the reverse takeover shares in the private company are exchanged for shares in the listed company and no new capital is being raised, the process can be completed much quicker.
2. An IPO is an expensive process and can cost between 3% and 5% of the capital being raised due to involvement of various parties, such as investment banks, law firms, etc, and the need to make the IPO attractive through issuing a prospectus and marketing the issue. A reverse takeover does not require such costs to be incurred and therefore is considerably cheaper.
3. In periods of economic downturn, recessions and periods of uncertainty, an IPO may not be successful. A lot of senior managerial time and effort will be spent, as well as expenditure, with nothing to show for it. On the other hand, a reverse takeover would not face this problem as it does not need external investors and it is not raising external finance, but is being used to gain from the potential benefits of going public by getting a listing.

Disadvantages relative to an IPO

1. The 'shell' listed company being used in the reverse takeover may have hidden liabilities and may be facing potential litigation, which may not be obvious at the outset. Proper and full due diligence is necessary before the process is started. A company undertaking an IPO would not face such difficulties.
2. The original shareholders of the listed company may want to sell their shares immediately after the reverse takeover process has taken place and this may affect the share price negatively. A lock-up period during which shares cannot be sold may be necessary to prevent this. [NB: An IPO may need a lock-up period as well, but this is not usually the case.]
3. The senior management of an unlisted company may not have the expertise and/or understanding of the rules and regulations which a listed company needs to comply with. The IPO process normally takes longer and is more involved, when compared to a reverse takeover. It also involves a greater involvement from external experts. These factors will provide the senior management involved in an IPO, with opportunities to develop the necessary expertise and knowledge of listing rules and regulations, which the reverse takeover process may not provide.
4. One of the main reasons for gaining a listing is to gain access to new investor capital. However, a smaller, private company which has become public through a reverse takeover may not obtain a sufficient analyst coverage and investor following, and it may have difficulty in raising new finance in future. A well-advertised IPO will probably not face these issues and find raising new funding to be easier.

(b) Report to the board of directors (BoD), Chrysos Co

This report provides extracts from the financial position and an estimate of the value of Chrysos Co after it has undertaken a restructuring programme. It also contains an explanation of the process used in estimating the value and of the assumptions made. Finally, the report discusses the impact of the restructuring programme on the company and on venture capital organisations.

It is recommended that the manufacturing business unit is unbundled through a management buy-out, rather than the assets being sold separately, and it is estimated that Chrysos Co will receive \$3,289m from the unbundling of the manufacturing business unit (appendix one). This amount is recorded as a cash receipt in the extract of the financial position given below.

Extract of Chrysos Co's financial position following the restructuring programme

	\$m
Non-current assets	
Land and buildings (80% x \$7,500m)	6,000
Equipment ((80% x \$5,400m) + \$1,200m)	5,520
Current assets	
Inventory (80% x \$1,800m)	1,440
Receivables (80% x \$900m)	720
Cash (\$3,289m + \$400m – \$1,200m – \$1,050m)	1,439
Total assets	<u>15,119</u>
Equity	
Share capital (\$1,800 + \$600m)	2,400
Reserves **	10,319
Non-current liabilities	
Bank loan	1,800
Current liabilities	
Payables (80% x \$750m)	600
Total equity and liabilities	<u>15,119</u>

** Balancing figure

Estimate of Chrysos Co's equity value following the restructuring programme

It is estimated that Chrysos Co's equity value after the restructuring programme has taken place will be just over \$46 billion (appendix three).

Process undertaken in determining Chrysos Co's equity value

The corporate value is based on a growth rate of 4% on cash flows in perpetuity, which are discounted at Chrysos Co's cost of capital (appendix two). The cash flows are estimated by calculating the profit before depreciation and tax of the unbundled firm consisting of just the mining and shipping business unit and then deducting the depreciation and taxation amounts from this.

The bank loan debt is then deducted from the corporate value to estimate the value of the firm which is attributable to the equity holders (appendix three).

Assumptions made in determining Chrysos Co's equity value

It is assumed that Sidero Co's ungeared cost of equity is equivalent to Chrysos Co's ungeared cost of equity, given that they are both in the same industry and therefore face the same business risk. Modigliani and Miller's proposition 2 is used to estimate Chrysos Co's restructured cost of equity and cost of capital.

It is assumed that deducting depreciation and tax from the profit before depreciation, interest and tax provides a reasonably accurate estimate of the free cash flows (appendix three). Other adjustments such as changes in working capital are reckoned to be immaterial and therefore not considered. Depreciation is not added back because it is assumed to be the same as the capital needed for reinvestment purposes.

It is assumed that the cash flows will grow in perpetuity. The assumption of growth in perpetuity may be over-optimistic and may give a higher than accurate estimate of Chrysos Co's equity value.

(Note: Credit will be given for alternative and relevant assumptions)

Impact of the restructuring programme on Chrysos Co and on the venture capital organisations (VCOs)

By acquiring an extra 600 million equity shares, the proportion of the VCOs' equity share capital will increase to 40% ((600m + 20% x 1,800)/(1,800 + 600)) from 20%. Therefore, the share of the equity value the VCOs will hold in Chrysos Co will increase by \$9,229m, which is 77.5% more than the total of the value of bonds cancelled and extra payment made (appendix four). As long as the VCOs are satisfied that the equity value of Chrysos Co after the restructuring programme has been undertaken is accurate, the value of their investment has increased substantially. The VCOs may want undertake a feasibility study on the annual growth rate in cash flows of 4% and the assumption of growth in perpetuity. However, the extent of additional value created seems to indicate that the impact for the VCOs is positive.

By cancelling the VCOs' unsecured bonds and repaying the other debt in non-current liabilities, an opportunity has been created for Chrysos Co to raise extra debt finance for future projects. Based on a long-term capital structure ratio of 80% equity and 20% debt, and a corporate value of \$47,944m (appendix three), this equates to just under \$9,600m of possible debt finance which could be accessed. Since the bank loan has a current value of \$1,800m, Chrysos Co could raise just under an extra \$7,800m debt funding and it would also have \$1,439 million in net cash available from the sale of the machinery parts manufacturing business unit.

Chrysos Co's current value has not been given and therefore it is not possible to determine the financial impact of the equity value after the restructuring has taken place on the company as a whole. Nevertheless, given that the company has access to an extra \$7,800m debt funding to expand its investment into new value-creating projects, it is likely that the restructuring

programme will be beneficial. However, it is recommended that the company tries to determine its current equity value and compares this with the proposed new value. A concern may be that both the five senior equity holders' group and the 30 other equity holders group's proportion of equity shares will reduce to 30% from 40% each, as a result of the VCOs acquiring an additional 600 million shares. Both these shareholder groups need to be satisfied about the potential negative impact of these situations against the potential additional benefits accruing from the restructuring programme, before the company proceeds with the programme.

Conclusion

The restructuring programme creates an opportunity for Chrysos Co to have access to extra funding and additional cash for investment in projects in the future. The VCOs are likely to benefit financially from the restructuring programme as long as they are satisfied about the assumptions made when assessing the value created. However, Chrysos Co will need to ensure that all equity holder groups are satisfied with the change in their respective equity holdings.

Report compiled by:

Date:

(Note: Credit will be given for alternative and relevant points)

Appendices

Appendix One: Unbundling the manufacturing business unit

Option 1: Sale of assets

Net proceeds to Chrysos Co from net sale of assets of the manufacturing business unit are \$3,102 million.

Option 2: Management buy-out

	\$m
Sales revenue (20% x \$16,800m)	3,360
Operating costs (25% x 10,080m)	(2,520)
Profit before depreciation, interest and tax	840
Depreciation (12% x 20% x (\$7,500m + \$5,400m))	(310)
	530
Tax (18% x \$530m)	(95)
Cash flows	435

Estimated value = $(\$435m \times 1.08)/0.10 = \$4,698m$

Amount payable to Chrysos Co = $70\% \times \$4,698m = \$3,289m$

The option to unbundle through a management buy-out (option 2) is marginally better for Chrysos Co and it will opt for this.

Appendix Two: Calculation of cost of equity and cost of capital

Chrysos Co, estimate of cost of equity (Ke) and cost of capital (CoC)

$Ke = 12.46\% + [0.82 \times (12.46\% - 4.5\%) \times (0.2/0.8)]$

$Ke = 14.09\%$

$CoC = 0.8 \times 14.09\% + 0.2 \times 4.5\% \times 0.82 = 12.01$, say 12%

Appendix Three: Estimate of value

	\$m
Sales revenue (80% x \$16,800m)	13,440
Costs prior to depreciation, interest and tax (75% x 10,080m)	(7,560)
Profit before depreciation and tax	5,880
Depreciation (12% x (\$6,000m + \$5,520m))	(1,382)
	4,498
Tax (18% x \$4,498m)	(810)
Cash flows	3,688

Cost of capital to be used in estimating Chrysos Co's value is 12% (appendix two)

Estimated corporate value = $(\$3,688m \times 1.04)/(0.12 - 0.04) = \$47,944m$

Estimated equity value = $\$47,944m - \$1,800m = \$46,144m$

(Note: It is also acceptable to calculate cash flows after interest payment and use the cost of equity to estimate the equity value based on cash flows to equity instead of cash flows to firm)

Appendix Four: Value created for VCOs

Value attributable to the VCOs = $40\% \times \$46,144\text{m} = \$18,458\text{m}$

Value from increased equity ownership (this has doubled from 20% to 40%)

$50\% \times \$18,458\text{m} = \$9,229\text{m}$

Value of unsecured bonds foregone by the VCOs = $\$4,800\text{m}$

Additional capital invested by the VCOs = $\$400\text{m}$

Total of additional capital invested and value of bonds foregone = $\$5,200\text{m}$

Additional value = $(\$9,229\text{m} - \$5,200\text{m})/\$5,200\text{m} = 77.5\%$ (or $\$4,029\text{m}$)

- (c) As a private company, Chrysos Co is able to ensure that the needs of its primary stakeholder groups – finance providers, managers and employees – are taken into account through the supervisory board. The supervisory board has representatives from each of these groups and each group member has a voice on the board. Each stakeholder group should be able to present its position to the board through its representatives, and decisions will be made after agreement from all group representatives. In this way, no single stakeholder group holds primacy over any other group.

Once Chrysos Co is listed and raises new capital, it is likely that it will have a large and diverse range of equity shareholders, who will likely be holding equity shares in many other companies. Therefore there is likely to be pressure on Chrysos Co to engage in value creating activity aimed at keeping its share price buoyant and thereby satisfying the equity shareholders. It is, therefore, likely that the equity shareholders' needs will hold primacy over the other stakeholder groups and quite possibly the power of the supervisory board will diminish as a result of this.

- 2 (a) According to traditional finance theory, Bournelorth Co's directors will wish to strive for long-term shareholder wealth maximisation. The directors may not have been fully committed to long-term wealth maximisation, as they seemed to have focused on the development aspects which interested them most and left the original business mostly to others. However, now they are likely to come under pressure from the new external shareholders to maximise shareholder wealth and pay an acceptable level of dividend. To achieve this, it seems that Bournelorth Co will have to commit further large sums to investment in development of diagnostic applications (apps) in order to keep up with competitors.

Selling off the IT services business

At present the IT services business seems to be a reliable generator of significant profits. Selling it off would very likely produce a significant cash boost now, when needed. However, it would remove the safety net of reasonably certain income and mean that Bournelorth Co followed a much riskier business model. The IT services business also offers a possible gateway to reach customers who may be interested in the apps which Bournelorth Co develops.

Rights issue

If the executive directors wish to maintain their current percentage holdings, they would have to subscribe to 75% of the shares issued under the rights issue. Even though the shares would be issued at a discount, the directors might well not have the personal wealth available to subscribe fully. Previously they had to seek a listing to obtain enough funds for expansion, even though they were reluctant to bring in external investors, and this suggests their personal financial resources are limited.

However, the directors may need to take up the rights issue in order to ensure its success. If they do not, it may send out a message to external investors that the directors are unwilling to make a further commitment themselves because of the risks involved. There are also other factors which indicate that the rights issue may not be successful. The directors did not achieve the initial market price which they originally hoped for when Bournelorth Co was listed and shareholders may question the need for a rights issue soon after listing.

If the executive directors do not take up all of their rights, and the rights issue is still successful, this may have consequences for the operation of the business. The external shareholders would own a greater percentage of Bournelorth Co's equity share capital and may be in a position to reinforce the wishes of non-executive directors for improved governance and control systems and change of behaviour by the executive directors. Possibly they may also demand additional executive and non-executive directors, which would change the balance of power on the board.

The level of dividend demanded by shareholders may be less predictable than the interest on debt. One of the directors is also concerned whether the stock market is efficient or whether the share price may be subject to behavioural factors (discussed in (c) below).

Debt finance

Debt providers will demand Bournelorth Co commits to paying interest and ultimately repaying debt. This may worry the directors because of the significant uncertainties surrounding returns from new apps. Significant debt may have restrictive covenants built in, particularly if Bournelorth Co cannot provide much security. The directors may be faced with restrictions on dividends, for example, which may upset external shareholders.

Uncertainties surrounding funding may also influence directors' decisions. Loan finance may be difficult to obtain, but the amount and repayments would be fixed and could be budgeted, whereas the success of a rights issue is uncertain.

- (b) (i) The main risks connected with development work are that time and resources are wasted on projects which do not generate sales or are not in line with corporate strategy. Directors may choose apps which interest them rather than apps which are best for the business. There is also the risk that projects do not deliver benefits, take too long or are too costly.

Bournelorth Co's directors' heavy involvement in development activities may have made it easier to monitor them. However, the dangers with this are that the directors focus too much on their own individual projects, do not consider their projects objectively and do not step back to consider the overall picture.

The board must decide on a clear strategy for investment in development and needs to approve major initiatives before they are undertaken. There must be proper planning and budgeting of all initiatives and a structured approach to development. The board must regularly review projects, comparing planned and actual expenditure and resource usage. The board must be prepared to halt projects which are unlikely to deliver benefits. One director should be given responsibility for monitoring overall development activity without being directly involved in any of the work. Post-completion reviews should be carried out when development projects have been completed.

- (ii) Communication with shareholders and other important stakeholders, such as potential customers, may be problematic. Bournelorth Co faces the general corporate governance requirement of transparency and has to comply with the specific disclosure requirements of its local stock market.

However, governance best practice also acknowledges that companies need to be allowed to preserve commercial confidentiality if appropriate, and clearly it will be relevant for Bournelorth Co. However, the less that it discloses, the less information finance providers will have on which to base their decisions.

Another issue with disclosure is that product failures may be more visible now that Bournelorth Co has obtained a listing and may have to include a business review in its accounts.

- (c) (i) Sewell defines behavioural finance as the influence of psychology on the behaviour of financial practitioners and the subsequent effect on markets. Behavioural finance suggests that individual decision-making is complex and will deviate from rational decision-making. Under rational decision-making, individual preferences will be clear and remain stable. Individuals will make choices with the aim of maximising utility, and adopt a rational approach for assessing outcomes.

Under behavioural finance, individuals may be more optimistic or conservative than appears to be warranted by rational analysis. They will try to simplify complex decisions and may make different decisions based on the same facts at different times.

- (ii) Bournelorth Co's share price may be significantly influenced by the impact of behavioural factors, as it is a newly listed company operating in a sector where returns have traditionally been variable and unpredictable. The impact of behavioural factors may be complex, and they may exert both upward and downward pressures on Bournelorth Co's share price. Investors may, for example, compensate for not knowing much about Bournelorth Co by anchoring, which means using information which is irrelevant, but which they do have, to judge investment in Bournelorth Co.

The possibility of very high returns may add to the appeal of Bournelorth Co's shares. Some investors may want the opportunity of obtaining high returns even if it is not very likely that they will. The IT sector has also been subject to herd behaviour, notably in the dotcom boom. The herd effect is when a large number of investors have taken the same decision, for example to invest in a particular sector, and this influences others to conform and take the same decision.

However, even if Bournelorth Co produces high returns for some time, the fact that it is in a volatile sector may lead to investors selling shares before it appears to be warranted on the evidence, on the grounds that by the laws of chance Bournelorth Co will make a loss eventually (known as the gambler's fallacy).

Under behavioural finance, the possible volatility of Bournelorth Co's results may lead to downward pressure on its share price for various reasons. First some investors have regret aversion, a general bias against making a loss anyway. This, it is claimed, means that the level of returns on equity is rather higher than the returns on debt than is warranted by a rational view of the risk of equity.

Similarly under prospect theory, investors are more likely to choose a net outcome which consists entirely of small gains, rather than an identical net outcome which consists of a combination of larger gains and some losses. At present also, Bournelorth Co does not have much of a history of results for the market to analyse. Even when it has been listed for some time, however, another aspect of behavioural finance is investors placing excessive weight on the most recent results.

If the market reacts very well or badly to news about Bournelorth Co, the large rise or fall in the share price which results may also not be sustainable, but may revert back over time.

- 3 (a) The currency swap will involve Buryecs Co taking out a loan in € and making an arrangement with a counterparty in Wirtonia, which takes out a loan in \$. Buryecs Co will pay the interest on the counterparty's loan and vice versa.

Advantages

Payment of interest in \$ can be used to match the income Buryecs Co will receive from the rail franchise, reducing foreign exchange risk.

Buryecs Co will be able to obtain the swap for the amount it requires and may be able to reverse the swap by exchanging with the other counterparty. Other methods of hedging risk may be less certain. The cost of a swap may also be cheaper than other methods of hedging, such as options.

The swap can be used to change Buryecs Co's debt profile if it is weighted towards fixed-rate debt and its directors want a greater proportion of floating rate debt, to diversify risk and take advantage of probable lower future interest rates.

Drawbacks

The counterparty may default. This would leave Buryecs Co liable to pay interest on the loan in its currency. The risk of default can be reduced by obtaining a bank guarantee for the counterparty.

The swap may not be a worthwhile means of hedging currency risk if the exchange rate is unpredictable. If it is assumed that exchange rates are largely determined by inflation rates, the predicted inflation rate in Wirtonia is not stable, making it more difficult to predict future exchange rates confidently. If the movement in the exchange rate is not as expected, it may turn out to have been better for Buryecs Co not to have hedged.

Buryecs Co is swapping a fixed rate commitment in the Eurozone for a floating rate in Wirtonia. Inflation is increasing in Wirtonia and there is a risk that interest rates will increase as a result, increasing Buryecs Co's finance costs.

The swap does not hedge the whole amount of the receipt in Year 3. Another method will have to be used to hedge the additional receipt from the government in Year 3 and the receipts in the intervening years.

If the government decides to impose exchange controls in Wirtonia, Buryecs Co may not be able to realise the receipt at the end of Year 3, but will still have to fulfil the swap contract.

(b) (i)	Buryecs Co	Counterparty	Interest rate benefit
Eurozone	4.0%	5.8%	1.8%
Wirtonia	Bank rate + 0.6%	Bank rate + 0.4%	0.2%
Gain on swap (60:40)	1.2%	0.8%	2.0%
Bank fee (60:40)	(0.3%)	(0.2%)	(0.5%)
Gain on swap after bank fee	0.9%	0.6%	1.5%

The swap arrangement will work as follows:

	Buryecs Co	Counterparty
Buryecs Co borrows at	4.0%	
Counterparty borrows at		Bank rate + 0.4%
Swap		
Counterparty receives		(Bank rate)
Buryecs Co pays	Bank rate	
Counterparty pays		4.6%
Buryecs Co receives	(4.6%)	
Advantage	120 basis points	80 basis points
Net result	Bank rate – 0.6%	5.0%

After paying the 30 point basis fee, Buryecs Co will effectively pay interest at the bank rate – 0.3% and benefit by 90 basis points or 0.9%. The counterparty will effectively pay interest at 5.2% and benefit by 60 basis points or 0.6%.

(ii) Using the purchasing power parity formula to calculate exchange rates:

$$S_1 = S_0 \times (1 + h_c) / (1 + h_b)$$

Year	1	2	3
	0.1430 x	0.1472 x	0.1417 x
	1.06/1.03	1.04/1.08	1.03/1.11
	= 0.1472	= 0.1417	= 0.1315

At Year 3, \$5,000 million will be exchanged at the original spot rate as per the agreement and the remaining inflows will be exchanged at the Year 3 rate.

	0 \$m	1 \$m	2 \$m	3 \$m
Initial fee	(5,000)			
Payment at end of franchise				7,500
Annual income		600	600	600
Year 0 Exchange rate	0.1430			
Years 1–3 Exchange rates		0.1472	0.1417	0.1315
	€m	€m	€m	€m
Swap translated at 0.1430	(715)			715
Amount not covered by swap (7,500 – 5,000) translated at 0.1315				329
Annual income		88	85	79
Cash flows in home country	(715)	88	85	1,123
Discount factor 14%	1.000	0.877	0.769	0.675
Present value	(715)	77	65	758

The net present value of the project is €185 million, indicating that it should go ahead. However, the value is dependent on the exchange rate, which is worsening for the foreign income. If there are also uncertainties about the variability of returns during the three years, the directors may consider the project to be in excess of their risk appetite and decline the opportunity.

As a result of the exchange rates on the initial fee being fixed at the year 0 spot rate, Buryecs Co has gained \$5,000 million $\times (0.1430 - 0.1315) \times 0.675 = \text{€}39$ million.

(c) Receipt using swap arrangement = €715m + €329m = €1,044m

Receipt if transaction unhedged = \$7,500m $\times 0.1315 = \text{€}986\text{m}$

Predicted exchange rate at year 3 is €0.1315 = \$1 or \$7.6046 = €1

Options

Buy \$ put options as receiving \$.

\$7.75 exercise price

Do not exercise

Net receipt = €986m – (1.6% $\times$ \$7,500m $\times 0.1430$) = €969m

\$7.25 exercise price

Exercise

Receipt from government = \$7,500m/7.25 = €1,034m

Net receipt = €1,034m – (2.7% $\times$ \$7,500m $\times 0.1430$) = €1,005m

The \$7.25 option gives a better result than not hedging, given the current expectations of the exchange rate. However, it gives a worse result than the swap even before the premium is deducted, because of the exchange rate being fixed on the swap back of the original amount paid. These calculations do not take into account possible variability of the finance costs associated with the swap, caused by swapping into floating rate borrowing.

4 (a) The government yield curve can be estimated from the data available:

Bond 1: \$104 = \$109/(1 + r_1)

$r_1 = (\$109/\$104) - 1 = 4.81\%$

Bond 2: \$102 = \$7/1.0481 + \$107/(1 + r_2)²

$r_2 = [107/((102 - 6.68))^{1/2} - 1 = 5.95\%$

Bond 3: \$98 = \$6/1.0481 + \$6/1.0595² + \$106/(1 + r_3)³

$r_3 = [106/(98 - 5.72 - 5.35))^{1/3} - 1 = 6.83\%$

Year	Govt yield curve	Spread old rating	Toltuck Co spot old rating	Spread new rating	Toltuck Co spot new rating
	%	%	%	%	%
1	4.81	0.18	4.99	0.54	5.35
2	5.95	0.31	6.26	0.69	6.64
3	6.83	0.45	7.28	0.86	7.69

Valuation of bond under old credit rating

Year	Payment	Discount factor	Discounted cash flow
	\$		\$
1	8	1/1.0499	7.62
2	8	1/1.0626 ²	7.09
3	110	1/1.0728 ³	89.09
Bond valuation			<u>103.80</u>

Valuation of bond under new credit rating

Year	Payment	Discount factor	Discounted cash flow
	\$		\$
1	8	1/1.0535	7.59
2	8	1/1.0664 ²	7.03
3	110	1/1.0769 ³	88.08
Bond valuation			<u>102.70</u>

Yield to maturity under old credit rating

Year	Payment	Discount factor	Discounted cash flow	Discount factor	Discounted cash flow
	\$	8%	\$	7%	\$
0	(103·80)	1·000	(103·80)	1·000	(103·80)
1–3	8·00	2·577	20·62	2·624	20·99
3	102·00	0·794	80·99	0·816	83·23
			(2·19)		0·42

Using IRR approach, yield to maturity = $7 + ((0·42/(2·19 + 0·42)) \times (8 - 7)) = 7·16\%$

Yield to maturity under new credit rating

Year	Payment	Discount factor	Discounted cash flow	Discount factor	Discounted cash flow
	\$	8%	\$	7%	\$
0	(102·70)	1·000	(102·70)	1·000	(102·70)
1–3	8·00	2·577	20·62	2·624	20·99
3	102·00	0·794	80·99	0·816	83·23
			(1·09)		1·52

Using IRR approach, yield to maturity = $7 + ((1·52/(1·09 + 1·52)) \times (8 - 7)) = 7·58\%$

Market value of \$100 bond has fallen by \$1·10 and the yield to maturity has risen by 0·42%.

- (b) The credit agency will have taken the following criteria into consideration when assessing Toltuck Co's credit rating:

Country

Toltuck Co's debt would not normally be rated higher than the credit ratings of its country of origin, Arumland. Therefore the credit rating of Arumland should normally be at least AA. The rating will also have depended on Toltuck Co's standing relative to other companies in Arumland. The credit agency may have reckoned that Toltuck Co's recent poor results have weakened its position.

Industry

The credit agency will have taken account of the impact of the recession on property construction companies generally in Arumland. Toltuck Co's position within the industry compared with competitors will also have been assessed. If similar recent developments by competitors have been more successful, this is likely to have had an adverse impact on Toltuck Co's rating.

Management

The credit agency will have made an overall assessment of management and succession planning at Toltuck Co. It will have looked at business and financing strategies and planning and controls. It will also have assessed how successful the management has been in terms of delivering financial results. The credit agency may have believed the poor returns on recent developments show shortcomings in management decision-making processes and it may have rated the current management team poorly.

Financial

The credit agency will have analysed financial results, using measures such as return on capital employed. The agency will also have assessed possible sources of future earnings growth. It may have been sceptical about prospects, certainly for the short term, given Toltuck Co's recent problems.

The credit agency will also have assessed the financial position of Toltuck Co, looking at its gearing and working capital management, and considering whether Toltuck Co has enough cash to finance its needs. The agency will also have looked at Toltuck Co's relationship with its bankers and its debt covenants, to assess how flexible its sources of finances are if it comes under stress. It may well have been worried about Toltuck Co's gearing being higher than the industry average and concerned about the high levels of cash it needs to finance operations. It will also have assessed returns on developments-in-progress compared with commitments to repay loans. Greater doubt about Toltuck Co's ability to meet its commitments is likely to have been a significant factor in the fall in its rating.

The agency will also have needed reassurance about the quality of the financial information it was using, so it will have looked at the audit report and accounting policies.

- (c) Toltuck Co may not have increased problems raising debt finance if debtholders do not react in the same way as the credit rating agency. They may attach different weightings to the criteria which they use. They may also come to different judgements about the quality of management and financial stability. Debtholders may believe that the recent problems Toltuck Co has had generating returns may be due more to external factors which its management could not have controlled.

However, it is probable that the fall in Toltuck Co's credit rating will result in it having more difficulty raising debt finance. Banks may be less willing to provide loans and investors less willing to subscribe for bonds. Even if debt finance is available, it may come with covenants restricting further debt or gearing levels. This will mean that if Toltuck Co requires substantial

additional finance, it is more likely to have to make a rights issue or issue new equity on the stock market. Shareholders may be faced with the choice of subscribing large amounts for new capital or having their influence diluted. This may particularly worry the more cautious shareholders.

Even if Toltuck Co can obtain the debt it needs, the predicted increase in yield to maturity may be matched by debtholders demanding a higher coupon rate on debt. This will increase finance costs, and decrease profits and earnings per share, with a possible impact on share price. It will also mean that fewer funds are available for paying dividends. Toltuck Co has been faced with difficult decisions on balancing investment expenditure versus paying dividends and these difficulties may well increase.

Additional debt may have other restrictive covenants. They may restrict Toltuck Co's buying and selling of assets, or its investment strategy. Restrictions on Toltuck Co's decisions about the developments it undertakes may impact adversely on shareholder returns.

Loan finance or bonds will also come with repayment covenants. These may require Toltuck Co to build up a fund over time which will be enough to redeem the debt at the end of its life. Given uncertainties over cash flows, this commitment to retain cash may make it more difficult to undertake major developments or pay an acceptable level of dividend.

The fall in Toltuck Co's credit rating may result in its cost of equity rising as well as its cost of debt. In turn, Toltuck Co's weighted average cost of capital will rise. This will affect its investment choices and hence its ability to generate wealth for shareholders. It may result in Toltuck Co prioritising developments offering better short-term returns. This may suit the more cautious shareholders, but the current majority may worry that Toltuck Co will have to turn down opportunities which offer the possibility of high returns.

		Marks		
1	(a)	Explanation of what a reverse takeover involves	2	
		Advantages (up to 2 marks per well explained advantage)	Max 4	
		Disadvantages (up to 2 marks per well explained disadvantage)	Max 4	
			Max 9	
	(b)	(i)	Extract of financial position after restructuring programme	5
			Appendix One	
			Manufacturing business unit unbundled through an MBO	
			Estimate of cash flows	3
			Estimate of amount payable to Chrysos Co	2
			Selection of higher value unbundling option	1
			Appendix Two	
			Chrysos Co, cost of equity	1
			Chrysos Co, cost of capital	1
			Appendix Three	
			Estimate of cash flows	3
			Estimate of equity value	2
				18
		(ii)	Explanation of approach taken	1–2
			Explanation of assumptions made (up to 2 marks per assumption)	3–4
				Max 5
		(iii)	Appendix Four	
			Value from increased ownership	1
			Additional value	1
			Discussion of restructuring programme on the VCOs	3–4
			Discussion of restructuring programme on Chrysos Co	4–5
				Max 10
		Professional marks for part (b)		
		Report format	1	
		Structure and presentation of the report	3	
			4	
	(c)	1–2 marks per relevant point	Max 4	
			Total 50	
2	(a)	Shareholder wealth maximisation and need for further investment	1–2	
		Sell-off of IT services business	1–2	
		Rights issue	2–3	
		Debt finance	2–3	
			Max 8	
	(b)	(i)	Risks associated with investment in development	up to 4
			Controls over development	up to 4
				Max 6
		(ii)	1 mark per point	3
	(c)	(i)	Explanation that behavioural finance departs from rational decision-making	3
		(ii)	Impacts of behavioural factors on share price – Up to 2 marks per well-explained point	Max 5
				Total 25

			<i>Marks</i>
3	(a)	Advantages	2–3
		Disadvantages	3–4
		Limit marks for (a) to 3 marks in total if answer does not mention Buryecs Co's situation	Max 6
	(b) (i)	Recognition that swap gives advantage	1
		Swap mechanism	2
			1
			<u>4</u>
	(ii)	Exchange rates	2
		Correct translation of amounts swapped	1
		Correct translation of other amounts	1
		Net present value	1
		Gain in € from the swap of the initial fee amount	1
		Comments	2–3
			<u>8</u>
	(c)	Put option	1
		\$7·25 option calculations	2
		\$7·75 option calculations	2
			2–3
			<u>7</u>
			Total 25
4	(a)	Government yield curve	2
		Toltuck Co spot-curve old and new	2
		Bond valuation – old and new	3
		Yield to maturity – old and new	3
			<u>10</u>
	(b)	Financial factors	4
		Other factors	4
		Limit marks for (b) to 3 marks in total if answer does not mention Toltuck Co's position and performance	8
	(c)	1–2 marks per impact discussed	Max 7
			Total 25

Professional Level – Options Module

Advanced Financial Management

September/December 2017 – Sample Questions



Time allowed: 3 hours 15 minutes

This question paper is divided into two sections:

Section A – This ONE question is compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Formulae and tables are on pages 10–14.

Do NOT open this question paper until instructed by the supervisor.

This question paper must not be removed from the examination hall.

P4 Paper

Think Ahead

ACCA

The Association of
Chartered Certified
Accountants

Section A – This ONE question is compulsory and MUST be attempted

- 1 Conejo Co is a listed company based in Ardilla and uses the \$ as its currency. The company was formed around 20 years ago and was initially involved in cybernetics, robotics and artificial intelligence within the information technology industry. At that time due to the risky ventures Conejo Co undertook, its cash flows and profits were very varied and unstable. Around 10 years ago, it started an information systems consultancy business and a business developing cyber security systems. Both these businesses have been successful and have been growing consistently. This in turn has resulted in a stable growth in revenues, profits and cash flows. The company continues its research and product development in artificial intelligence and robotics, but this business unit has shrunk proportionally to the other two units.

Just under eight years ago, Conejo Co was successfully listed on Ardilla's national stock exchange, offering 60% of its share capital to external equity holders, whilst the original founding members retained the remaining 40% of the equity capital. The company remains financed largely by equity capital and reserves, with only a small amount of debt capital. Due to this, and its steadily growing sales revenue, profits and cash flows, it has attracted a credit rating of A from the credit rating agencies.

At a recent board of directors (BoD) meeting, the company's chief financial officer (CFO) argued that it was time for Conejo Co to change its capital structure by undertaking a financial reconstruction, and be financed by higher levels of debt. As part of her explanation, the CFO said that Conejo Co is now better able to bear the increased risk resulting from higher levels of debt finance; would be better protected from predatory acquisition bids if it was financed by higher levels of debt; and could take advantage of the tax benefits offered by increased debt finance. She also suggested that the expected credit migration from a credit rating of A to a credit rating of BBB, if the financial reconstruction detailed below took place, would not weaken Conejo Co financially.

Financial reconstruction

The BoD decided to consider the financial reconstruction plan further before making a final decision. The financial reconstruction plan would involve raising \$1,320 million (\$1.32 billion) new debt finance consisting of bonds issued at their face value of \$100. The bonds would be redeemed in five years' time at their face value of \$100 each. The funds raised from the issue of the new bonds would be used to implement one of the following two proposals:

- (i) Proposal 1: Either buy back equity shares at their current share price, which would be cancelled after they have been repurchased; or
- (ii) Proposal 2: Invest in additional assets in new business ventures.

Conejo Co, Financial information

Extract from the forecast financial position for next year

	\$m
Non-current assets	1,735
Current assets	530
Total assets	2,265
Equity and liabilities	
Share capital (\$1 per share par value)	400
Reserves	1,700
Total equity	2,100
Non-current liabilities	120
Current liabilities	45
Total liabilities	165
Total liabilities and capital	2,265

Conejo Co's forecast after-tax profit for next year is \$350 million and its current share price is \$11 per share.

The non-current liabilities consist solely of 5·2% coupon bonds with a face value of \$100 each, which are redeemable at their face value in three years' time. These bonds are currently trading at \$107·80 per \$100. The bond's covenant stipulates that should Conejo Co's borrowing increase, the coupon payable on these bonds will increase by 37 basis points.

Conejo Co pays tax at a rate of 15% per year and its after-tax return on the new investment is estimated at 12%.

Other financial information

Current government bond yield curve

Year	1	2	3	4	5
	1·5%	1·7%	1·9%	2·2%	2·5%

Yield spreads (in basis points)

	1 year	2 years	3 years	4 years	5 years
A	40	49	59	68	75
BBB	70	81	94	105	112
BB	148	167	185	202	218

The finance director wants to determine the percentage change in the value of Conejo Co's current bonds, if the credit rating changes from A to BBB. Furthermore, she wants to determine the coupon rate at which the new bonds would need to be issued, based on the current yield curve and appropriate yield spreads given above.

Conejo Co's chief executive officer (CEO) suggested that if Conejo Co paid back the capital and interest of the new bond in fixed annual repayments of capital and interest through the five-year life of the bond, then the risk associated with the extra debt finance would be largely mitigated. In this case, it was possible that credit migration, by credit rating companies, from A rating to BBB rating may not happen. He suggested that comparing the duration of the new bond based on the interest payable annually and the face value in five years' time with the duration of the new bond where the borrowing is paid in fixed annual repayments of interest and capital could be used to demonstrate this risk mitigation.

Required:

- (a) **Discuss the possible reasons for the finance director's suggestions that Conejo Co could benefit from higher levels of debt with respect to risk, from protection against acquisition bids, and from tax benefits.** (7 marks)
- (b) **Prepare a report for the board of directors of Conejo Co which:**
 - (i) **Estimates, and briefly comments on, the change in value of the current bond and the coupon rate required for the new bond, as requested by the CFO;** (6 marks)
 - (ii) **Estimates the Macaulay duration of the new bond based on the interest payable annually and face value repayment, and the Macaulay duration based on the fixed annual repayment of the interest and capital, as suggested by the CEO;** (6 marks)
 - (iii) **Estimates the impact of the two proposals on how the funds may be used on next year's forecast earnings, forecast financial position, forecast earnings per share and on forecast gearing;** (11 marks)
 - (iv) **Using the estimates from (b)(i), (b)(ii) and (b)(iii), discusses the impact of the proposed financial reconstruction and the proposals on the use of funds on:**
 - Conejo Co;
 - Possible reaction(s) of credit rating companies and on the expected credit migration, including the suggestion made by the CEO;
 - Conejo Co's equity holders;
 - Conejo Co's current and new debt holders.
(16 marks)

Professional marks will be awarded in part (b) for the format, structure and presentation of the report.

(4 marks)

(50 marks)

Section B – TWO questions ONLY to be attempted

- 2 Eview Cinemas Co is a long-established chain of cinemas in the country of Taria. Twenty years ago Eview Cinemas Co's board decided to convert some of its cinemas into sports gyms, known as the EV clubs. The number of EV clubs has expanded since then. Eview Cinemas Co's board brought in outside managers to run the EV clubs, but over the years there have been disagreements between the clubs' managers and the board. The managers have felt that the board has wrongly prioritised investment in, and refurbishment of, the cinemas at the expense of the EV clubs.

Five years ago, Eview Cinemas Co undertook a major refurbishment of its cinemas, financing this work with various types of debt, including loan notes at a high coupon rate of 10%. Shortly after the work was undertaken, Taria entered into a recession which adversely affected profitability. The finance cost burden was high and Eview Cinemas Co was not able to pay a dividend for two years.

The recession is now over and Eview Cinemas Co has emerged in a good financial position, as two of its competitors went into insolvency during the recession. Eview Cinemas Co's board wishes to expand its chain of cinemas and open new, multiscreen cinemas in locations which are available because businesses were closed down during the recession.

In two years' time Taria is due to host a major sports festival. This has encouraged interest in sport and exercise in the country. As a result, some gym chains are looking to expand and have contacted Eview Cinemas Co's board to ask if it would be interested in selling the EV clubs. Most of the directors regard the cinemas as the main business and so are receptive to selling the EV clubs.

The finance director has recommended that the sales price of the EV clubs be based on predicted free cash flows as follows:

1. The predicted free cash flow figures in \$millions for EV clubs are as follows:

Year	1	2	3	4
	390	419	455	490

2. After Year 4, free cash flows should be assumed to increase at 5.2% per annum.
3. The discount rate to be used should be the current weighted average cost of capital, which is 12%.
4. The finance director believes that the result of the free cash flow valuation will represent a fair value of the EV clubs' business, but Eview Cinemas Co is looking to obtain a 25% premium on the fair value as the expected sales price.

Other information supplied by the finance director is as follows:

1. The predicted after-tax profits of the EV clubs are \$454 million in Year 1. This can be assumed to be 40% of total after-tax profits of EV Cinemas Co.
2. The expected proceeds which Eview Cinemas Co receives from selling the EV clubs will be used firstly to pay off the 10% loan notes. Part of the remaining amount from the sales proceeds will then be used to enhance liquidity by being held as part of current assets, so that the current ratio increases to 1.5. The rest of the remaining amount will be invested in property, plant and equipment. The current net book value of the non-current assets of the EV clubs to be sold can be assumed to be \$3,790 million. The profit on the sale of the EV clubs should be taken directly to reserves.
3. Eview Cinemas Co's asset beta for the cinemas can be assumed to be 0.952.
4. Eview Cinemas Co currently has 1,000 million \$1 shares in issue. These are currently trading at \$15.75 per share. The finance director expects the share price to rise by 10% once the sale has been completed, as he thinks that the stock market will perceive it to be a good deal.
5. Tradeable debt is currently quoted at \$96 per \$100 for the 10% loan notes and \$93 per \$100 for the other loan notes. The value of the other loan notes is not expected to change once the sale has been completed. The overall pre-tax cost of debt is currently 9% and can be assumed to fall to 8% when the 10% loan notes are redeemed.
6. The current tax rate on profits is 20%.
7. Additional investment in current assets is expected to earn a 7% pre-tax return and additional investment in property, plant and equipment is expected to earn a 12% pre-tax return.
8. The current risk-free rate is 4% and the return on the market portfolio is 10%.

Eview Cinemas Co's current summarised statement of financial position is shown below. The CEO wants to know the impact the sale of the EV clubs would have immediately on the statement of financial position, the impact on the Year 1 forecast earnings per share and on the weighted average cost of capital.

	\$m
Assets	
Non-current assets	15,621
Current assets	2,347
Total assets	17,968
Equity and liabilities	
Called-up share capital	1,000
Retained earnings	7,917
Total equity	8,917
Non-current liabilities	
10% loan notes	3,200
Other loan notes	2,700
Bank loans	985
Total non-current liabilities	6,885
Current liabilities	2,166
Total equity and liabilities	17,968

Required:

(a) Calculate the expected sales price of the EV clubs and demonstrate its impact on Eview Cinemas Co's statement of financial position, forecast earnings per share and weighted average cost of capital. (17 marks)

(b) Evaluate the decision to sell the EV clubs. (8 marks)

(25 marks)

3 High K Co is one of the three largest supermarket chains in the country of Townia. Its two principal competitors, Dely Co and Leminster Co, are of similar size to High K Co. In common with its competitors (but see below), High K Co operates three main types of store:

- Town centre stores – these sell food and drink and a range of small household items. High K Co's initial growth was based on its town centre stores, but it has been shutting them over the last decade, although the rate of closure has slowed in the last couple of years.
- Convenience stores – these are smaller and sell food and drink and very few other items. Between 2003 and 2013, High K Co greatly expanded the number of convenience stores it operated. Their performance has varied, however, and since 2013, High K Co has not opened any new stores and closed a number of the worst-performing stores.
- Out-of-town stores – these sell food and drink and a full range of household items, including large electrical goods and furniture. The number of out-of-town stores which High K Co operated increased significantly until 2010, but has only increased slightly since.

The majority of town centre and out-of-town stores premises are owned by High K Co, but 85% of convenience stores premises are currently leased.

High K Co also sells most of its range of products online, either offering customers home delivery or 'click and collect' (where the customer orders the goods online and picks them up from a collection point in one of the stores).

High K Co's year end is 31 December. When its 2016 results were published in April 2017, High K Co's chief executive emphasised that the group was focusing on:

- Increasing total shareholder return by improvements in operating efficiency and enhancement of responsiveness to customer needs
- Ensuring competitive position by maintaining flexibility to respond to new strategic challenges
- Maintaining financial strength by using diverse sources of funding, including making use in future of revolving credit facilities

Since April 2017, Dely Co and Leminster Co have both announced that they will be making significant investments to boost online sales. Dely Co intends to fund its investments by closing all its town centre and convenience stores, although it also intends to open more out-of-town stores in popular locations.

The government of Townia was re-elected in May 2017. In the 18 months prior to the election, it eased fiscal policy and consumer spending significantly increased. However, it has tightened fiscal policy since the election to avoid the economy overheating. It has also announced an investigation into whether the country's large retail chains treat their suppliers unfairly.

Extracts from High K Co's 2016 financial statements and other information about it are given below:

High K Co statement of profit or loss extracts

Year ending 31 December (all amounts in \$m)

	2014	2015	2016
Sales revenue	23,508	23,905	24,463
Gross profit	1,018	1,211	1,514
Operating profit	204	407	712
Finance costs	(125)	(115)	(100)
Profit after tax	52	220	468
Dividends	150	170	274

High K Co statement of financial position extracts

Year ending 31 December (all amounts in \$m)

Non-current assets	10,056	9,577	8,869
Cash and cash equivalents	24	709	1,215
Other current assets	497	618	747
Total non-current and current assets	10,577	10,904	10,831

Equity

Ordinary shares (\$1)	800	800	800
Reserves	7,448	7,519	7,627
Total equity	8,248	8,319	8,427
Non-current liabilities	1,706	1,556	1,246
Current liabilities	623	1,029	1,158

Other information

Market price per share (in \$, \$3.89 at end of 2013, \$3.17 currently)	3.54	3.34	3.23
Staff working in shops ('000)	78	75	72

Segment information

Revenue (\$m)

Town centre stores	5,265	5,189	5,192
Convenience stores	3,786	3,792	3,833
Out-of-town stores	10,220	10,340	10,547
Store revenue	19,271	19,321	19,572
Online sales	4,237	4,584	4,891

Number of stores

Town centre stores	165	157	153
Convenience stores	700	670	640
Out-of-town stores	220	224	227

Required:

- (a) Evaluate High K Co's financial performance. You should indicate in your discussion areas where further information about High K Co would be helpful. Provide relevant calculations for ratios and trends to support your evaluation.

Note: Up to 10 marks are available for calculations.

(21 marks)

- (b) Discuss how High K Co may seek to finance an investment programme.

(4 marks)

(25 marks)

- 4 Wardegul Co, a company based in the Eurozone, has expanded very rapidly over recent years by a combination of acquiring subsidiaries in foreign countries and setting up its own operations abroad. Wardegul Co's board has found it increasingly difficult to monitor its activities and Wardegul Co's support functions, including its treasury function, have struggled to cope with a greatly increased workload. Wardegul Co's board has decided to restructure the company on a regional basis, with regional boards and appropriate support functions. Managers in some of the larger countries in which Wardegul Co operates are unhappy with reorganisation on a regional basis, and believe that operations in their countries should be given a large amount of autonomy and be supported by internal functions organised on a national basis.

Assume it is now 1 October 2017. The central treasury function has just received information about a future transaction by a newly-acquired subsidiary in Euria, where the local currency is the dinar (D). The subsidiary expects to receive D27,000,000 on 31 January 2018. It wants this money to be invested locally in Euria, most probably for five months until 30 June 2018.

Wardegul Co's treasury team is aware that economic conditions in Euria are currently uncertain. The central bank base rate in Euria is currently 4.2% and the treasury team believes that it can invest funds in Euria at the central bank base rate less 30 basis points. However, treasury staff have seen predictions that the central bank base rate could increase by up to 1.1% or fall by up to 0.6% between now and 31 January 2018.

Wardegul Co's treasury staff normally hedge interest rate exposure by using whichever of the following products is most appropriate:

- Forward rate agreements (FRAs)
- Interest rate futures
- Options on interest rate futures

Treasury function guidelines emphasise the importance of mitigating the impact of adverse movements in interest rates. However, they also allow staff to take into consideration upside risks associated with interest rate exposure when deciding which instrument to use.

A local bank in Euria, with which Wardegul Co has not dealt before, has offered the following FRA rates:

4–9: 5.02%
5–10: 5.10%

The treasury team has also obtained the following information about exchange traded Dinar futures and options:

Three-month D futures, D500,000 contract size

Prices are quoted in basis points at 100 – annual % yield:

December 2017: 94.84
March 2018: 94.78
June 2018: 94.66

Options on three-month D futures, D500,000 contract size, option premiums are in annual %

Calls			Strike price	Put		
December	March	June		December	March	June
0.417	0.545	0.678	94.25	0.071	0.094	0.155
0.078	0.098	0.160	95.25	0.393	0.529	0.664

It can be assumed that futures and options contracts are settled at the end of each month. Basis can be assumed to diminish to zero at contract maturity at a constant rate, based on monthly time intervals. It can also be assumed that there is no basis risk and there are no margin requirements.

Required:

- (a) Recommend a hedging strategy for the D27,000,000 investment, based on the hedging choices which treasury staff are considering, if interest rates increase by 1·1% or decrease by 0·6%. Support your answer with appropriate calculations and discussion. (18 marks)
- (b) Discuss the advantages of operating treasury activities through regional treasury functions compared with:
- Each country having a separate treasury function.
 - Operating activities through a single global treasury function. (7 marks)

(25 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i(E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_o = \frac{D_o(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = br_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \qquad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model

$$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$$

Where:

$$d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$$

$$d_2 = d_1 - s\sqrt{t}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

		<i>Discount rate (r)</i>									
<i>Periods</i>											
(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

		<i>Discount rate (r)</i>									
<i>Periods</i>											
(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Answers

- 1 (a) Increasing the debt finance of a company relative to equity finance increases its financial risk, and therefore the company will need to be able to bear the consequences of this increased risk. However, companies face both financial risk, which increases as the debt levels in the capital structure increase, and business risk, which is present in a company due to the nature of its business.

In the case of Conejo Co, it could be argued that as its profits and cash flows have stabilised, the company's business risk has reduced, in contrast to early in its life, when its business risk would have been much higher due to unstable profits and cash flows. Therefore, whereas previously Conejo Co was not able to bear high levels of financial risk, it is able to do so now without having a detrimental impact on the overall risk profile of the company. It could therefore change its capital structure and have higher levels of debt finance relative to equity finance.

The predatory acquisition of one company by another could be undertaken for a number of reasons. One possible reason may be to gain access to cash resources, where a company which needs cash resources may want to take over another company which has significant cash resources or cash generative capability. Another reason may be to increase the debt capacity of the acquirer by using the assets of the target company. Where the relative level of debt finance is increased in the capital structure of a company through a financial reconstruction, like in the case of Conejo Co, these reasons for acquiring a company may be diminished. This is because the increased levels of debt would probably be secured against the assets of the company and therefore the acquirer cannot use them to raise additional debt finance, and cash resources would be needed to fund the higher interest payments.

Many tax jurisdictions worldwide allow debt interest to be deducted from profits before the amount of tax payable is calculated on the profits. Increasing the amount of debt finance will increase the amount of interest paid, reducing the taxable profits and therefore the tax paid. Modigliani and Miller referred to this as the benefit of the tax shield in their research into capital structure, where their amended capital proposition demonstrated the reduction in the cost of capital and increase in the value of the firm, as the proportion of debt in the capital structure increases.

(b) Report to the board of directors (BoD), Conejo Co

Introduction

This report discusses whether the proposed financial reconstruction scheme which increases the amount of debt finance in Conejo Co would be beneficial or not to the company and the main parties affected by the change in the funding, namely the equity holders, the debt holders and the credit rating companies. Financial estimates provided in the appendices are used to support the discussion.

Impact on Conejo Co

Benefits to Conejo Co include the areas discussed in part (a) above and as suggested by the CFO. The estimate in appendix 3 assumes that the interest payable on the new bonds and the extra interest payable on the existing bonds are net of the 15% tax. Therefore, the tax shield reduces the extra amount of interest paid. Further, it is likely that because of the large amount of debt finance which will be raised, the company's assets would have been used as collateral. This will help protect the company against hostile takeover bids. Additionally, proposal 2 (appendix 3) appears to be better than proposal 1, with a lower gearing figure and a higher earnings per share figure. However, this is dependent on the extra investment being able to generate an after-tax return of 12% immediately. The feasibility of this should be assessed further.

Conejo Co may also feel that this is the right time to raise debt finance as interest rates are lower and therefore it does not have to offer large coupons, compared to previous years. Appendix 1 estimates that the new bond will need to offer a coupon of 3.57%, whereas the existing bond is paying a coupon of 5.57%.

The benefits above need to be compared with potential negative aspects of raising such a substantial amount of debt finance. Conejo Co needs to ensure that it will be able to finance the interest payable on the bonds and it should ensure it is able to repay the capital amount borrowed (or be able to re-finance the loan) in the future. The extra interest payable (appendix 3) will probably not pose a significant issue given that the profit after tax is substantially more than the interest payment. However, the repayment of the capital amount will need careful thought because it is significant.

The substantial increase in gearing, especially with respect to proposal 1 (appendix 3), may worry some stakeholders because of the extra financial risk. However, based on market values, the level of gearing may not appear so high. The expected credit migration from A to BBB seems to indicate some increase in risk, but it is probably not substantial.

The BoD should also be aware of, and take account of, the fact that going to the capital markets to raise finance will require Conejo Co to disclose information, which may be considered strategically important and could impact negatively on areas where Conejo Co has a competitive advantage.

Reaction of credit rating companies

Credit ratings assigned to companies and to borrowings made by companies by credit rating companies depend on the probability of default and recovery rate. A credit migration from A to BBB means that Conejo Co has become riskier in that it is more likely to default and bondholders will find it more difficult to recover their entire loan if default does happen. Nevertheless, the relatively lower increase in yield spreads from A to BBB, compared to BBB to BB, indicates that BBB can still be considered a relatively safe investment.

Duration indicates the time it takes to recover half the repayments of interest and capital of a bond, in present value terms. Duration measures the sensitivity of bond prices to changes in interest rates. A bond with a higher duration would see a greater fluctuation in its value when interest rates change, compared to a bond with a lower duration. Appendix 2 shows that a bond which pays interest (coupon) and capital in equal annual instalments will have a lower duration. This is because a greater proportion of income is received earlier and income due to be received earlier is less risky. Therefore, when interest rates change, this bond's value will change by less than the bond with the higher duration. The CEO is correct that the bond with equal annual payments of interest and capital is less sensitive to interest rate changes, but it is not likely that this will be a significant factor for a credit rating company when assigning a credit rating.

A credit rating company will consider a number of criteria when assigning a credit rating, as these would give a more appropriate assessment of the probability of default and the recovery rate. These criteria include, for example, the industry within which the company operates, the company's position within that industry, the company's ability to generate profits in proportion to the capital invested, the amount of gearing, the quality of management and the amount of financial flexibility the company possesses. A credit rating company will be much less concerned about the manner in which a bond's value fluctuates when interest rates change.

Impact on equity holders

The purpose of the financial reconstruction would be of interest to the equity holders. If, for example, Conejo Co selects proposal 1 (appendix 3), it may give equity holders an opportunity to liquidate some of their invested capital. At present, the original members of the company hold 40% of the equity capital and proposal 1 provides them with the opportunity to realise a substantial capital without unnecessary fluctuations in the share price. Selling large quantities of equity shares in the stock exchange may move the price of the shares down and cause unnecessary fluctuations in the share price.

If, on the other hand, proposal 2 (appendix 3) is selected, any additional profits after the payment of interest will benefit the equity holders directly. In effect, debt capital is being used for the benefit of the equity holders.

It may be true that equity holders may be concerned about the increased risk which higher gearing will bring, and because of this, they may need higher returns to compensate for the higher risk. However, in terms of market values, the increased gearing may be of less concern to equity holders. Conejo Co should consider the capital structure of its competitors to assess what should be an appropriate level of gearing.

Equity holders will probably be more concerned about the additional restrictive covenants which will result from the extra debt finance, and the extent to which these covenants will restrict the financial flexibility of Conejo Co when undertaking future business opportunities.

Equity holders may also be concerned that because Conejo Co has to pay extra interest to debt holders, its ability to pay increasing amounts of dividends in the future could be affected. However, appendix 3 shows that the proportion of interest relative to after-tax profits is not too high and any concern from the equity holders is probably unfounded.

Impact on debt holders

Although the current debt holders may be concerned about the extra gearing which the new bonds would introduce to Conejo Co, appendix 1 shows that the higher coupon payments which the current debt holders will receive would negate any fall in the value of their bonds due to the credit migration to BBB rating from an A rating. Given that currently Conejo Co is subject to low financial risk, and probably lower business risk, it is unlikely that the current and new debt holders would be overly concerned about the extra gearing. The earnings figures in appendix 3 also show that the after-tax profit figures provide a substantial interest cover and therefore additional annual interest payment should not cause the debt holders undue concern either.

The current and new debt holders would be more concerned about Conejo Co's ability to pay back the large capital sum in five years' time. However, a convincing explanation of how this can be achieved or a plan to roll over the debt should allay these concerns.

The current and new debt holders may be concerned that Conejo Co is not tempted to take unnecessary risks with the additional investment finance, but sensible use of restrictive covenants and the requirement to make extra disclosures to the markets when raising the debt finance should help mitigate these concerns.

Conclusion

Overall, it seems that the proposed financial reconstruction will be beneficial, as it will provide opportunities for Conejo Co to make additional investments and/or an opportunity to reduce equity capital, and thereby increasing the earnings per share. The increased gearing may not look large when considered in terms of market values. It may also be advantageous to undertake the reconstruction scheme in a period when interest rates are low and the credit migration is not disadvantageous. However, Conejo Co needs to be mindful of how it intends to repay the capital amount in five years' time, the information it will disclose to the capital markets and the impact of any negative restrictive covenants.

Report compiled by:

Date

Appendices:

Appendix 1: Change in the value of the current bond from credit migration and coupon rate required from the new bond (Question (b) (i))

Spot yield rates (yield curve) based on BBB rating

1 year	2.20%
2 year	2.51%
3 year	2.84%
4 year	3.25%
5 year	3.62%

Bond value based on BBB rating

$$\$5.57 \times 1.0220^{-1} + \$5.57 \times 1.0251^{-2} + \$105.57 \times 1.0284^{-3} = \$107.81$$

Current bond value = \$107.80

Although the credit rating of Conejo Co declines from A to BBB, resulting in higher spot yield rates, the value of the bond does not change very much at all. This is because the increase in the coupons and the resultant increase in value almost exactly matches the fall in value from the higher spot yield rates.

Coupon rate required from the new bond

Take R as the coupon rate, such that:

$$(\$R \times 1.0220^{-1}) + (\$R \times 1.0251^{-2}) + (\$R \times 1.0284^{-3}) + (\$R \times 1.0325^{-4}) + (\$R \times 1.0362^{-5}) + (\$100 \times 1.0362^{-5}) = \$100$$

$$4.5665R + 83.71 = 100$$

$$R = \$3.57$$

Coupon rate for the new bond is 3.57%.

If the coupon payments on the bond are at a rate of 3.57% on the face value, it ensures that the present values of the coupons and the redemption of the bond at face value exactly equals the bond's current face value, based on Conejo Co's yield curve.

Appendix 2: Macaulay durations (Question (b) (ii))

Macaulay duration based on annual coupon of \$3.57 and redemption value of \$100 in year 5:

$$[(\$3.57 \times 1.0220^{-1} \times 1 \text{ year}) + (\$3.57 \times 1.0251^{-2} \times 2 \text{ years}) + (\$3.57 \times 1.0284^{-3} \times 3 \text{ years}) + (\$3.57 \times 1.0325^{-4} \times 4 \text{ years}) + (\$103.57 \times 1.0362^{-5} \times 5 \text{ years})]/\$100$$

$$= [3.49 + 6.79 + 9.85 + 12.57 + 433.50]/100 = 4.7 \text{ years}$$

Macaulay duration based on fixed annual repayments of interest and capital:

Annuity factor: (3.57%, 5 years) = $(1 - 1.0357^{-5})/0.0357 = 4.51$ approximately

Annual payments of capital and interest required to pay back new bond issue = $\$100/4.51 = \22.17 per \$100 bond approximately

$$[(\$22.17 \times 1.0220^{-1} \times 1 \text{ year}) + (\$22.17 \times 1.0251^{-2} \times 2 \text{ years}) + (\$22.17 \times 1.0284^{-3} \times 3 \text{ years}) + (\$22.17 \times 1.0325^{-4} \times 4 \text{ years}) + (\$22.17 \times 1.0362^{-5} \times 5 \text{ years})]/\$100$$

$$= [21.69 + 42.20 + 61.15 + 78.03 + 92.79]/100 = 3.0 \text{ years}$$

Appendix 3: Forecast earnings, financial position, earnings per share and gearing (Question (b) (iii))

Adjustments to forecast earnings

Amounts in \$ millions	Current	Proposal 1	Proposal 2
Forecast after-tax profit	350.00	350.00	350.00
Interest payable on additional borrowing (based on a coupon rate of 3.57%)			
3.57% x \$1,320m x (1 - 0.15)		(40.06)	(40.06)
Additional interest payable due to higher coupon 0.37% x \$120m x (1 - 0.15)		(0.38)	(0.38)
Return on additional investment (after tax) 12% x \$1,320m			158.40
Adjusted profit after tax	<u>350.00</u>	<u>309.56</u>	<u>467.96</u>

Forecast financial position

Amounts in \$ millions	Current	Proposal 1	Proposal 2
Non-current assets	1,735.00	1,735.00	3,055.00
Current assets	530.00	489.56	647.96
Total assets	2,265.00	2,224.56	3,702.96
Equity and liabilities			
Share capital (\$1 per share par value)	400.00	280.00	400.00
Reserves	1,700.00	459.56	1,817.96
Total equity	2,100.00	739.56	2,217.96
Non-current liabilities	120.00	1,440.00	1,440.00
Current liabilities	45.00	45.00	45.00
Total liabilities	165.00	1,485.00	1,485.00
Total liabilities and capital	2,265.00	2,224.56	3,702.96
	Current	Proposal 1	Proposal 2
Gearing % (non-current liabilities/equity)	5.7%	194.7%	64.9%
Earnings per share (in cents)			
(adj profit after tax/no. of shares)	87.5c	110.6c	117.0c

Notes:

If gearing is calculated based on non-current liabilities/(non-current liabilities + equity) and/or using market value of equity, instead of as above, then this is acceptable as well.

Proposal 1

Additional interest payable is deducted from current assets, assuming it is paid in cash and this is part of current assets. Reserves are also reduced by this amount.

Shares repurchased as follows: \$1 x 120m shares deducted from share capital and \$10 x 120m shares deducted from reserves. \$1,320m, consisting of \$11 x 120m shares, added to non-current liabilities.

Proposal 2

Treatment of additional interest payable is as per proposal 1.

Additional debt finance raised, \$1,320 million, is added to non-current liabilities and to non-current assets, assuming that all this amount is invested in non-current assets to generate extra income.

It is assumed that this additional investment generates returns at 12%, which is added to current assets and to profits (and therefore to reserves).

(Explanations given in notes are not required for full marks, but are included to explain how the figures given in appendix 3 are derived)

(Note: Credit will be given for alternative relevant presentation of financial positions and discussion)

2 (a) Proceeds from sales of EV clubs

Year (all figures \$m)	1	2	3	4
Free cash flows	390	419	455	490
Discount factor 12%	0.893	0.797	0.712	0.636
Present value	348	334	324	312
Present value	1,318			

Present value in Year 5 onwards = $\$490m \times 1.052 / (0.12 - 0.052) \times 0.636 = \$4,821m$

Total present value = $\$1,318m + \$4,821m = \$6,139m$

Desired sales proceeds (25% premium) = $\$6,139m \times 1.25 = \$7,674m$

Impact on statement of financial position (\$m)

Profit on sale = $\$7,674m - \$3,790m = \$3,884m$

Current assets adjustment = $\$2,347m + \$7,674m - (\$2,166m \times 1.5) = \$6,772m$

PPE adjustment = $\$6,772m - \$3,200m = \$3,572m$

	Original \$m	Sales proceeds \$m	Adjustments \$m	Final \$m
Assets				
Non-current assets	15,621	(3,790)	3,572	15,403
Current assets	2,347	7,674	(6,772)	3,249
Total assets	17,968			18,652
Equity and liabilities				
Called-up share capital	1,000			1,000
Retained earnings	7,917	3,884		11,801
Total equity	8,917			12,801
Non-current liabilities				
10% loan notes	3,200		(3,200)	–
Other loan notes	2,700			2,700
Bank loans	985			985
Total non-current liabilities	6,885			3,685
Current liabilities	2,166			2,166
Total equity and liabilities	17,968			18,652

Impact on eps (\$m)

	Current forecast	Revised forecast
Predicted post tax profits (\$454m x 10/4)	1,135	1,135
Less: profits from EV clubs		(454)
Add: interest saved, net of tax (\$3,200m x 10% x (1 – 0.2))		256
Add: return on additional non-current assets (\$3,572m x 12% x (1 – 0.2))		343
Add: return on additional current assets (\$902m x 7% x (1 – 0.2))		51
Adjusted profits	1,135	1,331
Number of shares	1,000 million	1,000 million
Adjusted eps	\$1.135	\$1.331

Impact on WACC

Equity beta

$$V_e = \$15,750\text{m} \times 1.1 = \$17,325\text{m}$$

$$V_d = (\$2,700 \times 0.93) + \$985\text{m} = \$3,496\text{m}$$

$$\beta_e = 0.952 ((17,325 + 3,496 (1 - 0.2)) / 17,325) = 1.106$$

Revised cost of equity

$$k_e = 4 + (10 - 4)1.106 = 10.64\%$$

Revised WACC

$$\text{WACC} = 10.64 (17,325 / (17,325 + 3,496)) + 8 (1 - 0.2) (3,496 / (17,325 + 3,496)) = 9.93\%$$

- (b) Shareholders would appear to have grounds for questioning the sale of the EV clubs. It would mean that Eview Cinemas Co was no longer diversified into two sectors. Although shareholders can achieve diversification themselves in theory, in practice transaction costs and other issues may mean they do not want to adjust their portfolio.

The increase in gym membership brought about by the forthcoming sports festival could justify the predicted increases in free cash flows made in the forecasts. Although increased earnings per share are forecast once the EV clubs are sold, these are dependent on Eview Cinemas Co achieving the sales price which it desires for the EV clubs and the predicted returns being achieved on the remaining assets.

The proposed expansion of multiscreen cinemas may be a worthwhile opportunity, but the level of demand for big cinema complexes may be doubtful and there may also be practical problems like negotiating change of use. In Year 1 the EV clubs would be forecast to make a post-tax return on assets of $(454/3,790) = 12.0\%$ compared with 9.6% ($12\% \times 0.8$) on the additional investment in the cinemas.

Investors may also wonder about the motives of Eview Cinemas Co's board. Selling the EV clubs offers the board a convenient way of resolving the conflict with the management team of the EV clubs and investors may feel that the board is trying to take an easy path by focusing on what they are comfortable with managing.

There may be arguments in favour of the sale, however. The lower WACC will be brought about by a fall in the cost of equity as well as the fall in the cost of debt. A reduction in the complexity of the business may result in a reduction in central management costs.

Eviev Cinemas Co may also be selling at a time when the EV clubs chain is at its most attractive as a business, in the period before the sports festival. The premium directors are hoping to obtain (on top of a valuation based on free cash flow figures which may be optimistic) suggest that they may be trying to realise maximum value while they can.

3 (a) Profitability

Revenues from the different types of store and online sales have all increased this year, despite a drop in store numbers. The increase in revenue this year may be largely due, however, to the government-induced pre-election boom in consumer expenditure, which appears unlikely to be sustained. Because the split of profits is not given, it is impossible to tell what has been the biggest contributor to increased profit. Profit as well as revenue details for different types of store would be helpful, also profit details for major product lines.

Improvements in return on capital employed derive from increases in profit margins and asset turnover.

The improvements in gross margins may be due to increased pressure being put on suppliers, in which case they may not be sustainable because of government pressure. The increased sales per store employee figures certainly reflects a fall in staff numbers, improving operating profit, although it could also be due to staff being better utilised or increased sales of higher value items in larger stores. If staff numbers continue to be cut, however, this could result in poorer service to customers, leading ultimately to decreased sales, so again it is questionable how much further High K Co can go.

The asset turnover shows an improvement which partly reflects the increase in sales. There have been only limited movements in the portfolio of the larger stores last year. The fall in non-current assets suggests an older, more depreciated, asset base. If there is no significant investment, this will mean a continued fall in capital employed and improved asset turnover. However, in order to maintain their appeal to customers, older stores will need to be refurbished and there is no information about refurbishment plans. Information about recent impairments in asset values would also be helpful, as these may indicate future trading problems and issues with realising values of assets sold.

Liquidity

The current ratio has improved, although the higher cash balances have been partly reflected by higher current liabilities. The increase in current liabilities may be due to a deliberate policy of taking more credit from suppliers, which the government may take measures to prevent. Being forced to pay suppliers sooner will reduce cash available for short-term opportunities.

Gearing

The gearing level in 2016 is below the 2014 level, but it would have fallen further had a fall in debt not been partly matched by a fall in High K Co's share price. It seems surprising that High K Co's debt levels fell during 2016 at a time of lower interest rates. Possibly lenders were (rightly) sceptical about whether the cut in central bank lending rate would be sustained and limited their fixed rate lending. Interest cover improved in 2016 and will improve further if High K Co makes use of revolving credit facilities. However, when High K Co's loans come up for renewal, terms available may not be as favourable as those High K Co has currently.

Investors

The increase in after-tax profits in 2015 and 2016 has not been matched by an increase in share price, which has continued to fall. The price/earnings ratio has been falling from an admittedly artificially high level, and the current level seems low despite earnings and dividends being higher. The stock market does not appear convinced by High K Co's current strategy. Return to shareholders in 2016 has continued to rise, but this has been caused by a significant % increase in dividend and hence increase in dividend yield. The continued fall in share price after the year end suggests that investors are sceptical about whether this increase can be maintained.

Revenue analysis

Town centre stores

High K Co has continued to close town centre stores, but closures have slowed recently and revenue increased in 2016. This suggests High K Co may have selected wisely in choosing which stores to keep open, although Dely Co believes there is no future for this type of store. Arguably though, town centre stores appeal to some customers who cannot easily get to out-of-town stores. Town centre stores may also be convenient collection points for customers using online click and collect facilities.

Convenience stores

High K Co has invested heavily in these since 2003. The figures in 2014 suggest it may have over-extended itself or possibly suffered from competitive pressures and saturation of the market. The 2016 results show an improvement despite closures of what may have been the worst-performing stores. The figures suggest Dely Co's decision to close its convenience stores may be premature, possibly offering High K Co the opportunity to take over some of its outlets. Maintaining its convenience store presence would also seem to be in line with High K Co's commitment to be responsive to customer needs. Profitability figures would be particularly helpful here, to assess the impact of rental commitments under leases.

Out-of-town stores

Although the revenue per store for out-of-town stores has shown limited improvement in 2016, this is less than might have been expected. The recent consumer boom would have been expected to benefit the out-of-town stores particularly, because expenditure on the larger items which they sell is more likely to be discretionary expenditure by consumers which will vary with the business cycle. Where Dely Co sites its new out-of-town stores will also be a major issue for High K Co, as it may find some of its best-performing stores face more competition. High K Co again may need to consider significant refurbishment expenditure to improve the look of these stores and customer experience in them.

Online sales

Online sales have shown steady growth over the last few years, but it is difficult to say how impressive High K Co's performance is. Comparisons with competitors would be particularly important here, looking at how results have changed over the years compared with the level of investment made. It is also impossible to tell from the figures how much increases in online sales have been at the expense of store sales.

Conclusion

If High K Co's share price is to improve, investors need it to make some sort of definite decision about strategy the way its competitors have since its last year end. What the chief executive has been saying about flexibility and keeping a varied portfolio has not convinced investors. If High K Co is to maintain its competitive position, it may well have no choice but to make a significant further investment in online operations. Possibly as well it could review where its competitor is closing convenience stores, as it may be able to open, with limited investment, new stores in locations with potential.

However, it also must decide what to do about the large out-of-town stores, as their performance is already stagnating and they are about to face enhanced competition. High K Co will also need to determine its dividend policy, with maybe a level of dividend which is considered the minimum acceptable to shareholders allowed for in planning cash outflows.

Appendix

	2014	2015	2016
Profitability			
Gross profit %	4.33	5.07	6.19
Operating profit %	0.87	1.70	2.91
Asset turnover (sales revenue/(total assets – current liabilities))	2.36	2.42	2.53
Return on capital employed % (operating profit % x asset turnover)	2.05	4.11	7.36
Liquidity			
Current ratio	0.84	1.29	1.69
Solvency			
Gearing (non-current liabilities/non-current liabilities + share capital)			
(Market values of share capital) %	37.6	36.8	32.5
Interest cover	1.63	3.54	7.12
Investors			
Dividend cover	0.35	1.29	1.71
Price/earnings ratio	54.46	12.15	5.52
Return to shareholders			
Dividend yield %	5.30	6.36	10.60
Share price gain/(loss) %	(9.00)	(5.65)	(3.29)
Total	(3.70)	0.71	7.31
Revenue/store (\$m)			
Town centre	31.91	33.05	33.93
Convenience	5.41	5.66	5.99
Out-of-town	46.45	46.16	46.46
Store revenue per store staff member (\$000)	247	258	272

(Note: Credit will be given for alternative relevant calculations and discussion. Candidates are not expected to complete all of the calculations or evaluation above to obtain the available marks.)

- (b) High K Co has not raised any equity finance over the last five years. Its falling share price means that a new share issue may not be successful. It may not only need debt finance to be renewed, but additional funding to be obtained.

High K Co intends to make more use of revolving credit facilities, which it need not draw on fully, rather than loans, which will mean that its finance costs are lower than on ordinary debt. However, these facilities are likely to be at floating rates, so if the government increases the central bank rate significantly, they could come at significant cost if High K Co decides to utilise them fully.

Finance costs on new debt, whatever form it takes, may therefore be significant and lower interest cover. High K Co may have to investigate selling some of the stores it owns either outright or on a sale or leaseback basis.

4 (a) Forward rate agreement

FRA 5.02% (4–9) since the investment will take place in four months' time for a period of five months.

If interest rates increase by 1.1% to 5.3%

	D
Investment return $5.0\% \times 5/12 \times \text{D}27,000,000$	562,500
Payment to bank $(5.3\% - 5.02\%) \times 5/12 \times \text{D}27,000,000$	(31,500)
Net receipt	<u>531,000</u>
Effective annual interest rate $531,000/27,000,000 \times 12/5$	4.72%

If interest rates fall by 0.6% to 3.6%

	D
Investment return $3.3\% \times 5/12 \times \text{D}27,000,000$	371,250
Receipt from bank $(5.02\% - 3.6\%) \times 5/12 \times \text{D}27,000,000$	159,750
Net receipt	<u>531,000</u>
Effective annual interest rate as above	4.72%

Futures

Go long in the futures market, as the hedge is against a fall in interest rates. Use March contracts, as investment will be made on 31 January.

Number of contracts = $\text{D}27,000,000/\text{D}500,000 \times 5 \text{ months}/3 \text{ months} = 90 \text{ contracts}$

Basis

Current price (1 October) – futures price = basis

$(100 - 4.20) - 94.78 = 1.02$

Unexpired basis on 31 January = $2/6 \times 1.02 = 0.34$

If interest rates increase by 1.1% to 5.3%

	D
Investment return as above	562,500
Expected futures price: $100 - 5.3 - 0.34 = 94.36$	
Loss on the futures market: $(0.9436 - 0.9478) \times \text{D}500,000 \times 3/12 \times 90$	(47,250)
Net return	<u>515,250</u>
Effective annual interest rate $515,250/27,000,000 \times 12/5$	4.58%

If interest rates fall by 0.6% to 3.6%

	D
Investment return as above	371,250
Expected futures price: $100 - 3.6 - 0.34 = 96.06$	
Profit on the futures market: $(0.9606 - 0.9478) \times \text{D}500,000 \times 3/12 \times 90$	144,000
Net receipt	<u>515,250</u>
Effective annual interest rate as above	4.58%

Options on futures

Buy call options as need to hedge against a fall in interest rates. As above, 90 contracts required.

If interest rates increase by 1.1% to 5.3%

Exercise price	94.25	95.25
Futures price as above	94.36	94.36
Exercise?	Yes	No
Gain in basis points	11	0
	D	D
Investment return (as above)	562,500	562,500
Gain from options $(0.0011 \times 500,000 \times 3/12 \times 90)$	12,375	0
Premium		
$0.00545 \times \text{D}500,000 \times 3/12 \times 90$	(61,313)	
$0.00098 \times \text{D}500,000 \times 3/12 \times 90$		(11,025)
Net return	<u>513,562</u>	<u>551,475</u>
Effective interest rate		
$513,562/27,000,000 \times 12/5$	4.56%	
$551,475/27,000,000 \times 12/5$		4.90%

If interest rates fall by 0·6% to 3·6%

Exercise price	94·25	95·25
Futures price as above	96·06	96·06
Exercise?	Yes	Yes
Gain in basis points	181	81
Investment return (as above)	371,250	371,250
Gain from options		
Gain from options: $0·0181 \times D500,000 \times 3/12 \times 90$	203,625	
Gain from options: $0·0081 \times D500,000 \times 3/12 \times 90$		91,125
Premium as above	(61,313)	(11,025)
Net return	<u>513,562</u>	<u>451,350</u>
Effective interest rate		
$513,562/27,000,000 \times 12/5$	4·56%	
$451,350/27,000,000 \times 12/5$		4·01%

Discussion

The forward rate agreement gives the highest guaranteed return. If Wardegul Co wishes to have a certain cash flow and is primarily concerned with protecting itself against a fall in interest rates, it will most likely choose the forward rate agreement. The 95·25 option gives a better rate if interest rates rise, but a significantly lower rate if interest rates fall, so if Wardegul Co is at all risk averse, it will choose the forward rate agreement.

This assumes that the bank which Wardegul Co deals with is reliable and there is no risk of default. If Wardegul Co believes that the current economic uncertainty may result in a risk that the bank will default, the choice will be between the futures and the options, as these are guaranteed by the exchange. Again the 95·25 option may be ruled out because it gives a much worse result if interest rates fall to 3·6%. The futures give a marginally better result than the 94·25 option in both scenarios but the difference is small. If Wardegul Co feels there is a possibility that interest rates will be higher than 5·41%, the point at which the 94·25 option would not be exercised, it may choose this option rather than the future.

(b) Regional functions compared with national functions

Organising treasury activities on a regional basis would be consistent with what is happening in the group overall. Other functions will be organised regionally. A regional treasury function may be able to achieve synergies with them and also benefit from information flows being organised based on the regional structure.

If, as part of a reorganisation, some treasury activities were to be devolved outside to a bank or other third party, it would be simpler to arrange for a single provider on a regional basis than arrange for separate providers in each country.

A regional function will avoid duplication of responsibilities over all the countries within a region. A regional function will have more work to do, with maybe a greater range of activities, whereas staff based nationally may be more likely to be under-employed. There may be enough complex work on a regional basis to justify employing specialists in particular treasury areas which will enhance the performance of the function. It may be easier to recruit these specialists if recruitment is done regionally rather than in each country.

Regional centres can carry out some activities on a regional basis which will simplify how funds are managed and mean less cost than managing funds on a national basis. These include pooling cash, borrowing and investing in bulk, and netting of foreign currency income and expenditure.

Regional centres could in theory be located anywhere in the region, rather than having one treasury function based in each country. This means that they could be located in the most important financial centres in each region or in countries which offered significant tax advantages.

From the point of view of Wardegul Co's directors and senior managers, it will be easier to enforce common standards and risk management policies on a few regional functions than on many national functions with differing cultures in individual countries.

Regional functions compared with global function

Wardegul Co is being reorganised on a regional basis because of the demands of its global expansion. As discussed above, reorganising treasury functions regionally will be consistent with the way other functions are organised. Reorganising the treasury function regionally will be one way of dealing with the problem of having a single, overstretched, global function.

A regional function could employ experts with knowledge of the regulations, practices and culture of the major countries within the region. It may be more difficult for a global function to recruit staff with local expertise.

There may be practical issues why individual countries prefer to deal with regional functions rather than a global function, for example, a regional function will be based in the same, or similar, time zone as the countries in its region.

A regional function may have better ideas of local finance and investment opportunities. There may, for example, be better alternatives for investment of the surplus funds than the centralised function has been able to identify.

		<i>Marks</i>
1	(a) Being able to bear higher levels of financial risk	Up to 3
	Better protection from predatory takeover bids	Up to 3
	Taxation benefit of higher levels of debt finance	Up to 2
		Max 7
	(b) (i) (Appendix 1)	
	Conejo Co's yield curve based on BBB rating	1
	Bond value based on BBB rating and spot yield rates	1
	Comment on reason for virtually no change in value	1
	Calculation of the coupon rate of the new bond	2
	Comment on coupon rate	1
		6
	(ii) (Appendix 2)	
	Duration based on annual coupon and balloon payment of \$100 in year 5	2
	Amount of fixed annual repayments of capital and interest	2
	Duration based on annual equivalent payments	2
		6
	(iii) (Appendix 3)	
	Financial position, proposal 1	3
	Financial position, proposal 2	3
	Interest payable on additional new debt finance	1
	Interest payable on higher coupon for current debt finance	1
	Return on additional investment	1
	Gearing calculations	1
	Earnings per share calculations	1
		11
	(iv) (Discussion in report)	
	Impact on Conejo Co	Up to 6
	Credit migration, credit rating agencies and CEO's opinion	Up to 6
	Impact on Conejo Co's equity holders	Up to 4
	Impact on Conejo Co's debt holders: current and new	Up to 3
		Max 16
	Professional marks for part (b)	
	Report format	1
	Structure and presentation of the report	3
		4
	Total	50

		<i>Marks</i>
2	(a) PV of free cash flows Years 1–4	1
	PV of free cash flows Years 5 onwards	2
	Desired sales proceeds	1
	Impact on statement of financial position	4
	Impact on eps	5
	Impact on WACC	
	Equity beta – cinemas	2
	Revised cost of equity – cinemas	1
	Revised WACC	1
		<u>17</u>
	(b) Arguments against sale	4–5
	Arguments for sale	4–5
	Max	<u>8</u>
	Total	<u>25</u>
3	(a) Ratios	
	Profitability	1–2
	Liquidity	1
	Solvency	1–2
	Investor	3–4
	Other ratios and trends	2–3
		<u></u>
	Max	<u>10</u>
	Discussion	
	Profitability	2–3
	Liquidity	1–2
	Gearing	1–2
	Investor	2–3
	Stores and online sales	3–4
	Conclusion	1–2
		<u></u>
	Max	<u>11</u>
	(b) 1 mark per relevant point	Max <u>4</u>
	Total	<u>25</u>
4	(a) Impact of FRA for rate increase and decrease	2
	Go long on futures	1
	Selection of March futures and options	1
	Number of contracts	1
	Basis calculation	1
	Impact of interest rate increase/decrease with futures	3
	Buy call options	1
	Premium calculations	1
	Exercise options?	1
	Impact of interest rate increase/decrease with options	3
	Discussion	3–4
		<u></u>
	Max	<u>18</u>
	(b) Regional functions compared with national functions	4–5
	Regional functions compared with global function	3–4
		<u></u>
	Max	<u>7</u>
	Total	<u>25</u>

Professional Level – Options Module

Advanced Financial Management

March/June 2018 – Sample Questions



Time allowed: 3 hours 15 minutes

This question paper is divided into two sections:

Section A – This ONE question is compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Formulae and tables are on pages 10–14.

Do NOT open this question paper until instructed by the supervisor.

This question paper must not be removed from the examination hall.

P4 Paper

Think Ahead

ACCA

The Association of
Chartered Certified
Accountants

Section A – This question is compulsory and MUST be attempted

- 1 Chikepe Co is a large listed company operating in the pharmaceutical industry with a current market value of equity of \$12,600 million and a debt to equity ratio of 30:70, in market value terms. Institutional investors hold most of its equity shares. The company develops and manufactures antibiotics and anti-viral medicines. Both the company and its products have an established positive reputation among the medical profession, and its products are used widely. However, its rate of innovation has slowed considerably in the last few years and it has fewer new medical products coming into the market.

At a recent meeting of the board of directors (BoD), it was decided that the company needed to change its current strategy of growing organically to one of acquiring companies, in order to maintain the growth in its share price in the future. The members of the BoD had different opinions on the type of acquisition strategy to pursue.

Director A was of the opinion that Chikepe Co should follow a strategy of acquiring companies in different business sectors. She suggested that focusing on just the pharmaceutical sector was too risky and acquiring companies in different business sectors will reduce this risk.

Director B was of the opinion that Director A's suggestion would not result in a reduction in risk for shareholders. In fact, he suggested that this would result in agency related issues with Chikepe Co's shareholders reacting negatively and as a result, the company's share price would fall. Instead, Director B suggested that Chikepe Co should focus on its current business and acquire other established pharmaceutical companies. In this way, the company will gain synergy benefits and thereby increase value for its shareholders.

Director C agreed with Director B, but suggested that Chikepe Co should consider relatively new pharmaceutical companies, as well as established businesses. In her opinion, newer companies might be involved in research and development of innovative products, which could have high potential in the future. She suggested that using real options methodology with traditional investment appraisal methods such as net present value could help establish a more accurate estimate of the potential value of such companies.

The company has asked its finance team to prepare a report on the value of a potential target company, Foshoro Co, before making a final decision.

Foshoro Co

Foshoro Co is a non-listed pharmaceutical company established about 10 years ago. Initially Foshoro Co grew rapidly, but this rate of growth slowed considerably three years ago, after a venture capital equity backer exited the company by selling its stake back to the founding directors. The directors had to raise substantial debt capital to buy back the equity stake. The company's current debt to equity ratio is 60:40. This high level of gearing means that the company will find it difficult to obtain funds to develop its innovative products in the future.

The following financial information relates to Foshoro Co:

Extract from the most recent statement of profit or loss

	\$ million
Sales revenue	878.1
Profit before interest and tax	192.3
Interest	78.6
Tax	22.7
Profit after tax	91.0

In arriving at the profit before interest and tax, Foshoro Co deducted tax allowable depreciation and other non-cash expenses totalling \$112.0 million. It requires a cash investment of \$98.2 million in non-current assets and working capital to continue its operations at the current level.

Three years ago, Foshoro Co's profit after tax was \$83.3 million and this has been growing steadily to their current level. Foshoro Co's profit before interest and tax and its cash flows grew at the same growth rate as well. It is likely that this growth rate will continue for the foreseeable future if Foshoro Co is not acquired by Chikepe Co. Foshoro Co's cost of capital has been estimated at 10%.

Combined company: Chikepe Co and Foshoro Co

Once Chikepe Co acquires Foshoro Co, it is predicted that the combined company's sales revenue will be \$4,200 million in the first year, and its operating profit margin on sales revenue will be 20% for the foreseeable future.

After the first year, the sales revenue is expected to grow at 7% per year for the following three years. It is anticipated that after the first four years, the growth rate of the combined company's free cash flows will be 5·6% per year.

The combined company's tax allowable depreciation is expected to be equivalent to the amount of investment needed to maintain the current level of operations. However, as the company's sales revenue increases over the four-year period, the combined company will require an additional investment in assets of \$200 million in the first year and then \$0·64 per \$1 increase in sales revenue for the next three years.

It can be assumed that the asset beta of the combined company is the weighted average of the individual companies' asset betas, weighted in proportion of the individual companies' value of equity. It can also be assumed that the capital structure of the combined company remains at Chikepe Co's current capital structure level, a debt to equity ratio of 30:70. Chikepe Co pays interest on borrowings at a rate of 5·3% per annum.

Chikepe Co estimates that it will be able to acquire Foshoro Co by paying a premium of 30% above its estimated equity value to Foshoro Co's shareholders.

Other financial information

	Equity beta	Asset beta
Chikepe Co	1·074	0·800
Foshoro Co	2·090	0·950

The current annual government borrowing base rate is 2% and the annual market risk premium is estimated at 7%.

Both companies pay tax at an annual rate of 20%.

Chikepe Co estimates equity values in acquisitions using the free cash flow to firm method.

Future acquisitions

The BoD agreed that in the future it is likely that Chikepe Co will target both listed and non-listed companies for acquisition. It is aware that when pursuing acquisitions of listed companies, the company would need to ensure that it complied with regulations such as the mandatory bid rule and the principle of equal treatment to protect shareholders. The BoD is also aware that some listed companies may attempt to defend acquisitions by employing anti-takeover measures such as poison pills and disposal of crown jewels.

Required:

- (a) Compare and contrast the reasons for the opinions held by Director A and by Director B, and discuss the types of synergy benefits which may arise from the acquisition strategy suggested by Director B. (9 marks)
- (b) Discuss how using real options methodology in conjunction with net present value could help establish a more accurate estimate of the potential value of companies, as suggested by Director C. (5 marks)
- (c) Prepare a report for the board of directors of Chikepe Co which:
 - (i) Estimates the current equity value of Foshoro Co; (6 marks)
 - (ii) Estimates the equity value arising from combining Foshoro Co with Chikepe Co; (11 marks)
 - (iii) Evaluates whether the acquisition of Foshoro Co would be beneficial to Chikepe Co's shareholders and discusses the limitations of the valuation method used in (c)(i) and (c)(ii) above. (7 marks)

Professional marks will be awarded in part (c) for the format, structure and presentation of the report. (4 marks)
- (d) Discuss how the mandatory bid rule and the principle of equal treatment protects shareholders in the event of their company facing a takeover bid, and discuss the effectiveness of poison pills and disposal of crown jewels as defensive tactics against hostile takeover bids. (8 marks)

(50 marks)

Section B – TWO questions ONLY to be attempted

- 2 Tipteline Co is based in Valliland. It is listed on Valliland's stock exchange but only has a small number of shareholders. Its directors collectively own 45% of the equity share capital.

Tipteline Co's growth has been based on the manufacture of household electrical goods. However, the directors have taken a strategic decision to diversify operations and to make a major investment in facilities for the manufacture of office equipment.

Details of investment

The new investment is being appraised over a four-year time horizon. Revenues from the new investment are uncertain and Tipteline Co's finance director has prepared what she regards as cautious forecasts. She predicts that it will generate \$2 million operating cash flows before marketing costs in Year 1 and \$14.5 million operating cash flows before marketing costs in Year 2, with operating cash flows rising by the expected levels of inflation in Years 3 and 4.

Marketing costs are predicted to be \$9 million in Year 1 and \$2 million in each of Years 2 to 4.

The new investment will require immediate expenditure on facilities of \$30.6 million. Tax allowable depreciation will be available on the new investment at an annual rate of 25% reducing balance basis. It can be assumed that there will either be a balancing allowance or charge in the final year of the appraisal. The finance director believes the facilities will remain viable after four years, and therefore a realisable value of \$13.5 million can be assumed at the end of the appraisal period.

The new facilities will also require an immediate initial investment in working capital of \$3 million. Working capital requirements will increase by the rate of inflation for the next three years and any working capital at the start of Year 4 will be assumed to be released at the end of the appraisal period.

Tipteline Co pays tax at an annual rate of 30%. Tax is payable with a year's time delay. Any tax losses on the investment can be assumed to be carried forward and written off against future profits from the investment.

Predicted inflation rates are as follows:

Year	1	2	3	4
	8%	6%	5%	4%

Financing the investment

Tipteline Co has been considering two choices for financing all of the \$30.6 million needed for the initial investment in the facilities:

- A subsidised loan from a government loan scheme, with the loan repayable at the end of the four years. Issue costs of 4% of the gross finance would be payable. Interest would be payable at a rate of 30 basis points below the risk free rate of 2.5%. In order to obtain the benefits of the loan scheme, Tipteline Co would have to fulfil various conditions, including locating the facilities in a remote part of Valliland where unemployment is high.
- Convertible loan notes, with the subscribers for the notes including some of Tipteline Co's directors. The loan notes would have issue costs of 4% of the gross finance. If not converted, the loan notes would be redeemed in six years' time. Interest would be payable at 5%, which is Tipteline Co's normal cost of borrowing. Conversion would take place at an effective price of \$2.75 per share. However, the loan note holders could enforce redemption at any time from the start of Year 3 if Tipteline Co's share price fell below \$1.50 per share. Tipteline Co's current share price is \$2.20 per share.

Issue costs for the subsidised loan and convertible loan notes would be paid out of available cash reserves. Issue costs are not allowable as a tax-deductible expense.

In initial discussions, the majority of the board favoured using the subsidised loan. The appraisal of the investment should be prepared on the basis that this method of finance will be used. However, the chairman argued strongly in favour of the convertible loan notes, as, in his view, operating costs will be lower if Tipteline Co does not have to fulfil the conditions laid down by the government of Valliland. Tipteline Co's finance director is sceptical, however, about whether the other shareholders would approve the issue of convertible loan notes on the terms suggested. The directors will decide which method of finance to use at the next board meeting.

Other information

Humabuz Co is a large manufacturer of office equipment in Valliland. Humabuz Co's geared cost of equity is estimated to be 10·5% and its pre-tax cost of debt to be 5·4%. These estimates are based on a capital structure comprising \$225 million 6% irredeemable bonds, trading at \$107 per \$100, and 125 million \$1 equity shares, trading at \$3·20 per share. Humabuz Co also pays tax at an annual rate of 30% on its taxable profits.

Required:

- (a) Calculate the adjusted present value for the investment on the basis that it is financed by the subsidised loan and conclude whether the project should be accepted or not. Show all relevant calculations. (17 marks)
- (b) Discuss the issues which Tippletine Co's shareholders who are not directors would consider if its directors decided that the new investment should be financed by the issue of convertible loan notes on the terms suggested.

Note: You are not required to carry out any calculations when answering part (b).

(8 marks)

(25 marks)

3 Arthuro Co group

Arthuro Co is based in Hittyland and is listed on Hittyland's stock exchange. Arthuro Co has one wholly-owned subsidiary, Bowerscots Co, based in the neighbouring country of Owlia. Hittyland and Owlia are in a currency union and the currency of both countries is the \$.

Arthuro Co purchased 100% of Bowerscots Co's share capital three years ago. Arthuro Co has the power under the acquisition to determine the level of dividend paid by Bowerscots Co. However, Arthuro Co's board decided to let Bowerscots Co's management team have some discretion when making investment decisions. Arthuro Co's board decided that it should receive dividends of 60% of Bowerscots Co's post-tax profits and has allowed Bowerscots Co to use its remaining retained earnings to fund investments chosen by its management. A bonus linked to Bowerscots Co's after-tax profits is a significant element of Bowerscots Co's managers' remuneration.

Bowerscots Co operates in a very competitive environment. Recently, a senior member of its management team has left to join a competitor.

Arthuro Co's dividend policy

Until three months ago, Arthuro Co had 90 million \$2 equity shares in issue and \$135 million 8% bonds. Three months ago it made a 1 for 3 rights issue. A number of shareholders did not take up their rights, but sold them on, so there have been changes in its shareholder base. Some shareholders expressed concern about dilution of their dividend income as a result of the rights issue. Therefore, Arthuro Co's board felt it had to promise, for the foreseeable future, at least to maintain the dividend of \$0.74 per equity share, which it paid for the two years before the rights issue.

Arthuro Co's board is nevertheless concerned about whether it will have sufficient funds available to fulfil its promise about the dividend. It has asked the finance director to forecast its dividend capacity based on assumptions about what will happen in a 'normal' year. The finance director has made the following assumptions in the forecast:

1. Sales revenue can be assumed to be 4% greater than the most recent year's of \$520 million.
2. The operating profit margin can be assumed to be 20%.
3. Operating profit can be assumed to be reported after charging depreciation of \$30 million and profit on disposal of non-current assets of \$5.9 million. The cost of the non-current assets sold can be assumed to be \$35 million and its accumulated depreciation to be \$24.6 million. Depreciation is allowable for tax and the profit on disposal is fully chargeable to tax.
4. The net book value of non-current assets at the year end in the most recent accounts was \$110 million. To maintain productive capacity, sufficient investment to increase this net book value figure 12 months later by 4% should be assumed, in line with the increase in sales. The calculation of investment required for the year should take into account the depreciation charged of \$30 million, and net book value of the non-current assets disposed of during the year.
5. A \$0.15 investment in working capital can be assumed for every \$1 increase in sales revenue.
6. Bowerscots Co's pre-tax profits can be assumed to be \$45 million.

Arthuro Co's directors have decided that if there is a shortfall of dividend capacity, compared with the dividends required to maintain the current dividend level, the percentage of post-tax profits of Bowerscots Co paid as dividend should increase, if necessary up to 100%.

Taxation

Arthuro Co pays corporation tax at 30% and Bowerscots Co pays corporation tax at 20%. A withholding tax of 5% is deducted from any dividends remitted by Bowerscots Co. There is a bilateral tax treaty between Hittyland and Owlia. Corporation tax is payable by Arthuro Co on profits declared by Bowerscots Co, but Hittyland gives full credit for corporation tax already paid in Owlia. Hittyland gives no credit for withholding tax paid on dividends in Owlia.

Required:

- (a) (i) **Estimate Arthuro Co's forecast dividend capacity for a 'normal' year;** (11 marks)
- (ii) **Estimate the level of dividend required from Bowerscots Co to give Arthuro Co sufficient dividend capacity to maintain its dividend level of \$0.74 per equity share.** (3 marks)

- (b) Arthuro Co has decided to increase its level of dividend from Bowerscots Co if its dividend capacity is insufficient.

Required:

- (i) From Arthuro Co's viewpoint, discuss the financial benefits of, and problems with, this decision; (5 marks)
- (ii) Discuss the agency problems, and how they might be resolved, with this decision. (6 marks)

(25 marks)

- 4 The Adverane Group is a multinational group of companies with its headquarters in Switzerland. The Adverane Group consists of a number of fully-owned subsidiaries and Elted Co, an associate company based in the USA in which Adverane Group owns 30% of the ordinary equity share capital. Balances owing between the parent, Adverane Co, and its subsidiaries and between subsidiaries are settled by multilateral netting. Transactions between the parent and Elted Co are settled separately.

Transactions with Elted Co

Adverane Co wishes to hedge transactions with Elted Co which are due to be settled in four months' time in US\$. Adverane Co will owe Elted Co US\$3.7 million for a major purchase of supplies and Elted Co will owe Adverane Co US\$10.15 million for non-current assets. Adverane Group's treasury department is considering whether to use money markets or exchange-traded currency futures for hedging.

Annual interest rates available to Adverane Co

	Investing rate	Borrowing rate
Switzerland	2.7%	3.9%
USA	2.5%	3.7%

Exchange traded currency futures

Contract size CHF125,000, price quotation US\$ per CHF1

Three-month expiry: 1.1213

Six-month expiry: 1.1204

Netting

The balances owed to and owed by members of Adverane Group when netting is to take place are as follows:

Owed by	Owed to	Local currency m
Adverane (Switzerland)	Bosha (Eurozone)	CHF15.90
Adverane (Switzerland)	Diling (Brazil)	CHF4.46
Bosha (Eurozone)	Cogate (USA)	€24.89
Bosha (Eurozone)	Diling (Brazil)	€18.57
Cogate (USA)	Adverane (Switzerland)	US\$27.08
Cogate (USA)	Diling (Brazil)	US\$5.68
Diling (Brazil)	Adverane (Switzerland)	BRL38.80
Diling (Brazil)	Bosha (Eurozone)	BRL51.20

Spot rates are currently as follows:

	CHF	€	US\$	BRL
1 CHF =	1.0000	0.9347–0.9369	1.1196–1.1222	3.1378–3.1760

The group members will make settlement in Swiss francs. Spot mid-rates will be used in calculations. Settlement will be made in the order that the company owing the largest net amount in Swiss francs will first settle with the company owed the smallest net amount in Swiss francs.

Transfer price arrangements

The Adverane Group board has been reviewing the valuation of inter-group transactions, as it is concerned that the current system is not working well. Currently inter-group transfer prices are mostly based on fixed cost plus a mark-up negotiated by the buying and selling divisions. If they cannot agree a price, either the sale does not take place or the central treasury department determines the margin. The board has the following concerns:

- Both selling and buying divisions have claimed that prices are unfair and distort the measurement of their performance.
- Significant treasury department time is being taken up dealing with disputes and then dealing with complaints that the price it has imposed is unfair on one or the other division.
- Some parts of the group are choosing to buy from external suppliers rather than from suppliers within the group.

As a result of the review, the Adverane Group board has decided that transfer prices should in future be based on market prices, where an external market exists.

Note: CHF is Swiss Franc, € is Euro, US\$ is United States dollar and BRL is Brazilian Real.

Required:

- (a) Advise Adverane Co on, and recommend, an appropriate hedging strategy for the US\$ cash flows it is due to receive from, or pay to, Elted Co.** (9 marks)
- (b) (i) Calculate the inter-group transfers which are forecast to take place.** (7 marks)
(ii) Discuss the advantages of multilateral netting by a central treasury function within the Adverane Group. (3 marks)
- (c) Evaluate the extent to which changing to a market-price system of transfer pricing will resolve the concerns of the Adverane Group board.** (6 marks)

(25 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i(E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_o = \frac{D_o(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = br_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \qquad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model

$$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$$

Where:

$$d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$$

$$d_2 = d_1 - s\sqrt{t}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

		<i>Discount rate (r)</i>									
<i>Periods</i>											
(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

		<i>Discount rate (r)</i>									
<i>Periods</i>											
(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Answers

- 1 (a) Director A's focus is on reducing the risk in the business through diversification and thereby increasing its value. A strategy of risk diversification resulting in greater value can work in situations where the equity holders are exposed to both unsystematic and systematic risks, for example, when their investment is concentrated in one company. In such situations, the shareholders would be subject to unsystematic risk and diversification would reduce this risk.

In the case of Chikepe Co, this is unlikely to be the case as a large proportion of shares are owned by institutional shareholders and it is likely that their investment portfolios are already well-diversified and therefore they are not exposed to unsystematic risk. Further diversification will be of no value to them. In fact, it may be construed that managers are only taking this action for their own benefit, as they may be closely tied to the company and therefore be exposed to total risk (both unsystematic and systematic risks). This may then become a source of agency related conflict between the management and the shareholders.

However, diversification overseas into markets which have some barriers to entry might reduce both systematic and unsystematic risks as well.

Director B, on the other hand, seems to be suggesting that Chikepe Co should focus on its core business and increase value through identifying areas of synergy benefits. It may be the case that Chikepe Co's management and directors are well placed to identify areas where the company can gain value by acquiring companies with potential synergy benefits.

The types of synergy benefits, which may arise in established pharmaceutical companies, can include:

Identifying undervalued companies, where the management is not effective in unlocking the true value of company. By replacing the existing management, Chikepe Co may be able to unlock the value of the company.

Acquiring companies which have strategic assets or product pipelines. Chikepe Co may be well placed to identify companies which have a number of product pipelines, which those companies are not exploiting fully. By acquiring such companies, Chikepe Co may be able to exploit the product pipelines.

Through acquisitions, Chikepe Co may be able to exploit economies of scope by eliminating process duplication or economies of scale where its size may enable it to negotiate favourable terms.

Foshoro Co may benefit if Chikepe Co acquires it because it is struggling to raise funding for its innovative products. Chikepe Co is an established company but has few new product innovations coming in the future. Therefore, it may have spare cash resources which Foshoro Co may be able to utilise.

(Note: Credit will be given for alternative valid discussion)

- (b) Traditional investment appraisal methods such as net present value assume that an investment needs to be taken on a now or never basis, and once undertaken, it cannot be reversed. Real options take into account the fact that in reality, most investments have within them certain amounts of flexibility, such as whether or not to undertake the investment immediately or to delay the decision; to pursue follow-on opportunities; and to cancel an investment opportunity after it has been undertaken. Where there is increasing uncertainty and risk, and where a decision can be changed or delayed, this flexibility has value, known as the time value of an option.

Net present value captures just the intrinsic value of an investment opportunity, whereas real options capture both the intrinsic value and the time value, to give an overall value for an opportunity. When a company still has time available to it before a decision needs to be made, it may have opportunities to increase the intrinsic value of the investment through the strategic decisions it makes.

Investing in new companies with numerous potential innovative product pipelines may provide opportunities for flexibility where decisions can be delayed and the intrinsic value can be increased through strategic decisions and actions taken by the company. Real options try to capture the value of this flexibility within companies with innovative product pipelines, whereas net present value does not.

- (c) Report to the board of directors (BoD), Chikepe Co

Introduction

This report evaluates whether the acquisition of Foshoro Co would be beneficial to Chikepe Co's shareholders by estimating the additional equity value created from the synergies resulting when the two companies are combined. The market values of equity of the two companies as separate entities are considered initially and then compared with the equity value of the two companies together. The free cash flow to firm valuation method is used to estimate the values of the companies and the limitations of this method are discussed.

The market value of equity of Chikepe Co is given as \$12,600 million.

Based on the free cash flow to firm valuation method:

The current market value of equity of Foshoro Co is estimated at \$986 million (appendix 1), and

The market value of equity of the combined company is estimated at \$14,993 million (appendix 2).

Therefore, the additional market value of equity arising from synergy benefits when the two companies are combined is estimated at \$1,407 million (appendix 3), which is then split between Foshoro Co's shareholders receiving \$296 million (a

30% premium) and Chikepe Co's shareholders receiving the balance of \$1,111 million, which is approximately 8·8% excess over the original equity value (appendix 3).

However, the valuation method used has a number of limitations, as follows:

- The values of both Foshoro Co and the combined company are based on estimations and assumptions, for example,
 - Foshoro Co's future growth rate of free cash flows is based on past growth rates and it is assumed that this will not change in the future;
 - It is not explained how Foshoro Co's cost of capital is estimated/calculated. Such an estimate may be more difficult to make for private companies;
 - The assumption of perpetuity is made when estimating the values of Foshoro Co and the combined company, and this may not be valid.
- The basis for the synergy benefits such as higher growth rates of sales revenue and profit margins needs to be explained and justified. It is not clear how these estimates have been made.
- Whereas it may be possible to estimate the asset beta of a listed company such as Chikepe Co, it may be more difficult to provide a reasonable estimate for the asset beta of Foshoro Co. Therefore, the estimate of the cost of capital of the combined company may not be accurate.
- The costs related to the acquisition process would need to be factored in.

Therefore, whereas the free cash flow method of estimating corporate values is theoretically sound, using it in practice to estimate values is open to errors and judgements.

Conclusion

The valuations indicate that Chikepe Co's shareholders would benefit from the acquisition of Foshoro Co and the value of their shares should increase by 8·8%. However, the method used to estimate the value created makes a number of estimates and assumptions. It is therefore recommended that a range of valuations is made under different assumptions and estimates, through a process of sensitivity analysis, before a final decision is made. As well as this, the limitations of the valuation method used should be well understood and taken into account.

Report compiled by:

Date

APPENDICES:

Appendix 1 (Part (c) (i)): Foshoro Co, estimate of current value

Cost of capital = 10%

Growth rate of profits and free cash flows = $(\$91·0m/\$83·3m)^{1/3} - 1 = 0·03 = 3\%$

Free cash flow to firm (FCFF) = PBIT + non-cash expenses – additional cash investment – tax

FCFF = \$192·3m + \$112·0m – \$98·2m – (20% x \$192·3m) = \$167·6m

Foshoro Co, estimated value = $(\$167·6m \times 1·03)/(0·10 - 0·03) = \$2,466·1m$

Current estimated market value of equity of Foshoro Co = \$2,466·1m x 40% = \$986·4m, say \$986m approximately.

Appendix 2 (Part (c) (ii)): Estimate of value created from combining Chikepe Co and Foshoro Co

Asset beta of combined company = $(0·800 \times \$12,600m + 0·950 \times \$986·4m)/(\$12,600m + \$986·4m) = 0·811$

Equity beta of combined company = $0·811 \times (0·70 + 0·30 \times 0·80)/0·70 = 1·089$

Cost of equity, combined company = 2% + 1·089 x 7% = 9·6% approx.

Cost of capital, combined company = 9·6% x 0·7 + 5·3% x 0·3 x 0·8 = 8% approx.

Combined company, free cash flows and value computation (\$ millions)

Sales growth rate, years 2 to 4 = 7% per annum; operating profit margin = 20%

Year	1	2	3	4
Sales revenue	4,200	4,494	4,809	5,146
Operating profit	840	899	962	1,029
Less tax (20%)	(168)	(180)	(192)	(206)
Less additional investment in assets	(200)	(188)	(202)	(216)
Free cash flows	472	531	568	607
PV of free cash flows (8%)	437	455	451	446

	\$ millions
Present value (first four years)	1,789
Present value (after four years)	
$607 \times 1.056 / (0.08 - 0.056) \times 0.735$	19,630
Estimated market value of the combined company	<u>21,419</u>

Market value of equity of the combined company = $70\% \times \$21,419\text{m} = \$14,993\text{m}$

Appendix 3: Synergy benefits and their distribution

Additional market equity value created by combining the two companies

$\$14,993\text{m} - (\$12,600\text{m} + \$986\text{m}) = \$1,407\text{m}$

Therefore, synergy benefits resulting from combining the two companies: \$1,407m

Premium payable to Foshoro Co shareholders: $30\% \times \$986\text{m} = \296m

Balance of synergy benefits going to Chikepe Co's shareholders: \$1,111m

As a percentage of current value: $\$1,111\text{m} / \$12,600\text{m} \times 100\% = 8.8\%$

- (d) Both the mandatory bid rule and the principle of equal treatment are designed to protect minority shareholders, where an acquirer has obtained a controlling interest of the target company. The mandatory bid rule provides minority shareholders with the opportunity to sell their shares and exit the target company at a specified fair share price. This price should not be lower than the highest price paid for shares, which have already been acquired within a specified period. The principle of equal treatment requires the acquiring company to offer the same terms to minority shareholders as were offered to the earlier shareholders from whom the controlling interest was acquired. Both these regulatory devices are designed to ensure that the minority shareholders are protected financially and are not exploited by the acquirer.

The purpose of both poison pills and disposal of crown jewels is to make the target company unattractive to the acquirer. Poison pills give existing shareholders in the target company the right to buy additional shares in their company at a discount once the acquiring company has bought a certain number of shares in the target company. The aim is to make the target company more expensive to purchase, as the acquirer needs to buy more shares. Disposal of crown jewels involves selling the target company's most valuable assets, and therefore making the target company less attractive to the acquirer. The effectiveness of either defence tactic can be limited, as the company's management would need its shareholders to authorise such moves (although there are ways in which poison pills can be incorporated without gaining prior authorisation from shareholders). Shareholders may not be willing to do this as they normally get premiums on their shares during takeover battles. Additionally, disposing of key strategic assets could substantially weaken a company's competitive advantage and therefore its future potential. Such action may be detrimental to the company and therefore shareholders would probably not approve that course of action.

2 (a) Year	0 \$000	1 \$000	2 \$000	3 \$000	4 \$000	5 \$000
Operating cash flow excluding marketing costs		2,000	14,500	15,225	15,834	
Marketing costs		(9,000)	(2,000)	(2,000)	(2,000)	
Cash flow before tax		(7,000)	12,500	13,225	13,834	
Taxation (W1)					(310)	(4,328)
Investment	(30,600)				13,500	
Working capital (W2)	(3,000)	(240)	(194)	(172)	3,606	
Cash flows	(33,600)	(7,240)	12,306	13,053	30,630	(4,328)
Discount factor 9% (W3)	1.000	0.917	0.842	0.772	0.708	0.650
Discounted cash flows	(33,600)	(6,639)	10,362	10,077	21,686	(2,813)
Base case NPV	(927)					

1 Taxation

Year	TAD = Tax-allowable depreciation	Balance \$000
1	Investment TAD 25% reducing balance	30,600 (7,650)
2	TAD 25% reducing balance	22,950 (5,738)
3	TAD 25% reducing balance	17,212 (4,303)
4	Balancing charge	12,909 591
		13,500

Year	1 \$000	2 \$000	3 \$000	4 \$000
Cash flow before tax	(7,000)	12,500	13,225	13,834
Tax-allowable depreciation	(7,650)	(5,738)	(4,303)	591
Adjusted cash flow	(14,650)	6,762	8,922	14,425
Offset against previous losses		(14,650)	(7,888)	
Losses carried forward	(14,650)	(7,888)		
Taxable cash flow			1,034	14,425
Taxation at 30%			310	4,328
Year			4	5

2 Working capital

Year	1 \$000	2 \$000	3 \$000	4 \$000
	$3,000 \times 0.08 = 240$	$(3,000 + 240) \times 0.06 = 194$	$(3,000 + 240 + 194) \times 0.05 = 172$	$3,000 + 240 + 194 + 172 = 3,606$

3 Ungeared cost of equity

Humabuz Co

MV debt = \$225 million $\times$ 1.07 = \$240.8 million

MV equity = 125 million $\times$ \$3.20 = \$400 million

Ungeared cost of equity

$$k_e = k_e^i + (1 - t) (k_e^i - k_d) V_d/V_e$$

$$10.5\% = k_e^i + (1 - 0.3) (k_e^i - 5.4\%) (240.8/400)$$

$$10.5\% + 2.28\% = 1.42 k_e^i$$

$$k_e^i = 9\%$$

4 Issue costs

Debt: $(\$30,600,000/0.96) = \$31,875,000$

Debt issue costs: $\$31,875,000 \times 0.04 = \$1,275,000$

5 Tax shield on loan

Use PV of an annuity (PVA) for years 2 – 5 at 5% (assume 5% is cost of debt).

(Note: The risk-free rate of 2.5% could also be used for discounting.)

Subsidised loan: $\$30,600,000 \times (0.025 - 0.003) \times 0.3 \times (4.329 - 0.952) = \$682,000$

6 Subsidy

Benefit = $\$30,600,000 \times (0.05 - 0.022) \times 3.546 = \$3,038,000$

Tax relief lost = $\$30,600,000 \times (0.05 - 0.022) \times 0.3 \times (4.329 - 0.952) = \$868,000$

7 Financing side effects

	\$000
Issue costs (W4)	(1,275)
Tax shield on loan (W5)	682
Subsidy benefit (W6)	3,038
Tax relief lost on subsidy benefit (W6)	(868)
Total benefit of financing side effects	<u>1,577</u>

Conclusion

If base case net present value is used, the project has a negative net present value of \$927,000 and on that basis should be rejected. However, the financing side effects add \$1,577,000 to the value of the project, giving a positive adjusted present value of \$650,000. On that basis the project should be accepted. The revenues from the project appear to be uncertain and the realisable value at the end of the project may be optimistic. It would be useful to have an indication of the range of outcomes and an idea of the probability that the project will have a negative APV.

(b) Advantages of convertible loan notes

The investors may be happy that directors are demonstrating their commitment to the company by subscribing to convertible loan notes. The conversion rights mean that these directors will benefit if the share price increases, aligning their interests with shareholders.

The conversion terms also mean that the loan notes will not necessarily have to be repaid in a few years' time. This may be significant if Tippletine Co does not have the cash available for redemption then.

Drawbacks of convertible loan notes

The convertible loan notes would be treated as debt, increasing Tippletine Co's gearing, which may concern the other shareholders. The interest on the convertible loan notes will be payable before dividends and may leave less money for distribution to shareholders. Shareholders may doubt whether the higher interest burden on the convertible loan notes compared with the subsidised loan is compensated for by the lower costs of Tippletine Co not having to fulfil the government's requirements.

The other shareholders may be concerned by the interest rate on the convertible notes being Tippletine Co's normal cost of borrowing. The option to convert is an advantage for convertible loan note holders. They would often effectively pay for this option by receiving a lower rate of interest on the loan notes.

Shareholders would want to assess how likely conversion would be, that is how likely it would be the share price will rise above \$2.75. The option to convert may also change the balance of shareholdings, giving the directors who held the notes a greater percentage of share capital and possibly more influence over Tippletine Co. The other shareholders may be unhappy with this.

The shareholders may also have reservations about the loan note holders having the option to redeem if Tippletine Co's share price is low. This reduces the risk of providing the finance from the loan note holders' viewpoint. However, if the share price is low, Tippletine Co's financial results and cash flows may be poor and it may struggle to redeem the loan notes. Shareholders may also be concerned that there is no cap the other way, allowing Tippletine Co to force conversion if the share price reaches a high enough level.

(Note: Credit will be given for alternative relevant discussion)

3 (a) (i) Forecast dividend capacity is as follows:

	\$000
Operating profit ($20\% \times 1.04 \times \520 million)	108,160
Less: Interest ($8\% \times \$135$ million)	(10,800)
Less: Taxation ($30\% \times (\$108.16 \text{ million} - \$10.8 \text{ million})$)	(29,208)
Add: Depreciation	30,000
Less: Profit on disposal of NCA	(5,900)
Add: Cash received on disposal of NCA (W1)	16,300
Less: Investment in new NCA (W2)	(44,800)
Less: Investment in working capital ($15\% \times 0.04 \times \520 million)	(3,120)
Add: Dividend remittance from Bowerscots Co (W3)	20,520
Less: Additional tax on Bowerscots Co's profits ($10\% \times \$45$ million)	(4,500)
Forecast dividend capacity	<u>76,652</u>

Workings

1 Disposal of non-current assets

	\$000
Profit on disposal	5,900
Cost	35,000
Less: Depreciation	(24,600)
Cash received on disposal	<u>16,300</u>

2 Investment in non-current assets

	\$000
Net book value at end of most recent year (start of 'normal' year)	110,000
Less: Depreciation in 'normal' year	(30,000)
Less: Net book value of assets disposed (\$35 million – \$24.6 million)	(10,400)
Net book value before investment in non-current assets	<u>69,600</u>
Required level of non-current assets (\$110 million x 1.04)	<u>114,400</u>
Investment in non-current assets	<u>44,800</u>

3 Dividend remittance from Bowerscots

	\$000
Profit before tax	45,000
Less: Tax at 20%	(9,000)
Profit after tax	<u>36,000</u>
Remitted to Arthuro Co (36,000 x 60% x 0.95)	<u>20,520</u>

(ii) Dividend capacity required = $(90 \times 4/3)$ million x \$0.74 = \$88.8 million

Increase in dividend remittances required = \$88.8 million – \$76.652 million = \$12.148 million

Total dividend remittance required = $(\$20.52 \text{ million} + \$12.148 \text{ million})/0.95 = \34.387 million

Distribution % required = $(\$34.387 \text{ million}/\$36 \text{ million}) \times 100\% = 95.5\%$

(b) (i) Benefits of policy

The change of policy appears to be viable. Arthuro Co would have had some slack if it had not undertaken the rights issue. The new policy takes up this slack and effectively tops up the amount required with an increase in dividends.

The new policy appears to ensure that Arthuro Co will have sufficient funds to pay the required level of dividends and fulfil its own investment requirements. It will mean that Bowerscots Co has less retained funds available for investment, but Arthuro Co's investment opportunities may be more profitable.

Problems with policy

Arthuro Co is now close to taking all of Bowerscots Co's post-tax earnings as dividends. Only a limited fall in Bowerscots Co's earnings would be needed for its dividends not to be enough to sustain Arthuro Co's dividend level. A fall could easily happen given the highly competitive environment in which Bowerscots Co operates. If Arthuro Co wanted to increase its dividends over time, it could not do so by receiving extra dividends from Bowerscots Co.

As mentioned, an increase in dividend will leave Bowerscots Co's management with less retained earnings to invest. The amount of investment they can undertake with the reduced funds available may be insufficient to sustain earnings levels and hence dividends for Arthuro Co.

The tax regime between the two countries means that the group will suffer more tax. The amount of additional tax payable by Arthuro Co on Bowerscots Co's profits will remain unchanged, but the increase in dividends will mean an increase in withholding tax, for which Arthuro Co will receive no credit. Given the lower tax rate in Owlia, for tax purposes higher retained earnings for Bowerscots Co would be preferable, possibly with funds loaned to Arthuro Co rather than paid as dividends.

(ii) Agency problems

An agency situation arises between Arthuro Co's board (the principal) and Bowerscots Co's management (the agent). The proposals are likely to involve agency costs.

The policy limits the discretion of Bowerscots Co's management by restricting the amounts of retained funds available. However, this seems an inefficient way of exercising closer control, with agency costs including the increased liability for withholding tax. If Arthuro Co's board has concerns about Bowerscots Co's management, it would be better to make changes in the management team.

Even if Arthuro Co's board has confidence in Bowerscots Co's management team, it may nevertheless wish to oversee Bowerscots Co more closely, given the dependence of its dividend capacity on the amount received from the subsidiary. Again, increased supervision will involve increased agency costs in terms of time spent by Arthuro Co's management.

Bowerscots Co's management may feel that the new policy threatens their remuneration, as the limited funds available for investment will adversely affect the company's ability to maintain its profit levels. The managers may seek to join competitors, disrupting Bowerscots Co's management, jeopardising its ability to achieve its profit forecasts.

Resolving agency problems

Ways of motivating Bowerscots Co's management include making their remuneration less dependent on Bowerscots Co's results, for example, allowing them share options in Arthuro Co. If more of their remuneration depends on the group's results, Bowerscots Co's management may be happier with the suggested arrangement if they feel it will benefit the group. However, this motivational effect will be limited if Bowerscots Co's management feels that the group results are not influenced much by what they do.

Alternatively, a greater proportion of Bowerscots Co's management's remuneration could be by methods which are not dependent on its results, for example, increased salary or better benefits. However, by weakening the link between results and remuneration, it lessens their incentive to strive to produce the results needed to maintain the required level of dividend.

The decision-making on investments at group level may also have to change. Bowerscots Co will, under the new policy, have insufficient funds for major investments. Its management team should have the opportunity to make a case for retaining a greater percentage of funds, as they may have better investment opportunities than those available to the parent.

- 4 (a) Net receipt = \$10,150,000 – \$3,700,000 = \$6,450,000

Adverane Co will have a net dollar receipt in four months' time and needs to hedge against the Swiss Franc strengthening.

Money market

Borrow US\$: $\text{US\$}6,450,000 / (1 + [0.037/3]) = \text{US\$}6,371,419$

Convert into CHF at spot rate: $\text{US\$}6,371,419 / 1.1222 = \text{CHF}5,677,615$

Invest in CHF: $\text{CHF}5,677,615 \times (1 + [0.027/3]) = \text{CHF}5,728,714$

Futures

Buy Swiss Franc futures and use six-month futures contracts.

Basis

Assume that basis reduces to zero at contract maturity in a linear fashion.

Using spot rate, predicted futures rate = $1.1222 - ([1.1222 - 1.1204] \times 4/6) = 1.1210$

Alternatively, predicted futures rate = $1.1213 - ([1.1213 - 1.1204] \times 1/3) = 1.1210$

Expected receipt = $\$6,450,000 / 1.1210 = \text{CHF}5,753,791$

Number of contracts = $\text{CHF}5,753,791 / 125,000 = 46.03$ contracts, approximately 46 contracts

On the basis that futures give the higher expected receipt, they should be chosen, but Adverane Co should assess whether basis risk is likely to be significant. Adverane Co should also consider, as regards money market hedging, that CHF receipts could be used to pay off any existing CHF loans, or for other investment purposes, in which case the benefit to Adverane Co could be greater than hedging using futures.

- (b) (i) Use mid-spot rates to translate amounts.

Owed by	Owed to	Local currency m	CHF m
Adverane (Switzerland)	Bosha (Eurozone)	CHF15.90	15.90
Adverane (Switzerland)	Diling (Brazil)	CHF4.46	4.46
Bosha (Eurozone)	Cogate (USA)	€24.89	26.60
Bosha (Eurozone)	Diling (Brazil)	€18.57	19.84
Cogate (USA)	Adverane (Switzerland)	US\$27.08	24.16
Cogate (USA)	Diling (Brazil)	US\$5.68	5.07
Diling (Brazil)	Adverane (Switzerland)	BRL38.80	12.29
Diling (Brazil)	Bosha (Eurozone)	BRL51.20	16.22

Owed to	Owed by				Total
	Adverane (Sw) CHFm	Bosha (Eu) CHFm	Cogate (US) CHFm	Diling (Br) CHFm	
Adverane (Sw)			24.16	12.29	36.45
Bosha (Eu)	15.90			16.22	32.12
Cogate (US)		26.60			26.60
Diling (Br)	4.46	19.84	5.07		29.37
Owed by	(20.36)	(46.44)	(29.23)	(28.51)	
Owed to	36.45	32.12	26.60	29.37	
Net	16.09	(14.32)	(2.63)	0.86	

Under the terms of the arrangement, Bosha, the company with the largest debt, will pay Diling, the company with the smallest amount owed to it, CHF0.86 million. Bosha will pay Adverane CHF13.46 million and Cogate will pay Adverane CHF2.63 million.

- (ii) The advantage of using a central treasury for multilateral netting is that the central treasury can coordinate the information about inter-group balances. There will be a smaller number of foreign exchange transactions, which will mean lower commission and transmission costs. There will be less loss of interest through money being in transit. The foreign exchange rates available may be more advantageous as a result of large transaction sizes resulting from consolidation. The netting arrangements should make cash flow forecasting easier in the group.

- (c) Setting the transfer price at market price should enable a fair assessment of the performance of both the buying and selling divisions. Both internal and external sales will be accounted for at the same price. However, this may distort performance in that the costs of internal sales may be lower than external sales. For example, administration costs should be lower and there should be no costs of bad debts. These cost savings should be shared between the two divisions to give a fair picture. If the selling division has spare capacity, selling at incremental cost rather than market price may provide greater certainty that the buying division will use the selling division.

In theory, using market price should mean that the central treasury function has to intervene less. Simple market price provides an objective measure over which the divisions should agree. However, in reality, there may be complications that require central intervention. The market price may be difficult to determine or may fluctuate wildly, and central treasury may have to decide which price to use. If it is decided that an allowance should be made for costs of internal transfer being lower, central treasury may have to determine what this should be as it may vary significantly between products and divisions.

Specifying the transaction takes place at market price is designed to ensure that the buying division buys from the selling division, rather than an external supplier if the buying and selling division have failed to agree a price. The implicit assumption is that the buying division will use the selling division because of better service from, and greater dependability of, dealing within the group. This may not necessarily be the case. If the buying division previously purchased internally as a result of a low transfer price, forcing it to pay market price may mean it chooses an external supplier for non-price reasons.

	<i>Marks</i>
1 (a) Compare and contrast the two directors' opinions	4–5
Discussion of types of synergy benefits	4–5
Max	9
(b) 1–2 marks per point	Max 5
(c) (i) (Appendix 1)	
Estimate of future growth rate	1
Exclude interest from free cash flows	1
Estimate of free cash flows	2
Estimate of Foshoro Co's value	1
Estimate of equity value of Foshoro Co	1
	6
(ii) (Appendix 2)	
Combined company asset beta	1
Combined company equity beta	1
Combined company cost of equity	1
Combined company cost of capital	1
Combined company sales revenue (or operating profits) (years 1 to 4)	1
Combined company taxation amounts (years 1 to 4)	1
Combined company additional asset investment (years 1 to 4)	1
Combined company total value (years 1 to 4)	1
Combined company value after first four years	1
Combined company total market value	1
Combined company market value of equity	1
	11
(iii) (Discussion in report and appendix 3)	
Evaluation of benefit to Chikepe Co shareholders	3–4
Discussion of the limitations of the valuation method used	4
Max	7
Professional marks for part (c)	
Report format	1
Structure and presentation of the report	3
	4
(d) Discussion of mandatory-bid rule and principle of equal treatment	3–4
Discussion of effectiveness of poison pills and disposal of crown jewels	4–5
Max	8
Total	50

		<i>Marks</i>
2	(a) Operating cash flow excluding marketing costs	1
	Tax allowable depreciation	1
	Taxation	2
	Working capital	2
	Discount factor	2
	Base case NPV	1
	Issue costs	1
	Tax shield on loan	2
	Subsidy	1
	Tax shield on subsidy	1
	Adjusted present value	1
	Comments and conclusion	2
		17
(b) 1–2 marks per point		Max 8
		Total 25
3	(a) (i) Operating profit	1
	Interest payable	1
	Tax payable for normal activities	1
	Depreciation	1
	Profit on disposal of non-current assets	1
	Cash from disposal of non-current assets	1
	Investment in new non-current assets	2
	Investment in working capital	1
	Dividend remittance from Bowerscots Co	1
	Additional tax payable on Bowerscots Co profits	1
		11
	(ii) Dividend capacity required	1
	Increase in dividend remittance	1
	Revised level of dividend required (% level or absolute amount)	1
		3
	(b) Benefits of new policy	2–3
	Problems of new policy	2–3
		Max 5
	(c) Agency problems	3–4
	Solutions to problems	3–4
		Max 6
	Total	25

		<i>Marks</i>
4	(a) Calculation of net US\$ receipt	1
	Money market hedge	2
	Futures	
	Buy futures	1
	Predicted futures rate based on basis reduction	2
	Expected receipt	1
	Number of contracts	1
	Conclusion	1–2
	Max	9
	(b) (i) CHF amounts owed and owing	2
	Totals owed and owing	2
	Net amounts owed	1
	Payments and receipts	2
		7
	(ii) Advantages – 1 mark each	Max 3
	(c) Performance assessment	2–3
	Work of central treasury	2–3
	Buying internally	2–3
	Max	6
	Total	25

Strategic Professional – Options

Advanced Financial Management

Friday 7 September 2018



Time allowed: 3 hours 15 minutes

This question paper is divided into two sections:

Section A – This ONE question is compulsory and **MUST** be attempted

Section B – BOTH questions are compulsory and **MUST** be attempted

Formulae and tables are on pages 10–14.

Do NOT open this question paper until instructed by the supervisor.

This question paper must not be removed from the examination hall.

Paper AFM

Think Ahead

ACCA

The Association of
Chartered Certified
Accountants

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The question paper begins on page 3.

Section A – This ONE question is compulsory and MUST be attempted

- 1 Washi Co is a large, unlisted company based in Japan and its local currency is the Japanese Yen (JPY). It manufactures industrial equipment and parts. Initially Washi Co's customers consisted of other Japanese companies, but over the last 12 years it has expanded into overseas markets and also sources its materials from around the world. The company's board of directors (BoD) believes that the strategy of overseas investments, through subsidiary companies, branches and joint ventures, has directly led to the company's substantial increase in value in the past few years.

Washi Co's BoD is considering investing in a project based in Airone, whose currency is the Airone Rand (ARD). It believes that the project will be an important addition to the company's portfolio of investments, because Washi Co does not currently have a significant presence in the part of the world where Airone is located. It is intended that the project will commence in one year's time. Details of the project are given below.

Washi Co intends to finance the project through proceeds from an agreed sale of a small European subsidiary, with any remaining funding requirement being met by additional debt finance issued in Japanese Yen. The company is due to receive the proceeds from the sale of a European subsidiary company in six months' time and it will then invest these funds in short-dated Japanese treasury bills for a further six months before they are needed for the project. Washi Co has a centralised treasury department, which hedges expected future cash flows against currency fluctuations.

Funding and financial information

The agreed proceeds from the sale of the European subsidiary company receivable in six months' time are Euro (EUR) 80 million. The BoD is concerned about a negative fluctuation in EUR/JPY rate between now and in six months when the EUR 80 million will be received. Therefore, it has asked Washi Co's treasury department to hedge the expected receipt using one of currency forwards, currency futures or exchange traded currency options. Washi Co's treasury department has obtained the following information:

	JPY per EUR 1	ARD per EUR 1
Spot	129·2–132·4	92·7–95·6
Six-month forward rate	125·3–128·6	

Currency futures (contract size EUR 125,000, quotation JPY per EUR 1)

Four-month expiry	126·9
Seven-month expiry	125·2

Currency options (contract size EUR 125,000, exercise price quotation: JPY per EUR 1, premium quotation: JPY per EUR 1)

At an exercise price of JPY 126·0 per EUR 1

	Four-month expiry	Seven-month expiry
Calls	2·3	2·6
Puts	3·4	3·8

Annualised yield on short-dated Japanese treasury bills 1·20%

Airone's annual inflation rate is 9% currently, but has fluctuated markedly in the last five years. The Japanese annual inflation rate is 1·5% and has been stable for many years.

Pato Bank has offered Washi Co the possibility of using over-the-counter options to hedge the EUR receipt instead of exchange traded currency options.

Airone project information

A member of Washi Co's finance team has produced the following estimates of the Airone project which is expected to last for four years. The estimates are based on the notes given below but not on the further information. The estimates have been checked and verified independently for their numerical accuracy.

All figures are in ARD millions

Project year	0	1	2	3	4
Sales revenue		13,000	30,800	32,300	4,500
Costs		(10,200)	(24,200)	(24,500)	(3,200)
Tax allowable depreciation		(1,000)	(1,000)	(1,000)	(1,000)
Pre-tax profits		1,800	5,600	6,800	300
Tax at 15%		(270)	(840)	(1,020)	(45)
Tax allowable depreciation		1,000	1,000	1,000	1,000
Working capital	(400)				400
Investment in buildings	(5,750)				
Investment in machinery	(4,000)				
Cash flows in ARD	(10,150)	2,530	5,760	6,780	1,655

Notes (incorporated into the estimates above):

1. The estimates are based on using the end of the first year, when the project commences, as the start of the project (year 0). The numbers are given in ARD million (m).
2. The total investment required for the project is ARD 10,150m and separated into buildings, machinery and working capital in the table above. The machinery is eligible for tax allowable depreciation on a straight-line basis and the working capital is redeemable at the end of the project.
3. The impact of inflation has been incorporated into the sales revenue and cost figures, at Airone's current annual inflation figures.
4. Corporation tax has been included based on Airone's annual rate of 15%. The tax is payable in the year that the tax liability arises.

Further information (not incorporated into the estimates above):

1. Undertaking the Airone project will result in lost sales for Washi Co. These sales would have generated a pre-tax contribution of JPY 110m in the first year of the project, rising by the Japanese rate of inflation in the following years 2 to 4 of the project.
2. The Airone project costs include components which are made in Japan by Washi Co and would be imported to the Airone project. The pre-inflation revenues generated from the sale of the components are estimated to be as follows:

In JPY millions

Project year	1	2	3	4
Components revenue	1,200	2,400	2,500	300

These revenues are expected to increase by the Japanese inflation rate in years 2 to 4 of the project. The contribution which Washi Co expects to earn on these components is 25% of revenue.

3. The Japanese annual corporation tax rate is 30% and tax is payable in the year that the tax liability arises. A bilateral tax treaty exists between Japan and Airone, which permits offset of overseas tax against any Japanese tax liability on overseas earnings.
4. Washi Co's finance department has estimated a cost of capital of 12% to be used as a discount rate for the project.

Required:

- (a) Discuss how investing in overseas projects may enable Washi Co to gain competitive advantage over its competitors, who only invest in domestic projects. (5 marks)
- (b) Discuss the advantages and drawbacks of exchange traded option contracts compared with over-the-counter options. (5 marks)
- (c) Prepare a report for the board of directors of Washi Co which:
 - (i) Estimates the expected amount of JPY receivable under each hedge choice and the additional debt finance needed to fund the Airone project for the preferred hedge choice; (12 marks)
 - (ii) Estimates the net present value of the Airone project in Japanese Yen, based on the end of year one being the start of the project (year 0); (9 marks)
 - (iii) Evaluates the preferred hedge choice made, the debt finance needed and whether the Airone project should be undertaken, considering both financial and non-financial factors. (8 marks)Professional marks will be awarded in part (c) for the format, structure and presentation of the report. (4 marks)
- (d) Washi Co's chief operations officer (COO) has suggested that it would be more beneficial for the company to let its major subsidiary companies have their own individual treasury departments, instead of having one centralised treasury department for the whole company.

Required:

Discuss the validity of the COO's suggestion. (7 marks)

(50 marks)

Section B – BOTH questions are compulsory and MUST be attempted

2 Tillinton Co

Tillinton Co is a listed company which has traditionally manufactured children's clothing and toys with long lives. Five years ago, it began manufacturing electronic toys and has since made significant investment in development and production facilities. The first electronic toys which Tillinton Co introduced into the market were received very well, partly as it was seen to be ahead of its competitors in making the most of the technology available.

The country where Tillinton Co is listed has seen a significant general increase in share prices over the last three years, with companies in the electronic goods sector showing particularly rapid increases.

Statement by Tillinton Co's chief executive

Assume it is now September 20X3. Tillinton Co's annual report for the year ended 31 March 20X3 has just been published. Its chief executive commented when announcing the company's results:

'I am very pleased to report that revenue and gross profits have shown bigger increases than in 20X2, resulting in higher post-tax earnings and our company being able to maintain increases in dividends. The sustained increase in our share price clearly demonstrates how happy investors are with us. Our cutting-edge electronic toys continue to perform well and justify our sustained investment in them. Our results have also benefited from improvements in operational efficiencies for our older ranges and better working capital management. We are considering the development of further ranges of electronic toys for children, or developing other electronic products for adults. If necessary, we may consider scaling down or selling off our operations for some of our older products.'

Steph Slindon represents an institutional investor who holds shares in Tillinton Co. Steph is doubtful whether its share price will continue to increase, because she thinks that Tillinton Co's situation may not be as good as its chief executive suggests and because she believes that current share price levels generally may not be sustainable.

Financial information

Extracts from Tillinton Co's financial statements for the last three years and other information about it are given below.

Tillinton Co statement of profit or loss in years ending 31 March (all amounts in \$m)

	20X1	20X2	20X3
Sales revenue	1,385	1,636	1,914
Gross profit	381	451	528
Operating profit	205	252	300
Finance costs	(46)	(50)	(66)
Profit before tax	159	202	234
Taxation	(40)	(51)	(65)
Profit after tax	119	151	169
Dividends	(60)	(72)	(84)

Tillinton Co statement of financial position in years ending 31 March
(all amounts in \$m)

	20X1	20X2	20X3
Non-current assets	2,070	2,235	2,449
Cash and cash equivalents	10	15	15
Other current assets	150	130	125
Total non-current and current assets	2,230	2,380	2,589
Equity			
Ordinary shares (\$0.50)	400	400	400
Reserves	805	884	969
Total equity	1,205	1,284	1,369
Non-current liabilities	920	970	1,000
Current liabilities	105	126	220
Total equity and liabilities	2,230	2,380	2,589
Other information			
Market price per \$0.50 share (in \$, \$2.50 at 31 March 20X0, \$5.06 in September 20X3)	2.76	3.49	4.44
Earnings per share (\$)	0.15	0.19	0.21
Dividend per share (\$)	0.075	0.09	0.105
Analysis of revenue			
Electronic toys	249	319	390
Non-electronic toys	302	350	404
Clothing	834	967	1,120
	1,385	1,636	1,914
Analysis of gross profit			
Electronic toys	100	112	113
Non-electronic toys	72	88	105
Clothing	209	251	310
	381	451	528

Note: None of Tillinton Co's loan finance in 20X3 is repayable within one year.

Required:

- (a) **Evaluate Tillinton Co's performance and business prospects in the light of the chief executive's comments and Steph Slindon's concerns. Provide relevant calculations for ratios and trends to support your evaluation.**

Note: 10 marks are available for the calculations.

(20 marks)

- (b) **Discuss how behavioural factors may have resulted in Tillinton Co's share price being higher than is warranted by a rational analysis of its position.**

(5 marks)

(25 marks)

3 Selorne Co

Selorne Co is one of the biggest removal companies in Pauland, offering home and business removals. It has a number of long-term contracts with large businesses, although it has not won any new major contracts in the last two years. Selorne Co is listed on Pauland's stock market for smaller companies. Selorne Co is financed by a mixture of equity and short and long-term debt, but its gearing level is below the average for its sector.

Selorne Co has four executive directors, who each own 20% of the company's share capital, with the other 20% owned by external shareholders. Selorne Co has paid a constant dividend since it has been listed and its share price has risen slightly over the last three years.

Selorne Co is based in a number of the large cities and towns in Pauland and owns the majority of the sites where it is located. Many of its employees have worked for the company for a long time. Drivers of the lorries used by Selorne Co are required to have a special, heavy vehicles licence. Salary levels at Selorne Co are relatively high compared with other companies in the sector.

Chawon Co

Selorne Co is currently considering making a bid for Chawon Co, an unlisted company specialising in distribution and delivery services. Chawon Co is owned 100% by its founder, Chris Chawon. Chawon Co has built up a portfolio of small contracts over time. It has made unsuccessful bids for two larger contracts over the last 12 months, the bids being rejected primarily because Chawon Co was not felt to be big enough to be able to guarantee the level of service required.

Chawon Co is based in many of the same cities and towns where Selorne Co is located, although Chawon's premises are all rented. The drivers of Chawon's vehicles do not require a heavy vehicles licence. Chawon Co has a few long-serving employees who are mostly centre managers. Most of its drivers and staff, however, stay at Chawon Co for only a short time. Salary levels are low, although Chawon Co pays high levels of overtime and high bonuses if target profit levels are achieved. Chawon Co is highly geared, leading to recent media speculation about its financial viability.

Terms of bid for Chawon Co

In initial discussions about the acquisition, Chris Chawon indicated that he would prefer the consideration to be a share-for-share exchange, the terms being one Chawon Co share for five Selorne Co shares.

Chawon Co has 2 million \$1 shares in issue, and Selorne Co has 50 million \$0.50 shares in issue. Each Selorne Co share is currently trading at \$6.50, which is a multiple of 8 of its free cash flow to equity. The multiple of 8 can be assumed to remain unchanged if the acquisition takes place. Chawon Co's free cash flow to equity is currently estimated at \$7 million, with an expected annual growth rate of 3%, and it is expected to generate a return on equity of 15%.

Chris Chawon expects that the total free cash flows to equity of the combined company will increase by \$5 million due to synergy benefits. He believes that Selorne Co will be able to win more contracts because it is larger and because it will be diversifying the services which it offers. He also believes that significant operational synergies can be achieved, pointing out the time Selorne Co drivers spend idle during the winter months when removal activity is traditionally lower. Chris Chawon believes that he can achieve the synergies if he is given management responsibility for the operational reorganisation, including dealing with the staff employment and retention issues. Chris Chawon thinks that synergies could also be achieved in central administration and in premises costs.

The chief executive and the finance director of Selorne Co are in favour of bidding for Chawon Co. However, one of the other executive directors is opposed to the bid. He is sceptical about the level of synergies which can be achieved and does not want Chris Chawon to be brought into the management of Selorne Co. He suggests that if the bid is to go ahead, it should be a cash offer rather than a share exchange. Selorne Co's chief executive has responded that Chris Chawon is likely to ask for a higher equivalent price if the purchase is for cash.

Financing the bid for Chawon Co

Selorne Co's finance director has pointed out that Selorne Co will need additional funding if Chawon Co is purchased for cash. He has suggested that there may be a number of possible sources of finance:

- A rights issue
- A fixed rate, long-term, bank loan
- A three-year, unsecured, mezzanine loan facility
- Convertible debt, with conversion rights being exercisable in five years' time

Required:

- (a) (i) Estimate the equity value of the combined company and the expected additional value arising from the combination of Selorne Co and Chawon Co. (6 marks)
- (ii) Estimate the share of the gain from the combination created for Chris Chawon and the share of the gain created for Selorne Co's shareholders and comment on your results. (6 marks)
- (b) Evaluate how reliable the estimates of the synergies for the combined company are likely to be and discuss the factors which may prevent the forecast synergies from being achieved. (7 marks)
- (c) Discuss the factors which Selorne Co's board will consider when determining which source or sources of finance are chosen to finance a possible cash bid for the share capital of Chawon Co. (6 marks)

(25 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d^i) \frac{V_d}{V_e}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i(E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_o = \frac{D_o(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = br_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \qquad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model

$$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$$

Where:

$$d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$$

$$d_2 = d_1 - s\sqrt{t}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

		<i>Discount rate (r)</i>									
<i>Periods</i>											
(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

		<i>Discount rate (r)</i>									
<i>Periods</i>											
(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Answers

- 1 (a) Washi Co may want to invest in overseas projects for a number of reasons which result in competitive advantage for it, for example:

Investing overseas may give Washi Co access to new markets and/or enable it to develop a market for its products in locations where none existed before. Being involved in marketing and selling products in overseas markets may also help it gain an understanding of the needs of customers, which it may not have had if it merely exported its products.

Investing overseas may give Washi Co easier and cheaper access to raw materials it needs. It would therefore make good strategic sense for it to undertake the overseas investment.

Investing in projects internationally may give Washi Co access to cheaper labour resources and/or access to expertise which may not be readily available in Japan. This could therefore lead to reduction in costs and give Washi Co an edge against its competitors.

Closer proximity to markets, raw materials and labour resources may enable Washi Co to reduce its costs. For example, transportation and other costs related to logistics may be reduced if products are manufactured close to the markets where they are sold.

Risk, such as economic risk resulting from long-term currency fluctuations, may be reduced where costs and revenues are matched and therefore naturally hedged.

Washi Co may increase its reputation because it is based in the country within which it trades leading to a competitive edge against its rivals.

International investments might reduce both the unsystematic and systematic risks for Washi Co if its shareholders only hold well diversified portfolios in domestic markets, but not internationally.

(Note: Credit will be given for alternative valid areas of discussion)

(b) Advantages

Exchange traded options are readily available on the financial markets, their price and contract details are transparent, and there is no need to negotiate these. Greater transparency and tight regulations can make exchange traded options less risky. For these reasons, exchange traded options' transaction costs can be lower.

The option buyer can sell (close) the options before expiry. American style options can be exercised any time before expiry and most traded options are American style options, whereas over-the-counter options tend to be European style options.

Disadvantages

The maturity date and contract sizes for exchange traded options are fixed, whereas over-the-counter options can be tailored to the needs of parties buying and selling the options.

Exchange traded options tend to be of shorter terms, so if longer term options are needed, then they would probably need to be over-the-counter.

A wider range of products (for example, a greater choice of currencies) is normally available in over-the-counter options markets.

(c) Report to the board of directors (BoD), Washi Co

Introduction

This report evaluates whether or not Washi Co should invest in the Airone project and the amount of debt finance required of JPY 3,408.6 million (appendix 1) to fund the project. The evaluation considers both the financial and the non-financial factors.

Evaluation of the preferred hedge choice and debt finance required

The income from the sale of the European subsidiary is maximised when futures contracts are used. Therefore, these are chosen as Washi Co will borrow the least amount of debt finance as a result. However, compared to the forward contract, futures are marked-to-market daily and require a margin to be placed with the broker. This could affect Washi Co's liquidity position. The assumption has been made that basis reduces proportionally as the futures contracts approach expiry, but there is no guarantee that this will be the case. Therefore, basis risk still exists with futures contracts. Although forward contracts give a smaller return, there is no basis risk and margin requirements. However, they do contain a higher risk of default as they are not market traded. Options give the lowest return but would give Washi Co the flexibility of not exercising the option should the Euro strengthen against the Yen.

Although the EUR 80 million receipt from the sale of the subsidiary has been agreed, there may be a risk that the sale may fall through and/or the funds or some proportion of the funds are not received. Washi Co may need to assess and factor in this risk, however small it may be.

The amount of interest on deposit is based on the current short-dated Japanese treasury bills and the estimate of the borrowing requirement is computed from the predicted exchange rate between ARD and JPY in a year's time, based on the purchasing power parity. Both these estimates could be inaccurate if changes occur over the coming months. Although insufficient

information is provided for a financial assessment, Washi Co should explore the possibility of converting the EUR into ARD immediately on receipt and keeping it in an ARD bank account until needed, instead of first converting EUR into JPY and then into ARD.

Using debt finance to make up any shortfall in the funding requirement may be appropriate for Washi Co given that it is an unlisted company and therefore access to other sources of funding may be limited. Nevertheless, Washi Co should assess how the extra borrowing would affect any restrictive covenants placed on it and the impact on its cost of capital. Since the amount seems to be small in the context of the project as a whole, this may not be a major problem.

Washi Co should also explore whether or not investing in the Airone project restricts its ability to fund other projects or affects its ability to continue normal business activity, especially if Washi Co is facing the possibility of hard capital rationing.

Evaluation of the Airone project

The net present value of the Airone project is estimated to be JPY (457) million (appendix 2). Given the negative net present value, the initial recommendation would be to reject the project. However, given that the result is marginal, Washi Co should consider the following factors before rejecting the project.

At present, Washi Co does not have a significant presence in the part of the world where Airone is located. Taking on the project may make good strategic sense and provide a platform for Washi Co to establish its presence in that part of the world.

Furthermore, once Washi Co has established itself in Airone, it may be able to develop further opportunities and new projects. The value of these follow-on options has not been incorporated into the financial assessment. Washi Co should explore the possibility of such opportunities and their possible value.

The financial assessment ends abruptly at the end of the four years. No indication is given on what would happen to the project thereafter. It may be sold as a going concern or, if closed, its land and assets may be sold. The cash flows from these possible courses of action need to be incorporated into the assessment, and these could make the project worthwhile.

A number of assumptions and estimates would have been made in the financial assessment. For example, the rate of inflation used for future figures is the current rate and the tax rate used is the current rate, these may well change in the coming years. Therefore, it is best to undertake sensitivity analysis and produce a number of financial assessments before making any firm commitment to proceed with the project or deciding to reject it.

Conclusion and recommendation

The income from the sale of the European subsidiary is maximised when futures contracts are used, but Washi Co should weigh this against the benefits and drawbacks of all hedging instruments before making a final decision.

Although the project is currently giving a negative net present value, rejecting it at the outset is premature. A number of factors, discussed above, need to be considered and assessed before a final decision is made. Sensitivity analysis would be very helpful in this respect.

Finally, Washi Co should consider alternative uses for the funding which will be dedicated to the project. These alternative uses for the finance need to be considered before any decision is made, especially if Washi Co is facing the possibility of hard capital rationing.

Report compiled by:

Date

APPENDICES:

Appendix 1 (Part (c) (i)): Japanese Yen receivable from sale of European subsidiary under each hedging choice and the additional debt finance needed to fund the Airone project

Forward rate

Since it is a EUR receipt, the lock-in rate of JPY125·3 per EUR will be used.

Expected receipt from sale: EUR 80m x 125·3 = JPY 10,024m

Futures contracts

The futures contracts need to show a gain when the Euro depreciates against the Yen, therefore a short position is needed, using the seven-month contracts. It is assumed that basis will depreciate proportionally to the time expired.

Predicted futures rate

$125·2 + 1/3 \times (126·9 - 125·2) = 125·8$

[Or: $125·2 + 1/7 \times (129·2 - 125·2) = 125·8$]

Number of contracts sold = EUR 80,000,000/EUR 125,000 = 640 contracts

Expected receipt from sale: EUR 125,000 x 640 x 125·8 = JPY 10,064m

Options contracts

640 seven-month put options contracts will be purchased to protect against a depreciation of Euro.

If options are exercised:

EUR 125,000 x 640 x 126 = JPY 10,080m

Premium payable = JPY 3·8 x 125,000 x 640 = JPY 304m

Net income = JPY 10,080m – JPY 304m = JPY 9,776m

Conclusion

Futures contracts give the highest receipt and will, therefore, be used to hedge the expected Euro receipt.

Receipt invested

Invested for further six months till needed for the Airone project.

$\text{JPY } 10,064\text{m} \times (1 + (0.012/2)) = \text{JPY } 10,124.4\text{m}$

Spot cross rates: 0.70 – 0.74 ARD per JPY 1

$[92.7/132.4 = 0.70 \text{ and } 95.6/129.2 = 0.74]$

Expected ARD/JPY conversion spot rate in 12 months = $0.70 \times 1.09/1.015 = 0.75$

Additional debt finance needed to fund Airone project

Investment amount required = $\text{ARD } 10,150\text{m}/0.75 = \text{JPY } 13,533\text{m}$

Debt finance required = $\text{JPY } 13,533\text{m} - \text{JPY } 10,124.4\text{m} = \text{JPY } 3,408.6\text{m}$

Appendix 2 (Part (c) (ii): Airone project net present value

Project year	0	1	2	3	4
Cash flows in ARD (millions)	(10,150)	2,530	5,760	6,780	1,655
Future exchange rate ARD/JPY (W1)	0.75	0.81	0.87	0.93	1.00

In JPY million

Project year	0	1	2	3	4
Cash flows	(13,533)	3,123	6,621	7,290	1,655
Lost contribution		(110)	(112)	(113)	(115)
Tax saving on lost contribution (30%)		33	34	34	35
Contribution: components (W2)		300	609	644	79
Tax on components cont. (30%)		(90)	(183)	(193)	(24)
Additional tax payable (15%) (W3)		(333)	(966)	(1,097)	(45)
Net cash flows	(13,533)	2,923	6,003	6,565	1,585
Present value (discounted at 12%)	(13,533)	2,610	4,784	4,674	1,008

Expected net present value is JPY (457)m

Workings:

W1: Predicted future exchange rate (ARD/JPY)

Project year	0	1	2	3	4
	0.70 x	0.75 x	0.81 x	0.87 x	0.93 x
	1.09/1.015	1.09/1.015	1.09/1.015	1.09/1.015	1.09/1.015
Exchange rate ARD/JPY	0.75	0.81	0.87	0.93	1.00

W2: Components contribution

Project year	1	2	3	4
In JPY millions				
Components revenue (post inflation)	1,200	2,436	2,576	314
Contribution (25%)	300	609	644	79

W3: Additional tax payable

Project year	1	2	3	4
Pre-tax profits (ARD m)	1,800	5,600	6,800	300
Additional tax payable at 15% (JPY m)	$1,800/0.81 \times 0.15 = 333$	$5,600/0.87 \times 0.15 = 966$	$6,800/0.93 \times 0.15 = 1,097$	$300/1.0 \times 0.15 = 45$

- (d) It is difficult to conclude definitively whether a centralised treasury department is beneficial or not in all circumstances and for all companies. It depends on each company itself and the circumstances it faces. Washi Co should take this into account before making a final decision.

Benefits of a centralised treasury department

Having a centralised treasury management function avoids the need to have many bank accounts and may therefore reduce transactions costs and high bank charges.

Large cash deposits may give Washi Co access to a larger, diverse range of investment opportunities and it may be able to earn interest on a short-term basis, to which smaller cash deposits do not have access. On the other hand, if bulk borrowings are required, it may be possible for Washi Co to negotiate lower interest rates, which it would not be able to do on smaller borrowings.

A centralised treasury function can offer the opportunity for Washi Co to match income and expenditure and reduce the need for excessive risk management, and thereby reduce costs related to this.

A centralised treasury management department could hire experts, which smaller, diverse treasury management departments may not have access to.

A centralised treasury function may be better able to access what is beneficial for Washi Co as a whole, whereas local treasury functions may lead to dysfunctional behaviour.

Benefits of separate (decentralised) treasury departments

It could be argued that decentralised treasury departments are better able to match and judge the funding required with the need for asset purchases for investment purposes on a local level. Therefore, they may be able to respond quicker when opportunities arise and so could be more effective and efficient.

Individual departments within a subsidiary may have better relationships with the treasury departments of that subsidiary and are therefore able to present their case without lengthy bureaucratic delays.

Ultimately, the benefits may be implicit rather than explicit. Having decentralised treasury departments may make the subsidiary companies' senior management and directors more empowered and have greater autonomy. This in turn may increase their levels of motivation, as they are more in control of their own future, resulting in better decisions being made.

2 (a) Profitability

Tillinton Co's chief executive is correct in saying that the absolute increase in revenue and gross profits on all products was greater in 20X3 than 20X2, but the % increase in revenue was smaller on all products, and the % increase in gross profit on toys was also lower. The % increase on the electronic toys shows the biggest fall, possibly indicating greater competition.

The improvements in operations mentioned by the chief executive seem to have maintained gross and operating profit margins and resulted in the absolute overall increases in gross and operating profits. However, this aspect of performance is almost all attributable to Tillinton Co's older products. The gross profit on electronic toys has hardly increased and the gross profit margin has fallen over the last two years. Although the margin remains higher than on the other products, even the 20X3 margin may not be sustainable. If competitors are starting to catch up with Tillinton Co, then the profit margin on the current range of electronic toys may continue to fall in future years, as prices fall to maintain market share.

Despite the emphasis on developing the products, the revenue generated by electronic toys is still below the revenue generated by non-electronic toys.

Asset turnover and return on capital employed have risen significantly over the last two years. However, part of the reason for the 20X3 increases was the significant increase in current liabilities. The further amount of investment which the chief executive appears to be contemplating suggests that asset turnover and return on capital employed may fall in future years, particularly if profit margins on electronic toys cannot be sustained.

Liquidity

The figures for other current assets seem to support the chief executive's contention that working capital is being managed better, as other current assets are falling as revenue and gross profits are rising.

However, the fall of the current ratio from 1.52 to 0.64 is significant, and the biggest reason for the fall in 20X3 was the large increase in current liabilities. Cash balances have remained at a low level, despite higher revenues and profit. Possibly there is now a bank overdraft, which could have contributed to the significant increase in finance costs between 20X2 and 20X3. It would seem that cash reserves have been exhausted by the combination of investment in non-current assets and the payments to finance providers (both interest and dividends), and Tillinton Co is more dependent on short-term liability finance. Slowdown in any product area, particularly electronic toys, may result in significant liquidity problems.

Solvency

Gearing has fallen over the last two years, but this is due to share price increases which may not be sustainable. If book values rather than market values are used to calculate gearing, the fall in gearing is much smaller. More information is needed about why finance costs have increased so much, leading to the deterioration in interest rate cover. Tillinton Co has only taken out an additional \$30 million in long-term loans. Although costs on these may be higher than on its current loans, this would not account for all the increase in finance costs. As discussed above, Tillinton Co may be making use of overdraft finance. The fact that current liabilities have increased much more than non-current liabilities could be an indication that Tillinton Co is having problems raising all the longer term loan finance which it requires.

The figures suggest that Tillinton Co's board needs to review future financing carefully if the company wants to make further investment in electronic products. At some stage, the board will have to consider raising further finance, either through an issue of shares or through selling off parts of its operations.

Investor ratios

Both earnings and dividends per share have risen since 20X1, which could help explain the significant increase in share price. Dividend cover has remained around 2.0 despite an increase in earnings. Although dividends have increased, dividend yield has fallen since 20X1. The increase in total shareholder return is due solely to the increases in share price, which have also resulted in the price-earnings ratio increasing in 20X3. The current rate of share price increase does not appear to be warranted by the most recent results and may be partly due to generous dividend levels, which may not be sustainable if more cash is required for investment.

Conclusion

Despite the chief executive's optimistic message in the annual report, the benefits from the electronic toys development may be short-lived. There appears to be a mismatch between investment, dividend and financing policies. As discussed, margins on current products may fall further and there is no guarantee that margins on new electronic toys or other products will be higher if competition generally is increasing.

Further significant investment in electronic toys or other goods may be difficult to finance. Increased reliance on short-term finance is clearly not sustainable, but obtaining more debt may be problematic, particularly if gearing levels rise as share prices fall. Tillinton Co seems reluctant to take advantage of high share price levels to issue equity capital. This, plus the increase in dividends, may indicate Tillinton Co's board is unwilling to risk upsetting shareholders, despite the large increases in share price. The chief executive may be right in saying that funds may have to be obtained by selling off one of the other parts of the business, but revenue and profits from the older products may be more sustainable. An increased concentration on electronic products may be a high-risk strategy. Possibly, if investors become less positive towards the electronic goods sector, they may realise this, resulting in an increase in cost of capital and a fall in share price.

Note: Credit will be given for relevant, alternative approaches to the calculations and discussion.

Appendix

	20X1	20X2	20X3
Profitability			
% increase in revenue		18.1	17.0
Gross profit %	27.5	27.6	27.6
% increase in gross profit		18.4	17.1
Operating profit %	14.8	15.4	15.7
% increase in operating profit		22.9	19.0
Asset turnover (revenue/(total assets – current liabilities))	0.65	0.73	0.81
Return on capital employed % (operating profit % x asset turnover)	9.6	11.2	12.7
Liquidity			
Current ratio	1.52	1.15	0.64
Solvency			
Gearing (non-current liabilities/(non-current liabilities + market value of share capital)) %	29.4	25.8	22.0
Gearing (non-current liabilities/(non-current liabilities + book value of share capital + reserves)) %	43.3	43.0	42.2
Interest cover	4.5	5.0	4.5
Investors			
Earnings per share (\$)	0.15	0.19	0.21
Dividend per share (\$)	0.075	0.09	0.105
Dividend cover	1.98	2.10	2.01
Market price per \$0.50 share	2.76	3.49	4.44
Price/earnings ratio	18.4	18.4	21.1
Dividend yield % (dividend per share/share price)	2.72	2.58	2.36
Share price gain/(loss) %	10.40	26.45	27.22
Total shareholder return %	13.12	29.03	29.58
Types of product			
Electronic toys			
% increase in revenue		28.1	22.3
Gross profit %	40.2	35.1	29.0
% increase in gross profit		12.0	1.0
Other toys			
% increase in revenue		15.9	15.4
Gross profit %	23.8	25.1	26.0
% increase in gross profit		22.2	19.3
Clothing			
% increase in revenue		15.9	15.8
Gross profit %	25.1	26.0	27.7
% Increase in gross profit		20.1	23.5

(Note: Credit will be given for alternative, relevant, calculations and discussion. Candidates will not be expected to complete all the calculations above to obtain 10 marks.)

- (b) Tillinton Co's shares may be overvalued because share prices generally are too high. The situation may be a stock market bubble. Share prices have been rising consistently recently and this could be encouraging investors to buy more shares, further increasing share prices.

The bubble could be more localised. Tillinton Co seems to be positioning itself as much in terms of producing technologically-advanced electronic products as manufacturing toys. The electronic goods sector may be more likely than other sectors to attract investors on the basis of future profit potential, with investors possibly following a herd instinct, investing because others have been investing in the expectation of future gains.

Possibly, investors are more persuaded by the chairman's confident language and future promises than they are by the concerns the figures suggest. They may also be paying excessive attention to the most recent set of results, rather than seeing them in the context of whether they can be sustained in the future.

If investors are attempting to make a valuation, they could prefer using a model which confirms what they believe the shares are worth (confirmation bias), rather than one which gives a more reliable indication of value. As discussed above, shareholders may be basing their estimates of value on the recent increases in dividend, even though it may be doubtful whether this is sustainable.

- 3 (a) (i)** Selorne Co current equity value = 50m shares x \$6.50 = \$325m
 Chawon Co current equity value = \$7 million x 1.03/(0.15 – 0.03) = \$60.1m
 Selorne Co free cash flow to equity = \$325m/8 = \$40.6m
 Combined company valuation = (\$40.6m + \$7m + \$5m) x 8 = \$420.8m
 Additional value created = \$420.8m – \$325m – \$60.1m = \$35.7m
- (ii)** Chris Chawon will hold 2m x 5 = 10m shares in combined company
 Value per share in combined company = \$420.8m/(50m + 10m) = \$7.01
 Value of Chris Chawon's shareholding = 10m x \$7.01 = \$70.1m
 Gain created for Chris Chawon = \$70.1m – \$60.1m = \$10m
 Gain created for Selorne Co shareholders = \$35.7m – \$10m = \$25.7m
 Chris Chawon will have a 16.7% (10m/(50m + 10m)) shareholding in the combined company but 28.0% (\$10m/\$35.7m) of the gain on the combination will be attributable to him. Shareholders who are doubtful about the merger may question whether this is excessive, as possibly Chawon Co's desire to sell is being prompted by the company struggling to remain solvent.

(b) Reliability of synergy estimates

The reliability of the estimates may vary depending on the synergies involved. The synergies relating to size and services offered will depend on the ability to gain large contracts and neither company has had recent success in doing this. However, the contracts recently bid for by Chawon Co might have been won if the larger combined company had bid. The synergies relating to operations and working practices may be difficult to obtain if it is difficult to change the employment conditions of Selorne Co drivers. Claims that improved driver utilisation may reduce spare capacity may be true, but there is likely to be less spare capacity anyway if more contracts are won.

Other synergies may be easier to obtain. Duplication of premises in some locations should be eliminated easily, providing Chawon Co does not have onerous rental contracts and there is space on Selorne Co's sites. Combining central administrative functions should reduce some staffing costs, although these are likely to be smaller synergies than the potential operational synergies.

Problems with achieving synergies

A significant problem may be lack of unity at the top of the company. Selorne Co's directors are not all keen on the acquisition and this may spill over into being unable to agree on a clear post-acquisition plan. If lack of unity at board level becomes apparent to staff, it may be difficult to achieve unity at employee level.

Chris Chawon's role in the combined company may also make synergies difficult to achieve. He will have a significant shareholding and a place on the board, so it will be difficult for him not to be involved. Possibly he has the abilities and desire to achieve changes in operational practices which other board members lack. However, if Chris is given the leading role he requires, there may be a change in management style which may upset long-serving Selorne Co staff. Some may leave, jeopardising the continuity which seems to have been an important part of Selorne Co's success.

Another reason for possible problems with staff is the differing remuneration arrangements. Selorne Co's staff may have stayed with the company because both their job prospects and their remuneration have been safe. Attempts to change their employment conditions may lead to resistance and employee departures. Ex-Chawon Co employees who have been with the company for a while may expect salaries to be increased to be more in line with Selorne Co's employees, particularly if bonus arrangements become less generous.

The success of the acquisition may also depend on how well the staff of the two businesses integrate. Integration may be difficult to achieve. Many of Chawon Co's staff will not have the necessary licence to drive the Selorne Co lorries and may not wish to go through the process of obtaining this licence. Selorne Co drivers may be reluctant to drive the smaller vehicles. Staff sticking to what they have been used to driving is likely to prolong a 'them and us' culture.

(c) Availability

Although the finance director has identified possible sources of finance, there is no guarantee that they will necessarily be available. The success of a rights issue may well depend on the willingness and ability of the director-shareholders to subscribe. It may be difficult to find others willing to take up the directors' rights if they do not subscribe, as the directors' unwillingness may be seen as indicating a lack of confidence in the business. A rights issue may also take longer to arrange than other methods, which may be significant if Selorne Co needs the finance quickly to complete the acquisition.

Obtaining a bank loan or mezzanine finance may be difficult if Selorne Co takes on Chawon Co's debt and is viewed as too highly geared as a result. The success of a convertible debt issue may depend on the terms, also how possible subscribers view the future prospects of Selorne Co and the marketability of the shares.

Cost

Cost will be another significant factor. The cost of equity will normally be viewed as higher anyway than the cost of debt. Issue costs of equity are likely to be higher than those of debt. As Selorne Co's share price is stable, its current external shareholders appear content with the dividends paid, so there does not appear to be pressure to increase them. In any case, the board is not required to pay dividends every year.

Fixed interest cost on the bank loan may become a burden if interest rates fall, but the cost can be forecast with certainty. Because the mezzanine finance is unsecured, it is likely to have a higher interest cost than the bank loan. The rights of conversion to shares attaching to the convertible debt will mean a lower rate can be set for this, but the cost will depend on how appealing the possibility of conversion is. Again, the finance cost of debt will depend on the finance providers' attitude towards the increased debt burden resulting from the acquisition of Chawon Co.

Director preferences

The choice will also be determined by Selorne Co's board's attitude to gearing as well as how the possible finance providers view the company's gearing level. The board may feel that Selorne Co has reached, or exceeded, the gearing level which it would regard as desirable by taking on Chawon Co's debt. If this is the case, the board would have to use equity finance. The board may also be influenced by how gearing is likely to change over time. Over the next few years gearing may fall as Selorne Co makes profits and (hopefully) its share price increases. Chawon Co's debt may be repaid and not replaced. The convertible debt and mezzanine finance will also not be long-term sources of debt finance.

Control of Selorne Co

Selorne Co's board decision may also be determined by the implications of the different sources of finance for control of the company. The directors' control of the company will not be diminished if a rights issue is used and they take up their rights. An issue of shares arising from the convertible debt would change the balance of shareholdings, so the directors would have to decide how significant this would be. Mezzanine finance may also offer conversion rights, but possibly these could only be exercised if Selorne Co defaulted, which the board may view as unlikely.

Using a bank loan will have no impact on share capital, but the bank may impose restrictions which the directors are unwilling to bear, particularly if high gearing is an issue. These conditions could include restrictions on the sale of assets, limitations of dividends, or requiring accounting figures, for example, liquidity or solvency ratios, not to go beyond certain levels.

Mix of finance

Ultimately the board may also consider the possibility of a mix of finance. The offer could be backed by a core of equity finance from a rights issue, but if Selorne Co has to pay a higher price than expected, the difference could be made up by mezzanine finance.

Note: Credit will be given for alternative, relevant answers.

		<i>Marks</i>
1	(a) 1–2 marks per valid point	Max <u>5</u>
	(b) Advantages	2–3
	Disadvantages	<u>2–3</u>
	Max	5
(c)	(i) (Appendix 1)	
	Amount to be received based on forward rate	1
	Decision to go short on futures	1
	Estimate of futures rate in six months based on basis	1
	Amount to be received based on futures market	1
	Decision to purchase put options	1
	Premium payable	1
	Amount to be received based on options market	1
	Decision: select appropriate hedge instrument	1
	JPY receivable following further six months of investment	1
	Estimate of current cross rate(s)	1
	Estimate of ARD/JPY rate in one year's time	1
	Debt borrowing required	1
		<u>12</u>
	(ii) (Appendix 2)	
	Estimate of future ARD/JPY rates	1
	Lost contribution	1
	Tax saving on lost contribution	1
	Contribution from sales of components	2
	Tax on contribution from components sales	1
	Additional tax payable in Japan	2
	Present values and net present value	1
		<u>9</u>
	(iii) (Report on project and funding evaluation)	
	Evaluation of hedge choice and debt finance required	3–4
	Evaluation of Airone project	3–4
	Conclusion	1–2
	Max	8
	Professional marks for part (c)	
	Report format	1
	Structure and presentation of the report	3
		<u>4</u>
(d)	Benefits of a centralised treasury department	3–4
	Benefits of decentralised treasury departments	2–3
	Over-arching commentary	1
	<i>(Note: Max 6 marks if no over-arching commentary)</i>	
	Max	7
	Total	50

		<i>Marks</i>
2	(a) Ratios	
	Profitability	2–3
	Liquidity	1
	Solvency	1–2
	Investor	3–4
	Other ratios and trends including product type split	2–3
	Max	10
	Discussion	
	Profitability	3–4
	Liquidity	2–3
	Solvency	2–3
	Investor	2–3
	Conclusion	2–3
	Max	10
	(b) Up to 2 marks per relevant point	Max 5
	Total	25
3	(a) (i)	
	Valuation of Selorne Co	1
	Valuation of Chawon Co	2
	Valuation of Selorne Co FCFE	1
	Valuation of combined company	1
	Additional value created	1
		6
	(ii)	
	Value per share combined company	1
	Value of Chris Chawon's shareholding in combined company	1
	Share of gain created for Chris Chawon	1
	Share of gain created for Selorne Co shareholders	1
	Comments	2
		6
	(b) Up to 2 marks per relevant point discussed. Discussion must be related to Selorne Co to obtain 2 marks for a point.	
	Reliability of estimates	3–4
	Problems with achieving synergies	3–4
	Max	7
	(c) Up to 2 marks per relevant factor discussed	Max 6
	Total	25

Strategic Professional – Options

Advanced Financial Management (AFM)

Friday 7 December 2018



AFM

Time allowed: 3 hours 15 minutes

This question paper is divided into two sections:

Section A – This ONE question is compulsory and MUST be attempted

Section B – BOTH questions are compulsory and MUST be attempted

Formulae and tables are on pages 6–10.

Do NOT open this question paper until instructed by the supervisor.

This question paper must not be removed from the examination hall.

Think Ahead

ACCA

The Association of
Chartered Certified
Accountants

Section A – This ONE question is compulsory and MUST be attempted

- 1 Around seven years ago, Opao Co, a private conglomerate company involved in many different businesses, decided to obtain a listing on a recognised stock exchange by offering a small proportion of its equity shares to the public. Before the listing, the company was owned by around 100 shareholders, who were all closely linked to Opao Co and had their entire shareholding wealth invested in the company. However, soon after the listing these individuals started selling their shares in Opao Co, and over a two-year period after the listing, its ownership structure changed to one of many diverse individual and institutional shareholders.

As a consequence of this change in ownership structure, Opao Co's board of directors (BoD) commenced an aggressive period of business reorganisation through portfolio and organisational restructuring. This resulted in Opao Co changing from a conglomerate company to a company focusing on just two business sectors: financial services and food manufacturing. The financial press reported that Opao Co had been forced to take this action because of the change in the type of its shareholders. The equity markets seem to support this action, and Opao Co's share price has grown strongly during this period of restructuring, after growing very slowly initially.

Opao Co recently sold a subsidiary company, Burgut Co, through a management buy-in (MBI), although it also had the option to dispose of Burgut Co through a management buy-out (MBO). In a statement, Opao Co's BoD justified this by stating that Burgut Co would be better off being controlled by the MBI team.

Opao Co is now considering acquiring Tai Co and details of the proposed acquisition are as follows:

Proposed acquisition of Tai Co

Tai Co is an unlisted company involved in food manufacturing. Opao Co's BoD is of the opinion that the range of products produced by Tai Co will fit very well with its own product portfolio, leading to cross-selling opportunities, new innovations, and a larger market share. The BoD also thinks that there is a possibility for economies of scale and scope, such as shared logistic and storage facilities, giving cost saving opportunities. This, the BoD believes, will lead to significant synergy benefits and therefore it is of the opinion that Opao Co should make a bid to acquire Tai Co.

Financial information related to Opao Co, Tai Co and the combined company

Opao Co

Opao Co has 2,000 million shares in issue and are currently trading at \$2.50 each.

Tai Co

Tai Co has 263 million shares in issue and the current market value of its debt is \$400 million. Its most recent profit before interest and tax was \$132.0 million, after deducting tax allowable depreciation and non-cash expenses of \$27.4 million. Tai Co makes an annual cash investment of \$24.3 million in non-current assets and working capital. It is estimated that its cash flows will grow by 3% annually for the foreseeable future. Tai Co's current cost of capital is estimated to be 11%.

Combined company

If Opao Co acquires Tai Co, it is expected that the combined company's sales revenue will be \$7,351 million in the first year and its annual pre-tax profit margin on sales will be 15.4% for the foreseeable future. After the first year, sales revenue will grow by 5.02% every year for the next three years. It can be assumed that the combined company's annual depreciation will be equivalent to the investment required to maintain the company at current operational levels. However, in order to increase the sales revenue levels each year, the combined company will require an additional investment of \$109 million in the first year and \$0.31 for every \$1 increase in sales revenue for each of the next three years.

After the first four years, it is expected that the combined company's free cash flows will grow by 2.4% annually for the foreseeable future. The combined company's cost of capital is estimated to be 10%. It is expected that the combined company's debt to equity level will be maintained at 40:60, in market value terms, after the acquisition has taken place.

Both Opao Co and Tai Co pay corporation tax on profits at an annual rate of 20% and it is expected that this rate will not change if Opao Co acquires Tai Co. It can be assumed that corporation tax is payable in the same year as the profits it is charged on.

Possible acquisition price offers

Opao Co's BoD is proposing that Tai Co's acquisition be made through one of the following payment methods:

- (i) A cash payment offer of \$4.40 for each Tai Co share, or
- (ii) Through a share-for-share exchange, where a number of Tai Co shares are exchanged for a number of Opao Co shares, such that 55.5% of the additional value created from the acquisition is allocated to Tai Co's shareholders and the remaining 44.5% of the additional value is allocated to Opao Co's shareholders, or
- (iii) Through a mixed offer of a cash payment of \$2.09 per share and one Opao Co share for each Tai Co share. It is estimated that Opao Co's share price will become \$2.60 per share when such a mixed offer is made.

Similar acquisitions in the food manufacturing industry have normally attracted a share price premium of between 15% and 40% previously.

Required:

- (a) **Distinguish between a management buy-out (MBO) and a management buy-in (MBI), and discuss why Opao Co's board of directors (BoD) might have sold Burgut Co through an MBI.** (4 marks)
- (b) **Explain what portfolio restructuring and organisational restructuring involve, and discuss possible reason(s) why the change in the type of shareholders may have made Opao Co change from being a conglomerate to one focusing on just two business sectors.** (5 marks)
- (c) **Prepare a report for the board of directors of Opao Co which:**
 - (i) **Estimates the value of equity of Opao Co and of Tai Co before the acquisition, and of the combined company after the acquisition;** (10 marks)
 - (ii) **Estimates the percentage gain in value for each Opao Co share and Tai Co share, under each of the cash, the share-for-share, and the mixed offers;** (12 marks)
 - (iii) **Evaluates the likely reaction of Opao Co's and Tai Co's shareholders to the acquisition offers.** (7 marks)

Professional marks will be awarded in part (c) for the format, structure and presentation of the report. (4 marks)
- (d) Following the MBI, the BoD of Burgut Co announced that its intention was to list the company on a recognised stock exchange within seven years. The BoD is discussing whether to obtain the listing through an initial public offering (IPO) or through a reverse takeover, but it does not currently have a strong preference for either option.

Required:

Distinguish between an IPO and a reverse takeover, and discuss whether an IPO or a reverse takeover would be an appropriate method for Burgut Co to obtain a listing. (8 marks)

(50 marks)

Section B – BOTH questions are compulsory and MUST be attempted

- 2 Nutourne Co is a company based in the USA, supplying medical equipment to the USA and Europe.

It is 30 November 20X8. Nutourne Co's treasury department is currently dealing with a sale to a Swiss customer of CHF12.3 million which has just been agreed, where the customer will pay for the equipment on 31 May 20X9. The treasury department intends to hedge the foreign exchange risk on this transaction using traded futures or options as far as possible. Any amount not hedged by a futures or option contract will be hedged on the forward market.

Exchange rates (quoted as US\$/CHF 1)

Spot	1.0292–1.0309
Three months forward	1.0327–1.0347
Six months forward	1.0358–1.0380

Currency futures (contract size CHF125,000, futures price quoted as US\$ per CHF1)

	Futures price
December	1.0318
March	1.0345
June	1.0369

Currency options (contract size CHF125,000, exercise price quotation US\$ per CHF1, premium: US cents per CHF1)

	Calls			Puts		
Exercise price	December	March	June	December	March	June
1.0375	0.47	0.50	0.53	0.74	0.79	0.86

Futures and options contracts mature at the month end.

Non-executive director's comments

A new non-executive director has recently been briefed about the work of the treasury department and has a number of questions about hedging activities. He wants to understand the significance of basis risk in relation to futures. He also wants to know the significant features of over-the-counter forward contracts and options, and why Nutourne Co prefers to use exchange-traded derivatives for hedging.

The non-executive director has also heard about the mark-to-market process and wants to understand the terminology involved, and how the process works, using the transaction with the Swiss customer as an example. The treasury department has supplied relevant information to answer his query. The contract specification for the CHF futures contract states that an initial margin of US\$1,450 per contract will be required and a maintenance margin of US\$1,360 per contract will also be required. The tick size on the contract is US\$0.0001 and the tick value is US\$12.50. You can assume that on the first day when Nutourne Co holds the futures contracts, the loss per contract is US\$0.0011.

Required:

- Evaluate which of the exchange-traded derivatives would give Nutourne Co the higher receipt, considering scenarios when the options are and are not exercised.** (12 marks)
- Discuss the benefits and drawbacks for Nutourne Co in using forward contracts compared with using over-the-counter currency options, and explain why Nutourne Co may prefer to use exchange-traded derivatives rather than over-the-counter derivatives to hedge foreign currency risk.** (7 marks)
- Explain to the non-executive director how the mark-to-market process would work for the CHF futures, including the significance of the data supplied by the treasury department. Illustrate your explanation with calculations showing what would happen on the first day, using the data supplied by the treasury department.** (6 marks)

(25 marks)

- 3** Amberle Co is a listed company with divisions which manufacture cars, motorbikes and cycles. Over the last few years, Amberle Co has used a mixture of equity and debt finance for its investments. However, it is about to make a new investment of \$150 million in facilities to produce electric cars, which it proposes to finance solely by debt finance.

Project information

Amberle Co's finance director has prepared estimates of the post-tax cash flows for the project, using a four-year time horizon, together with the realisable value at the end of four years:

Year	1	2	3	4
	\$m	\$m	\$m	\$m
Post-tax operating cash flows	28.50	36.70	44.40	50.90
Realisable value				45.00

Working capital of \$6 million, not included in the estimates above and funded from retained earnings, will also be required immediately for the project, rising by the predicted rate of inflation for each year. Any remaining working capital will be released in full at the end of the project.

Predicted rates of inflation are as follows:

Year	1	2	3	4
	8%	6%	5%	4%

The finance director has proposed the following finance package for the new investment:

	\$m
Bank loan, repayable in equal annual instalments over the project's life, interest payable at 8% per year	70
Subsidised loan from a government loan scheme over the project's life on which interest is payable at 3.1% per year	80
	<u>150</u>

Issue costs of 3% of gross proceeds will be payable on the subsidised loan. No issue costs will be payable on the bank loan. Issue costs are not allowable for tax.

Financial information

Amberle Co pays tax at an annual rate of 30% on profits in the same year in which profits arise.

Amberle Co's asset beta is currently estimated at 1.14. The current return on the market is estimated at 11%. The current risk-free rate is 4% per year.

Amberle Co's chairman has noted that all of the company's debt, including the new debt, will be repayable within three to five years. He is wondering whether Amberle Co needs to develop a longer term financing policy in broad terms and how flexible this policy should be.

Required:

- (a) Calculate the adjusted present value (APV) for the project and conclude whether the project should be accepted or not. (15 marks)

- (b) Discuss the factors which may determine the long-term finance policy which Amberle Co's board may adopt and the factors which may cause the policy to change. (10 marks)

(25 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d^i) \frac{V_d}{V_e}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i(E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_o = \frac{D_o(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = br_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \qquad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model

$$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$$

Where:

$$d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$$

$$d_2 = d_1 - s\sqrt{t}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

		<i>Discount rate (r)</i>									
<i>Periods</i>											
(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

		<i>Discount rate (r)</i>									
<i>Periods</i>											
(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Answers

- 1 (a) A management buy-out (MBO) involves the purchase of a company by the management running that company. Hence Burgut Co's current management team would be buying Burgut Co from Opao Co. A management buy-in (MBI) involves selling Burgut Co to a management team brought in from outside the company.

Opao Co may have sold Burgut Co through a MBI for the following reasons. Opao Co's BoD may have felt that Burgut Co's current management team lacked fresh ideas and strategies which could have driven Burgut Co forward successfully. Instead, it may have felt that a fresh team, with skills and expertise gained externally, would have had the required innovative ideas and skills. It may be that the external team of managers may have had the finance available to move quickly, whereas the internal team of managers may not have had the finance in place to purchase Burgut Co at that time. It is also possible that the management teams within Burgut Co and Opao Co had disagreements in the past, and Opao Co's BoD may have believed the two management teams would not be able to work together in the future, if needed. Thus, the BoD may have felt that a fresh management team was the better option going forwards.

- (b) Portfolio restructuring involves the acquisition of companies, or disposals of assets, business units and/or subsidiary companies through divestments, demergers, spin-offs, MBOs and MBIs. Organisational restructuring involves changing the way a company is organised. This may involve changing the structure of divisions in a business, business processes and other changes such as corporate governance.

The aim of either type of restructuring is to increase the performance and value of the business.

Opao Co, in going from a conglomerate business to one focusing on just two business areas, can be seen as restructuring its portfolio, as businesses and assets which are not part of financial services and food manufacturing are disposed of, and businesses focusing on these areas are acquired. Financial markets may take the view that focusing on food manufacturing and financial services has enabled Opao Co's senior management to concentrate on areas in which they have expertise. Whereas other businesses in which the senior management are not experts are disposed of. This activity leads to the maximisation of business value.

Shareholders are interested in maximising returns from their investments, which companies achieve through maximising business value, whilst minimising the risks inherent in their investment activity. Shareholders who are closely linked to a particular business do not hold diversified investment portfolios, and therefore benefit from diversification of risk undertaken by a company, investing in many different areas. On the other hand, institutional shareholders and other shareholders, who hold diversified portfolios, would not benefit from a company undertaking risk management through diversification by becoming a conglomerate. Instead, such companies would increase value by focusing on areas in which they have relative expertise, as Opao Co seems to do. So Opao Co's changing owner clientele has forced it to change its overall strategy. This strategy change was implemented through portfolio restructuring.

- (c) **Report to the board of directors (BoD), Opao Co**

Introduction

This report provides an estimate of the additional value created if Opao Co were to acquire Tai Co, and the gain for each company's shareholders based on a cash offer, a share-for-share offer and a mixed offer. It evaluates the likely reaction of the two companies' shareholders to each payment method.

Summary of the estimates from the appendices

From appendix 1

Opao Co equity value pre-acquisition: \$5,000m

Tai Co equity value pre-acquisition: \$1,000m

Combined company equity value post-acquisition: \$6,720m

From appendix 2

Therefore, additional value based on synergy benefits is \$720m or 12% (\$720m/\$6,000m)

Estimated percentage gain in value

	Opao Co	Tai Co
Cash offer	11.2%	15.8%
Share-for-share offer	6.4%	40.0%
Mixed offer	9.7%	23.4%

Likely reactions

Tai Co's shareholders are likely to consider all the offers made, because they all fall within the range of premiums paid in previous acquisitions of 15% to 40%. The cash offer is at the lower end of the range, the share-for-share offer at the top end of the range and the mixed offer in between. It is likely that Tai Co's shareholders will be more attracted to the share-for-share offer as it maximises their return. However, this offer is reliant on the fact that the expected synergy benefits will be realised and Tai Co will probably need to analyse the likelihood of this. Cash payment, although much lower, gives a certainty of return. The mixed offer provides some of the certainty of a cash payment, but also offers a higher return compared to the cash offer. This return is roughly in the middle of the premium range. It may therefore prove to be the better option for Tai Co's shareholders.

Opao Co's shareholders benefit less from the acquisition compared to Tai Co's shareholders. In each case, they get less than the additional value created of 12%, with the cash payment offering the highest return of 11.2%, which is just below the 12% overall return. The share-for-share offer gives the least return at just over half (6.4%) of the overall return of 12%. Nevertheless, with this option, cash is retained within Opao Co and can be used for other value creating projects. Opao Co's shareholders may also prefer the mixed offer, because the return they are expecting to receive is between the cash and share-for-share offers. Also, less cash resources are used compared to the cash offer, and they still benefit from a significant proportion of the additional value created.

Conclusion

Based on the benefits accruing to both sets of shareholders, it is not possible to conclusively say that one method of acquisition payment would be acceptable to both sets of shareholders. However, both sets of shareholders may be persuaded that the mixed offer provides a reasonable compromise between the wholly cash and the wholly share-for-share prices. Given that synergy benefits are shared (even if not equally), both companies' share prices should increase if the acquisition proceeds, as long as the estimates when estimating the valuations are reasonably accurate.

Report compiled by:

Date

APPENDICES:

Appendix 1 (Part (c) (i)):

Equity value of Opao Co prior to acquisition

$\$2.50/\text{share} \times 2,000\text{m shares} = \$5,000\text{m}$

Equity value of Tai Co prior to acquisition

Free cash flows to firm = $\$132.0\text{m} + \$27.4\text{m} - \$24.3\text{m} - (\$132.0\text{m} \times 0.2) = \108.7m

Company value = $\$108.7\text{m} \times 1.03/(0.11 - 0.03) = \$1,399.5\text{m}$, say $\$1,400\text{m}$

Equity value = $\$1,400\text{m} - \$400\text{m} = \$1,000\text{m}$

Equity value of combined company post acquisition

All amounts in \$ millions

Year	1	2	3	4
Sales revenue (5.02% growth, yrs 2 to 4)	7,351	7,720	8,108	8,515
Pre-tax profit (15.4% of sales revenue)	1,132	1,189	1,249	1,311
Less: Tax (20%)	(226)	(238)	(250)	(262)
Less: Additional investment (\$0.31 per \$1, yrs 2 to 4)	(109)	(114)	(120)	(126)
Free cash flows	797	837	879	923
Present value of free cash flows (10%)	724	691	660	630

Combined company value: years 1 to 4 = $\$2,705\text{m}$

Combined company value: after year 4 = $923 \times 1.024/(0.1 - 0.024) \times 1.1^{-4} = \$8,494\text{m}$

Total combined company value = $\$11,199\text{m}$

Equity value ($60\% \times \$11,199\text{m}$) = $\$6,719.4\text{m}$, say $\$6,720\text{m}$

Appendix 2 (Part (c) (ii): Percentage gains for Tai Co and Opao Co shareholders under each payment method

Estimate of additional value created from acquisition due to synergy benefits

$\$6,720\text{m} - (\$5,000\text{m} + \$1,000\text{m}) = \720m

Tai Co, value per share = $\$1,000\text{m}/263\text{m shares} = \$3.80/\text{share approx.}$

Cash offer

Tai Co shareholders, percentage gain

$(\$4.40 - \$3.80)/\$3.80 = \$0.60/\$3.80 = 15.8\%$

Opao Co shareholders, percentage gain

Amount of additional value created going to Tai Co shareholders = $\$0.60 \times 263\text{m shares} = \157.8m

Amount of additional value created going to Opao Co shareholders = $\$720\text{m} - \$157.8\text{m} = \$562.2\text{m}$

As a percentage = $(\$562.2\text{m}/2,000\text{m shares})/\$2.50 = 11.2\%$

Share-for-share offer

Share of additional value to Tai Co shareholders = $\$720\text{m} \times 0.555 = \399.6m

Share of additional value to Opao Co shareholders = $\$720\text{m} \times 0.445 = \320.4m

Opao Co equity value after acquisition = $\$5,320.4\text{m}$

Opao Co, estimated share price after acquisition = $\$5,320.4\text{m}/2,000\text{m shares} = \$2.66/\text{share}$

Opao Co shares to be allocated to Tai Co shareholders = $(\$1,000\text{m} + \$399.6\text{m})/\$2.66 = 526\text{m shares approximately}$

Therefore, share-for-share offer will be 2 Opao Co shares for 1 Tai Co share [$526/263 = 2$]

Tai Co shareholders, percentage gain

$$(\$2.66 \times 2 \text{ shares} - \$3.80 \times 1 \text{ share}) / (\$3.80 \times 1 \text{ share}) = 40\%$$

Opao Co shareholders, percentage gain

$$(\$2.66 - \$2.50) / \$2.50 = 6.4\%$$

Mixed offer**Tai Co shareholders, percentage gain**

$$((\$2.60 + \$2.09) - \$3.80) / \$3.80 = \$0.89 / \$3.80 = 23.4\%$$

Opao Co shareholders, percentage gain

$$\text{Amount of additional value going to Tai Co shareholders} = \$0.89 \times 263\text{m} = \$234.1\text{m}$$

$$\text{Amount of additional value created going to Opao Co shareholders} = \$720\text{m} - \$234.1\text{m} = \$485.9\text{m}$$

$$\text{As a percentage} = (\$485.9\text{m} / 2,000\text{m shares}) / \$2.50 = 9.7\%$$

- (d) The initial public offering (IPO) is the conventional way to obtain a listing where a company issues and offers shares to the public. When doing this, the company will follow the normal procedures and processes required by the stock exchange regarding a new issue of shares and will comply with the regulatory requirements.

Undertaking a reverse takeover enables a company to obtain a listing without going through the IPO process. The BoD of Burgut Co would initially take control of a 'shell' listed company by buying some shares in that company and taking over as its BoD. The 'shell' listed company was probably a normal listed company previously, but is no longer trading. New equity shares in the listed company would then be exchanged for Burgut Co's shares, with the external appearance that the listed company has taken over Burgut Co. But in reality Burgut Co has now effectively got a listing, having taken control of the listed company previously. Normally, the name of the original listed company would then be changed to Burgut Co.

Compared with an IPO, the main benefits of undertaking a reverse takeover are that it is cheaper, takes less time and ensures that Burgut Co will obtain a listing on a stock exchange. An IPO can cost between 3% and 5% of the capital being raised because it involves investment banks, lawyers, and other experts. A marketing campaign and issuing a prospectus are also needed to make the offering attractive and ensure shares to the public do get sold. A reverse takeover does not need any of these and therefore avoids the related costs. The IPO process can typically take one or two years to complete due to hiring the experts, the marketing process and the need to obtain a value for the shares. Additionally, the regulatory process and procedures of the stock exchange need to be complied with. With a reverse takeover, none of these are required and therefore the process is quicker. Finally, there is no guarantee that an IPO will be successful. In times of uncertainty, economic downturn or recession, it may not attract the attention of investors and a listing may not be obtained. With reverse takeover, because the transaction is an internal one, between two parties, it will happen and Burgut Co will be listed.

However, obtaining a listing through a reverse takeover can have issues attached to it. The listed 'shell' company may have potential liabilities which are not transparent at the outset, such as potential litigation action. A full due diligence of the listed company should be conducted before the reverse takeover process is started. The IPO process is probably better at helping provide the senior management of Burgut Co with knowledge of the stock exchange and its regulatory environment. The involvement of experts and the time senior management need to devote to the listing process will help in this regard. Due to the marketing effort involved with an IPO launch, it will probably have an investor following, which a reverse takeover would not. Therefore, a company which has gone through an IPO would probably find it easier to raise extra funds, whilst a company which has gone through a reverse takeover may find it more difficult to raise new funding.

Overall, neither option of obtaining a listing has a clear advantage over the other. The choice of listing method depends on the company undertaking the listing and the purpose for which it is doing so.

(Note: Credit will be given for alternative valid areas of discussion)

- 2 (a) Nutourne Co will have a Swiss Franc receipt in six months' time and needs to hedge against the dollar strengthening.

Futures

Sell Swiss futures and use June futures contracts.

$$\text{No. of contracts} = \text{CHF}12,300,000 / 125,000 = 98.4, \text{ say } 98, \text{ hedging } \text{CHF}12,250,000$$

$$\text{Remainder to be hedged on the forward market is } \text{CHF}12,300,000 - \text{CHF}12,250,000 = \text{CHF } 50,000$$

$$\text{Receipt} = \text{CHF}50,000 \times 1.0358 = \$51,790$$

Calculation of futures price

Assume that basis reduces to zero at contract maturity in a linear fashion.

Estimate from March and June futures contract rates.

$$\text{Predicted futures rate at the end of May} = 1.0345 + ([1.0369 - 1.0345] \times 2/3) = 1.0361$$

$$\text{Expected receipt} = \text{CHF}12,250,000 \times 1.0361 = \$12,692,225$$

Outcome

	\$
Futures	12,692,225
Remainder on forward market	51,790
	12,744,015

Or

Calculation of futures price

Alternatively, use spot rate = 1.0292

Predicted futures rate at the end of May = $1.0292 + (6/7 \times (1.0369 - 1.0292)) = 1.0358$ (when the June futures contract is closed out in May).

Expected receipt = CHF12,250,000 x 1.0358 = \$12,688,550

Outcome

	\$
Futures	12,688,550
Remainder on forward market	51,790
	12,740,340

Options contract

Nutourne Co would purchase CHF June put options.

Number of contracts 98, as before.

Amount not hedged, hedged by forward contract CHF translated as \$51,790 as before.

Assuming the options are exercised:

	\$
Receipt (W1)	12,709,375
Premium (W2)	(105,350)
Forward contract	51,790
	12,655,815

Workings

1 Receipt

$$\text{CHF}125,000 \times 98 \times 1.0375 = \$12,709,375$$

2 Premium

$$1.0375 \text{ options} = 98 \times 125,000 \times 0.0086 = \$105,350$$

The options would give the higher receipt if they were not exercised and the spot rate moved sufficiently in Nutourne Co's favour. If Nutourne Co allowed the option to lapse, it would obtain the same receipt as under the futures if the US\$/CHF spot rate was x, such that:

$$12,692,225 = 12,250,000x - 105,350$$

$$12,250,000x = 12,692,225 + 105,350$$

$$\text{so that } x \text{ is } \text{US\$}1.0447 = \text{CHF}1.$$

Or

$$12,688,550 = 12,250,000x - 105,350$$

$$12,250,000x = 12,688,550 + 105,350$$

$$\text{so that } x \text{ is } \text{US\$}1.0444 = \text{CHF}1.$$

Comments

If the options are exercised, the futures would give the higher receipt. The options give a lower receipt because of the premium which Nutourne Co has to pay. The futures will be subject to the risk that basis (the difference between the futures price and the spot price) may not decrease linearly as the futures approach maturity as assumed in the above calculations. This will mean that the hedge of the CHF 12,250,000 is imperfect, and the receipt may be unpredictable despite a futures hedge being taken out.

The options can also be allowed to lapse if for some reason the contract is not completed. If this happens, Nutourne Co will only have to settle the forward contract.

(b) Benefits of a forward contract

A forward contract would not involve payment of a large premium upfront to the counterparty.

A forward contract is a simple arrangement to understand, whereas the basis of calculation of the premium for an over-the-counter (OTC) option may be unclear.

A forward contract gives a certain receipt for the purposes of budgeting.

Drawbacks of a forward contract

A forward contract has to be fulfilled, even if the transaction which led to the forward contract being purchased is cancelled. Exchange rate movements may mean that the contract has to be fulfilled at an unfavourable rate. An OTC option can be allowed to lapse if it is not needed.

A forward contract does not allow the holder to take advantage of favourable exchange rate movements. An OTC option need not be exercised if the exchange rate moves in the holder's favour.

A forward contract may only be available for a short time period, depending on what currencies are involved. An OTC option may be purchased for a longer time period, over a year.

The rate offered on a forward contract will be determined by a prediction based on expected interest rates. The rate offered on an OTC option may be more flexible. This may suit a holder who is prepared to tolerate the risk of some loss in order to have the opportunity to take advantage of favourable exchange rate movements, but who wishes to use the option to set a limit to possible losses.

Reasons why exchange-traded derivatives are used

One of the main reasons why the treasury function uses exchange-traded derivatives is that the contracts can be bought and sold as required. Also, because the markets are regulated by an exchange, counterparty risk (the risk of the other party to the transaction defaulting) should be minimised.

- (c)** The mark-to-market process begins with Nutourne Co having to deposit an amount (the initial margin) in a margin account with the futures exchange when it takes out the futures. The margin account will remain open as long as the futures are open. The profit or loss on the futures is calculated daily and the margin account is adjusted for the profit or loss.

The maintenance margin is the minimum balance which has to be maintained on the margin account.

If the losses on the futures are so large that the balance on the margin account is less than the maintenance margin, then the futures exchange will make a demand (a margin call) for an extra payment (the variation margin) to increase the balance on the account back to the maintenance margin.

In the example, initial margin = $\$1,450 \times 98 = \$142,100$

Maintenance margin = $\$1,360 \times 98 = \$133,280$

Loss in ticks = $0.0011 / 0.0001 = 11$

Total loss = $11 \text{ ticks} \times \$12.50 \times 98 = \$13,475$

Balance on margin account = $\$142,100 - \$13,475 = \$128,625$

This is less than the maintenance margin, so Nutourne Co would have to deposit an extra $(\$133,280 - \$128,625) = \$4,655$ (the variation margin) to bring the balance on the margin account up to the maintenance margin.

Alternative solution

In some exchanges, a variation margin may be required to increase the balance on the account back to its initial margin level. Therefore, in this case, the variation margin amount would be $\$13,475$ (i.e. $\$142,100 - \$128,625$).

3 (a) Year	0 \$m	1 \$m	2 \$m	3 \$m	4 \$m
Post-tax operating cash flows		28.50	36.70	44.40	50.90
Investment	(150.00)				
Realisable value					45.00
Working capital (W1)	(6.00)	(0.48)	(0.39)	(0.34)	7.21
Cash flows	(156.00)	28.02	36.31	44.06	103.11
Discount factor 12% (W2)	1.000	0.893	0.797	0.712	0.636
Present value	(156.00)	25.02	28.94	31.37	65.58
Base case net present value	(5.09)				

Base case net present value is approximately (\$5.09 million) and on this basis, the investment should be rejected.

Workings

1 Working capital

Year	0 \$m	1 \$m	2 \$m	3 \$m	4 \$m
Working capital		6.00	6.48	6.87	7.21
Required/(released)	6.00	0.48	0.39	0.34	(7.21)

2 Discount rate

Using asset beta

All-equity financed discount rate = $4\% + (11\% - 4\%) 1.14 = 12\%$

3 Issue costs

$\$80 \text{ million} / 0.97 = \$82,474,227$

Issue costs = $3\% \times \$82,474,227 = \$2,474,227$

There will be no issue costs for the bank loan.

4 Tax shield on subsidised loan

Use PV of an annuity (PVA) years 1 to 4 at 8% (normal borrowing rate)

$\$80\text{m} \times 0.031 \times 30\% \times 3.312 = \$2,464,128$

Note to markers

Full credit should be given if tax shield is discounted at the government interest rate of 3.1% rather than the normal borrowing rate of 8%.

5 Tax shield on bank loan

Annual repayment = $(\$70\text{m} / \text{PVA } 8\% \text{ Yr } 1 - 4) = (\$70\text{m} / 3.312) = \$21,135,266$

Year	1 \$000	2 \$000	3 \$000	4 \$000
Opening balance	70,000	54,465	37,687	19,567
Interest at 8%	5,600	4,357	3,015	1,565
Repayment	(21,135)	(21,135)	(21,135)	(21,135)
Closing balance	54,465	37,687	19,567	(3)

Year	1 \$000	2 \$000	3 \$000	4 \$000
Interest cost	5,600	4,357	3,015	1,565
Tax relief at 30%	1,680	1,307	905	470
Discount factor 8%	0.926	0.857	0.794	0.735
Present value	1,556	1,120	719	345
Net present value	3,740			

6 Subsidy benefit

Benefit = $\$80\text{m} \times (0.08 - 0.031) \times 70\% \times 3.312 = \$9,088,128$

7 Financing side effects

	\$000
Issue costs (W3)	(2,474)
Tax shield on subsidised loan (W4)	2,464
Tax shield on bank loan (W5)	3,740
Subsidy benefit (W6)	9,088
Total benefit of financing side effects	12,818

Financing the project in this way would add around \$12.82 million to the value of the project.

The adjusted present value of the project is around \$7.73 million and so the project should be accepted. Sensitivity analysis should be undertaken on all the significant variables. Further analysis may be needed, particularly of the assumptions which lie behind the post-tax cash flows, such as sales and the tax rate. The realisable value of \$45 million may be questionable. On the other hand, the time horizon of four years seems low and analysis should be done of potential cash flows beyond that time.

- (b) Amberle Co's board can use various principles to determine its long-term finance mix. The directors may aim to follow consistent long-term policies, or they may have preferences which change as circumstances change.

Long-term policy factors

At present Amberle Co is using a mix of finance, raising the question of whether the directors are aiming for an optimal level of gearing, or there is a level which they do not wish gearing to exceed. If the board wishes to maintain gearing at an optimal level, this is likely to be determined by a balance of risks and advantages. The main risks are not being able to maintain the required level of payment to finance providers, interest to debt providers or required level of dividend to shareholders. Advantages may include lower costs of debt, tax relief on finance costs as shown in the APV calculation or, on the other hand, not being legally required to pay dividends in a particular year.

Another issue is whether Amberle Co's board has preferences about what source of finance should be used and in what order. One example of this is following the pecking order of retained earnings, then debt, then equity. The board may prefer this pecking order on the grounds that avoiding a new equity issue means that the composition of shareholdings is unchanged, or because retained earnings and longer term debt are judged low risk, or because the market will assume that an equity issue is being made because directors want to take advantage of Amberle Co's shares being over-priced. Other specific sources of finance may have benefits which attract the directors or drawbacks which deter them.

This investment highlights the aspect of whether the board prefers to match sources of finance with specific investments. Matching arguably gives greater flexibility and avoids committing Amberle Co to a long-term interest burden. However, to adopt this approach, the board will need assurance either that the investment will be able to meet finance costs and ultimately repayment burdens, or these can be met from surpluses from other operations.

Changing long-term financing policy

As well as deciding what financing mix or sources of finance they desire to use, the directors will also need to consider what factors would cause this decision to change.

A major change in the scope of the operations, with investment requirements being paramount, may cause a change in financing policy. Here the \$150 million investment has been financed entirely by medium-term debt. Amberle Co may have chosen solely to use debt if it has made a recent equity issue and does not feel it can make another one so soon afterwards. In addition, if Amberle Co expands its manufacture of electric cars, it may decide to sell off its motorbike or cycles divisions if they are performing less well. If part of the business is sold, the sale proceeds could help finance new investment in the cars division.

The board may also be flexible at times and take advantage of whatever source of finance seems to be offering the best terms for Amberle Co. Here the board is taking advantage of loan finance being available at a low cost, thanks to the government loan scheme.

A change in the business or economic environment may also lead to the board rethinking how the company is financed. An economic recession, leading to falling share prices, may mean that the results of a share issue are uncertain. On the other hand, an increase in economic or business risk may mean that lenders are less likely to lend at acceptable rates or will impose greater restrictions. If the directors are risk-averse, they may not seek new finance during a recession but instead rely on retained earnings to finance any expansion.

		<i>Marks</i>
1	(a) Distinguishing between MBO and MBI	1–2
	Discussion of choice of MBI	2–3
	Max	4
	(b) Explanation of portfolio restructuring and organisational restructuring	2
	Discussion of reason(s) for change in business focus	3
		5
	(c) (i) (Appendix 1)	
	Equity value of Opao Co	1
	Tai Co, free cash flow to firm	2
	Estimate of value of Tai Co	1
	Estimate of equity value of Tai Co	1
	Combined company, free cash flows	2
	Value of combined company, years 1 to 4	1
	Value of combined company, after year 4	1
	Equity value of combined company	1
		10
	(ii) (Appendix 2)	
	Cash offer, percentage gain, Tai Co	1
	Cash offer, percentage gain, Opao Co	2
	Share-for-share offer, share of additional value	1
	Share-for-share offer, Opao Co share value	1
	Share-for-share offer, total shares allocated to Tai Co	1
	Share-for-share offer, 2 Opao Co shares for 1 Tai Co share	1
	Share-for-share offer, percentage gain, Tai Co	1
	Share-for-share offer, percentage gain, Opao Co	1
	Mixed offer, percentage gain, Tai Co	1
	Mixed offer, percentage gain, Opao Co	2
		12
	(iii) (Report on proposed acquisition)	
	Evaluation: Opao Co	3–4
	Evaluation: Tai Co	3–4
	Max	7
Professional marks for part (c)		
Report format		1
Structure and presentation of the report		3
		4
(d)	Explanation of difference between an IPO and reverse takeover	3
	Discussion of using an IPO or reverse takeover to obtain a listing	5
		8
Total		50

		<i>Marks</i>
2	(a) Futures	
	Sell futures now	1
	Number of contracts	1
	Forward hedge	1
	Predicted futures rate using basis	1
	Overall expected receipt	1
	Options	
	Purchase June put	1
	Premium	1
	Overall expected receipt	1
	Calculation of when option is a better choice	2
	Comments	2
		12
	(b) Advantages of forward contract	2–3
	Disadvantages of forward contract	2–3
	Reasons for using exchange-traded derivatives	1–2
	Max	7
	(c) Significance of initial and maintenance margins	2
	Mark-to-market explanation	1
	Numerical illustration using Nutourne Co's figures	3
		6
	Total	25
3	(a) Working capital	2
	Discount rate	1
	Base case net present value	2
	Issue costs	1
	Tax shield benefit – subsidised loan	1
	Tax shield benefit – bank loan	4
	Subsidy benefit	1
	Adjusted present value	1
	Comments and conclusion	2
		15
	(b) Factors determining long-term finance policy	5–6
	Factors which cause policy to change	5–6
	Max	10
	Total	25

Strategic Professional – Options

Advanced Financial Management (AFM)

March/June 2019 – Sample Questions



Time allowed: 3 hours 15 minutes

This question paper is divided into two sections:

Section A – This ONE question is compulsory and MUST be attempted

Section B – BOTH questions are compulsory and MUST be attempted

Formulae and tables are on pages 8–12.

Do NOT open this question paper until instructed by the supervisor.

This question paper must not be removed from the examination hall.

AFM

Think Ahead

ACCA

The Association of
Chartered Certified
Accountants

Section A – This ONE question is compulsory and MUST be attempted

- 1 Talam Co, a listed company, aims to manufacture innovative engineering products which are environmentally friendly and sustainable. These products have been highly marketable because of their affordability. Talam Co's mission statement also states its desire to operate to the highest ethical standards. These commitments have meant that Talam Co has a very high reputation and a high share price compared to its competitors.

Talam Co is considering a new project, the Uwa Project, to manufacture drones for use in the agricultural industry, which are at least 50% biodegradable, at competitive prices. The drones will enable farmers to increase crop yields and reduce crop damage. Manufacture of drones is a new business area for Talam Co. The project is expected to last for four years.

Talam Co will also work on the Jigu Project (a follow-on project to the Uwa Project) to make 95%+ biodegradable drones. It is expected that the Jigu Project will last for a further five years after the Uwa Project has finished. If the Uwa Project is discontinued or sold sooner than four years, the Jigu Project could still be undertaken after four years.

Uwa Project

The following number of drones are expected to be produced and sold:

Year	1	2	3	4
Number of drones produced and sold	4,300	19,200	35,600	25,400

In the first year, for each drone, it is expected that the selling price will be \$1,200 and the variable costs will be \$480. The total annual direct fixed costs will be \$2,700,000. After the first year, the selling price is expected to increase by 8% annually, the variable costs by 4% annually and the fixed costs by 10% annually, for the next three years. Training costs are expected to be 200% of the variable costs in year 1, 60% in year 2, and 10% in each of years 3 and 4. There is substantial uncertainty about the drones produced and sold, and Talam Co estimates the project to have a standard deviation of 30%.

At the start of every year, the Uwa Project will need working capital. In the first year, this will be 20% of sales revenue. In subsequent years, the project will require additional or a reduction in working capital of 10% for every \$1 increase or decrease in sales revenue respectively. The working capital is expected to be fully recovered when the Uwa Project ceases.

The Uwa Project will need \$35,000,000 of machinery to produce the drones at the start of the project. Tax allowable depreciation is available on the machinery at 15% per year on a straight-line basis. The machinery is expected to be sold for \$7,000,000 (post-inflation) at the end of the project. Talam Co makes sufficient profits from its other activities to take advantage of any tax loss relief. Tax is paid in the year it falls due.

Jigu Project as a real option

Talam Co estimates that Jigu Project's cash flows are highly uncertain and its standard deviation is 50%. It is estimated that \$60,000,000 will be required at the start of the project in four years' time. Using conventional net present value, Talam Co's best estimate is that net present value of the project will be \$10,000,000 at the start of the project.

The following figures were estimated for the Jigu Project using the real options method.

Asset value (P_a) = \$46,100,000 (to nearest 100,000)

Exercise price (P_e) = \$60,000,000

Exercise date (t) = 4 years

Risk-free rate (r) = 2.30%

Volatility (s) = 50%

$$d_1 = 0.329$$

$$d_2 = -0.671$$

$$N(d_1) = 0.6288$$

$$N(d_2) = 0.2510$$

Call option value: \$15,258,399

It can be assumed that the call option value is accurate.

Talam Co's finance director wants to know how the asset value of \$46,100,000 has been estimated.

Honua Co's offer

Honua Co, whose main business is drone production, has approached Talam Co with an offer to buy the Uwa Project in its entirety from Talam Co, for \$30,000,000 at the start of the third year of the project's life.

Talam Co has calculated some figures to assess the value of Honua Co's offer using the real options method, as follows:

$$d_1 = 0.779$$

$$d_2 = 0.355$$

$$N(d_1) = 0.7821$$

$$N(d_2) = 0.6387$$

Talam Co's finance director has requested that the value of Honua Co's offer is estimated using the real options method. She has also requested to know the amounts of the initial variables which would have been used to calculate the d_1 , d_2 , $N(d_1)$ and $N(d_2)$ figures.

It can be assumed that the d_1 , d_2 , $N(d_1)$ and $N(d_2)$ figures are accurate.

Additional information

Both Honua Co and Talam Co pay corporation tax at an annual rate of 20%. Talam Co has estimated Uwa Project's and Jigu Project's risk-adjusted cost of capital at 11%, based on Honua Co's asset beta. Talam Co believes that LIBOR, which is currently 2.30%, provides a good estimate of the risk-free rate of interest.

Required:

- (a) **Discuss how incorporating real options into net present value decisions may help Talam Co with its investment appraisal decisions.** (5 marks)

- (b) **Prepare a report for the board of directors (BoD) of Talam Co which:**

- (i) **Estimates, showing all relevant calculations, the net present value of the Uwa Project before considering the offer from Honua Co and the Jigu Project;** (12 marks)
- (ii) **Addresses the requests made by the finance director about the initial variables and estimated value of the offer from Honua Co using the real options method;** (9 marks)
- (iii) **Assesses whether the Uwa Project should be undertaken, using the results from, and discusses the assumptions made in, the calculations in (b)(i) and (b)(ii) above.** (10 marks)

Professional marks will be awarded in part (b) for the format, structure and presentation of the report.

(4 marks)

- (c) At a recent trade show, the biodegradable drones attracted considerable interest from organisations worldwide. Nevertheless, some expressed concern about the drone price, which they felt was too high.

Talam Co estimates that even a modest reduction in each drone's price would make the projects unprofitable. Therefore, the operations director suggested that costs could be reduced if drone components were produced in Dunia, a country where Talam Co already gets some of its other products made.

However, the public relations director brought up an issue concerning Dunia. He said that several companies in Dunia, which Talam Co trades with, employ young teenage children. These companies pay the education fees for the teenagers and the companies argued that stopping this practice would harm the teenagers' families financially.

Required:

Discuss the impact on Talam Co and its aims arising from the possible sustainability and ethical issues above, and advise on how these issues may be addressed. (10 marks)

(50 marks)

Section B – BOTH questions are compulsory and MUST be attempted

- 2 Lurgshall Co is a listed electronics company. Lurgshall Co has recently appointed a new chief executive, who has a number of plans to expand the company. The chief executive also plans to look carefully at the costs of all departments in Lurgshall Co's head office, including the centralised treasury department.

The first major investment which the chief executive will oversee is an investment in facilities to produce applications-specific components. To finance the planned investment, it is likely that Lurgshall Co will have to borrow money. It is now 1 May. At present, it seems that Lurgshall Co will need to borrow \$84 million on 1 September, for a period of six months, though both the amount and the period of borrowing are subject to some uncertainty. The treasurer plans to borrow the funds at a variable rate of LIBOR plus 50 basis points. LIBOR is currently 4.5% but is expected to rise by up to 0.6% between now and 1 September.

So far, the possibility of hedging a rise in LIBOR of 0.6% using a forward rate agreement or September \$ futures has been investigated. The results of the calculations for these instruments were as follows:

4–10 Forward rate agreement from Birdam Bank: 5.38%

Three-month traded September \$ futures: 5.36%

Lurgshall Co's treasurer also wants to consider using options on futures to hedge loans.

Although Lurgshall Co has not previously used swaps for hedging purposes, the treasurer has asked Birdam Bank to find a counterparty for a potential swap arrangement.

Relevant information about options and swaps is as follows:

Options

The current price for three-month \$ September futures, \$2 million contract size is 95.05. The price is quoted in basis points at 100 – annual % yield.

Options on three-month September \$ futures, \$2 million contract size, option premiums are in annual %

September calls	Strike price	September puts
0.132	95.25	0.411

It can be assumed that futures and options contracts are settled at the end of each month. Basis can be assumed to diminish to zero at contract maturity at a constant rate, based on monthly time intervals. It can also be assumed that there is no basis risk and there are no margin requirements.

Swap

Birdam Bank has found a possible counterparty to enter into a swap with Lurgshall Co. The counterparty can borrow at an annual floating rate of LIBOR + 1.5% or a fixed rate of 6.1%. Birdam Bank has quoted Lurgshall Co a notional fixed rate of 5.6% for it to borrow. Birdam Bank would charge a fee of 10 basis points to each party individually to act as the intermediary of the swap. Both parties would share equally the potential gains from the swap contract.

Treasury staffing

Lurgshall Co's new chief executive has made the following comments: 'I understand that the treasury department has a number of day-to-day responsibilities, including investing surplus funds for the short-term liquidity management and hedging against currency and interest rates. However, these tasks could all be carried out by the junior, less experienced, members of the department. I do not see why the department needs to employ experienced, expensive staff, as it does not contribute to the strategic success of the company.'

Required:

- (a) Compare the results of hedging the \$84 million, using the options and the swap, with the results already obtained using the forward rate agreement and futures, and comment on the results. Show all relevant calculations, including how the interest rate swap would work. (15 marks)
- (b) Discuss the advantages and disadvantages of using swaps as a means of hedging interest rate risk for Lurgshall Co. (5 marks)
- (c) Criticise the views of the chief executive about the work carried out by the treasury department and the staff required to do this work. (5 marks)

(25 marks)

- 3 Newimber Co is a listed company which has always manufactured formal clothing for adults and children. It obtained a listing ten years ago after years of steady growth. 70% of shares in the company are owned by its directors or their relatives, with the remaining 30% owned by external investors, including institutional investors.

Sportswear division

Eight years ago it set up a division to manufacture sportswear. This investment has been very successful and the sportswear division now accounts for 40% of total group revenue, having grown much quicker than the original formal clothing division.

Newimber Co's board has given divisional management at the sportswear division more authority over time, although the board has continued to make major policy and investment decisions relating to the division. Initially, relations between Newimber Co's board and management of the sportswear division were good, but there have been problems over the last couple of years. The sportswear division's management has been frustrated by the board's refusal to approve their recent investment plans on the grounds that they were too risky. In order to achieve operational efficiencies, the sportswear division's management would also like to pursue stricter policies for managing operational staff and suppliers than Newimber Co's board has so far allowed.

In addition, Newimber Co started to prepare an integrated report three years ago, but Newimber Co's board has had difficulties in obtaining all the information it requires for the report from the sportswear division.

Restructuring

A few months ago, the management of the sportswear division approached Newimber Co's board with a proposal for a management buyout of the sportswear division. However, the price the sportswear division's management was able to offer was insufficient to persuade Newimber Co's board to sell the sportswear division to them.

Newimber Co's board has, subsequently, decided that the sportswear division should be demerged into a new company, Poykins Co. The shareholders and proportion of shares held would be the same for Poykins Co as currently for Newimber Co. The sportswear division's senior management team would become the board of Poykins Co and Poykins Co would seek an immediate listing on the same stock exchange as Newimber Co.

Financial information

The market capitalisation of Newimber Co's share capital is currently \$585 million. Newimber Co also currently has \$200 million 5·9% loan notes. The loan notes are redeemable in five years' time at a premium of 5%. Newimber Co's equity beta is currently estimated at 1·4. Newimber Co's current cost of equity is 11·8% and its current before-tax cost of debt is 4·5%.

The asset beta of the formal clothing division is estimated to be 1·21. The weighting in estimating Newimber Co's overall asset beta is 60% for the formal clothing division to 40% for the sportswear division. The debt beta can be assumed to be zero.

In return for 40% of the issued share capital of Newimber Co, its current shareholders will receive 100% of the issued share capital of Poykins Co, corresponding to the assets and liabilities being transferred. The shares in Newimber Co which shareholders have given up will be cancelled. After the demerger, Newimber Co's new market capitalisation can be assumed to be \$351 million. Poykins Co will have no long-term debt, the liability for the \$200 million loan notes remaining with Newimber Co.

The current risk-free rate of return is estimated to be 3·4%. The market risk premium is estimated to be 6%. A tax rate of 28% is applicable to all companies.

The sportswear division currently has \$36 million operating cash flows. Its managers believe that operating cash flows can increase by the following rates once Poykins Co has been listed:

Year	%
1	25
2	20
3	15
4 onwards	2

The sportswear division's managers believe that Poykins Co will require a \$20 million investment of additional assets in Year 1, rising to \$22 million in each of Years 2 and 3, and to \$25 million annually from Year 4 onwards.

Required:

- (a) Discuss the advantages and disadvantages of demerging the sportswear division into a new company. (5 marks)
- (b) Calculate:
- The change in the weighted average cost of capital of Newimber Co if the demerger of the sportswear division takes place;
 - The valuation of Poynins Co using free cash flows, based on the information and assumptions given and briefly discuss your results. (15 marks)
- (c) Discuss the factors which may determine the policies Poynins Co should adopt for communication of information to its shareholders and other significant stakeholders. (5 marks)

(25 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d^i) \frac{V_d}{V_e}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i(E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_o = \frac{D_o(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = br_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \qquad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model

$$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$$

Where:

$$d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$$

$$d_2 = d_1 - s\sqrt{t}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

		<i>Discount rate (r)</i>									
<i>Periods</i>											
(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

		<i>Discount rate (r)</i>									
<i>Periods</i>											
(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Answers

- 1 (a) When making decisions, following investment appraisals of projects, net present value assumes that a decision must be made immediately or not at all, and once made, it cannot be changed. Real options, on the other hand, recognise that many investment appraisal decisions have some flexibility.

For example, decisions may not have to be made immediately and can be delayed to assess the impact of any uncertainties or risks attached to the projects. Alternatively, once a decision on a project has been made, to change it, if circumstances surrounding the project change. Finally, to recognise the potential future opportunities, if the initial project is undertaken, like the Jigu Project.

Real options give managers choices when making decisions about whether or not to undertake projects, by estimating the value of this flexibility or choice. Real options take into account the time available before a decision, on a project, has to be made, and the risks and uncertainties attached to the project. It uses these factors to estimate an additional value which can be attributable to the project. Real options view risks and uncertainties as opportunities, where upside outcomes can be exploited, and a company has the option to disregard any downside impact.

By incorporating the value of any real options available into an investment appraisal decision, Talam Co will be able to assess the full value of a project.

- (b) **Report to the board of directors (BoD), Talam Co**

Introduction

This report assesses whether or not the Uwa Project should be undertaken based on its value from an initial net present value (NPV) calculation, and then taking into account the options provided by the offer from Honua Co and the Jigu Project. As part of the assessment, a discussion of the assumptions and their impact on the assessment is provided.

Assessment

The value of the Uwa Project based on just the initial NPV is a small negative amount of \$(6,000) approximately (appendix 1). This would indicate that the project is not worth pursuing, although the result is very marginal. The offer from Honua Co, and the Jigu Project, using the real options method, gives an estimated value of \$17,668,000 (appendix 2), which is positive and substantial. This indicates that the Uwa Project should be undertaken.

Assumptions

The following assumptions have been made when calculating the values in appendices 1 and 2.

- Since the Uwa Project is in a different industry to Talam Co's current activities, the project-specific, risk-adjusted cost of capital of 11% based on Honua Co's asset beta is used. It is assumed that Honua Co's asset beta would provide a good approximation of the business risk inherent in drone production.
- It is assumed that all the variables used to calculate the values of the projects in appendices 1 and 2 are correct and accurate. Furthermore, it is assumed all the variables such as inflation rates, tax rates, interest rates and volatility figures, remain as forecast through the period of each project. It is also assumed that the time periods related to the projects and the offer from Honua is accurate and/or reasonable.
- The Black-Scholes option pricing (BSOP) model is used to estimate the real option values of the Jigu Project and the Honua Co offer. The BSOP model was developed for financial products and not for physical products, on which real options are applied. The BSOP model assumes that a market exists to trade the underlying project or asset without restrictions, within frictionless financial and product markets.
- The BSOP model assumes that the volatility or risk of the underlying asset can be determined accurately and readily. Whereas for traded financial assets this would most probably be reasonable, as there is likely to be sufficient historical data available to assess the underlying asset's volatility, this is probably not going to be the case for real options. For large, one-off projects, there would be little or no historical data available. Volatility in such situations would need to be estimated using simulation models, such as the Monte-Carlo simulation, with the need to ensure that the model is developed accurately and the data input used to generate outcomes reasonably reflects what is likely to happen in practice.
- The BSOP model assumes that the real option is a European-style option which can only be exercised on the date when the option expires. In some cases, it may make more strategic sense to exercise an option earlier. The real option is more representative of an American-style option which can be exercised before expiry. Therefore, the BSOP model may underestimate the true value of an option.
- Real options models assume that any contractual obligations involving future commitments made between parties will be binding, and will be fulfilled. For example, it is assumed that Honua Co will fulfil its commitment to purchase the project from Talam Co at the start of the third year for \$30 million and there is therefore no risk of non-fulfilment of that commitment.
- The BSOP model does not take account of behavioural anomalies which may be displayed by managers when making decisions.

Conclusion

The initial recommendation is that the Uwa Project should be undertaken when the offer from Honua Co and going ahead with the Jigu Project are included. Taken together, these result in a significant positive NPV. However, one or more of the above assumptions may not apply and therefore NPV value is not a 'correct' value. Instead, the appendices provide indicative value which can be attached to the flexibility of a choice of possible future actions which are embedded with the Uwa Project and indicate that it should be undertaken.

Report compiled by:

Date

(Note: Credit will be given for alternative and valid discussion comments)

APPENDICES:

Appendix 1 (Part (b) (i)):

Net present value computation of the Uwa Project before incorporating the offer from Honua Co and the financial impact of the Jigu Project. All figures are in \$000s.

Year	0	1	2	3	4
Sales revenue (w1)		5,160	24,883	49,840	38,405
Less:					
Variable costs (w2)		2,064	9,581	18,476	13,716
Fixed costs		2,700	2,970	3,267	3,594
Training costs		4,128	5,749	1,848	1,372
Cash flows before tax		(3,732)	6,583	26,249	19,723
Tax (w3)		1,796	(267)	(4,200)	(1,495)
Working capital	(1,032)	(1,972)	(2,496)	1,144	4,356
Machinery purchase and sale	(35,000)				7,000
Net cash flows	(36,032)	(3,908)	3,820	23,193	29,584
Present value of cash flows (discounted at 11%)	(36,032)	(3,521)	3,100	16,959	19,488

Approximate net present value of the project = \$(6,000)

Workings:

Working 1 (w1): Sales revenue

Year	1	2	3	4
Units produced and sold	4,300	19,200	35,600	25,400
Selling price (\$) (inflated at 8%)	1,200	1,296	1,400	1,512
Sales revenue (\$000s)	5,160	24,883	49,840	38,405

Working 2 (w2): Variable costs

Year	1	2	3	4
Units produced and sold	4,300	19,200	35,600	25,400
Variable costs per unit (\$) (inflated at 4%)	480	499	519	540
Total variable costs (\$000s)	2,064	9,581	18,476	13,716

Working 3 (w3): Tax

Year	1	2	3	4
Cash flows before tax	(3,732)	6,583	26,249	19,723
Tax allowable depreciation	(5,250)	(5,250)	(5,250)	(12,250)
Taxable cash flows	(8,982)	1,333	20,999	7,473
Tax payable (20%)	(1,796)	267	4,200	1,495

Appendix 2 (Part (b) (ii)):

Jigu Project: Asset value

Asset value of Jigu Project of \$46,100,000 is estimated as present value of future cash flows related to the project: $\$70,000,000 \times 1.11^{-4}$, where $\$70,000,000 = \$60,000,000 + \$10,000,000$.

Honua Co offer, initial variables used to calculate the d_1 , d_2 , $N(d_1)$ and $N(d_2)$ figures:

Asset value (P_a) = $\$16,959,000 + \$19,488,000 = \$36,447,000$ (cash flows foregone)

Exercise price (P_e) = $\$30,000,000$

Exercise date (t) = 2 years

Risk-free rate (r) = 2.30%

Volatility (s) = 30%

Value of Honua Co offer

Value of Honua Co's offer

Call value: $\$36,447,000 \times 0.7821 - \$30,000,000 \times 0.6387 \times e^{(-0.023 \times 2)} = \$10,205,640$

Honua Co's offer is equivalent to a put option.

Put value: $\$10,205,640 - \$36,447,000 + \$30,000,000 \times e^{(-0.023 \times 2)} = \$2,409,899$ **Estimated total value arising from the two real options**

Value of Jigu Project: \$15,258,399

Value of Honua Co's offer: \$2,409,899

Estimated total value from the two real options: $\$2,409,899 + \$15,258,399 = \$17,668,298$

- (c) The overarching issue is that of conflict between the need to satisfy shareholders and the financial markets, and Talam Co's stated aims of bringing affordable environmentally friendly products to market and maintaining high ethical standards. This overarching issue can be broken down into smaller related issues.

Producing profitable products will presumably result in positive NPV projects, thus ensuring a continued strong share price performance. This should satisfy the markets and shareholders. However, if the products cannot be sold at a reasonable selling price because some farmers are not able to afford the higher prices, then this may compromise Talam Co's aim of bringing environmentally friendly products to market and making them affordable.

A possible solution is to lower production costs, by shifting manufacturing to locations where such costs are lower. Talam Co's BoD thus considered the move to Dunia, to lower production costs. This presumably would allow Talam Co to reduce prices and make the drones more affordable, but at the same time ensure that the projects result in positive NPVs. However, the issue here is that supplier companies in Dunia whom Talam Co trades with use young teenage children as part of their workforce. This may impact negatively on Talam Co's stated aim of maintaining high ethical standards. In fact, Talam Co may need to rethink its links with companies it trades with in Dunia entirely. Otherwise there is a real risk that Talam Co could suffer from long-term loss of reputation, and this may cause substantial and sustained financial damage to the company.

Talam Co may decide that maintaining its share price and its reputation should take the highest priority and therefore it may reach a decision that the best way to address the issue(s) is to not try to reduce costs, and to withdraw from Dunia completely. But this would prevent many agriculturalists from taking advantage of the biodegradable drones. Therefore, Talam Co may want to explore alternative ways to meet all the aims.

Talam Co could consider moving to another location, if this was feasible. It is not known from the narrative whether or not viable alternatives are available, but Talam Co would need to ensure that possible alternative locations would have the infrastructure to produce the components at the same or lower costs. Talam Co may also want to consider the softer issues; for example, it will want a good working relationship and network in the new locations which it has with the companies in Dunia. These may need to be developed and would take time and probably incur additional costs.

For these reasons, Talam Co may decide to explore the existing production facilities in Dunia further. It is possible that the supplier companies are not exploiting the young teenage children, but are supporting their education and their families in a positive way. Stopping the relationship may jeopardise this support. Talam Co would need to investigate the working conditions of the children and the manner in which they are rewarded and supported. It may want to consult the guardians of the young teenage children and see if there are other feasible solutions. For example, could the guardians be employed instead of the young teenage children or are they already engaged in alternative employment?

After all factors are considered, Talam Co may conclude that the best way to achieve all its aims is to continue in Dunia and also have the production of drone components located there. If this is the case and young teenage children continue to be employed there, then Talam Co would need a sustained public relations campaign to defend its position and demonstrate how it ensures that the teenage children have not been exploited, but are gainfully employed and receiving a good education to help them progress in life.

(Note: Credit will be given for alternative and valid discussion comments)

2 (a) Options

Buy put options as need to hedge against a rise in interest rates.

Number of contracts required: $\$84,000,000 / \$2,000,000 \times 6/3 = 84$

Total basis = current price (1 May) – futures price = $(100 - 4.50) - 95.05 = 0.45$

Unexpired basis on 1 September = $0.45 \times 1/5 = 0.09$

Expected futures price = $100 - 5.1 - 0.09 = 94.81$

Exercise price	95.25
Futures price as above	94.81
Exercise?	Yes
Gain in basis points	44

	\$
Interest paid ($\$84,000,000 \times 5.6\% \times 6/12$)	2,352,000
Gain from options $0.0044 \times \$2,000,000 \times 3/12 \times 84$	(184,800)
Premium $0.00411 \times \$2,000,000 \times 3/12 \times 84$	172,620
Net payment	<u>2,339,820</u>
Effective annual interest rate $2,339,820/84,000,000 \times 12/6$	5.57%

Swaps

	Lurgshall Co	Counterparty	Interest rate differential
Fixed rate	5.60%	6.10%	0.50%
Floating rate	LIBOR + 0.50%	LIBOR + 1.50%	1.00%

Lurgshall Co has an advantage in borrowing at both fixed and floating rates, but the floating rate advantage is larger.

Gain % for Lurgshall Co = $50\% (1 - 0.5 - 0.2) = 0.15$

	Lurgshall Co	Counterparty
Rate without swap	(5.60%)	(LIBOR + 1.50%)
Benefit	0.15%	0.15%
Net result	(5.45%)	(LIBOR + 1.35%)

Swap

Borrows at	(LIBOR + 0.50%)	(6.10%)
Lurgshall Co pays	(4.85%)	4.85%
Counterparty pays	LIBOR	(LIBOR)
Bank fee	(0.10%)	(0.10%)
Net result	(5.45%)	(LIBOR + 1.35%)

Comments

The swap gives a result which is marginally worse than the forward rate agreement and the futures. The options give a worse result than the other choices.

Risks which might be considered include counterparty risk for the forward rate agreement and swap. Using Birdam Bank should mean that this risk is low for forward rate agreements, and also for swaps, assuming that the bank bears the risk of the counterparty defaulting.

Basis risk should be considered for the traded futures. Here, because the differences between the instruments are small, a failure to estimate basis accurately may mean that futures are chosen when they do not offer the lowest borrowing cost. For the swaps, if Lurgshall Co swaps into fixed rate debt, it faces the market risk of an unexpected fall in interest rates.

Other factors to consider include the possibility that rates will increase rather less than forecast, meaning that the option would not be exercised and at some point would be the lowest cost choice. The length of time of the swap also needs to be considered. Although it commits Lurgshall Co to the fixed rate, if the borrowing turns out to be longer than the six months, the swap may provide a better time match than the other hedging opportunities.

(b) Advantages of swaps

Transaction costs are generally relatively low. If Lurgshall Co arranged the swap itself, the costs would be limited to legal fees. The transaction costs may also be lower than the costs of terminating one loan and arranging another.

Lurgshall Co can, as here, swap a commitment to pay a variable rate of interest which is uncertain with a guaranteed fixed rate of interest. This allows Lurgshall Co to forecast finance costs on the loan with certainty.

Swaps are over-the-counter arrangements. They can be arranged in any size and for whatever time period is required, unlike traded derivatives. The period available for the swap may be longer than is offered for other interest rate derivatives.

Swaps make use of the principle of comparative advantage. Lurgshall Co can borrow in the market where the best deal is available to it, and then use the swap to access the loan finance it actually wants at an overall cheaper cost.

Disadvantages of swaps

Swaps are subject to counterparty risk, the risk that the other party to the arrangement may default on the arrangement. This would apply in particular if Lurgshall Co arranged the swap itself. If it is arranged through a bank, the bank can provide a guarantee that the swap will be honoured.

If Lurgshall Co swaps into a fixed rate commitment, it cannot then change that commitment. This means it cannot take advantage of favourable interest rate changes as it could if it used options. This may be a particular problem if the swap period is more than a few months and interest rates are expected to be volatile.

As swaps are over-the-counter instruments, they cannot be easily traded or allowed to lapse if they are not needed or become no longer advantageous. It is possible that a bank may allow a reswapping arrangement to reverse a swap which is not required, but this will incur further costs.

- (c) The chief executive appears to underestimate the degree of knowledge required for day-to-day work. Less experienced staff may be able to arrange borrowing if the lender has already been chosen or, for example, arrange forward rate agreements to be used if they are prescribed.

However, if judgement is required as to, for example, which lender or hedging instrument to use, using less experienced staff may mean that a sub-optimal decision is taken. Poor decisions may result in opportunity costs, for example, not using the lender who gives the best deal or being committed to a fixed forward rate agreement when an option would have allowed the business to take advantage of favourable rate movements. These opportunity costs may not be as clear as the salary costs of experienced staff.

As the business operates internationally, the treasury department will need to monitor financial market conditions and exchange rates, and other issues which may be significant such as political developments. Because of their previous experiences, longer-serving staff are more likely to appreciate the implications of developments and whether treasury policies and decisions need to change in response to changes in risk. Senior staff are also needed to manage the work of less experienced staff to prevent or mitigate the effect of mistakes which may be costly.

Experienced staff are also needed to establish overall guidelines and policies for treasury activities. Their judgement will be required to establish principles which will mean that actions taken by staff are in line with the risk appetite of the business and are sufficiently prudent from the viewpoint of risk management. Experienced staff will also have greater knowledge of law, accounting standards and tax regulations, which can help the business avoid penalties and perhaps structure its dealings so that it can, for example, minimise the level of tax paid.

The chief executive has plans for a major expansion of the business, involving significant investment and financing decisions. Advice from experienced treasury staff will be invaluable in supporting the decisions required. If Lurgshall Co is planning a major acquisition, the treasury function can provide advice on the structure of consideration and financing implications. If, as here, a major investment is being contemplated, experienced staff can advise on translating views on risk into a relevant cost of capital, which will help ensure that the financial appraisal of the investment is realistic.

3 (a) Advantages of demerger

If the managers of the sportswear division's belief that they can run the division better without the interventions of senior management at Newimber Co is well-founded, the business may be able to achieve operational efficiencies and increases in value.

The new company is not tied to the financial commitments associated with the formal clothing division in terms of finance cost and loan repayment. Its management will have the ability to determine the finance structure which best suits the new business.

Newimber Co's shareholders will continue to own both companies. If shareholders are concerned about the diversification of their portfolio, this will remain unchanged.

The demerger may allow Newimber Co's management team to focus on the formal clothing division. They should not need to spend time dealing with disagreements with the sportswear division's management team.

Disadvantages of demerger

There will be legal costs associated with the demerger, such as the cost of obtaining a listing for the new company arising out of the sportswear division. Also setting up the new company and establishing the new structure looks likely to take up significant management time. This may mean that neither company is focused on external opportunities and challenges for some time, maybe impacting results and competitive position.

Both the new companies may suffer adverse effects through being smaller entities. Economies of scale may be lost and the companies may find it less easy to raise new finance. Looking at the position across both companies in total, distributable profits may fall because of a rise of overheads as each company will need its separate infrastructure and service departments.

The current arrangement may frustrate the management of the sportswear division but the command structure is clear. Once the director-shareholders of Newimber Co merely become shareholders of the new company, they will not be able to intervene actively in its management and overrule its management team. Agency problems may arise if these shareholders have different attitudes to risk to Poyntins Co's board or different views on the importance of short-term versus long-term objectives.

(b) Current WACC Newimber Co

k_e is 11.8% and k_d is 4.5%

Annuity factor 4.5% for 5 years = $1 - (1 + 0.045)^{-5} / 0.045 = 4.390$

Loan value per \$100 = $(\$5.90 \times 4.390) + (\$105.00 \times 1.045^{-5}) = \110.16

$MV_d = \$110.16 / 100 \times \$200 \text{ million} = \$220 \text{ million}$

WACC = $((585 \times 11.8\%) + (220 \times 4.5\% \times 0.72)) / 805 = 9.5\%$

New WACC Newimber Co

MV_e is \$351 million

$$\beta_e = 1.21 ((351 + (220 (1 - 0.28)))/351) = 1.76$$

$$k_e = 3.4\% + (1.76 \times 6\%) = 14.0\%$$

$$WACC = ((351 \times 14.0\%) + (220 \times 4.5\% \times 0.72))/571 = 9.9\%, \text{ an increase of } 0.4\%$$

WACC Poykins Co

$$\text{Current } \beta_a \text{ of Newimber Co} = 1.4(585/(585 + (220 (1 - 0.28)))) = 1.10$$

$$\beta \text{ Poykins Co} = (1.10 - (0.6 \times 1.21))/0.4 = 0.935$$

$$WACC \text{ Poykins Co} = 3.4\% + (0.935 \times 6\%) = 9.0\%$$

Free cash flows Poykins Co

Year	1 \$m	2 \$m	3 \$m
Operating cash flows	45.0	54.0	62.1
Tax	(12.6)	(15.1)	(17.4)
Post-tax cash flows	32.4	38.9	44.7
Investment in assets	(20.0)	(22.0)	(22.0)
Free cash flows	12.4	16.9	22.7
Discount factor (9%)	0.917	0.842	0.772
Discounted cash flows	11.4	14.2	17.5

Discounted free cash flows Years 1 to 3 = \$43.1m

Discounted post-tax cash flows Year 4 onwards = $(\$44.7m (1 + 0.02)/0.09 - 0.02) = \$651.3m \times 0.772 = \$502.8m$

Discounted investment in assets Year 4 onwards = $(25/0.09) = \$277.8m \times 0.772 = \$214.5m$

Poykins Co's valuation = \$43.1m + \$502.8m - \$214.5m = \$331.4m

Discussion

If the managers' estimates of the sportswear division's future free cash flows are realistic, then the valuation using free cash flows (\$331.4m) exceeds the current valuation (\$585m - \$351m = \$234m).

The valuation is dependent upon achieving ambitious growth targets in Years 1 to 3, particularly given the loss of economies of scale discussed above. The board and shareholders of Newimber Co would want details about the assumptions behind these figures, particularly as growth after that is only assumed to be 2%. The valuation is also dependent upon the investment figures being accurate, so directors and shareholders would again need more detail of these so that they can decide whether the extra investment is likely to generate the increased cash flows predicted.

They would also want to determine how the managers of the sportswear division plan to fund the investments, particularly if initial operating cash flows are not as high as expected.

The restructuring will lead to a marginal increase in the WACC of Newimber Co, as its financial risk increases with more gearing. The directors may be worried that Newimber Co's credit rating will fall.

(c) Requirement for business review

The directors of Poykins Co will have to fulfil the same statutory and listing requirements as Newimber Co currently fulfils. These are likely to include the requirements for a business review.

Investors are likely to be particularly interested in how future strategies for Poykins Co may differ from those which have been pursued recently. They are also likely to want to know about attitudes to risk management and risk management policies, as the new company appears to be likely to be more risk-seeking than the old division. They will also want to know about changes in finance policy, particularly if dividend policies are likely to differ.

Communication with stakeholders

Poykins Co's directors are likely to communicate with major shareholders on a regular basis, more than once a year. These will include the director-shareholders actively involved in Newimber Co and external investors. Poykins Co's directors will need to ensure that what they communicate keeps both sets of shareholders happy if the two groups have different priorities.

Poykins Co's directors will also have to be mindful of the need to communicate what their plans are to other important stakeholders. Employees and suppliers are particularly important here, as Poykins Co's board has plans for operational efficiencies. Employees may be interested in being informed about changes in working conditions. Attempts to impose tougher conditions on employees without communication or consultation may lead to employee departures or other disruptions. Suppliers will be interested in changes to payment arrangements. Suppliers may be concerned anyway about dealing with a new, smaller company, so may seek to impose shorter credit periods or lower credit limits if they do not have sufficient information.

Use of integrated reporting

In particular, Poykins Co's directors will have to consider if, and how, they use integrated reporting. They are not bound by the decision of Newimber Co's directors to prepare an integrated report. However, if they do not do so, it may suggest to investors and other stakeholders that the directors are not keen to disclose information about how the business is using its resources and maintaining relationships. This may affect their confidence in how the directors are running the company.

If the directors decide to prepare an integrated report for Poykins Co, readers of the accounts are likely to look for significant differences between this report and the integrated report of Newimber Co, and how the differences are justified. These may include differences in the approach to value creation, the outlook for the business and whether the information has been prepared and presented in different ways.

(Note: Credit will be given for alternative, valid comments)

		<i>Marks</i>
1	(a) 1–2 marks per well-discussed comment	Max <u>5</u>
	(b) (i) (Appendix 1)	
	Sales revenue	2
	Variable costs	2
	Fixed costs	1
	Training costs	2
	Tax	2
	Working capital	2
	Uwa Project net present value	<u>1</u>
		12
	(ii) (Appendix 2)	
	Jigu Project: underlying asset value	2
	Honua Co offer: exercise price	1
	Honua Co offer: underlying asset value	2
	Honua Co offer: other variables used in option calculation	1
	Honua Co offer: call value	1
	Honua Co offer: put value	<u>2</u>
		9
	(iii) Initial assessment of value of Uwa Project	2–3
	Up to 2 marks per well-discussed assumption	
	(Max 4 marks if assumptions related to real options are not discussed)	<u>8</u>
		Max <u>10</u>
	Professional marks for part (b)	
	Report format	1
	Structure and presentation of the report	<u>3</u>
		4
	(c) Discussion of the issues	4–5
	Discussion of how the issues may be addressed	<u>5–6</u>
		Max <u>10</u>
		Total <u>50</u>

		<i>Marks</i>
2	(a) Options	
	Buy put options	1
	Number of contracts	1
	Basis calculation	1
	Premium calculation	1
	Exercise option?	1
	Final outcome	1
	Swaps	
	Comparative advantage and recognition of benefit	2
	Initial decision to borrow floating by Lurgshall Co and fixed by counterparty	1
	Swap impact	2
	Net benefit after bank charges	1
	Comments	3–4
	Max	15
	(b) Advantages of swaps	2–3
	Disadvantages of swaps	2–3
	Max	5
	(c) 1–2 marks per relevant point	Max 5
	Total	25
3	(a) Advantages of demerger	2–3
	Disadvantages of demerger	2–3
	Max	5
	(b) MV_d	2
	Pre demerger WACC	1
	New β_e and k_e Newimber Co	1
	New WACC Newimber Co	1
	Pre demerger β_a	1
	β Poynins Co	1
	WACC Poynins Co	1
	Discounted free cash flows Poynins Co Years 1 to 3	3
	Discounted free cash flow Poynins Co Year 4 onwards	2
	Discussion	2–3
	Max	15
	(c) 1–2 marks per relevant point	Max 5
	Total	25

Strategic Professional – Options

Advanced Financial Management (AFM)

September/December 2019 –
Sample Questions



Time allowed: 3 hours 15 minutes

This question paper is divided into two sections:

Section A – This ONE question is compulsory and MUST be attempted

Section B – BOTH questions are compulsory and MUST be attempted

Formulae and tables are on pages 8–12.

Do NOT open this question paper until instructed by the supervisor.

This question paper must not be removed from the examination hall.

AFM

Think Ahead

ACCA

The Association of
Chartered Certified
Accountants

Section A – This ONE question is compulsory and MUST be attempted

- 1 Okan Co, a large listed company located in Yasailand whose currency is the Y\$, manufactures engines and engine parts. It is considering whether or not to invest in one of two new four-year projects: Project Alpha or Project Beta. Details of both projects are given separately. Previously, Okan Co has used relevant risk-adjusted discount rates to calculate the net present value (NPV) of projects. However, the finance director believes that calculating adjusted present values (APV) of projects would be more appropriate. Okan Co wants to base its decision on which project to invest in, on the returns generated by the projects, the projects' risk as measured by their project durations, and important non-financial aspects. Both projects are due to commence in six months' time.

Funding for projects Alpha and Beta

Project Alpha or Project Beta will each require the same amount of initial funding of Y\$50,000,000.

Proceeds from the sale of a factory based in Europe in six months' time, for Euro (€)10,000,000, will provide part of the funding and the balance will be financed by debt borrowing.

Okan Co expects to hedge the €10,000,000 using either forward markets or money markets. The following information is available on these markets:

Foreign exchange rates

	Y\$/€1
Spot	2.5210–2.5862
Six months forward	2.5462–2.6121

Bank interest rates

	Investing	Borrowing
Yasailand	2.40%	5.00%
Eurozone	1.05%	2.20%

The balance of funding raised by domestic debt borrowing will be through a four-year subsidised loan on which interest is payable at 2.1%, although Okan Co's normal borrowing rate is 5%.

Issue costs related to raising this finance will be 3% of the gross proceeds.

Project Alpha details

Project Alpha's base case NPV and APV, in six months' time when the project will commence, should be estimated using the following information.

The sales revenues and production costs related to Project Alpha in six months' time, before any annual price or cost increases, are estimated as follows:

Year	1	2	3	4
Sales revenue (Y\$ 000s)	15,750	28,350	47,250	23,100
Production cost (Y\$ 000s)	6,120	10,710	21,420	8,160

It is expected that the sales price will increase at an annual inflation rate of 10%. Domestic production costs are likely to increase at Yasailand's annual inflation rate.

In addition to the above, components will be imported from the UK (currency £), at the following current cost:

Year	1	2	3	4
Component costs (£ 000s)	1,200	1,800	3,700	1,400

The costs of components from the UK are fixed and not subject to inflation.

The funds of Y\$50,000,000 for Project Alpha will be used to purchase plant and equipment needed for manufacturing purposes. Tax allowable depreciation is available on the value of the plant and equipment at 25% per year on a reducing balance basis, with a balancing allowance or charge applicable at the end of the project. The plant and equipment is expected to be sold for Y\$10,000,000 (post-inflation) at the end of the project.

At the start of every year, Project Alpha will require working capital. In the first year this will be 10% of the estimated year 1 sales revenue. In subsequent years, the project will require an increase or a reduction in working capital of 15% for every \$1 increase or decrease in sales revenue respectively. The working capital is expected to be fully released when Project Alpha ceases.

The expected spot exchange rate between the Y\$ and the £, in six months' time, is expected to be Y\$3.03 per £1. The annual inflation rates are currently 2% in the UK and 4% in Yasailand. It can be assumed that these inflation rates will not change for the foreseeable future.

The cost of capital for appraising the base case net present value of Project Alpha is 10%. Okan Co pays tax at an annual rate of 20%. Tax is payable in the same year as the profits it is based on. Okan Co makes sufficient profits from its other activities to take advantage of any tax loss relief.

Project Beta details

Given below are Project Beta's base case present values, based on the project start date in six months' time, discounted at the project's relevant risk-adjusted all-equity financed discount rate:

Year	1	2	3	4
Present values (Y\$ 000s)	8,450	19,360	22,340	4,950

It can be assumed that any working capital requirements for Project Beta are included in the annual cash flows.

Project Beta's duration has been calculated as 2.43 years, based on its base case present values.

Economic risk and risk categories

One of Okan Co's subsidiary companies in Yasailand, which produces and sells all its products domestically, has still found that it is exposed to economic risk (economic exposure). The directors of the subsidiary believe that this is because Yasailand's government has maintained comparatively higher interest rates, even though the inflation in Yasailand is now under control.

Okan Co categorises the risks inherent in its projects according to the severity of their impact and the frequency of their occurrence, as follows: (i) severe and frequent, (ii) not severe but frequent, (iii) severe but not frequent, and (iv) neither severe nor frequent.

Required:

(a) Discuss why a company may prefer to use the adjusted present value (APV) method, rather than the net present value (NPV) method. (4 marks)

(b) Prepare a report for the board of directors (BoD) of Okan Co which:

(i) Estimates the minimum amount of debt borrowing Okan Co would require; (4 marks)

(ii) Estimates

– Project Alpha's and Project Beta's base case NPV, in six months' time, before considering the financing side effects, (12 marks)

– Project Alpha's and Project Beta's APV, in six months' time, and (6 marks)

– Project Alpha's duration based on its base case present values of cash flows; (2 marks)

(iii) Evaluates and justifies which project Okan Co should choose, basing the decision on the factors Okan Co considers to be important. The evaluation should include a discussion of the assumptions made. (8 marks)

Professional marks will be awarded in part (b) for the format, structure and presentation of the report. (4 marks)

(c) Discuss why Okan Co's subsidiary company may be exposed to economic risk (economic exposure) and how it may be managed. (4 marks)

(d) Discuss how each category of risk, in terms of severity and frequency, may be managed. (6 marks)

(50 marks)

Section B – BOTH questions are compulsory and MUST be attempted

2 Cadnam Co is a large company in the support services sector.

Cadnam Co's most recent annual report, for the year ended 31 December 20X5, acknowledged challenges for the company, including financing the major investment programme required to meet its clients' increasing expectations. Cadnam Co also faced upward pressure on employment costs, fuelled by a 'fair wage' campaign which adversely compared wage rises in the support services sector with increases in dividends and directors' remuneration, and a consequent government enquiry into low pay in the sector.

Cadnam Co's board, however, was confident that the company would be able to renew a number of large contracts which were coming up for review. The report stressed the strength of Cadnam Co's senior management team as a vital success factor. Directors' remuneration packages thus reflected the need to retain its directors in a competitive labour market at senior level.

In the stakeholder engagement section of its annual report, Cadnam Co highlighted that it had fulfilled its aim of guaranteeing investors a consistent rise in dividends and its board was confident that Cadnam Co would be able to maintain the recent rate of dividend increase. The report also stated that Cadnam Co was looking to publish a full integrated report over the next couple of years.

Dividend policy

At Cadnam Co's last annual general meeting, there were no questions about the level of profits, dividends or directors' remuneration. However, a recent investment analysts' report on the support services sector highlighted Cadnam Co as a company which might have problems in the next few years. The report suggested that Cadnam Co's investment and dividend policies could not both be maintained. It highlighted one of Cadnam Co's principal competitors, Holmsley Co, as a company whose policies it believed would sustain long-term growth. It highlighted directors' remuneration as an area where Holmsley Co's policies were more likely to encourage long-term value creation and share price increases than Cadnam Co's policies.

Cadnam Co's board is currently considering the comments made by the investment analysts, and also assessing what the dividend for 20X6 should be.

Cadnam Co

	20X2	20X3	20X4	20X5
	\$m	\$m	\$m	\$m
Profit after tax	1,380	1,490	1,550	1,580
Dividends	765	840	925	1,020
Investment in additional assets	282	312	584	864
Share price (\$)	\$4.88	\$5.35	\$5.61	\$5.75
Gearing (debt/(debt + equity)) (market value) x 100%	33.0%	33.2%	35.0%	38.8%

Holmsley Co

	20X2	20X3	20X4	20X5
	\$m	\$m	\$m	\$m
Profit after tax	1,485	1,590	1,700	1,830
Dividends	560	590	621	654
Investment in additional assets	595	625	660	690
Share price	\$5.04	\$5.23	\$5.55	\$5.93
Gearing (debt/(debt + equity)) (market value) x 100%	35.1%	35.2%	34.9%	34.7%

Average gearing in the support services sector since 20X1 has been stable at around 34%. There have been no changes in the issued share capital of Cadnam Co and Holmsley Co since 20X1.

Directors' remuneration

	Cadnam Co	Holmsley Co
Average salary executive director	\$550,000	\$550,000
Performance bonus	Maximum 25% of salary	Maximum 30% of salary
Loyalty bonus	Maximum 10% of salary	None
Share options	None	Options to be exercised on 31 December 20X8 at an exercise price of \$7.00

Cadnam Co 20X6 forecast

Forecasts prepared by Cadnam Co's finance director for 20X6 predict that:

- Cadnam Co's pre-tax operating profit for 20X6 will be \$2,678m, an increase of 3% compared with 20X5. The operating profit margin will be 2%, the same as for 20X5.
- The tax rate will be 30%.
- Average debt in 20X6 will be \$10,250m and predicted year-end gearing will be 41.3%. The average pre-tax interest rate on the debt will be 8%.
- The investment required to keep the non-current asset base at its present productive capacity in 20X6 will be \$2,430m, which has been included in the calculation of operating profit as depreciation.
- Investment required in additional assets in 20X6 will be \$0.25 for every \$1 increase in revenue.

Required:

(a) Calculate the forecast dividend capacity of Cadnam Co for 20X6. (5 marks)

(b) Discuss the viability and financial impacts of Cadnam Co seeking to maintain its current dividend policy, supporting your answers with relevant calculations.

Note: 6 marks are available for calculations in part (b). (12 marks)

(c) Discuss the governance and ethical issues associated with Cadnam Co's dividend and directors' remuneration policies. (8 marks)

(25 marks)

- 3 A new client has approached you for advice on a potential acquisition. Kerrin Co is a consumer electronics manufacturer and retailer. The company obtained a listing eight years ago with the founders retaining a 20% stake in the business. Whilst Kerrin Co had previously experienced rapid growth in earnings before tax, problems arose soon after the listing as competition intensified. Although the company remains profitable, annual growth has declined significantly and is currently 3%.

The board is concerned by the lack of future growth opportunities. The current share price reflects these concerns, trading well below the offer price of eight years ago. In response, the directors have decided to invest in a market development strategy for future growth, utilising significant cash reserves to acquire companies in other areas of the country where competition is less intense. The board has identified a potential target, Danton Co.

Danton Co

Danton Co is a privately owned consumer electronics company, established ten years ago. Significant unrelieved losses were incurred in the early years of development although the company is now profitable and achieving growth in earnings before tax of 6% per year. However, cash reserves are low. Access to capital has acted as a severe constraint on Danton Co's reinvestment potential throughout this period. The founders and their families own 60% of the shares with the balance held by a venture capitalist organisation, which acquired its equity stake around six years ago.

Acquisition information

Kerrin Co's board is keen to ensure that Danton Co's founders remain as directors after the acquisition and the company has sufficient cash reserves to purchase Danton Co outright.

Early discussions between the directors of both companies suggest Danton Co's shareholders would approve a cash offer of \$13.10 per share. As an alternative, the board is considering a share-for-share exchange to fund the acquisition in order to preserve cash for future acquisitions and dividend payments. Recent mergers have attracted an acquisition premium of around 25%–30% and Danton Co's directors indicated their shareholders would be expecting a premium towards the higher end of this scale for a share-for-share offer. Kerrin Co has therefore asked you to design a share-for-share offer scheme which will allow for a 30% acquisition premium. You have been provided with extracts from the latest financial statements for both companies.

Extracts from the most recent financial statements

	Kerrin Co	Danton Co
	\$m	\$m
Operating profit	448.6	201.8
Earnings before tax	381.9	116.3

Additional financial information

The book value of Kerrin Co's \$0.50 ordinary shares is \$375m. These shares are currently trading at \$5.28 and the finance director expects the price earnings (PE) ratio to increase by 10% if the acquisition proceeds.

Danton Co upgraded its main manufacturing facility during the previous year and expects to make annual pre-tax cost savings of \$2.5m from the start of the current financial year. The book value of Danton Co's \$0.25 ordinary shares is \$35m. Based on an analysis of companies of a comparable size and cost structure, it is estimated that Danton Co's PE ratio is 20% higher than Kerrin Co's current PE ratio.

Kerrin Co's chief executive officer estimates annual pre-tax revenue and cost synergies of \$15.2m to arise as a result of the acquisition. In addition, the finance director anticipates annual pre-tax financial synergies of \$5.3m although she insists this is a cautious estimate after reading an article on recent merger and acquisition activity where post-acquisition synergies have either been overestimated or failed to materialise.

The rate of corporation tax relevant to both companies is 20%.

Required:

- (a) Discuss possible sources of financial synergy arising from Kerrin Co's acquisition of Danton Co and comment on the finance director's concern that synergy is often overestimated, including any steps which could be taken by Kerrin Co's board to address this problem. (8 marks)
- (b) Advise the directors on a suitable share-for-share exchange offer which meets the criteria specified by Danton Co's shareholders and calculate the effect of the cash and share-for-share offers on the post-acquisition wealth of both Kerrin Co's and Danton Co's shareholders. (13 marks)
- (c) Discuss the likely reaction of Kerrin Co's and Danton Co's shareholders to the cash and share-for-share offers. (4 marks)

(25 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d^i) \frac{V_d}{V_e}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i(E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_o = \frac{D_o(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = br_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \qquad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model

$$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$$

Where:

$$d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$$

$$d_2 = d_1 - s\sqrt{t}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

		<i>Discount rate (r)</i>									
<i>Periods</i>											
(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

		<i>Discount rate (r)</i>									
<i>Periods</i>											
(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Answers

- 1 (a) Adjusted present values (APVs) separate out a project's cash flows and allocate a specific discount rate to each type of cash flow, dependent on the risk attributable to that particular type of cash flow. Net present value (NPV) discounts all cash flows by the average discount rate attributable to the average risk of a project.

One reason why APV may be preferable to NPV is because by separating out different types of cash flows, the company's managers will be able to see which part of the project generates what proportion of the project's value. Furthermore, allocating a specific discount rate to a cash flow part helps determine the value added or destroyed. In this example, Okan Co is able to determine how much value is being created by the investment and how much by the debt financing. For complex projects, investment related cash flows could be further distinguished by their constituent risk factors, where applicable.

(b) Report to the board of directors (BoD), Okan Co

Introduction

This report evaluates, and provides a justification and decision, on whether Okan Co should pursue Project Alpha or Project Beta, based on the important factors identified by the company, namely the returns generated by the projects, the projects' risks and non-financial aspects.

Evaluation

Financing

Using forward markets to hedge the expected receipt in six months' time results in the higher receipt equalling Y\$25,462,000 approximately. If the money markets hedge is used the receipt is Y\$25,234,936 (appendix 1).

Using forward markets to hedge the expected receipt would therefore minimise the amount of debt borrowing. However, the amount receivable from the money markets hedge is based on the annual bank investment rate available to Okan Co of 2.4%. Okan Co may be able to use the funds borrowed to generate a higher return than the bank investment and therefore using money markets to undertake the hedge may be financially advisable. Okan Co should investigate any opportunities for higher income, but based on the current results, the forward market hedge is recommended to minimise the amount of debt finance needed.

Minimum amount of debt borrowing required is Y\$24,538,000 approximately.

Project returns and risk

	Project Alpha	Project Beta
Base case net present value (NPV)		
(in six months' time)	Y\$5,272,000 (appendix 2a)	Y\$5,100,000 (appendix 2a)
Adjusted present value (APV)	Y\$6,897,218 (appendix 2b)	Y\$6,725,218 (appendix 2b)
Project duration	3.04 years (appendix 2c)	2.43 years (given)

Project Alpha's and Project Beta's base case NPVs and APVs are similar to each other, with Project Alpha expected to yield a small amount in excess to the yield expected from Project Beta. However, Project Beta's project duration is significantly lower. This is because a higher proportion of Project Beta's cash flows come earlier in the project's life, compared to Project Alpha. There is more certainty to earlier cash flows and this is reflected in the lower duration for Project Beta. Project Beta's risk is lower than Project Alpha.

In estimating the base case NPV and APV for Project Alpha, it is assumed that the cash flows are known with reasonable certainty and the inflation rates will not change during the life of the project. It is also assumed that the future exchange rate between the Y\$ and the £ will change in accordance with the purchasing power parity differential. Furthermore, it is assumed that the prices and costs related to Project Alpha will increase in line with inflation during the six months before the project starts. For Project Alpha, it is assumed that the initial working capital requirement is funded by the company and not from the funds raised from the subsidised loan, similar to the assumption made for Project Beta. However, for both projects, Okan Co needs to consider, and take account of, the opportunity costs related to this.

In terms of the Project Alpha's discount rate, it is assumed that the given discount rate accurately reflects the business risk of the project.

Whilst this level of detail is not provided for Project Beta, it is assumed that similar assumptions will have been made for Project Beta as well. In the case of both projects, Okan Co should assess the accuracy or reasonableness of the assumptions, and if necessary, conduct sensitivity analysis to observe how much the projects' values change if input variables are altered.

Notwithstanding the assumptions and caveats made above, it would appear that Project Beta would be preferable to Project Alpha, given that it has a similar APV but a significantly lower risk.

Nevertheless, there may be good strategic reasons why Okan Co may select Project Alpha over Project Beta. For example, these reasons may include providing access to new markets, enabling Okan Co to erect barriers to entry against competitors or looking at follow-on opportunities as possible real options.

Justification

Due to the substantially lower risk (as measured by the project duration) and similar APV, it is recommended that Project Beta be selected by Okan Co. It is also recommended that forward markets are used to hedge the income expected in six months' time to part fund the project. This would minimise the debt borrowing needed.

However, this decision is predicated on the fact that the implications of the assumptions and the wider strategic reasons discussed above have been carefully considered by Okan Co.

Report compiled by:

Date

(Note: Credit will be given for alternative and valid evaluative comments)

Appendices:

Appendix 1 (Part (b) (i)):

Expected receipt in six months' time, using forward markets
 $\text{€}10,000,000 \times 2.5462 = \text{Y\$}25,462,000$

Expected receipt in six months' time, using money markets
 $\text{€}10,000,000 / (1 + 0.022/2) = \text{€}9,891,197$
 $\text{€}9,891,197 \times 2.5210 = \text{Y\$}24,935,708$
 $\text{Y\$}24,935,708 \times (1 + 0.024/2) = \text{Y\$}25,234,936$

Minimum amount of debt borrowing Okan Co would require
 $\text{Y\$}50,000,000 - \text{Y\$}25,462,000 = \text{Y\$}24,538,000$

Appendix 2a (Part (b) (ii)): Projects Alpha and Beta, base case net present value, in six months' time

Base case net present value before considering financing side effects. All figures are in Y\$000s.

Year	0	1	2	3	4
Sales revenue (w1)		17,325	34,304	62,890	33,821
Less:					
Production costs (w2)		(6,365)	(11,584)	(24,095)	(9,546)
Component costs (w3)		(3,708)	(5,670)	(11,877)	(4,578)
Cash flows before tax		7,252	17,050	26,918	19,697
Tax (w4)		1,050	(1,535)	(3,977)	(1,721)
Working capital	(1,733)	(2,547)	(4,288)	4,360	4,208
Plant purchase and sale	(50,000)				10,000
Net cash flows	(51,733)	5,755	11,227	27,301	32,184
Base case present value of cash flows (discounted at 10%)	(51,733)	5,232	9,279	20,512	21,982

Approximate, base case net present value (NPV) of Project Alpha = Y\$5,272,000.

Base case net present value (NPV) of Project Beta = Y\$(8,450,000 + 19,360,000 + 22,340,000 + 4,950,000) – Y\$50,000,000 = Y\$ 5,100,000

Workings:

Working 1 (w1): Sales revenue

Year	1	2	3	4
Pre-inflated revenues (Y\$ 000s)	15,750	28,350	47,250	23,100
Inflation	$\times 1.1^1$	$\times 1.1^2$	$\times 1.1^3$	$\times 1.1^4$
Post-inflated revenues (Y\$ 000s)	17,325	34,304	62,890	33,821

Working 2 (w2): Production costs

Year	1	2	3	4
Pre-inflated production costs (Y\$ 000s)	6,120	10,710	21,420	8,160
Inflation	$\times 1.04^1$	$\times 1.04^2$	$\times 1.04^3$	$\times 1.04^4$
Post-inflated production costs (Y\$ 000s)	6,365	11,584	24,095	9,546

Component costs are not inflated, but future exchange rates are based on purchasing power parity (PPP).

Working 3 (w3): Component cost

Year	1	2	3	4
PPP multiplier	3.03 x	3.09 x	3.15 x	3.21 x
	1.04/1.02	1.04/1.02	1.04/1.02	1.04/1.02
Forecast Y\$ per £1	3.09	3.15	3.21	3.27
Component cost (£)	1,200	1,800	3,700	1,400
Component cost (Y\$)	3,708	5,670	11,877	4,578

Working 4 (w4): Tax

Year	1	2	3	4
Cash flows before tax	7,252	17,050	26,918	19,697
Tax allowable depreciation	(12,500)	(9,375)	(7,031)	(11,094)
Taxable cash flows	(5,248)	7,675	19,887	8,603
Tax payable (20%)	(1,050)	1,535	3,977	1,721

Appendix 2b (Part (b) (ii)): Projects Alpha and Beta, adjusted present value (APV), in six months' time

Issue costs = $3/97 \times \text{Y\$}24,538,000 = \text{Y\$}758,907$

Annual tax shield = $2.1\% \times \text{Y\$}24,538,000 \times 20\% = \text{Y\$}103,060$

Annual interest saved on subsidised loan = $2.9\% \times \text{Y\$}24,538,000 \times 80\% = \text{Y\$}569,282$

Annuity factor, years 1 to 4 at 5% interest = 3.546

Present value of the tax shield and loan subsidy benefit = $(\text{Y\$}103,060 + \text{Y\$}569,282) \times 3.546 = \text{Y\$}2,384,125$

Project Alpha APV

	Y\$
Base case NPV of Project Alpha (appendix 2a)	5,272,000
Issue costs	(758,907)
Present value of the tax shield and loan subsidy benefit	2,384,125
APV	6,897,218

Project Beta APV

	Y\$
Base case NPV of Project Beta (appendix 2a)	5,100,000
Issue costs	(758,907)
Present value of the tax shield and loan subsidy benefit	2,384,125
APV	6,725,218

Appendix 2c (Part (b) (ii)): Project Alpha's duration based on its base case present values of cash flows**Project Alpha**

Year	1	2	3	4
PVs x years	5,232,000 x 1	9,279,000 x 2	20,512,000 x 3	21,982,000 x 4
	= 5,232,000	= 18,558,000	= 61,536,000	= 87,928,000

Total PVs x time = 173,254,000 approximately

Total PVs = 57,005,000 approximately

Project Alpha duration = $173,254,000/57,005,000 = 3.04$ years

(c) Explanation of why the subsidiary company may be exposed to economic risk and how it may be managed

Companies face economic exposure when their competitive position is affected due to macroeconomic factors such as changes in currency rates, political stability, or changes in the regulatory environment. Long-term economic exposure or economic shocks can cause a permanent shift in the purchasing power and other parity conditions. Normally, companies face economic exposure when they trade internationally. However, even companies which do not trade internationally nor rely on inputs sourced internationally may still face economic exposure.

In the case of Okan Co's subsidiary company, economic risk may have occurred because interest rates have been kept at a high level, causing the original parity conditions to break down. High interest rates will be attractive to international investors, as they can get higher returns and may lead to the Y\$ becoming stronger relative to other currencies. This in turn would allow international competitors to produce goods more cheaply than the subsidiary company and thereby enhance their competitive position relative to the subsidiary company.

Managing economic exposure is difficult due to its long-lasting nature and because it can be difficult to identify. Financial instruments, such as derivatives, and money markets cannot normally be used to manage such risks. Okan Co's subsidiary company can try tactics such as borrowing in international or eurocurrency markets, sourcing input products from overseas suppliers and ultimately shifting production facilities overseas. None of these are easy or cheap, and can expose the company to new types of risks. Okan Co would also need to assess that any action it takes to manage economic risk fits into its overall risk management strategy.

(d) How each category or risk may be managed

Risks which fall into the severe and frequent category need immediate attention, as they could threaten the company's survival or derail its long-term strategy. The aim here would be to reduce the severity of the risks and the frequency with which they occur quickly. It may mean avoiding certain actions or abandoning certain projects, even if they could be profitable in the long term. Where a company has a real option, and does not need to take action which will result in high frequency and high severity of risk, it may prefer to wait and see what happens.

Where the frequency of risks occurring is high but their impact is not severe, action needs to be taken so that such risks do not become severe in the future. For example, the company could put systems into place to detect these risks early and plans to deal with them if they do occur. Where the same kind of risks occur often, the company may decide to have set processes for dealing with them. For example, where there is a loss of relatively unskilled staff, the company may decide to replace staff quickly with casual workers, but also have appropriate training facilities in place.

If there are risks which are severe but only happen occasionally or infrequently, the company should try to insure against these. Contingency plans could also be put into place to mitigate the severity. For example, if the consequences of IT failure are high when a business decides to move to a new system, it could put appropriate contingencies into place. These may include secondary backup IT systems or initially trialling the new system on a few business units before undertaking a complete role out.

Risks which are neither severe, nor frequent, should be monitored and kept under review, but no significant action should be taken. It is possible that any significant action would incur costs which would likely be higher than the benefits derived from eliminating such risks. Monitoring such risks will ensure that should they move out of this category into the more severe/frequent categories, the company can start to take appropriate action.

(Note: Credit will be given for alternative and valid explanatory and discussion comments)

2 (a) Dividend capacity

	\$m
Operating profit	2,678
Less: Interest (8% x \$10,250m)	(820)
Less: Taxation (30% x (\$2,678m – \$820m))	(557)
Less: Investment in additional assets (25% x 0.03 x \$2,678m/(1.03 x 0.02))	(975)
Forecast dividend capacity	326

(b) Growth in profit after tax

	20X3	20X4	20X5	Geometric mean annual growth rate
	%	%	%	%
Cadnam	8.0	4.0	1.9	4.6
Holmsley	7.1	6.9	7.6	7.2

Dividend payout ratios

	20X2	20X3	20X4	20X5
	%	%	%	%
Cadnam	55.4	56.4	59.7	64.6
Holmsley	37.7	37.1	36.5	35.7

Growth in dividends

	20X3	20X4	20X5	Geometric mean annual growth rate
	%	%	%	%
Cadnam	9.8	10.1	10.3	10.1
Holmsley	5.4	5.3	5.3	5.3

Residual profit (after tax-profit for the year – dividend – new investment)

	20X2	20X3	20X4	20X5
	\$m	\$m	\$m	\$m
Cadnam	333	338	41	(304)
Holmsley	330	375	419	486

Growth in share price

	20X3	20X4	20X5	Geometric mean annual growth rate
	%	%	%	%
Cadnam	9.6	4.9	2.5	5.6
Holmsley	3.8	6.1	6.8	5.6

Comments

Dividends

Both companies have shown fairly consistent increases over the last three years, with Cadnam Co's dividends increasing at around 10% each year, and Holmsley Co's dividends increasing at around 5% over the last three years. However, Holmsley Co's policy appears to be sustainable at present, whereas it is doubtful whether Cadnam Co's policy is sustainable.

Holmsley Co has managed to increase dividends, gradually also increasing investment in additional assets and residual profits, whilst at the same time having a decreasing dividend pay-out ratio.

In order to maintain its rate of dividend increase, however, Cadnam Co has had to pay out an increasing proportion of earnings each year. It seems that Cadnam Co may have sustained dividend increases up to 20X3 at the expense of additional investment, and is now having to increase additional investment significantly in order to make up for previous under-investment. In 20X5, Cadnam Co's residual profits became negative, and the dividend capacity calculation for 20X6 suggests that a much lower level of dividends would be appropriate.

Gearing

Holmsley Co's gearing appears to be stable at around the average for the industry, suggesting that perhaps it has found its optimum level. Up to 20X4, Cadnam Co could perhaps have been taking advantage of debt capacity to increase its debt towards this level. However, its gearing now appears to be on a rising trend, of necessity increasing significantly in 20X5 to fund both additional investment and increasing dividends, despite the rise in its share price. Cadnam Co's gearing is predicted to increase further in 20X6, but how long this is sustainable is uncertain.

Share price

The average increase for both companies over the last four years has been the same. However, the percentage increases over the last two years for Holmsley Co has been higher than for Cadnam Co. This suggests that the market has placed more significance on the higher % growth in Holmsley Co's after-tax profits than in Cadnam Co's higher % growth in dividends, maybe seeing this as an indication that Holmsley Co's strategy has been more successful and is more likely in future to deliver higher share price growth.

(c) Dividend policy

One possible question is whether the statement in the annual report fairly reflects the likely future dividend policy of Cadnam Co. The report gives the impression that the current dividend policy will be sustained, whereas the figures suggest that this may not be the case. If the policy proves not to be sustainable, it would suggest a failure either of integrity (if the directors made a statement with a high risk that it would not be true) or due care (that they failed to take into account indicators which suggested their policy is not sustainable). The directors may be questioned by the auditors about whether this statement is true and fair.

There is also the question of balancing the interests of different stakeholders. To some degree, criticism of rises in dividend and director remuneration levels versus increases in employee salary levels could be said to be a matter of opinion.

However, the fact that there is a government enquiry into low pay in the sector suggests that pay levels are lower than society deems desirable. The force of the criticisms may be enhanced by the statements which Cadnam Co has made about developing an integrated reporting approach. Integrated reporting is not just about extra details in the annual report, but also reflects an underlying policy of carrying out business which includes responsiveness to the needs of different stakeholders.

The dividend capacity and gearing figures may also call into question whether Cadnam Co's board is taking excessive risks. Does paying out increasing dividends in future mean that the company is likely to have inadequate resources to sustain its business, and may be jeopardising the interests of lenders and employees as well as stakeholders? Certainly there appears to be doubts about maintenance of future income levels with a number of contracts coming up for renewal and terms possibly being tightened by clients. Clients may be doubtful about renewing contracts if Cadnam Co's solvency appears doubtful.

Directors' remuneration

The directors' remuneration packages also raise concerns. Comparison with Holmsley Co shows that salary, which is not dependent on performance, is a more significant element of the remuneration packages at Cadnam Co than Holmsley Co. Both companies have bonuses which depend to some degree on performance. However, Cadnam Co's directors are also rewarded by loyalty bonuses, which again do not depend on performance but staying with the company. Holmsley Co has a share option scheme in place, which would seem to reward longer-term good performance, although it cannot be determined how significant a part of remuneration share options will be. Cadnam Co's remuneration scheme appears only to reward short-term profitability, possibly meaning that the directors may neglect the longer-term success and possibly even viability of the company.

3 (a) Financial synergies

Many acquisitions are justified on the basis that the combined organisation will be more profitable or grow at a faster rate than the companies operating independently. The expectation is that the acquisition will generate higher expected cash flows or a lower cost of capital, creating value for shareholders. The additional value created is known as synergy, the sources of which can be categorised into three types: revenue, cost and financial synergies.

Based on the scenario, there are a number of possible sources of financial synergy. As a private company, Danton Co is experiencing a funding constraint whereas Kerrin Co has significant cash reserves but limited growth opportunities. The combination of the two can create additional value since Danton Co may be able to utilise Kerrin Co's cash resources to fund its expansion in a way which would not have been possible otherwise, leading to an increase in the expected cash flows.

Assuming both companies' cash flows are less than perfectly correlated, those of the combined company will be less volatile than the individual companies operating independently. This reduction in volatility enables the combined company to borrow more and possibly cheaper financing than would otherwise have been possible. This increase in debt capacity, and therefore the present value of the tax shield, increases the value of the combined company in the form of a lower cost of capital.

Further benefits may arise if Kerrin Co is able to utilise Danton Co's unrelieved tax losses. Whilst Danton is no longer loss making and could offset these tax losses independently, the combined company may be able to obtain tax relief earlier since the acquisition increases the availability of profits against which carried forward tax losses can be offset. The present value of the tax saved will therefore be greater in the combined company.

If both companies were publicly traded, there would be no benefit from diversification since investors are capable of diversifying at a lower cost and with greater ease than the company. However, Danton Co is privately owned and the shareholders are therefore exposed to diversifiable unsystematic risk. Therefore the acquisition may lead to potential diversification and risk reduction benefits. The reduction in the cost of the capital increases the value of the combined company.

Overestimation of synergy value

There is evidence that bidding companies often overestimate the value of synergy arising from a potential acquisition with the result that companies pay too much for their target. When this happens, there is destruction in wealth for the bidding company's shareholders. There are a number of possible explanations for this problem.

First, merger and acquisition activity tends to be driven by the availability of cheap credit. At the peak of a wave of activity, there may be competition for targets, thereby increasing acquisition premiums.

Second, conflicts of interest may lead to a biased evaluation process. Deal advisers such as investment banks earn a large proportion of their fees from mergers and acquisitions. Their advice on whether an acquisition makes sense is potentially biased if they do not look after their clients' interests.

Third, management overconfidence may explain why this occurs. Acquiring companies may overestimate the acquisition synergy and/or underestimate the time it will take to deliver. Management may then be reluctant to admit mistakes when the facts change, even when there is still time to back out of a deal. Agency costs may be also a factor if managers are more interested in pursuing personal goals than maximising shareholder wealth.

Finally, there may be difficulties integrating the companies due to different work cultures and conflicts of interest.

Steps to address this problem

Kerrin Co's board needs to plan for synergy and take active steps to ensure that it is delivered. This responsibility needs to be allocated to someone who can ensure spare cash is utilised to invest in new growth opportunities, that tax losses are offset as efficiently as possible and that the combined company avails itself of cheaper financing. Companies which allocate this responsibility and monitor and review performance tend to be more successful in creating value. In order to avoid any bias, the deal advisers who stand to profit from an acquisition need to be separate from the evaluation process. Effective due diligence ensures the financial documents which form the basis of a valuation are scrutinised and inspected.

(b) Pre-acquisition valuations

Kerrin Co number of shares = $375\text{m}/0.5 = 750\text{m}$

Kerrin Co market value = $750\text{m} \times \$5.28 = \$3,960\text{m}$

Future maintainable earnings (FME) = $(\$381.9) \times 0.8 = \305.5m

Price earnings (PE) ratio = $\$3,960\text{m}/\$305.5\text{m} = 12.96$

Danton Co future maintainable earnings = $(\$116.3\text{m} + \$2.5\text{m}) \times 0.8 = \95.0m

Danton Co PE ratio = $12.96 \times 1.20 = 15.55$

Danton Co PE valuation = $15.55 \times \$95\text{m} = \$1,477.3\text{m}$

Combined Co pre-acquisition valuation = $\$3,960\text{m} + \$1,477.3\text{m} = \$5,437.3\text{m}$

Post-acquisition valuation including synergies

Combined Co FME = $\$305.5\text{m} + \$95\text{m} + (\$20.5\text{m} \times 0.8) = \416.9m

Combined Co PE ratio $12.96 \times 1.1 = 14.3$

Combined Co post-merger valuation = $14.3 \times \$416.9\text{m} = \$5,961.7\text{m}$

Value created based on synergies = $\$5,961.7\text{m} - \$5,437.3\text{m} = \$524.4\text{m}$

Share-for-share offer

	Kerrin Co \$m	Danton Co \$m
Pre-acquisition valuation	3,960.0	1,477.3
Add premium (\$1,477.3m x 0.3)		443.2
Balance of excess value to Kerrin Co: \$524.4m – \$443.2m	81.2	
Post-merger valuation	4,041.2	1,920.5
Relative valuation	2.1	1

Kerrin Co new share issue = $750\text{m}/2.1 = 357.14\text{m}$

Number of existing Danton Co shares = $35\text{m} \times 4 = 140\text{m}$

Therefore share-for-share offer = $357.14/140$, i.e. approximately 18 Kerrin Co shares for every 7 Danton Co old shares

Advice on terms of share-for-share offer

Danton Co shareholders would receive $140\text{m} \times (18/7) = 360\text{m}$ new Kerrin Co shares.

Total Kerrin Co shares = $750\text{m} + 360\text{m} = 1,110\text{m}$.

Kerrin Co shareholders own 67.6% ($750\text{m}/1,110\text{m}$) and Danton Co shareholders 32.4% ($360\text{m}/1,110\text{m}$) of the post-acquisition company.

Impact on shareholder wealth

	Kerrin Co \$m	Danton Co \$m
Pre-acquisition valuation	3,960.0	1,477.3
Cash offer		
Danton Co shareholders cash received: \$13.10 x 140m shares		1,834.0
Kerrin Co post-acquisition equity valuation: \$5,961.7m less acquisition cost of \$1,834.0m	4,127.7	
Increase in shareholder wealth	4.2%	24.1%

Share-for-share offer

Post-acquisition value

Kerrin Co: $(750/1,110) \times \$5,961.7$	4,028.2	
Danton Co $(360/1,110) \times \$5,961.7$		1,933.5
Increase in shareholder wealth	1.7%	30.9%

The terms of the share-for-share offer meet the criteria specified by Danton Co's directors.

(c) Cash offer

The main advantage of a cash offer is that it provides Danton Co's shareholders with a certain and immediate return. However, the premium is lower compared to the share-for-share offer and may be reduced even further if the realised gain gives rise to a tax liability. By indicating their preferred premium under both offers, it is possible the shareholders have priced in the risks associated with an uncertain share-for-share offer and on this basis may be indifferent between the two. The cash offer may give rise to agency issues since Danton Co's founders no longer have a stake in the business even though Kerrin Co's board is keen to ensure the founders remain in position after the acquisition. The information provided is too limited to read too much into the intentions of the venture capitalist. However, typically a venture capitalist would be expected to exit within three to five years. In this case, they may prefer the certainty of the cash offer.

The cash offer transfers more of the added value to Kerrin Co without the need for dilution, which may appeal to the shareholders. As indicated in the question, Kerrin Co's existing reserves are sufficient to fund the cash offer although this may constrain future dividends and/or investment decisions.

Share-for-share offer

Both sets of shareholders benefit from increased wealth as a result of the share offer, albeit only marginally so in the case of Kerrin Co's shareholders. Another drawback is that Kerrin Co's shareholders' percentages are also diluted under this method. However, a share-for-share offer would ensure that Danton Co's founders' interests are aligned with Kerrin Co's shareholders, reducing possible agency costs. It also provides Danton Co's shareholders with the right to participate in the future growth of the larger company, which the cash offer would prevent.

No basis has been provided for the synergistic benefits; the increase in shareholder wealth is so marginal even a minor deviation from the estimates could result in a reduction in shareholder wealth for the owners of Kerrin Co. On this basis, it is quite likely they will not approve a share-for-share offer without further negotiation around the acquisition premium.

(Note: Credit will be given for alternative and valid comments)

		<i>Marks</i>
1	(a) 1–2 marks per well-discussed comment	Max <u>4</u>
	(b) (i) (Appendix 1)	
	Forward market hedge	1
	Money markets hedge	2
	Minimum borrowing required	1
		<u>4</u>
	(ii) (Appendix 2a)	
	Sales revenue	2
	Production costs	1
	Component costs	3
	Tax	2
	Working capital	2
	Project Alpha base case NPV	1
	Project Beta base case NPV	1
		<u>12</u>
	(Appendix 2b)	
	Issue costs	1
	Annual tax shield	1
	Annual subsidy	1
	Present value of tax shield and subsidy	1
	Project Alpha adjusted present value	1
	Project Beta adjusted present value	1
		<u>6</u>
	(Appendix 2c)	
	Project Alpha duration	2
	(iii) Discussion of the assumptions made	4–5
	Evaluation and justification	3–4
	(Maximum 7 marks if no considered justification given)	Max <u>8</u>
	Professional marks for part (b)	
	Report format	1
	Structure and presentation of the report	3
		<u>4</u>
	(c) Explanation of economic risk faced by Okan Co	2–3
	Discussion of management of economic risk	1–2
		Max <u>4</u>
	(d) Discussion of management of each of the four risk categories	Max <u>6</u>
	(Maximum 2 marks per risk category discussed. Maximum of 5 marks if not all categories discussed)	
	Total	<u>50</u>

		<i>Marks</i>
2	(a) Interest	1
	Tax	1
	Investment in additional assets	2
	Depreciation	1
		5
	(b) Calculations	
	Growth in PAT	1–2
	Dividend payout ratios	1–2
	Growth in dividends	1–2
	Residual profit	1–2
	Growth in share price	1–2
	Calculations	Max 6
	Discussion	
	Dividends	2–3
	Gearing	2
	Share price	2
		Max 12
	(c) Dividend policy statement	Up to 5
	Directors' remuneration	Up to 5
		Max 8
		Total 25
3	(a) Financial synergies	4–5
	Overestimation of synergies	2–3
	Proposed steps	1–2
		Max 8
	(b) Kerrin PE ratio	2
	Danton valuation	2
	Post-acquisition valuation	3
	Share-for-share offer terms	2
	Advice	1
	Impact on shareholder wealth	3
		13
	(c) Cash offer	2–3
	Share-for-share offer	2–3
		Max 4
		Total 25

Strategic Professional – Options

Advanced Financial Management (AFM)

Friday 6 March 2020



AFM

Time allowed: 3 hours 15 minutes

This question paper is divided into two sections:

Section A – This ONE question is compulsory and MUST be attempted

Section B – BOTH questions are compulsory and MUST be attempted

Formulae and tables are on pages 10–14.

Do NOT open this question paper until instructed by the supervisor.

This question paper must not be removed from the examination hall.

Think Ahead

ACCA

The Association of
Chartered Certified
Accountants

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The question paper begins on page 3.

Section A – This ONE question is compulsory and MUST be attempted

1 Introduction

Westparley Co is a listed retailer, mainly selling food and small household goods. It has outperformed its competitors over the last few years as a result of providing high quality products at reasonable prices, and also having a stronger presence online. It has kept a control on costs, partly by avoiding operating large stores on expensive city centre sites. Instead, it has had smaller stores on the edge of cities and towns, and a limited number of larger stores on convenient out-of-town sites, aiming at customers who want their journeys to shops to be quick. One of its advertising slogans has been: 'We are where you want us to be.'

Westparley Co's share price has recently performed better than most companies in the retail sector generally. Share prices in the retail sector have been relatively low as a result of poor results due to high competition, large fixed cost base and high interest rates. The exception has been shares in retailers specialising in computer and high-technology goods. These shares appear to have benefited from a boom generally in share prices of high-technology companies. Some analysts believe share prices of many companies in the high-technology sector are significantly higher than a rational analysis of their future prospects would indicate.

Matravers Co

Westparley Co has identified the listed retailer Matravers Co as an acquisition target, because it believes that Matravers Co's shares are currently undervalued and part of Matravers Co's operations would be a good strategic fit for Westparley Co.

Matravers Co operates two types of store:

Matravers Home mainly sells larger household items and home furnishings. These types of retailer have performed particularly badly recently and one major competitor of Matravers Home has just gone out of business. Matravers Home operates a number of city centre sites but has a much higher proportion of out-of-town sites than its competitors.

Matravers Tech sells computers and mobile phones in much smaller outlets than those of Matravers Home.

Extracts from Matravers Co's latest annual report are given below:

	\$m
Pre-tax profit	1,950
Long-term loan	6,500
Share capital (\$1 shares)	5,000

The share of pre-tax profit between Matravers Home and Matravers Tech was 80:20.

The current market value of Matravers Co's shares is \$12,500m and its debt is currently trading at its book value. Westparley Co believes that it will have to pay a premium of 15% to Matravers Co's shareholders to buy the company.

Westparley Co intends to take advantage of the current values attributed to businesses such as Matravers Tech by selling this part of Matravers Co at the relevant sector price earnings ratio of 18, rather than a forecast estimate of Matravers Tech's present value of future free cash flows of \$4,500m.

The company tax rate for both companies is 28% per year.

Post-acquisition cost of capital

The post-acquisition cost of capital of the combined company will be based on its cost of equity and cost of debt. The asset beta post-acquisition can be assumed to be both companies' asset betas weighted in proportion to their current market value of equity.

Westparley Co has 4,000 million \$1 shares in issue, currently trading at \$8.50. It has \$26,000m debt in issue, currently trading at \$105 per \$100 nominal value. Its equity beta is 1.02.

Matravers Co's asset beta is 0.75. The current market value of Matravers Co's shares is \$12,500m and its long-term loan is currently trading at its book value of \$6,500m.

The risk-free rate of return is estimated to be 3.5% and the market risk premium is estimated to be 8%.

The pre-tax cost of debt of the combined company is expected to be 9.8%. It can be assumed that the debt:equity ratio of the combined company will be the same as Westparley Co's current debt:equity ratio in market values.

The company tax rate for both companies is 28% per year.

Plans for Matravers Co

The offer for Matravers Co will be a cash offer. Any funding required for this offer will be a mixture of debt and equity. Although for the purposes of the calculation it has been assumed that the overall mix of debt and equity will remain the same, the directors are considering various plans for funding the purchase which could result in a change in Westparley Co's gearing.

As soon as it acquires all of Matravers Co's share capital, Westparley Co would sell Matravers Tech as it does not fit in with Westparley Co's strategic plans and Westparley Co wishes to take advantage of the large values currently attributed to high-technology businesses. Westparley Co would then close Matravers Home's worst-performing city centre stores. It anticipates the loss of returns from these stores would be partly compensated by higher online sales by Matravers Co, generated by increased investment in its online operations. The remaining city centre stores and all out-of-town stores would start selling the food and household items currently sold in Westparley Co's stores, and Westparley Co believes that this would increase profits from those stores.

Westparley Co also feels that reorganising Matravvers Co's administrative functions and using increased power as a larger retailer can lead to synergies after the acquisition.

Post-acquisition details

Once Matravers Tech has been sold, Westparley Co estimates that sales revenue from the Matravers Home stores which remain open, together with the online sales from its home business, will be \$43,260m in the first year post-acquisition, and this figure is expected to grow by 3% per year in years 2 to 4.

The profit margin before interest and tax is expected to be 6% of sales revenue in years 1 to 4.

Tax allowable depreciation is assumed to be equivalent to the amount of investment needed to maintain existing operations. However, an investment in assets (including working capital) will be required of \$630m in year 1. In years 2 to 4, investment in assets each year will be \$0.50 of every \$1 increase in sales revenue.

After four years, the annual growth rate of free cash flows is expected to be 2% for the foreseeable future.

As well as the free cash flows from Matravers Co, Westparley Co expects that post-tax synergies will arise from its planned reorganisation of Matravers Co as follows in the next three years:

Year	1	2	3
	\$m	\$m	\$m
Free cash flows	700	750	780

The current market value of Matravers Co's shares is \$12,500m and its debt is currently trading at its book value of \$6,500m.

Required:

- (a) Discuss the behavioural factors which may have led to businesses such as Matravers Tech being valued highly. (6 marks)
- (b) Prepare a report for the board of directors of Westparley Co which:
- (i) compares the additional value which Westparley Co believes can be generated from the sale of Matravers Tech based on the P/E ratio, with that of the projected present value of its future free cash flows; (4 marks)
 - (ii) calculates the weighted average cost of capital for the combined company; (6 marks)
 - (iii) estimates the total value which Westparley Co's shareholders will gain from the acquisition of Matravers Co; and (10 marks)
 - (iv) assesses the strategic and financial value to Westparley Co of the acquisition, including a discussion of the estimations and assumptions made. (12 marks)
- Professional marks will be awarded in part (b) for the format, structure and presentation of the report. (4 marks)
- (c) Discuss the factors which may determine how the offer for Matravers Co will be financed and hence the level of gearing which Westparley Co will have. (8 marks)

(50 marks)

Section B – BOTH questions are compulsory and MUST be attempted

- 2 Boullain Co is based in the Eurozone and manufactures components for agricultural machinery. The company is financed by a combination of debt and equity, having obtained a listing five years ago. In addition to the founder's equity stake, the shareholders consist of pension funds and other institutional investors. Until recently, sales have been generated exclusively within the Eurozone area but the directors are keen to expand and have identified North America as a key export market. The company recently completed its first sale to a customer based in the United States, although payment will not be received for another six months.

Hedging policy and key stakeholders

At a recent board meeting, Boullain Co's finance director argued that the expansion into foreign markets creates the need for a formal hedging policy and that shareholder value would be enhanced if this policy was communicated to the company's other stakeholders. However, Boullain Co's chief executive officer disagreed with the finance director on the following grounds. First, existing shareholders are already well diversified and would therefore not benefit from additional risk reduction hedging strategies. Second, there is no obvious benefit to shareholder value by communicating the hedging policy to other stakeholders such as debt providers, employees, customers and suppliers. You have been asked to provide a rationale for the finance director's comments in advance of the next board meeting.

Hedging products

Assume today's date is 1 March 20X0. Boullain Co is due to receive \$18,600,000 from the American customer on 31 August 20X0. The finance director is keen to minimise the company's exposure to foreign exchange risk and has identified forward contracts, exchange traded futures and options as a way of achieving this objective.

The following quotations have been obtained.

Exchange rates (quoted as €/US\$1)

Spot	0.8707–0.8711
Six months forward	0.8729–0.8744

Currency futures (contract size €200,000; exercise price quoted as US\$ per €1)

	Exercise price
March	1.1476
June	1.1449
September	1.1422

Currency options (contract size €200,000; exercise price quoted as US\$ per €1, premium: US cents per €1)

	Calls			Puts		
Exercise price	March	June	September	March	June	September
1.1420	0.43	0.59	0.77	0.62	0.78	0.89

Assume futures and options contracts mature at the month end and that there is no basis risk. The number of contracts to be used should be rounded down to the nearest whole number in calculations. If the full amount cannot be hedged using an exact number of futures or options contracts, the balance is hedged using the forward market.

Margin information

Once the position is open, the euro futures contract outlined above will be marked-to-market on a daily basis. The terms of the contract require Boullain Co to deposit an initial margin of \$3,500 per contract with the clearing house. Assume the maintenance margin is equivalent to the initial margin. The tick size on the contract is \$0.0001.

Your manager is concerned about the impact of an open futures position on Boullain Co's cash flow and has asked you to calculate and explain the impact of the following hypothetical changes in the closing settlement price in the first three days of the contract.

Closing settlement prices (US\$ per €1)

Date	Settlement price
1 March	1·1410
2 March	1·1418
3 March	1·1433

Required:

- (a) Explain the rationale for the policy of hedging Boullain Co's foreign exchange risk and the potential benefits to shareholder value if that policy is effectively communicated to the company's key stakeholders. (7 marks)
- (b) Recommend a hedging strategy for Boullain Co's foreign currency receipt in six months' time based on the hedging choices the finance director is considering. Support your recommendation with appropriate discussion and relevant calculations. (11 marks)
- (c) Calculate and explain the impact of the open futures position on Boullain Co's US\$ cash flow, based on the settlement prices provided. (7 marks)

(25 marks)

- 3** Hathaway Co operates in the aviation industry, manufacturing safety equipment for commercial aircraft. The company has a policy of carefully appraising new investment opportunities, including the detailed analysis of all cost and revenue assumptions prior to their approval.

Project chi

Hathaway Co's board is reviewing a potential investment, project chi. The company's engineers have developed a new technology which can detect the potential for mechanical failure with a greater degree of accuracy than has previously been the case. Early test results have been extremely encouraging.

If the board accepts the engineers' proposal, Hathaway Co would need to submit an application to the relevant regulatory authority. It is expected regulatory approval would be granted in one year's time. Manufacturing and sales would commence immediately after being granted regulatory approval. Hathaway Co's chief engineer presented an investment case for project chi to the board, including a summary of the following cost and revenue forecasts and assumptions.

Hathaway Co is expected to sell 3,000 units in the first year of production with demand increasing by 5% in each subsequent year of its four-year life. These sales forecasts are based on a contribution of \$5,000 per unit in the first year of production and increasing at 2% per year in subsequent years. Annual fixed costs of \$8.7m are expected in the first year of production, increasing at 3% per year throughout the life of the project.

An investment in plant and machinery of \$12m will be required as soon as regulatory approval has been granted. Tax allowable depreciation is available on the plant and machinery at an annual rate of 20% on a straight-line basis. A balancing adjustment is expected at the end of the project when the plant and machinery will be scrapped.

Tax is payable at 20% in the year in which profits are made. The relevant cost of capital to be used in the appraisal is 12%.

Project chi extra information

The finance director, however, raised the following objections and consequences to the chief engineer's presentation.

The chief engineer's cost and revenue assumptions ignore the possibility of a recession, which has a 20% probability of occurring. In a recession, the total present values for the four years of production are likely to be 40% lower. The finance director also believes there is an alternative, mutually exclusive, development opportunity based on the new technology although this would still depend on it being granted regulatory approval. This alternative option would incur an identical investment cost of \$12m but generate annual, inflation adjusted, post-tax cash flows of \$3.43m over its seven-year life from year two onwards.

The investment case assumes regulatory approval is certain whereas historically only 70% of Hathaway Co's applications have been approved. In one year's time, if the regulatory application is not approved, it is assumed that the concept can be sold to Gepe Co for \$1.0m at that time. If the board rejects the proposal now, assume the concept can be sold for \$4.3m immediately.

Projects lambda and kappa

A recent board meeting discussed two recent investments, projects lambda and kappa, both involving the construction of new manufacturing plants for safety equipment. Both projects are now operational although project lambda experienced significant time delays and cost overruns while project kappa was under budget and within schedule.

On closer examination, the directors noticed that project lambda's revenue far exceeded initial expectations, whereas project kappa's revenue was much less than originally expected. On balance, Hathaway Co's chief executive officer suggested that each project's successes compensated for their respective failings and that this was to be expected when making predictions about the future in an investment plan. However, one of the directors suggested the company could benefit from the introduction of a capital investment monitoring system and post-completion audit. The directors agreed to discuss this in greater depth at the next board meeting.

Required:

- (a) (i) Evaluate the financial acceptability of the project chi investment proposal based on the chief engineer's forecasts, assuming regulatory approval is granted in one year's time. (6 marks)
- (ii) Calculate the expected net present value of the proposal based on the finance director's assumptions about the likelihood of a recession and the potential impact on project chi's cash flows. (2 marks)
- (iii) Calculate the net present value of the finance director's alternative option for the technology and advise the board whether this is worth pursuing. (3 marks)
- (iv) Recommend whether the board should proceed with the application for regulatory approval after taking into consideration Hathaway Co's 70% approval rate with its regulatory applications or to sell the concept now to Gepe Co. Include in your analysis any comments on your findings. (6 marks)
- (b) Explain the rationale for implementing capital investment monitoring systems and post-completion audits. Suggest ways in which Hathaway Co may have benefited if these procedures had been applied to projects lambda and kappa. (8 marks)

(25 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d^i) \frac{V_d}{V_e}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i(E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_o = \frac{D_o(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = br_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \qquad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model

$$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$$

Where:

$$d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$$

$$d_2 = d_1 - s\sqrt{t}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

		<i>Discount rate (r)</i>									
<i>Periods</i>		1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
(n)											
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

		<i>Discount rate (r)</i>									
<i>Periods</i>											
(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Answers

1 (a) Individual business

A number of behavioural factors, to do with the individual company as well as the sector as a whole, may lead to Matravers Tech being valued higher than appears to be warranted by rational analysis of its future prospects. One possible factor is the asking price, even if it is not a fair one, may provide a reference point which significantly influences the purchaser's valuation of the business.

The fact that Matravers Tech is available for purchase may help raise its price. Purchasers may see this as a rare opportunity to buy an attractive business in this retail sector. This will be made more likely if investors have loss aversion bias, a desire to buy Matravers Tech now because otherwise the opportunity will be lost.

Matravers Tech being offered for sale will mean that information about the company, showing it in a positive light, will be available for purchasers. This could result in availability bias, investors taking particular note of this information because they can readily obtain it, rather than other information which may be more difficult or costly to find.

Sector

There are a number of possible behavioural reasons why share prices in this sector appear generally higher than rational analysis indicates. One is the herd instinct, investing in the sector because other investors have also been buying shares, not wishing to make judgements independently of other investors.

The herd instinct may be generated by previous share price movements. Investors may believe once prices start rising in the sector, they will continue to do so indefinitely.

Following fashion may also be a factor. Fund managers who wish to give the impression that they are actively managing their portfolio by making regular changes to it, may have a preference for companies which appear up-to-date and are currently popular. This may be linked to an expectation that sales of technologically-advanced goods are likely to generate high returns.

There is also confirmation bias, the idea that investors will pay attention to evidence which confirms their views that the sector is a good one in which to invest, and ignore evidence which contradicts their beliefs. In the past, technology companies have been valued using methods which support the beliefs of investors that they are of high value, rather than traditional methods, such as cash flow analysis, which suggest a lower business value is more realistic.

(b) (iv) Report to the board of directors, Westparley Co

This report evaluates whether the acquisition of Matravers Co would be beneficial to Westparley Co's shareholders by estimating the future value generated by Matravers Co (i.e. Matravers Home currently), the proceeds from selling Matravers Tech and the additional value created from synergies immediately after the companies are combined.

Strategic fit

The strategic case for taking over the business appears to be strongest for the out-of-town stores and the online business. The acquisition would provide an additional out-of-town presence for Westparley Co. Better usage in the out-of-town stores could generate higher returns. Having the food and home businesses on the same site could generate some cross-sales between the two. Possibly combining the two companies' online presence and investing further could mean Matravers Home benefiting from the factors which have driven strong performance by Westparley Co.

Taking over the city centre stores, even the successful ones, seems to have less strategic logic, however. Westparley Co would be taking on a high cost burden. The success of the food business in city centres is doubtful, as food shops sited there will be less convenient for customers who do not live in the city centres, and Westparley Co has marketed itself as being easily accessible for customers. There is, perhaps, wider incompatibility between the two businesses. The food business is characterised by quick shopping for often a limited number of items, whereas purchases in the home business, particularly of larger items, are likely to take longer and site convenience be less of an issue.

Financial aspects

Based on the predictions for future cash flows and required premiums from Matravers Co's shareholders, the acquisition would add value to Westparley Co's shareholders, if, and only if, the excess value on selling Matravers Tech and the synergies are both largely achieved. Together they add up to \$2,400m (\$558m + \$1,842m) compared with total added value of \$1,897m. There are questions about the estimates for these figures and also the estimates for the future free cash flows of the current Matravers Home business.

Synergies

Most of the additional value is due to synergies and it is difficult to see how the synergies are calculated. There is likely to be scope for some administrative savings. However, operational cost synergies appear less obvious as the two companies are operating in different retail sectors. Any synergy figures will also have to take account of costs in achieving synergies, such as store closure costs, and also commitments such as leases which may be a burden for some time. Synergies may also not be achieved because of lack of co-operation by staff or problems integrating the two businesses.

Current Matravers Home business

The suggested increase in cash flows appears doubtful for a number of reasons. If stores being closed are making positive cash flow contributions, these will have to be replaced. Whether they can be is doubtful given the problems in this part of the retail sector. It may be a more profitable use of store space to have an area for food sales, but the food sales generated in Matravers Co's shops may take business from Westparley Co's existing shops. Similarly, increased online sales may be at the expense of sales in stores.

Sale of Matravers Tech

There is no indication of how interested buyers will be in the business. The industry price-earnings (P/E) ratio used may be an average which does not reflect Matravers Tech's circumstances. It would be better to find a P/E ratio for a proxy company with similar financial and business risk. As Matravers Tech would not be listed, this would suggest a discount to the P/E ratio should be applied. Since also Westparley Co has an estimate of future free cash flow, potential buyers may be able to come up with their own estimates and base the price they are prepared to pay on their estimates.

Other assumptions

One important assumption is the 15% premium expected to be required by Matravers Co's shareholders. Other assumptions made in the calculations include operating profit margin and tax rates remaining constant and cash flows being assumed to increase to perpetuity. Incremental capital investment is assumed to be accurate. It is assumed that the cost of debt will remain unchanged and that the asset beta, cost of equity and cost of debt can be determined accurately. Given all the assumptions, Westparley Co should carry out sensitivity analysis using different assumptions and obtaining a range of values.

Conclusion

On the assumptions made, the acquisition appears to add financial value for the shareholders of Westparley Co. However, the figures are subject to a significant number of uncertainties and the strategic logic for buying the whole Matravers business appears unclear. On balance, Westparley Co may want to consider a more limited acquisition of just the out-of-town stores if these are available, as their acquisition appears to make better strategic sense.

Report compiled by:

Date

Appendix 1 Estimate of additional value created from sell-off of Matravers Tech (b) (i)

Share of pre-tax profit = $20\% \times \$1,950\text{m} = \390m

After-tax profit = $\$390\text{m} \times (1 - 0.28) = \281m

Proceeds from sell-off based on P/E ratio = $\$281\text{m} \times 18 = \$5,058\text{m}$

Excess value from sell-off = $\$5,058\text{m} - \$4,500\text{m} = \$558\text{m}$

Appendix 2 Estimate of combined company cost of capital (b) (ii)

Matravers Co asset beta = 0.75

Westparley Co asset beta

Market value of debt = $1.05 \times \$26,000\text{m} = \$27,300\text{m}$

Market value of equity = $4,000 \text{ million} \times \$8.50 = \$34,000\text{m}$

Asset beta = $1.02 \times (34,000) / (34,000 + (27,300 \times 0.72)) = 0.65$

Combined company, asset beta

Market value of Matravers Co equity = $\$12,500\text{m}$

Asset beta = $((0.75 \times 12,500) + (0.65 \times 34,000)) / (12,500 + 34,000) = 0.68$

Equity beta = $0.68 ((34,000 + (27,300 \times 0.72)) / 34,000) = 1.07$

Combined company cost of equity = $3.5\% + (1.07 \times 8\%) = 12.1\%$

Combined company cost of capital = $((34,000 \times 12.1\%) + (27,300 \times 9.8\% \times 0.72)) / (34,000 + 27,300) = 9.9\%$, say 10%

Appendix 3 Estimate of the value created for Westparley Co's shareholders (b) (iii)

Cash flows, years 1 to 4

Year	1 \$m	2 \$m	3 \$m	4 \$m
Sales revenue	43,260	44,558	45,895	47,272
Profit before interest and tax	2,596	2,673	2,754	2,836
Tax	(727)	(748)	(771)	(794)
Additional capital investment	(630)	(649)	(669)	(689)
Free cash flows	1,239	1,276	1,314	1,353
Discount factor	0.909	0.826	0.751	0.683
Present value of free cash flows	1,126	1,054	987	924

Present value years 1 to 4 = \$4,091m

Present value year 5 onwards $((\$1,353\text{m} \times 1.02)/(0.1 - 0.02)) \times 1 \cdot 10^{-4} = \$11,781\text{m}$

Total present value = \$4,091m + \$11,781m = \$15,872m

Synergies

Year	1 \$m	2 \$m	3 \$m
Free cash flows	700	750	780
Discount factor	0.909	0.826	0.751
Present value of cash flows	636	620	586

Present value of synergies = \$1,842m

Amount payable for Matravers Co's shares = \$12,500m $\times 1.15 = \$14,375\text{m}$

Value attributable to Matravers Co's investors = \$14,375m + \$6,500m = \$20,875m

Value attributable to Westparley Co shareholders = present value of cash flows + proceeds from sell-off + value of synergies – value to Matravers Co's investors
= \$15,872m + \$5,058m + \$1,842m – \$20,875m = \$1,897m

(c) Calculation of gearing

If gearing is calculated on the basis of market values, a fall in the share price will result in the level of gearing rising. Westparley Co's board may be worried about a fall in the share price given the problems affecting many companies' share prices in the retail sector and the possibility that the stock market may react adversely to the acquisition.

Directors' preferences

Directors may have their own preferences about financing. They may be able to choose a mix of sources and a level of gearing which reflects these preferences. Directors may be concerned about too high a burden of payment to finance providers, in terms of cost or ultimately repayment of debt. They may not wish to commit the company to conditions imposed by finance providers. By contrast, they may be concerned about how a change in the shareholder base as a result of a share issue may impact upon their own position. Directors may also be concerned about the impression given by their choice of finance. Pecking order theory states that equity issue is seen as the last resort for financing, so if the purchase is financed by an equity issue, it may be seen as a sign of a lack of confidence by directors that Westparley Co can sustain its current share price.

Costs and cash flows

Gearing decisions may not just be determined by their own preferences but by external conditions or constraints. Choosing more debt could lower the overall cost of capital, due to lower cost and tax relief, making investments such as Matravers Co appear more profitable. Higher levels of debt may result in the cost of equity rising, reducing the overall impact on the cost of capital. Against that, higher levels of debt mean increased finance cost commitments, even though Westparley Co may need further cash for investment in stores. This may be an important concern if interest rates are high. By contrast, dividends to shareholders do not have to be paid when returns are low or money is required for investment, although failure to meet dividend expectations may result in the board being pressurised by shareholders.

Availability

The availability of finance may also be a significant issue, particularly if an acquisition has to be completed quickly. An equity issue may take time to arrange and require shareholder approval. Sufficient debt finance may be difficult to obtain if lenders feel that Westparley Co already has significant commitments to debt finance providers. The timescale over which finance is available may be significant. Westparley Co may seek longer-term finance if existing debt finance is due to be repaid soon or if significant cash is needed for short-term investment, not just in Matravers Co's stores, but also in Westparley Co's existing stores.

Mix

Other external factors may influence the mix of finance chosen. Westparley Co's directors may be concerned about keeping the level of gearing at or below the industry average, because of finance providers becoming worried if gearing exceeds industry

levels. Keeping debt as a significant element in overall finance may act as a deterrent to acquirers becoming interested in making a bid for Westparley Co. Directors may also not have a target figure in mind but be content if gearing is within a range of values.

2 (a) Rationale for hedging policy

Within the framework of Modigliani and Miller, Boullain Co's CEO is correct in stating that a company's hedging policy is irrelevant. In a world without transaction or agency costs, and where markets are efficient and information symmetrical, hedging creates no value if shareholders are well diversified. Shareholder value may even be destroyed if the costs associated with hedging exceed the benefits.

However, in the real world where market imperfections exist, including the transaction costs of bankruptcy and other types of financial distress, hedging protects shareholder value by avoiding the distress costs associated with potentially devastating foreign exchange fluctuations.

Active hedging may also benefit debt-holders by reducing the agency costs of debt. A clearly defined hedging policy acts as a signalling tool between shareholders and debt-holders. In this sense, hedging allows for higher leverage and a lower cost of debt and reduces the need for restrictive covenants.

Communication of policy with stakeholders

Even when foreign exchange risks are hedged, the funding of variation margin payments on exchange traded futures can create financial distress. A well communicated hedging strategy allows debt providers to make informed decisions about Boullain Co's ability to service its debt.

Agency costs and the risk of financial distress also impact the expected wealth of employees who, unlike shareholders, may not enjoy the risk reduction benefits of a diversified portfolio. A consistent hedging policy reduces the risks faced by employees which may serve to benefit Boullain Co in the form of motivational and productivity improvements.

Customers and suppliers have claims on a company which create shareholder value but are conditional upon Boullain Co's survival. Suppliers may invest in production systems which create value in the form of lower costs. For customers, these claims reflect promises of quality and after-sales service levels which enable Boullain Co to charge higher prices. In both cases, shareholder value is created as long as the customers and suppliers believe these claims will be honoured. One way of achieving this is by implementing a hedging strategy and communicating it to stakeholders.

In conclusion, management should attempt to communicate the principles underlying its hedging strategy and the benefits to shareholder value in the form of reduced agency and distress costs. In this way, stakeholders can make informed decisions about the potential risks and impact on their expected wealth.

(b) Forward contract

$$\$18,600,000 \times 0.8729 = \text{€}16,235,940$$

Futures

Buy September € futures

Calculation of futures price

$$\text{Spot rate (US$/€1)} = 1/0.8707 = 1.1485$$

$$\text{Predicted futures using spot rate} = 1.1422 + ((1.1485 - 1.1422) \times 1/7) = 1.1431$$

$$\text{Or using futures: } 1.1422 + ((1.1449 - 1.1422) \times 1/3) = 1.1431$$

Number of contracts

$$\text{Expected receipt} = \$18,600,000 / 1.1431 = \text{€}16,271,542$$

$$\text{Number of contracts} = \text{€}16,271,542 / \text{€}200,000 = 81.4, \text{ say } 81 \text{ contracts}$$

$$\text{Amount underhedged} = \$18,600,000 - (81 \times \text{€}200,000 \times 1.1431\$/\text{€}) = \$81,780$$

$$\text{Receipt at forward rate} = \$81,780 \times 0.8729\text{€}/\$ = \text{€}71,386$$

Outcome

	€
Futures (81 x 200,000)	16,200,000
Forward market	71,386
	<u>16,271,386</u>

Options

September € call options

Number of contracts

$$\text{Payment} = \$18,600,000 / 1.1420\$/\text{€} = \text{€}16,287,215$$

$$\text{Number of contracts} = \text{€}16,287,215 / \text{€}200,000 = 81.4, \text{ say } 81 \text{ contracts}$$

Premium

$$\text{Premium} = 81 \times \text{€}200,000 \times 0.0077\$/\text{€} = \$124,740$$

$$\text{Translate at spot} = \$124,740 \times 0.8711\text{€}/\$ = \text{€}108,661$$

Amount underhedged = \$18,600,000 – (81 x €200,000 x 1.1420\$/€) = \$99,600

Receipt at forward rate = \$99,600 x 0.8729€//\$ = €86,941

Outcome

	€
Options (81 x €200,000)	16,200,000
Premium	(108,661)
Forward market	86,941
	<u>16,178,280</u>

Recommendation

The forward and futures contracts fix the exchange rate with the futures contract generating a slightly higher euro receipt compared to the forward. However, the futures contract is exposed to basis risk and is marked-to-market daily. The initial margin and variation margins need to be funded and would impact cash flow in the short term.

The option outcome of €16,178,280 provides a worst-case scenario based on the option being exercised. The option premium is expensive which results in a lower receipt if the option is exercised. Unlike the forward and futures contracts, however, the option allows Boullain Co to retain the upside whilst also protecting against the downside risk. Based on the forward and futures markets, the dollar is expected to strengthen and it is therefore unlikely the option would be exercised.

The final hedging choice depends on the board's attitude to risk. However, assuming there is no default risk associated with the forward contract, this may be the best choice under the circumstances. The board may also wish to consider the possibility of not hedging since the dollar is expected to strengthen.

(c) Marking-to-market

Initial margin = maintenance margin = \$3,500 x 81 = \$283,500

1 March:

$((1.1410 - 1.1422)/0.0001) \times \$20 \times 81 = \$19,440$ loss

Maintenance margin is 100% of initial margin:

Therefore variation margin = \$19,440

2 March:

$((1.1418 - 1.1410)/0.0001) \times \$20 \times 81 = \$12,960$ profit

3 March:

$((1.1433 - 1.1418)/0.0001) \times \$20 \times 81 = \$24,300$ profit

In order to reduce counter-party risk, Boullain Co deposits the initial margin of \$283,500 with the clearing house when the futures position is opened. The notional profit or loss at each day's closing settlement price is added to or subtracted from the margin account balance. If the margin account balance falls below the level of the maintenance margin, Boullain Co is required to deposit additional funds to top up the margin account.

Boullain Co makes a notional loss at the end of the first day and would therefore pay a variation margin to return the margin account to the level of the specified maintenance margin. Since Boullain Co makes a notional profit on the subsequent two days, the amount in the margin account will be greater than the specified maintenance margin and no variation margin is required. The profit on each of those days may be withdrawn in cash.

3 (a) (i) Net present value (NPV): All figures are in \$ms unless otherwise indicated

Year	1	2	3	4	5
Contribution (w1)		15.00	16.07	17.21	18.43
Fixed costs		(8.70)	(8.96)	(9.23)	(9.51)
Tax allowable depreciation		(2.40)	(2.40)	(2.40)	(2.40)
Balancing adjustment					(2.40)
Taxable profits		3.90	4.71	5.58	4.12
Taxation (20%)		(0.78)	(0.94)	(1.12)	(0.82)
Add back depreciation		2.40	2.40	2.40	4.80
Investment cost	(12.00)				
Cash flows	(12.00)	5.52	6.17	6.86	8.1
Discount factors (12%)	0.893	0.797	0.712	0.636	0.567
Discounted cash flows	(10.72)	4.40	4.39	4.36	4.59
NPV	7.02				

Workings

Working 1 (w1): Contribution

Year	2	3	4	5
Volume (000s)	3.00	3.15	3.31	3.47
Contribution per unit (\$000s)	5.00	5.10	5.20	5.31
Contribution (\$m)	15.00	16.07	17.21	18.43

(ii) Incorporating finance director's objections

Expected NPV of chief engineer's proposal:

PV of years 2–5 = \$17.74m

60% PV of years 2–5 = \$10.64m

Expected PV of years 2–5 = $(0.8 \times \$17.74m + 0.2 \times \$10.64m) = \$16.32m$

Expected NPV = \$16.32m – \$10.72m = \$5.6m

(iii) Alternative option NPV

Annuity factor (12%, $t_2 - t_8$) = $4.968 - 0.893 = 4.075$

PV of years 2–8 = $4.075 \times \$3.43m = \$13.98m$

NPV = \$13.98m – \$10.72m = \$3.26m

Therefore, it is more beneficial to follow the chief engineer's proposal.

(iv) Apply for regulatory approval or sell

Next, consider decision to sell to Gepe Co now or continue with application for regulatory approval.

NPV of sale to Gepe Co now = \$4.3m

Expected NPV = $0.7 \times \$5.6m + 0.3 \times (0.893 \times \$1.0m) = \$4.19m$

Therefore, more beneficial to sell immediately to Gepe Co.

Recommendation:

Immediate sale to Gepe Co for \$4.3 million.

Comments

Based on the chief engineer's assumptions, the project generates a positive NPV of \$7.02 million and should therefore be accepted in preference to the option to sell the concept for \$4.3 million. On the other hand, when the finance director's objections are incorporated into the appraisal, the expected NPV is only \$4.19 million and should therefore be rejected in favour of the option to sell.

It should be noted that the expected NPV of \$4.19 million is an average. In other words, it is the average NPV if the project is carried out repeatedly which may not be useful in the case of a one-off development opportunity. Based on the calculations above, there is a 30% chance that the NPV will be only \$893,000, which may pose a risk the directors are not prepared to take. The directors' attitude to risk will be an important factor in the final decision.

Furthermore, the analysis largely depends upon the values of the probabilities prescribed, the range of possible outcomes and the accuracy of the revenue and cost assumptions. Sensitivity analysis may be useful in testing the impact of variations in each of these variables on the final outcome.

- (b) Whilst Hathaway Co's investment plans are based on a detailed analysis of all cost and revenue assumptions, projects lambda and kappa highlight failings in the appraisal and implementation phases.

Capital investment monitoring

Capital investment monitoring involves reviewing the implementation of an investment project to ensure it progresses according to the original investment plan, timescale and budget. This involves assessing the risks associated with the implementation phase and identifying deviations from the investment plan so that remedial action can be taken where necessary. Controls should be established to ensure effective delivery of the project.

Effective investment monitoring may have avoided the cost overruns and time delays experienced by Hathaway Co's project lambda. The appointment of a project manager would have ensured ownership of the project and provided accountability for time delays and cost increases. The original investment plan, including cost estimates, provides a benchmark against which actual performance can be assessed. An effective monitoring system would ensure that any changes to the cost estimates would be justified and authorised. Where significant deviations are encountered, it may be necessary to terminate the project. A project steering committee would ensure greater scrutiny and accountability and oversee the implementation phase.

Post-completion audit

A post-completion audit is an objective, after the fact, appraisal of all phases of the capital investment process regarding a specific project. Each project is examined from conception until as much as a few years after it has become operational. It examines the rationale behind the initial investment decision, including the strategic fit, and the efficiency and effectiveness of the outcome. The key objective is to improve the appraisal and implementation of future capital investment projects by learning from past mistakes and successes.

An effective post-completion audit may have identified the reasons behind the failure of Hathaway Co's project kappa to achieve its forecast revenues. By comparing the actual project outcome with the original projections, an audit will examine whether the benefits claimed prior to approval ever materialise. The audit is not an academic exercise; an effective audit would identify failings and help Hathaway Co learn from past mistakes as well replicate its successes. Project kappa's principal failing seems to be the inaccuracy of the revenue assumptions. An audit would establish the reasons behind that failing and identify ways in which this can be addressed. For example, it is possible the initial assumptions failed to predict future competitor actions or the full range of potential economic scenarios. In this way, Hathaway Co's managers benefit by learning how to appraise investment proposals more accurately and implement them more efficiently than before.

(Note: Credit will be given for alternative and valid discursive comments)

		<i>Marks</i>	
1	(a) 1–2 marks per relevant point (examples may include asking price, opportunity to purchase and information available is positive, herd instinct, following fashion and confirmation bias)	Max	6
	(b) (i) Share of pre-tax profit		1
	After-tax profit		1
	Proceeds from sell-off		1
	Comparison with free cash flow valuation		1
			4
	(ii) Westparley Co MV debt and equity		1
	Westparley Co asset beta		1
	Combined company asset beta		1
	Combined company equity beta		1
	Combined company cost of equity		1
	Combined company cost of capital		1
			6
	(iii) Sales revenue		1
	Profit before interest and tax		1
	Tax		1
	Additional capital investment		1
	PV of free cash flows years 1–4		1
	PV of free cash flows year 5 onwards		2
	Present value of synergies		1
	Premium payable		1
	Value attributable to Westparley Co's shareholders		1
			10
	(iv) Strategic value		3–4
	Financial value		2–3
	Estimations made		3–4
	Assumptions made		3–4
		Max	12
	Professional marks for part (b)		
	Report format		1
	Structure and presentation of the report		3
			4
	(c) 1–2 marks per relevant point (examples may include available security, available tax relief, shareholder attitude to debt, industry norms)	Max	8
		Total	50

		<i>Marks</i>
2	(a) Rationale for hedging policy	3–4
	Communication of policy with stakeholders	3–4
	Max	7
	(b) Forward	1
	Buy September futures	1
	Number of futures contracts	1
	Predicted futures rate	1
	Underhedge futures	1
	Buy September calls	1
	Option premium	1
	Futures outcome	1
	Outcome	1
	Discussion and recommendation	2–3
	Max	11
	(c) Initial and/or maintenance margins	1
	Daily profit or loss calculations	3
	Explanation	3
		7
	Total	25
3	(a) (i) Contribution	2
	Fixed costs	1
	Tax (including tax allowable depreciation)	2
	NPV	1
		6
	(ii) Expected NPV	2
	(iii) Alternative option NPV	2
	Decision outcome	1
		3
	(iv) Decision	2
	Comments	4
		6
	(b) Capital investment monitoring:	
	Rationale	2
	Benefits	2–3
	Post-completion audit:	
	Rationale	2
	Benefits	2–3
	Max	8
	Total	25

Strategic Professional – Options

Advanced Financial Management (AFM)

September/December 2020 –
Sample Questions



AFM

Time allowed: 3 hours 15 minutes

This question paper is divided into two sections:

Section A – This ONE question is compulsory and MUST be attempted

Section B – BOTH questions are compulsory and MUST be attempted

Formulae and tables are on pages 7–11.

Do NOT open this question paper until instructed by the supervisor.

This question paper must not be removed from the examination hall.

Think Ahead

ACCA

The Association of
Chartered Certified
Accountants

Section A – This ONE question is compulsory and MUST be attempted

1 Kingtim Co

Kingtim Co is a nationwide chain of garden centres, selling products such as plants, fertilisers, tools and garden furniture. It was established 20 years ago by the current team of executive directors, and achieved a listing on its local stock market four years ago. Since listing, the company has made consistent profits and has been able to increase its dividends each year. The executive directors collectively own between them 25% of issued share capital. The remaining shares are held by a number of investors, with none of them owning more than 10% of issued share capital.

Two of Kingtim Co's major competitors have been taken over in the last two years. Media coverage suggests that further takeovers are possible. Potential acquirers include other chains of garden centres, property developers and supermarket chains looking to diversify their business into the profitable garden centre sector. As yet, no potential acquirer has approached Kingtim Co to buy the whole chain, although Kingtim Co has had enquiries from other businesses wanting to purchase individual garden centres. Kingtim Co can sell a number of individual garden centres without threatening its continued existence.

Takeover defences

Kingtim Co's executive directors remain committed to the business. They are fearful of a takeover, believing that the new owners will want some, or all of them, to leave the company. They have therefore been considering possible defences against a takeover bid. Kingtim Co's chief executive has received two proposals from directors:

- Sell individual garden centres which would be particularly attractive to purchasers. Disposal of these centres would make Kingtim Co, overall, a less attractive purchase.
- Pay the executive directors higher remuneration and change their contracts so that they would receive much higher compensation for loss of office if their contracts were terminated early.

Kingtim Co's chief executive believes, however, that any defence Kingtim Co adopts should also strengthen the company's future. She is therefore proposing to expand the company's current limited sales of camping products in its garden centres by establishing a chain of Kingtim outdoor shops, selling camping, walking and other outdoor equipment. The outdoor retail sector is competitive, but the chief executive believes that Kingtim Co will be successful. The establishment of the chain of outdoor shops would be funded solely by debt, the idea being that changing Kingtim Co's finance structure by having significantly more debt would make it less attractive to acquirers.

Financial details

Kingtim Co currently has 25 million \$1 shares in issue, with a current share price of \$5.56 per share. It also has 0.45 million 6.5% bonds in issue. Each 6.5% bond has a nominal value of \$100, and is currently trading at \$104 per \$100. The premium on redemption of the bonds in three years' time is 2%. Based on a yield to maturity approach, the after-tax cost of the bonds is 4.1%.

Kingtim Co's quoted equity beta for its existing garden centre business is 0.9.

Kingtim Co plans to issue 0.6 million, 7.5%, new bonds, each with a nominal value of \$100. These bonds will be redeemable in four years' time at a premium of 8%. The coupon on these bonds will be payable on an annual basis. These bonds are anticipated to have a credit rating of BBB-. The issue of the new 7.5% bonds will not affect the market value of Kingtim Co's shares or the existing 6.5% bonds.

The market value of the new bonds will be determined by using information relating to Kingtim Co's credit rating and the four bonds which the government has issued to estimate Kingtim Co's yield curve. All the bonds are of the same risk class. Details of the bonds are as follows:

Bond	Annual yield (based on spot rate)	Redeemable in
Ga	4%	1 year
Th	4.3%	2 years
De	4.7%	3 years
Ro	5.2%	4 years

Credit spreads, shown in basis points, are as follows:

Rating	1 year	2 years	3 years	4 years
BBB-	56	78	106	135

Kingtim Co plans to invest \$60m in non-current assets for the outdoor shops (working capital requirements can be ignored). Currently, Kingtim Co's non-current assets have a net book value of \$150m. It is assumed that the proportion of the book value of non-current assets which will be invested in the outdoor shops and the garden centres will give a fair representation of the size of each business within Kingtim Co. The asset beta of similar companies in the outdoor retail sector is assumed to be 0.88.

Before taking into consideration the impact of this new investment, Kingtim Co's forecast pre-tax earnings for the coming year is \$24m. It is estimated that the new investment will make a 10% pre-tax return.

The corporation tax rate applicable to all companies is 25% per year. The current risk-free rate of return is estimated to be 4% and the market risk premium is estimated to be 9%.

Employee remuneration

Kingtim Co's annual report contains a general commitment to act with social responsibility, in line with society's expectations. It also commits to paying its staff fairly in accordance with their responsibilities and states that its staff are vital to its success.

To try to improve the situation of low-paid employees, the government has recommended a basic hourly wage as the minimum level employees should be paid, although this minimum is not legally enforceable. A newspaper investigation has revealed that some staff in Kingtim Co's garden centres in the northern region of the country are paid up to 15% less per hour than the recommended minimum wage. Most of these staff are part-time staff, working limited hours each week.

The manager of Kingtim Co's northern region centres, when asked to comment, stated that Kingtim Co had obligations to its shareholders to control staff costs. Lower pay levels were necessary to differentiate between staff, ensuring that managers and staff with experience and expertise were appropriately rewarded. The manager commented that pay levels also reflected the lower commitment to Kingtim Co which part-time staff made compared with full-time staff.

Required:

(a) Discuss the feasibility and effectiveness of the defence strategies of selling off individual garden centres and enhancing directors' remuneration. (7 marks)

(b) Prepare a report for the board of directors of Kingtim Co which:

- (i) estimates the company's cost of capital before the new bonds are issued;** (4 marks)
- (ii) estimates the market value and yield to maturity of the new bonds;** (7 marks)
- (iii) estimates the revised cost of equity and revised cost of capital if the new bonds are issued;** (7 marks)
- (iv) estimates the impact of the chief executive's proposal on forecast after-tax earnings for the coming year; and** (3 marks)
- (v) discusses the impact on Kingtim Co's cost of capital and the reaction of equity and bond holders to the chief executive's proposal. The discussion should include an explanation of any assumptions made in the estimates in (b) (i) – (iv) above.** (10 marks)

Professional marks will be awarded in part (b) for the format, structure and presentation of the report.

(4 marks)

(c) Discuss the approach taken to employee remuneration by Kingtim Co's Northern region and the issues associated with it. (8 marks)

(50 marks)

Section B – BOTH questions are compulsory and MUST be attempted

2 Colvin Co

Colvin Co is based in the eurozone region and was established ten years ago to manufacture competition standard bicycles for professional road racers. When the company obtained a listing five years ago, the founder retained a small minority shareholding. The remaining shares are held by a number of institutional investors.

The board recently decided to expand the range of models and to look for new growth opportunities abroad. Whilst manufacturing is currently restricted to the eurozone, the board of directors has identified Canvia as a key growth market and is considering a potential investment project to manufacture and sell a new model there. This would involve establishing a subsidiary in Canvia.

Project information

The currency in Canvia is the Canvian lira (CL) and the current exchange rate is CL9·91 per euro (€). The annual rate of inflation in Canvia is expected to remain at 10% throughout the four-year duration of the project.

The finance director estimates the project's sales volumes, inflation-adjusted, pre-tax contribution and fixed costs as follows:

Year	1	2	3	4
Sales volume (units)	109,725	121,795	148,590	197,624
Pre-tax contribution (CLm)	419·4	500·2	671·3	961·2
Fixed costs (CLm)	270·0	291·6	314·9	340·1

The project will require an immediate investment of CL75m in land and buildings and CL700m in plant and machinery. Tax allowable depreciation is available on plant and machinery on a straight-line basis at an annual rate of 25% on cost. Colvin Co's finance director believes the plant and machinery will have a zero residual value at the end of the four years. The land and buildings will be disposed of at the end of the project and their tax exempt value is expected to increase at an annual rate of 30% throughout the four-year life of the investment.

The project will also require an immediate investment in working capital of CL25m. The annual working capital requirement is expected to increase in line with inflation in Canvia and will be released back in full at the end of the project. Colvin Co has a policy of extracting remittable cash flows as dividends at the earliest possible opportunity.

All components for the new bicycle will be produced or purchased in Canvia except for a gearing system component which will be manufactured by Colvin Co in the eurozone. The cost of acquiring this component from the eurozone is already included in the pre-tax contribution estimates, based on a transfer price of €10 per component. The finance director estimates a manufacturing cost of €2 per component. Both the transfer price and manufacturing cost are expected to increase in line with eurozone annual inflation of 4% in the first two years of the project and 2% in years three and four.

Corporation tax in Canvia is payable annually at 25% and companies are allowed to carry losses forward to be offset against future trading profits. Colvin Co pays corporation tax in its home country at an annual rate of 20%. Taxes are payable in both countries in the year the liability is incurred. A bi-lateral tax treaty exists between the two countries, which permits the offset of overseas tax against any domestic tax liability incurred on overseas earnings.

Discount rate

The board proposes financing the project with a mix of equity and debt in such a way that the existing capital structure remains unchanged. For the purposes of this project, the chief executive believes Colvin Co's weighted average cost of capital of 13% should be adjusted to include a country risk premium on the basis that Canvia is a developing economy and appears to be economically less stable than the eurozone countries. She made this decision after consulting a country risk index, which compares the standard deviation of market returns in various countries.

Additional factors taken into consideration include foreign exchange risk, the fact that there have been frequent changes of government in Canvia and the main opposition party has threatened to reintroduce controls on dividend remittances if elected. You have therefore been asked to use a discount rate of 16% to appraise this investment project.

Required:

- (a) Evaluate the suitability of the investment proposal in Canvia, including the impact of the country risk premium on the net present value of the project. (15 marks)
- (b) Explain possible strategies Colvin Co could adopt to avoid a block on dividend remittances if Canvia's opposition party was to win the election. (4 marks)
- (c) Discuss the validity of the chief executive's reasons for adjusting the discount rate used in appraising the project in Canvia. (6 marks)

(25 marks)

3 Fitzharris Co

Fitzharris Co is a large construction company. Its treasury department uses a variety of derivatives regularly to manage interest rate and commodity price risk.

Fitzharris Co's chief executive has recently been reviewing how the treasury department uses derivatives, in particular options, to hedge risk. She has raised queries about aspects of option pricing which she does not understand. In particular, she wants to know the impact upon option price of the time until expiry of the option and the interest rate.

Transaction to be hedged

Today's date is 1 August. Fitzharris Co plans to borrow an amount of \$48m on 1 December, to finance a major construction project, for a period of up to three years. Its treasury department has decided to hedge the risk associated with this borrowing, as there is some uncertainty about how interest rates will move over the rest of this year. The current central bank base rate is 3.7%, but predictions in the media suggest that it could rise or fall by 0.4% by 1 December. Fitzharris Co can currently borrow funds at a floating rate of central bank base rate plus 50 basis points.

Fitzharris Co's treasury department is considering hedging the interest rate risk by using:

- An interest rate swap arranged through Fitzharris Co's bank.
- A collar on options on interest rate futures.

Swap

Fitzharris Co's bank has found a possible counterparty for a swap with Fitzharris Co. The counterparty can borrow at an annual floating rate of base rate plus 130 basis points, or a fixed rate of 4.8%. Fitzharris Co's bank has quoted it a nominal fixed rate of 4.6% for it to borrow. The bank would charge a fee of 5 basis points to each party individually to act as the intermediary of the swap. Both parties would share equally the potential gains from the swap.

Collar

Options on three-month December \$ futures, \$1,000,000 contract size, option premiums are in annual %

Strike price	Calls	Puts
96.25	0.198	
95.75		0.211

The current three-month \$ futures price for December futures is 95.85.

Futures and options contracts are assumed to be settled at the end of each month. Basis is assumed to diminish to zero at contract maturity at a constant rate, based on monthly time intervals. It is also assumed that there is no basis risk and there are no margin requirements.

Required:

- Calculate, in % terms, the results of the hedging strategies that are being considered for the \$48m loan, if the central bank base rate increases to 4.1% or falls to 3.3%. Your calculations should demonstrate the rates at which payments between counterparties should be made.** (13 marks)
- Comment on the results of your calculations in (a), and discuss the advantages and drawbacks for Fitzharris Co of interest rate swaps compared with traded collars.** (8 marks)
- Explain the significance of the time until expiry and the interest rate in the context of option valuation.** (4 marks)

(25 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i(E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_o = \frac{D_o(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = br_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \quad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model

$$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$$

Where:

$$d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$$

$$d_2 = d_1 - s\sqrt{t}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

		<i>Discount rate (r)</i>									
<i>Periods</i>		1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
(n)											
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

		Discount rate (r)									
Periods											
(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Answers

1 (a) Sell-off of garden centres

deter acquirers:
ngan can nguoi thau tom

Selling some of the most desirable garden centres, known as selling the crown jewels, may deter some acquirers looking to buy the whole chain if Kingtim Co sells the assets they most desire. Kingtim Co could take this option if it is able to sell off individual centres without jeopardising its overall existence.

However if no particular use is made of the cash raised from the sales, Kingtim Co would still remain a tempting takeover target due to its cash surpluses. Returning the surplus cash to shareholders in the form of a one-off dividend might be popular with shareholders, but equally they might be concerned about their future returns given the sale of assets generating significant income. Shareholders and others interested in Kingtim Co might also question what future strategies the board had in mind if it did not use these cash surpluses for investment.

Also, if the money was distributed to shareholders, Kingtim Co would become a smaller company and perhaps more affordable to some potential acquirers.

Enhanced directors' remuneration and contracts

The enhanced commitments to the directors would represent an increased burden for acquirers, either the costs of honouring them, or the cost and the time involved in terminating the directors' employment and compensating them. This burden may deter acquirers, particularly if the decision to acquire is marginal.

However, enhancing the commitments to the directors could be ineffective. The acquirer could decide to keep the directors on and pay the increased remuneration. Alternatively, the acquirer may feel that buying out the directors' contracts and compensating them is a necessary cost that it is prepared to bear.

Corporate governance aspects are also important. As a listed company, Kingtim Co should have a remuneration committee made up of non-executive directors, who should be reviewing the executive directors' remuneration packages. Kingtim Co may have to publish a remuneration report to explain the rationale for directors' remuneration, and to allow shareholders to discuss and perhaps vote on the report.

Shareholders may believe that the directors are being given a better compensation package without having earned it, and for no other reason than to try to protect their own positions. They may doubt whether directors are acting in the best interests of the company and its shareholders.

(b) Report to board of directors, Kingtim Co

Introduction

This report indicates the impact of the proposed investment in outdoor shops and the consequent increase in debt finance. It also discusses the possible reactions of equity and bond holders to the proposals. Financial estimates provided in the appendices are used to support the discussion and assumptions underlying the estimates are set out below.

Cost of capital Debt tang co 2 anh huong nguoc nhau

There are two impacts, in opposite directions, on the weighted average cost of capital. Kingtim Co's cost of equity has risen significantly. This is due to increased business risk, resulting from the investment in the outdoor shops and increased financial risk from the additional debt. The increase in the cost of equity has pushed the weighted average cost of capital upwards. However, the higher proportion of debt in the company's finance structure, with debt having a lower cost than equity and also being tax-deductible, has pushed the weighted average cost of capital downwards.

Overall, however, the weighted average cost of capital has risen, meaning the increase in the cost of equity has had the greater impact.

anh huong cua Ke lon hon Kd

Assumptions

The assumptions about the returns from the new investment may depend on how much Kingtim Co can attract customers away from competitors rather than finding a new market niche itself. Competitor reaction may also impact upon returns.

The CAPM model used is assumed to be a good predictor of equity returns, although some published evidence suggests that it may not be.

The asset beta used for the outdoor shops is a representative beta for similar companies and may not be accurate for Kingtim Co. The asset beta used to calculate the revised cost of equity is a weighted average of the asset betas of the two businesses. The weighting used is the non-current assets in each business, which is assumed to approximate to the size of each business. This assumes that non-current assets currently held are valued fairly, and that their valuation represents their income-generating potential and the proportion of business risk that each business represents.

The share price and price of the existing bonds are assumed to remain unchanged when the new investment is made. As discussed below, there is a strong possibility of changes in the shareholder base leading to changes in the share price and hence in the cost of capital.

Equity holders

Equity holders may consider the returns from the new investment to be insufficient. The pre-tax return of 10% is lower than the 16% pre-tax return on the existing garden centres, and is not much above the 7.5% pre-tax finance cost of the bonds used to finance the investment.

Equity holders are likely to be concerned about the increases in both business and financial risks. The increase in business risk is due to the higher business risk for the outdoor shops, due to the competition in that sector. Equity holders will be concerned about the possible variability of returns and also of dividends, as the company is committed to an increased operating cost burden in terms of extra premises and increased finance costs. Variability of returns may also result in the share price becoming more volatile.

Other aspects concerning equity holders might be any restrictive covenants attached to the new bonds that affect payment of dividends and also the planned repayment of the bonds. Kingtim Co already has a significant commitment to repay the \$45m bonds in three years' time. The new bonds would mean an additional commitment to repay \$60m just a year later. The alternative is refinancing, but the terms that would be available are currently unknown.

These risks may mean that equity holders reconsider their investment in Kingtim Co, if they are risk-averse and do not feel that the additional returns compensate for the risk. They will take into account that the return on investment in the new business is lower than the current return on investment in the garden centres, although they are not required to make any additional investment themselves for the return on the outdoor shops. The share price will fall if a significant number of shareholders decide to sell their shares, although Kingtim Co may attract a new clientele of shareholders who are more risk-seeking.

Bond holders

Bond holders are likely to be most concerned about Kingtim Co's ability to meet its interest and repayment commitments. Holders of the new bonds are particularly likely to be concerned about the ability to repay their capital, given the commitment to repay existing bond holders. Bond holders may also be concerned about whether the financing of the investment allows Kingtim Co to take undue risks. They may wonder about the motivation for undertaking the new investment using debt finance, particularly if they are not convinced about its business case.

Conclusion

Assuming a strong business case can be made for the investment and the estimates are robust, Kingtim Co may be able to justify financing it solely by debt and claim that the increase in financial risk is within acceptable levels. However, before committing to further debt, Kingtim Co must provide a clear plan for repayment of both the current and new bonds, or offer sufficient assurance that it will be able to refinance its debt when it is due for repayment.

Appendices:

Appendix 1 Estimate of existing cost of capital (b) (i)

Cost of equity

$$k_e = 4.0\% + (0.9 \times 9.0\%) = 12.1\%$$

$$\text{Value of equity } (V_e) = \$5.56 \times 25 \text{ million shares} = \$139\text{m}$$

Value of existing bonds

$$V_d = \$104 \times 0.45 \text{ million} = \$46.8\text{m}$$

Current WACC

$$\text{WACC} = ((12.1\% \times 139) + (4.1\% \times 46.8)) / (139 + 46.8) = 10.1\%$$

Appendix 2 Estimate of cost of new bonds (b) (ii)

Annual yield curve

Bond	Government annual yield curve	Credit spread	Kingtim Co annual yield curve
Ga	4%	56	4.56%
Th	4.3%	78	5.08%
De	4.7%	106	5.76%
Ro	5.2%	135	6.55%

Value of new bonds based on annual yield curve

$$\$7.50 \times 1.0456^{-1} + \$7.50 \times 1.0508^{-2} + \$7.50 \times 1.0576^{-3} + \$115.50 \times 1.0655^{-4} = \$109.92$$

Market value of new bonds

$$\$109.92 \times 0.6\text{m} = \$65.952\text{m}$$

Yield to maturity of new bonds

Year		\$	5%	\$	3%	\$
0	Market value	(109.92)	1.000	(109.92)	1.000	(109.92)
1 – 4	Interest (post-tax)	5.63	3.546	19.96	3.717	20.93
4	Redemption	108.00	0.823	88.88	0.888	95.90
				(1.08)		6.91

$$\text{YTM } (k_d) = 3\% + ((6.91 / (6.91 + 1.08)) \times (5\% - 3\%)) = 4.7\%$$

Appendix 3 Revised cost of equity and WACC (b) (iii)

$$\beta_a \text{ garden centre business} = 0.9 \times (139 / (139 + (46.8 \times 0.75))) = 0.72$$

$$\text{Weighted average } \beta_a = (0.72 \times (150 / (150 + 60))) + (0.88 \times (60 / (150 + 60))) = 0.77$$

$$\beta_e = 0.77 \times ((139 + ((46.8 + 65.952) \times 0.75)) / 139) = 1.24$$

$$k_e = 4.0\% + (1.24 \times 9.0\%) = 15.2\%$$

$$\text{WACC} = ((15.2\% \times 139) + (4.1\% \times 46.8) + (4.7\% \times 65.952)) / (139 + 46.8 + 65.952) = 10.4\%$$

Appendix 4 Revised earnings forecast (b) (iv)

Forecast statement of profit or loss for the coming year

	\$000
Forecast after-tax earnings (\$24m x 0.75)	18,000
Additional finance cost (\$60m x 7.5% x 0.75)	(3,375)
Additional after-tax earnings due to new investment (\$60m x 10% x 0.75)	4,500
Revised forecast after-tax earnings	19,125
Increase in after-tax earnings	1,125

(c) Approach taken

The stated approach to **employee remuneration** has some **business logic**. Expertise, experience, seniority and commitment are all attributes that staff have that could be reflected in **extra rewards for them**, not only out of **fairness to the staff** but also **because of their value to the business**. If staff with these attributes believe they are not being rewarded fairly, they may leave and perhaps join a competitor.

Kingtim Co also has **a duty to enhance the wealth of its shareholders** and has raised expectations by recently increasing dividends. There is a **stakeholder conflict**, as increasing the wages of many employees would lead to lower profits and less money available for distribution to shareholders.

Issues with approach

The statement about **part-time staff not having the same level of commitment may well be unjust**, as they may be as committed as full-time staff during the hours they work.

The current approach **raises a number of ethical issues**, which may also **harm Kingtim Co's reputation**. It has made commitments to act in accordance with society's expectations and to treat its staff fairly. Although the basic wage is not legally enforceable, it does **represent society's expectations about what employees should be paid**. Limiting rewards to staff who may only be able to work part-time because of other commitments could also be something that society judges to be **discriminatory and may be against the law**.

In addition, if Kingtim Co's **directors are given more lucrative contracts as a takeover defence mechanism**, this undermines the argument for limiting staff costs in order to maintain shareholder returns.

The consequences of these threats to reputation might again be that lower-paid staff eventually decide to leave. A **high staff turnover** will mean few staff develop experience and expertise over time, which may **impact on customer quality**. Kingtim Co may also have **problems recruiting staff for its new outdoor business**. **Customers may also stop shopping** at Kingtim Co in protest at the poor treatment of staff.

2 (a) Project cash flows: All figures are in CL millions

Year	0	1	2	3	4
Contribution		419.4	500.2	671.3	961.2
Fixed costs		(270.0)	(291.6)	(314.9)	(340.1)
Tax allowable depreciation		(175.0)	(175.0)	(175.0)	(175.0)
Taxable profit/(loss)		(25.6)	33.6	181.4	446.1
Tax loss carried forward			(25.6)		
Adjusted taxable profit		(25.6)	8.0	181.4	446.1
Taxation (25%)			(2.0)	(45.4)	(111.5)
Add loss carried forward			25.6		
Add depreciation		175.0	175.0	175.0	175.0
Cash flows after tax		149.4	206.6	311.0	509.6
Working capital (w2)	(25.0)	(2.5)	(2.8)	(3.0)	33.3
Investment cost	(775.0)				214.2
Cash flows	(800.0)	146.9	203.8	308.0	757.1

Cash flows: All figures are in € millions

Year	0	1	2	3	4
Exchange rate (w1)	9.91	10.48	11.09	11.96	12.89
Total investment cost	(80.7)				
Remittable cash flows		14.0	18.4	25.8	58.7
Component contribution (w3)		0.9	1.1	1.3	1.8
Tax on net contribution (20%)		(0.2)	(0.2)	(0.3)	(0.4)
Cash flows	(80.7)	14.7	19.3	26.8	60.1

Net present value using 16% discount rate: All figures are in € millions

Year	0	1	2	3	4
Cash flows	(80.7)	14.7	19.3	26.8	60.1
Discount rate (16%)	1.000	0.862	0.743	0.641	0.552
Present values	(80.7)	12.7	14.3	17.2	33.2

Net present value (€3.3m)

Net present value using 13% discount rate: All figures are in € millions

Year	0	1	2	3	4
Cash flows	(80.7)	14.7	19.3	26.8	60.1
Discount rate (13%)	1.000	0.885	0.783	0.693	0.613
Present values	(80.7)	13.0	15.1	18.6	36.8

Net present value €2.8m

Workings

Working 1 (w1): Exchange rates

Year	1	2	3	4
CL/€	9.91 x 1.10/1.04 = 10.48	10.48 x 1.10/1.04 = 11.09	11.09 x 1.10/1.02 = 11.96	11.96 x 1.10/1.02 = 12.89

Working 2 (w2): Working capital (CLm)

Year	0	1	2	3	4
Inflation		10%	10%	10%	10%
Increase with inflation	(25.0)	(27.5)	(30.3)	(33.3)	
Incremental working capital	(25.0)	(2.5)	(2.8)	(3.0)	33.3

Working 3 (w3): Component contribution (€)

Year	1	2	3	4
Contribution	109,725	121,795	148,590	197,624
	x 8 x 1.04 = 0.9m	x 8 x 1.04 ² = 1.1m	x 8 x 1.04 ² x 1.02 = 1.3m	x 8 x 1.04 ² x 1.02 = 1.8m

Comment

The **decision** whether to accept or reject the project **critically depends on the discount rate**, switching from a negative net present value of €3.3m when the discount rate includes a country risk premium to a positive net present value of €2.8m when there is no premium. This adjustment to the weighted average cost of capital requires further investigation because it is possible Colvin Co could reject projects that increase shareholder wealth.

The outcome **assumes** the contribution and other cash flows are reliably estimated. Other critical inputs include the assumption that land and buildings will increase in value at an annual rate of 30% and that any disposal is tax exempt.

- (b) There are a number of possible strategies to avoid exchange controls on remittances. Colvin Co could **increase the transfer price** paid by the Canvian subsidiary for the gearing system component. This would increase Colvin Co's profit to the detriment of the subsidiary's profit. Alternatively, Colvin Co doesn't currently levy a **management charge or royalty** but if these were introduced it would transfer profit from the subsidiary without the need for a dividend.

These methods assume the exchange controls imposed in Canvia are not applied to repatriations in general. An alternative would be for Colvin Co's subsidiary to make a loan to the Canvian subsidiary of another Eurozone based parent company which, in return, would lend an equivalent amount in Euros to Colvin Co. The same objective could be achieved with a **currency swap** although counterparty risk would be a factor in both cases.

- (c) Colvin Co's investment in Canvia does not involve a change in business risk or capital structure. The company's weighted average cost of capital would normally be expected to provide a reasonable measure of risk for the new project. The chief executive's justification for a risk premium is based on the increased risk the company is exposed to in Canvia, a developing economy, compared to the company's existing business in the Eurozone. This perception of increased risk is based on a country risk index, which compares the standard deviation of market indices around the world. **The chief executive has incorporated other factors, such as political risk and foreign exchange risk in determining this premium.**

However, standard deviation is not the appropriate measure of risk for Colvin Co's investment since any portion of **total risk that is uncorrelated across different markets can be diversified** away at no cost to investors. For example, adverse political events in Canvia may be partially offset by more favourable events in other parts of the world. **No rational investor would pay a premium for risk that can be avoided.** In this sense, although Colvin Co's investment in Canvia is exposed to **foreign exchange risk, this too can be mitigated by an appropriate hedging policy.**

Furthermore, Colvin Co's institutional shareholders are likely to be well diversified across global markets and asset classes. The potential for further risk reduction by Colvin Co from diversifying operations globally is therefore limited when the shareholders can achieve this more efficiently on their own.

The only component of total risk that could **justify** a premium to Colvin Co's cost of capital is **market risk or undiversifiable risk.** This assumes returns across countries are significantly **positively correlated.** For example, there is a strong possibility that a recession in the Eurozone may lead to a downturn in Canvia too rather than offset it, transmitted through trade links and closer integration between markets. This tendency for markets across the world to move together means reduced risk reduction benefits from diversification, hence a higher cost of capital. **The key issue therefore is whether the risk of the new investment is diversifiable or not.** If returns across markets are significantly positively correlated and the risk undiversifiable, the new project in Canvia may therefore command a risk premium although **no justification is provided for the chief executive's premium of 3% which would require further investigation and analysis.**

3 (a) Swap

	Fitzharris Co	Counterparty	Interest rate differential
Fixed rate	4.60%	4.80%	0.20%
Floating rate	Base rate + 0.50%	Base rate + 1.30%	0.80%

Fitzharris Co has an advantage in borrowing at both fixed and floating rates, but the floating rate advantage is larger.

Gain % for Fitzharris Co = 50% (0.8 – 0.2 – 0.1) = 0.25

	Fitzharris Co	Counterparty
Rate without swap	(4.60%)	(Base rate + 1.30%)
Benefit	0.25%	0.25%
Net result	(4.35%)	(Base rate + 1.05%)

Swap

Borrows at	(Base rate + 0.50%)	(4.80%)
Fitzharris Co pays	(3.80%)	3.80%
Counterparty pays	Base rate	(Base rate)
Bank fee	(0.05%)	(0.05%)
Net result	(4.35%)	(Base rate + 1.05%)

Collar

Buy December put options at 95.75 for 0.211 and sell December call options at 96.25 for 0.198

Number of contracts = (\$48,000,000/\$1,000,000) x (36 months/3 months) = 576**

Basis = Current price (1 August) – futures price

(100 – 3.70) – 95.85 = 0.45

Unexpired basis on 1 December = 1/5 x 0.45 = 0.09

Premium = (0.00211 – 0.00198) = 0.013%

If base rate rises by 0.4% to 4.1%

Futures price = 100 – 4.1 – 0.09 = 95.81

	Buy put	Sell call
Exercise price	95.75	96.25
Futures price	95.81	95.81
Exercise?*	No	No
Loss in basis points	–	–

* The put option is not exercised, because Fitzharris Co can sell the futures at the futures market price of 95.81 rather than the option exercise price of 95.75. The call option is not exercised, as the option holder can buy the futures at the lower futures market price of 95.81 rather than the exercise price of 96.25.

	%
Borrowing cost (4.1% + 0.5%)	4.600
Premium	0.013
Effective annual interest rate	4.613

If base rate falls by 0.4% to 3.3%

Futures price = $100 - 3.3 - 0.09 = 96.61$

	Buy put	Sell call
Exercise price	95.75	96.25
Futures price	96.61	96.61
Exercise?*	No	Yes
Loss in basis points	–	36

* The put option is not exercised, as by not exercising the option Fitzharris Co can sell the futures at the higher futures market price of 96.61 rather than the lower exercise price of 95.75. The call option is exercised, because the option holder can buy the futures at the option exercise price of 96.25 rather than the futures market price of 96.61.

	%
Borrowing cost (3.3% + 0.5%)	3.800
Loss on options (0.0036 x 100)	0.360
Premium	0.013
Effective annual interest rate	4.173

**** Note for tutors**

It is possible to justify a range of different hedging periods for this situation. Any justified hedging period from the four-month period of uncertainty, outlined in the question, up to 36 months was awarded credit. It was recognised that in reality a collar for 36 months would not happen and instead there would be a rolling series of hedges.

Answers which calculated costs in dollar amounts based on their number of contracts and then calculated an effective annual rate, were also eligible for full credit.

(b) Comment

The calculations do **not give a clear indication** of which strategy should be chosen. The swap gives a better result if base rate rises by 0.4%, the options if base rate falls by 0.4%. The decision may be determined by **whether the company views a rise or fall in interest rates as being more likely, or how it views the advantages and disadvantages of the strategies.**

Advantages of swaps

As **swaps** are **over-the-counter arrangements**, they can be arranged in **any size**. The amount covered by **collars** based on traded options is determined by the **size of the option contract**. There may be over and under hedging.

The **traded options** available may last for a **short period**, perhaps **up to two years**, less maybe than the period of the loan. **Swaps** can be arranged for a **much longer period**.

Fitzharris Co is swapping here a commitment to pay a variable rate of interest that is uncertain with a guaranteed fixed rate of interest. This allows Fitzharris Co to **forecast finance costs on the loan with certainty**. The net payments on the collar will depend on how interest rates move.

Unlike collars, **swaps make use of the principle of comparative advantage**. Fitzharris Co **can borrow in the market where the best deal is available to it**.

Disadvantages of swaps

khi arranges itself not through the bank

Swaps are subject to counterparty risk, the risk that the other party may default on the arrangement. This should not generally be a problem if Fitzharris Co arranges the swap through the bank. It may, however, be a problem if it arranges the swap itself. As **the options that the collar is based on are traded on the derivatives markets, this should guarantee there will be no counterparty risk**.

As Fitzharris Co is swapping into a fixed rate commitment, it cannot take advantage of favourable interest rate changes as it could, to some extent, if it used collars. Here the swap results in a lower cost than the collar if interest rates rise, but the collar is better if interest rates fall.

As **swaps are over-the-counter instruments, they cannot be traded or allowed to lapse if they are not needed**. **The options can be traded on a derivative market**.

Theta (Hao mòn thời gian) là thước đo tỉ lệ suy giảm giá trị của hợp đồng quyền chọn theo thời gian. Theta tăng lên nhanh chóng khi ngày đáo hạn của quyền chọn đến gần vì càng có ít thời gian để thu lợi nhuận từ giao dịch hơn.

(c) Time

An option's price consists of two elements, its intrinsic value and its time premium. The time premium diminishes over time to zero at the point that the option expires. Theta measures how much time value is lost over time. It is generally expressed as an amount lost per day. Theta reduces the value of both put and call options for holders.

The change in theta for in the money and out of the money options is broadly linear. At the money options have the greatest time premium and greatest theta. Theta for at the money options does not change in a linear fashion, but changes more rapidly as the expiry date approaches.

Interest rate

Rho measures how the option price varies with changes in interest rates. An option's rho is the amount of change in value for a 1% change in the option's risk free interest rate. The rho is positive for call options if the risk-free interest rate increases and negative for put options.

Compared with other factors affecting option price, the interest rate is not a significant influence, as interest rates often move slowly. A change in interest rates will be more significant the longer the time until expiry of an option.

<https://vietnambiz.vn/rho-la-gi-cach-tinh-rho-20191218172207765.htm>

Rho là tỉ lệ mà giá của chứng khoán phải sinh thay đổi so với mức thay đổi của lãi suất phi rủi ro.
Rho lớn nhất khi các quyền chọn trong vùng có tiền (in the money) và giảm dần khi quyền chọn ở trong vùng mất tiền (out of the money). Ngoài ra, tăng khi thời gian đáo hạn tăng.

		Marks
1	(a) Sell-off assets (examples of points could include company less appealing, use of proceeds from sell-off, how Kingtim Co will be viewed, smaller company being more affordable) Onerous contracts (examples of points could include cost burden, acquirer may be prepared to bear it, corporate governance requirements, shareholder reaction)	3–4 3–4
	Max	7
	(b) (i) Cost of equity Value of equity Value of existing bonds WACC	1 1 1 1 4
	(ii) Annual spot yield curve Value of new bonds Market value of new bonds Yield to maturity of proposed bonds	1 3 1 2 7
	(iii) Asset beta garden centre business Weighted asset beta Revised equity beta Revised cost of equity Revised WACC	1 2 1 1 2 7
	(iv) Revised P/L	3 3
	(v) Cost of capital Assumptions Equity holders (examples of points could include inadequate returns, higher business and financial risks, threat to earnings/dividend/share price, covenant/repayment commitments, forecast extra earnings insufficient, sale of shares/changes in shareholder base) Bond holders	1–2 3–4 3–4 2–3 10
	Max	10
	Professional marks for part (b) Report format Structure and presentation of the report	1 3 4
	(c) Up to 2 marks for each well-explained issue (issues could include rewarding expertise/seniority fairly, balancing shareholder and employee interests, unfair to question staff's commitment, society's expectations/law, expectations raised by Kingtim Co's statements, employee/customer reaction to poor practices)	8
	Max	8
	Total	50

		Marks
2	(a)	
	Exchange rates	2
	Tax	2
	Working capital	2
	Land and buildings residual value	1
	Remittable cash flows in euros	1
	Contribution from component	2
	Tax on contribution	1
	Net present values	2
	Comment	2–3
	Max	15
	(b)	
	Up to 2 marks per point (e.g. transfer price, royalty)	4
	(c)	
	Up to 2 marks per point (e.g. argument for WACC, total risk v market risk, correlation across countries)	6
	Max	6
	Total	25
3	(a)	
	Swaps	
	Comparative advantage of 0·6%	1
	Initial decision to borrow floating by Fitzharris Co and fixed by counterparty	1
	Advantage of 0·25% per party after the bank fee	1
	Suitable swap rates	1
	Final rate to be paid by Fitzharris Co	1
	Collars	
	Number of contracts	1
	Basis calculation	1
	Buy put and sell call options	1
	Premium calculation	1
	Exercise options?	2
	Impact of interest rate increase/decrease with collars	2
		13
	(b)	
	Comment on calculations	1
	Advantages of swaps compared with collars (advantages could include flexibility, longer time period, certainty of finance costs, comparative advantage)	3–4
	Disadvantages of swaps compared with collars (disadvantages could include counterparty risk, inability take advantage of favourable rate movements, swaps cannot be traded)	3–4
	Max	8
	(c)	
	Time (Theta)	2–3
	Interest rate (Rho)	2–3
	Max	4
		25



Think Ahead

Advanced Financial Management (AFM)

March/June 2021 (20/21 Syllabus)

Get to know your exam

These graphical representations are intended to give an indication of past exam requirements and associated question content.

Please note that you will not be able to complete answers within these documents and in isolation they will not sufficiently prepare you for your exam.

We encourage you to visit the ACCA Practice Platform in order to attempt up to date practice exams within the computer-based exam environment. Further instructions on how to use the platform will be provided before you attempt the exam.

Introduction screen

AFM March/June 2021 (20/21 syllabus)



Introduction

This exam contains past exam content and indicates how the live exam will be structured and assessed, the likely layout and style of questions and the range of response options that could be presented to you in each question's workspace.

You should use this exam to become familiar with the workspace and the features and functionality contained within the live exam.

The past content exam is reflective of the live exam experience but has some differences:

- The live exam will be timed, however there is no time limit in the past content exam.
- If you want to sit this exam in exam-style conditions you should answer the questions presented within a **3 hour 15 minute time period**.
- Once you have started this past content exam, you are able to leave at any time by closing the browser window. When you return, anything you have entered into the response options will be saved and you can continue sitting the exam.
- In the live exam your answers entered into the workspace response options will be expert-marked. At the end of this exam, you should use the solution material provided to assess your performance.
- You will be able to access solution material at the end of this exam when using the Self Marking resources which include a Marking Guide and Sample Answer for each question. If you wish to access these without completing the questions, click on **End Exam** on the Item Review Screen and navigate to the Marking tab on the Dashboard to Self-Mark.

Exam summary screen

AFM March/June 2021 (20/21 syllabus)

Exam Summary

Time allowed: This exam is not timed.

This exam is divided into two sections:

Section A

- One question worth 50 marks.
- 50 marks in total.

Section B

- Two questions, each worth 25 marks.
- 50 marks in total.

All questions are compulsory.

Select **Next** to start your exam.

Exam summary screen (continued)

Section A

AFM March/June 2021 (20/21 syllabus)

Section A

This section of the exam contains **one question**.

This question is worth **50 marks** and is compulsory.

This exam section is worth **50 marks** in total.

Important:

In your live exam, you should:

- Indicate which requirement each of your responses relate to so that this is clear for markers.
- Show all notes/workings that you want the marker to see within your responses. Remember, any notes/workings made on the Scratch Pad or on your workings paper will not be marked.

Select **Next** to continue.

Scenario 1 – Initial screen

AFM March/June 2021 (20/21 syllabus)

\$ Symbol  ▼ Highlight  Strikethrough  Calculator  Scratch Pad

Exhibits

 1. Introduction

 2. Areas for further clarification

 3. Capital structure details

 4. Kawa Co as a demerged company

 5. Acquisition of Kawa Co by Lahla Co

Requirements

 Requirements (50 marks)

Response Options

 Word Processor

 Spreadsheet

The following **exhibits**, available on the left-hand side of the screen, provide information relevant to the question:

1. Introduction – about Chakula Co, the demerger of Kawa Co and Lahla Co a prospective buyer of Kawa Co
2. Areas for further clarification – requested by Lahla Co
3. Capital structure details – for all companies
4. Kawa Co as a demerged company
5. Acquisition of Kawa Co by Lahla Co

This information should be used to answer the question **requirements** within your chosen **response option(s)**.

Scenario 1: requirements

Requirements (50 marks)

(a) Explain why a regulatory framework related to mergers and acquisitions is necessary to protect the interests of shareholders and other stakeholders.

(5 marks)

(b) Discuss the two theoretical propositions, as raised by Lahla Co's board of directors (BoD), in relation to a company's capital structure.

(6 marks)

(c) Prepare a report for the BoD of Lahla Co which:

(i) Estimates the value of each Kawa Co share if it is demerged and listed as an independent company;

(12 marks)

(ii) Estimates;

- the additional equity value created when combining Lahla Co and Kawa Co;
- the percentage gain to each of Lahla Co's and Kawa Co's shareholder group under each payment method;
- the impact on Lahla Co's capital structure under each payment method; and

(12 marks)

(iii) Evaluates the financial and other factors that both Lahla Co's shareholders and Kawa Co's shareholders would consider prior to agreeing to the acquisition, and the impact on Lahla Co's capital structure under each payment method.

(11 marks)

Professional marks will be awarded in part (c) for the format, structure and presentation of the report.

(4 marks)

Scenario 1: Exhibits

Exhibit 1: Introduction

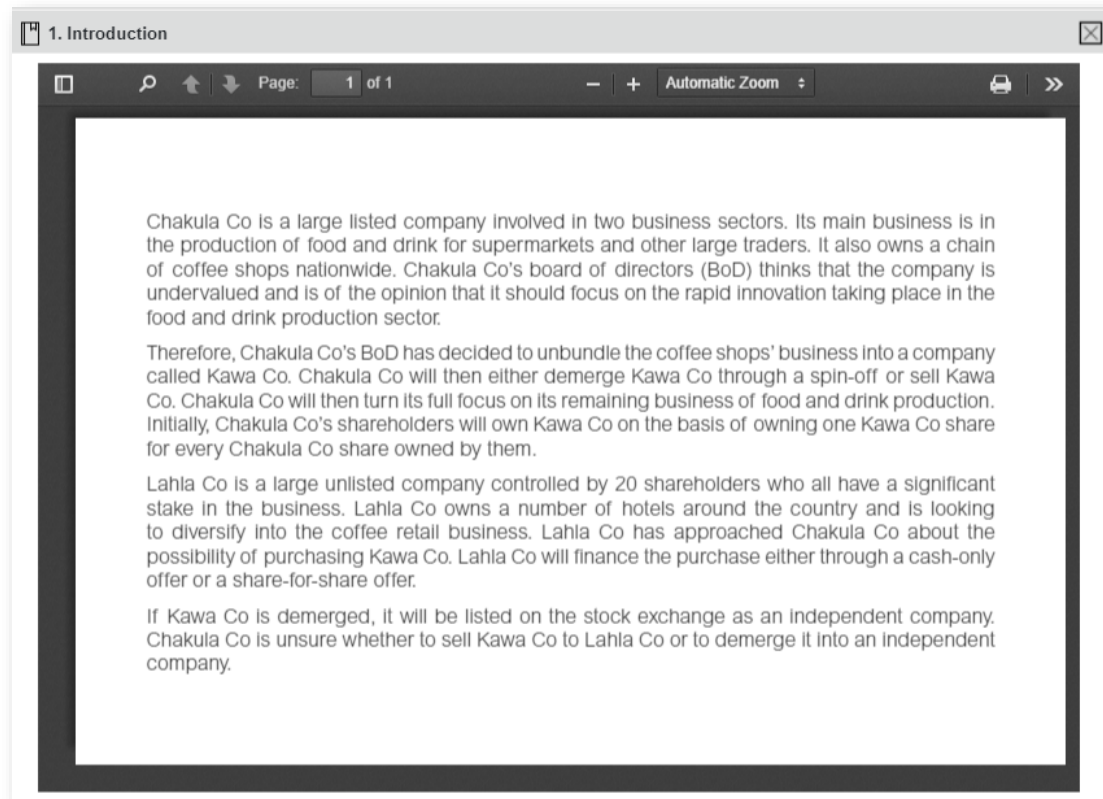


Exhibit 2: Areas for further clarification

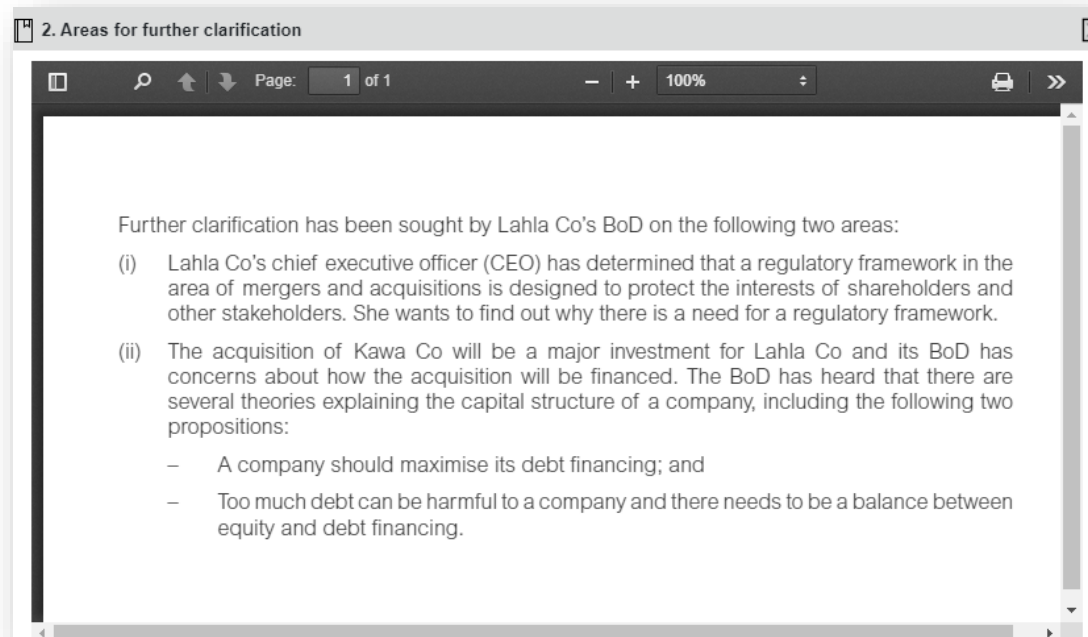


Exhibit 3: Capital structure details

3. Capital structure details

Page: 1 of 1

Automatic Zoom

Extracts from Chakula Co's financial statements are as follows:

	\$m
Assets less current liabilities	5,010
Financed by:	
Share capital (nominal value \$0.50 per share)	1,000
Reserves	1,180
Non-current liabilities: Loan notes A (nominal value \$100 per loan note)	2,470
Non-current liabilities: Loan notes B (nominal value \$100 per loan note)	360

Chakula Co's shares are trading at \$2.45 each and have an equity beta of 1.3. The part of the business which will become Kawa Co accounts for 24.5% of Chakula Co's total equity value.

Chakula Co's loan notes A currently have a total market value of \$2,100m. Loan notes B currently have a total market value of \$400m. The pre-tax cost of debt has been determined at 4.3% for loan notes A and 4.1% for loan notes B. After the unbundling, loan notes B will be serviced by Kawa Co and loan notes A will remain with Chakula Co, but the pre-tax cost of debt for both will increase by 0.3%. It is expected that Kawa Co will maintain its capital structure after the unbundling.

Lahla Co's debt to equity ratio is estimated to be 40:60 in equivalent market value terms and it has 1,200 million shares in issue.

The coffee retail industry is dominated by Buni Co, a large listed company which owns many coffee shops around the country. The average asset beta for the coffee retail industry is estimated at 1.15.

All companies pay corporation tax at a rate of 20% per year and tax is payable in the same year as the profits it is based on. The risk-free rate of return is estimated at 3% and the market risk premium at 7.2%.

Exhibit 4: Kawa Co as a demerged company

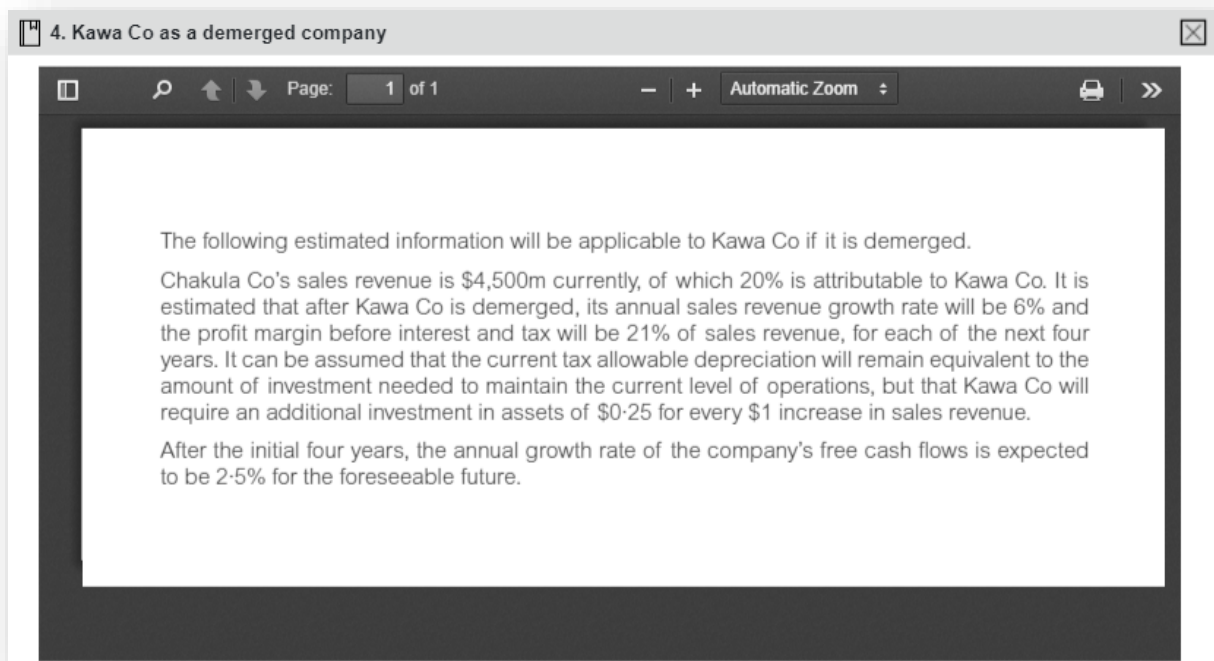


Exhibit 5: Acquisition of Kawa Co by Lahla Co

5. Acquisition of Kawa Co by Lahla Co

The following estimated information applies to the acquisition of Kawa Co by Lahla Co, if Kawa Co is acquired.

The average price to earnings (PE) ratio for the hotel industry is 15.61, however, Lahla Co's PE ratio is estimated to be 10% lower than this.

Extracts from the current statements of profit or loss applicable to Lahla Co and Kawa Co are as follows:

	Lahla Co \$m	Kawa Co \$m
Profit before interest and tax	305.0	161.2
Interest	(91.2)	(14.8)
Tax (20%)	(42.8)	(29.3)
Profit after tax	<u>171.0</u>	<u>117.1</u>

After the acquisition, it is expected that the PE ratio of the combined company will be the midpoint between the two individual companies' PE ratios. The annual after-tax profits will increase by \$62m due to combining the two companies.

Lahla Co has proposed to pay for acquiring Kawa Co either through a cash offer of \$0.66 for a Kawa Co share, or one Lahla Co share for every three Kawa Co shares. Lahla Co will borrow the money needed to pay for the acquisition.

Exam summary screen (continued)

Section B

AFM March/June 2021 (20/21 syllabus)

Section B

This section of the exam contains **two questions**.

Each question is worth **25 marks** and is compulsory.

This exam section is worth **50 marks** in total.

Important:

In your live exam, you should:

- Indicate which requirement each of your responses relate to so that this is clear for markers.
- Show all notes/workings that you want the marker to see within your responses. Remember, any notes/workings made on the Scratch Pad or on your workings paper will not be marked.

Select **Next** to continue.

Scenario 2 – Initial screen

AFM March/June 2021 (20/21 syllabus)

\$ Symbol  ▼ Highlight  Strikethrough  Calculator  Scratch Pad

Exhibits

-  1. Robson Co and project information
-  2. Further information on project financ

Requirements

-  Requirements (25 marks)

Response Options

-  Word Processor
-  Spreadsheet

The following **exhibits**, available on the left-hand side of the screen, provide information relevant to the question:

- 1. Robson Co and project information
- 2. Further information on project finance

This information should be used to answer the question **requirements** within your chosen **response option(s)**.

Scenario 2: requirements

🔍 Requirements (25 marks) 



(a) Calculate the adjusted present value of the investment and recommend whether the project should be accepted or not.

(14 marks)

(b) Discuss the factors the capital providers, excluding the bank, will consider before deciding whether or not to approve the funding decision for Robson Co's investment in a new manufacturing plant.

(11 marks)

Scenario 2: Exhibits

Exhibit 1: Robson Co and project information

1. Robson Co and project information

Page: 1 of 1

90%

Robson Co is a food manufacturer with a portfolio of well-known brands. The founding directors retain a significant minority shareholding in the company and continue to serve on the board following a successful listing ten years ago. After obtaining the listing, Robson Co's gearing ratio increased significantly above the sector average as the result of a poorly timed expansion strategy, mainly financed by debt. Earnings became increasingly volatile and the debt burden triggered a decline in the company's financial performance. The board responded to these problems five years ago by pursuing a debt-reduction turnaround strategy, which has been financed by a series of rights issues and asset disposals.

Even though this strategy successfully reduced the gearing ratio, which is now equal to the industry average, the share price remains depressed due to competitive pressures within the industry. The company's credit rating has recently been downgraded once again. Robson Co's chief executive officer (CEO) has identified an opportunity to relocate the manufacturing plant and develop a state-of-the-art automated production line, which will reduce the underlying cost base and be a source of competitive advantage.

Project information

Robson Co's finance director has prepared estimates of the free cash flows generated by the project, based on a four-year time horizon:

Year	0	1	2	3	4
	\$m	\$m	\$m	\$m	\$m
Free cash flows		20.9	20.6	28.7	104.6

The investment cost is \$120m, which Robson Co's CEO proposes to finance as follows:

	\$m
Disposal of existing manufacturing plant	20
Rights issue	10
Subsidised loan, 3.5% annual interest rate	40
Bank loan, 9% annual interest rate	50
Total	120

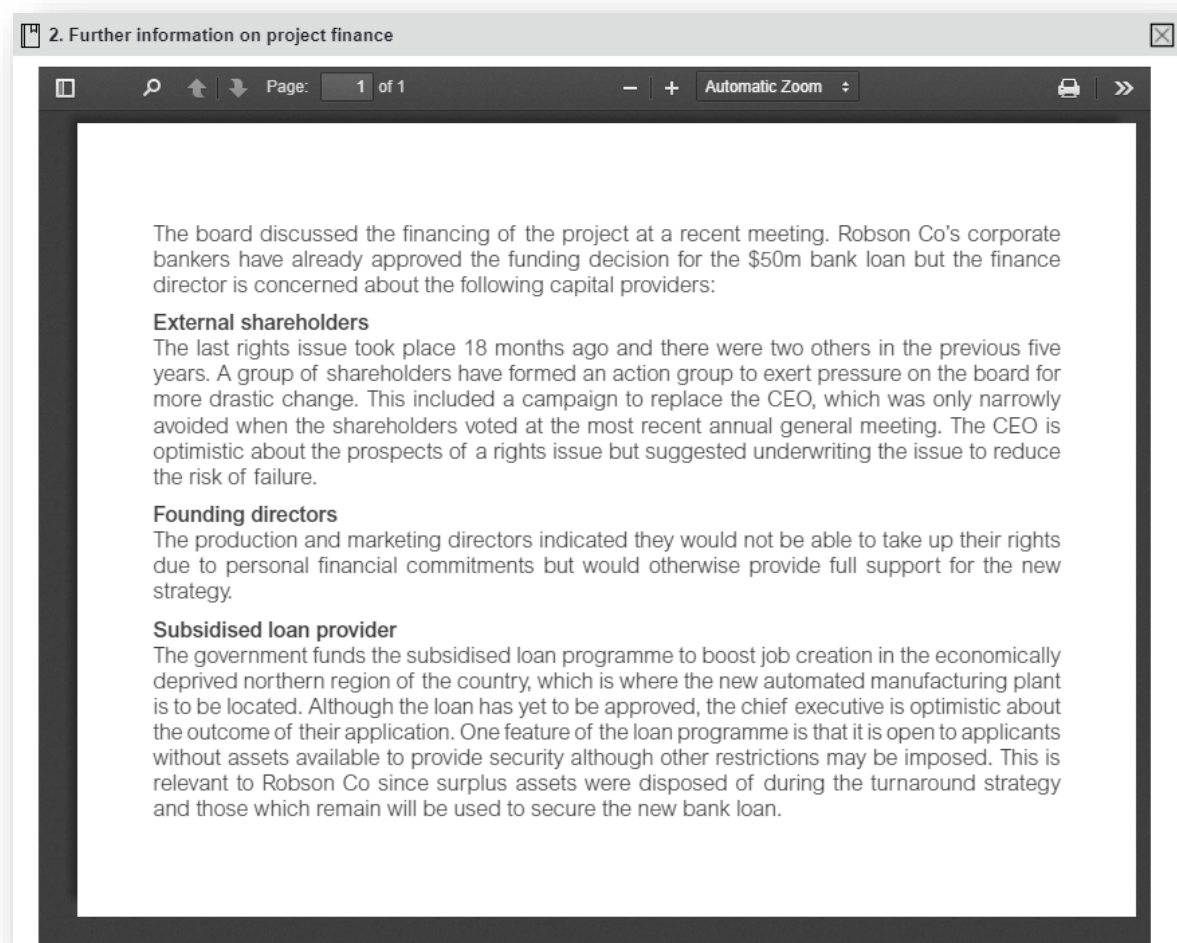
The bank loan is repayable in equal annual instalments over four years. Issue costs of 2% are payable on gross external financing and are not allowable for corporation tax. Issue costs are payable out of available cash reserves. The finance director has asked you to ignore underwriting costs relating to the rights issue.

Additional information

Robson Co's current equity beta is 1.418 and the debt:equity ratio is 1:5 based on market values. The risk free rate is 3% and the market risk premium is 9%. The CEO expects the business risk of the company to remain unchanged as a result of the investment.

Corporation tax is payable at an annual rate of 20%.

Exhibit 2: Further information on project finance



Scenario 3 – Initial screen

AFM March/June 2021 (20/21 syllabus)

\$ Symbol  ▼ Highlight  Strikethrough  Calculator  Scratch Pad

Exhibits

 1. Gogarth Co's currency risk management

 2. Board queries about risk management

Requirements

 Requirements (25 marks)

Response Options

 Word Processor

 Spreadsheet

The following **exhibits**, available on the left-hand side of the screen, provide information relevant to the question:

1. Gogarth Co's currency risk management
2. Board queries about risk management

This information should be used to answer the question **requirements** within your chosen **response option(s)**.

Scenario 3: requirements

Requirements (25 marks) 

(a) Advise Gogarth Co on, and recommend, an appropriate hedging strategy for its US\$ cash flows on 31 August. Include relevant calculations.

(15 marks)

(b) Discuss the advantages and disadvantages for Gogarth Co in using exchange-traded currency options compared with using over-the-counter currency options.

(5 marks)

(c) Discuss the role of Gogarth Co's treasury function in relation to the management of economic risk.

(5 marks)

Scenario 3: Exhibits

Exhibit 1: Gogarth Co's currency risk management

1. Gogarth Co's currency risk management

Page: 1 of 1

Automatic Zoom

Gogarth Co is an electrical equipment manufacturer, based in Malaysia, looking to develop its operations abroad. One of its biggest sales markets is the USA and Gogarth Co also imports components from the USA. Gogarth Co regularly hedges transactions in foreign currencies.

It is currently 1 May. On 31 August, Gogarth Co is due to pay \$14,500,000 to an American supplier and receive \$37,400,000 from an American customer.

The following quotations have been obtained:

Exchange rates (quoted as US dollar per Malaysian Ringgit US\$/MR1)

Spot	0.2355 – 0.2358
Four months forward	0.2370 – 0.2374

Currency futures (contract size MR500,000, futures price quoted as US\$/MR1)

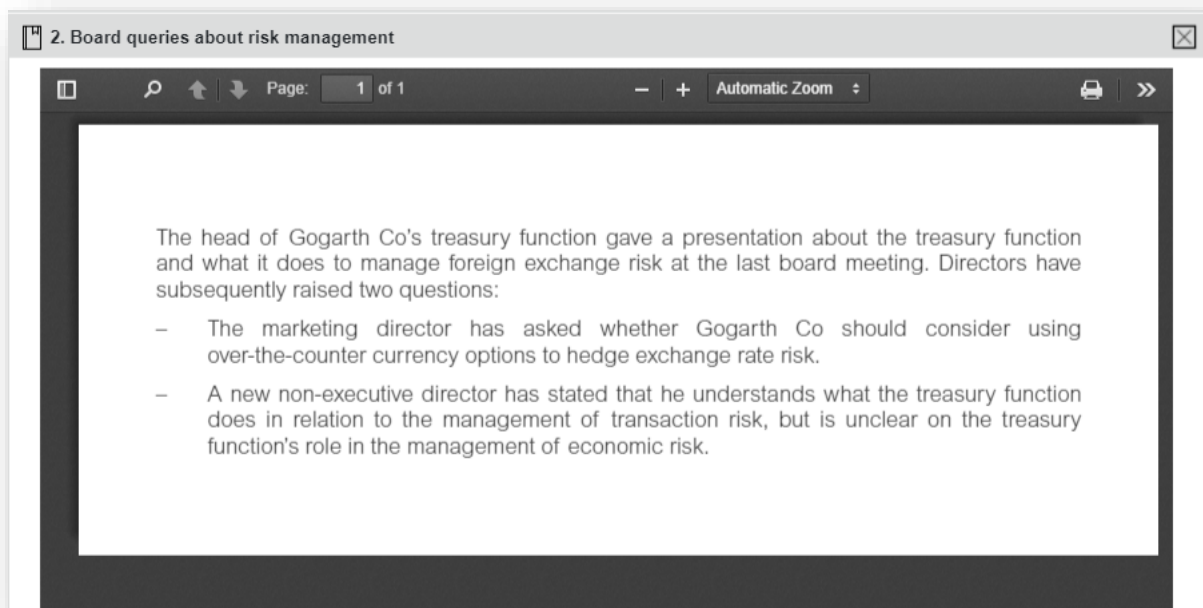
	Futures price
June	0.2366
September	0.2378

Currency options (contract size MR500,000, exercise price quoted as US\$/MR1, premium: US cents/MR1)

	Calls		Puts	
Exercise price	June	September	June	September
0.2368	0.11	0.14	0.19	0.23

Futures and options contracts mature at the month end. The number of contracts to be used should be rounded to the nearest whole number in calculations. If the amount cannot be hedged using an exact number of futures or options contracts, the amount unhedged or over-hedged should be hedged using the forward market. For the purposes of the calculations, it should be assumed that the options are exercised.

Exhibit 2: Board queries about risk management



Answers

1 Chakula

- (a) The key reason for a regulatory framework to exist in merger and acquisition (M&A) activity is to ensure that the interests of stakeholders are protected, and where the natural market forces may not be sufficient on their own to ensure that this happens. The regulatory framework aims to ensure a well-functioning market for corporate control.

With respect to shareholders, as a major stakeholder group, the regulatory framework aims to establish that shareholders of the target company are not affected negatively by ensuring that:

- Minority shareholders' rights are protected;
- The target company's management cannot block a M&A where it is in commercial and economic interest of shareholders; and
- Sufficient time is made available for a proposal to be properly scrutinised. The regulatory framework also aims to ensure that sufficient information is provided about the proposed M&A for all investor groups to evaluate the proposed deal properly.

With respect to other stakeholders, the aim of the regulatory framework is to ensure that there is not a substantial lessening of competition after the M&A has taken place. This will protect the choice that consumers, suppliers and employees have in engaging with a range of organisations in that business sector, and within a properly functioning economic market.

- (b) The two theoretical propositions are based on the opinion that a company's capital structure does matter to the value of the company. The first proposition posits that since debt is cheaper than equity and there is a 'tax shield' attached to debt finance, it is better for a company to be financed by as much debt as possible. This is the view presented by the Modigliani and Miller with taxes model. Since interest is paid before a company pays corporation tax, but dividends are not, a company does not have to pay taxes on profits used to pay interest. This is referred to as a tax shield. The presence of a tax shield results in the cost of capital reducing as the proportion of debt financing increases.

The second proposition builds on this by arguing that although debt carries with it the advantage of a tax shield, at high levels of gearing this position no longer holds true. Here, financial risk increases significantly and the company experiences increasing levels of financial distress, resulting in the cost of equity increasing significantly. This overrides the benefits gained from the tax shield. As a result, the cost of capital increases. At very high levels of gearing, even the cost of debt starts to increase significantly. Hence, there is a trade-off between the benefits of the tax shield and the costs related to financial distress, such that the cost of capital reduces initially but then rises, meaning that there is an optimal, minimum cost of capital where corporate value is maximised.

- (c) **Report to the board of directors (BoD), Lahla Co**

This report evaluates and discusses the financial and other factors that both Lahla Co's and Kawa Co's shareholders would consider prior to agreeing to the acquisition. It also evaluates and discusses the impact of the acquisition on Lahla Co's capital structure under the two payment methods.

Factors to consider

Demerger (Appendix 2)	Additional value created for Kawa Co's shareholders 18.3%	
Acquisition, cash payment (Appendix 3)	Additional value created for Kawa Co's shareholders 10.0%	Additional value created for Lahla Co's shareholders 22.0%
Acquisition, share-for-share exchange (Appendix 3)	Additional value created for Kawa Co's shareholders 26.7%	Additional value created for Lahla Co's shareholders 14.0%

The initial evaluation would indicate that the demerger is the better option for Kawa Co's shareholders, compared to the acquisition, if the acquisition is paid for by cash. However, the share-for-share exchange gives a higher return compared to the demerger and therefore, on purely financial grounds, this is the best option for Kawa Co's shareholders. Although Lahla Co's shareholders lose some additional value derived from the acquisition if the share-for-share option is chosen, they would probably still be in favour of the acquisition because the company's value will increase and so will the value of their shares.

However, the following additional factors also need to be considered in the evaluation:

The value estimates are based on predicted variables, both for the demerger valuation and for the acquisition valuations. It is likely that there will be changes to the actual variables, and it is recommended that Lahla Co undertake sensitivity analysis and assess the results of this before making the final acquisition decision.

Kawa Co's shareholders probably have three main areas they would want considered further with respect to the acquisition with the share-for-share exchange.

First, they would become part of a larger company with interests both in hotels and in coffee shops and they would own just under 36% (667m shares/1,867m shares) of the share capital of the new combined company. However, they would be minority shareholders. As such, they may feel that they do not have sufficient influence in the major decisions the company makes.

Therefore, Kawa Co's shareholders may be of the opinion that operating as a stand-alone demerged independent company may give them a better opportunity to shape the company's strategy. On the other hand, they may equally decide that they would need to be part of a large company to be able to compete effectively against Buni Co.

Second, Kawa Co's shareholders cannot be certain whether the 26.7% additional value is realistic or not. This may be especially pertinent because Lahla Co is an unlisted company and therefore may keep proprietary/strategic information private, limiting the ability for external parties to undertake a full and effective evaluation.

Third, because Lahla Co is an unlisted company, Kawa Co's shareholders may be concerned about how they would be able to exit the company, if they want to. For instance, if their investment portfolios become imbalanced when the companies are combined, they may need to sell some shares to rebalance it. Lahla Co should consider the possibility of undertaking a partial listing in order to make the deal more palatable for Kawa Co's shareholders.

In addition to ensuring that the acquisition is financially beneficial for them, Lahla Co's shareholders' main concern would be that Kawa Co's shareholders will own a significant portion of the combined company (just under 36%). This could mean that the new shareholders would have a significant influence on the way the company is run and its strategic direction, which may be different to what Lahla Co's current shareholders want.

Capital structure changes

Capital structure	Equity	Debt
Original: Lahla Co	60.0%	40.0%
Cash payment: Combined company (Appendix 3)	46.9%	53.1%
Share-for-share exchange: Combined company (Appendix 3)	68.0%	32.0%

The cash payment option means that the proportion of market value of debt increases significantly and is higher than the market value of equity. This would probably increase the costs related to financial distress and future borrowing costs would increase as a result.

On the other hand, the share-for-share exchange increases the proportion of equity compared to debt financing. This may reduce financial distress costs, but also reduce Lahla Co's ability to benefit from the tax shields.

On the face of it, it would appear that Lahla Co would find it difficult to raise the funds needed through just debt financing, although the BoD could explore this option further. Equity finance through a partial listing may be a necessary option which Lahla Co will need to explore as well, although this may require Lahla Co to disclose private information to the markets.

Tutorial note: Additional consideration which could be made

If the cash payment to Kawa Co's shareholders is increased to \$0.71/share, to bring it in line with the value obtained from the demerger, and the funding is sought from debt financing, then the debt percentage compared to total firm value will increase to 53.8% (as shown below):

[\$0.71 x 2,000m shares = \$1,420m.

Market value of equity: \$2,933.7m, 46.2%

Market value of debt = (\$1,601.7m + \$1,420m + \$400m) = \$3,421.7m, 53.8%]

Conclusion

The share-for-share exchange gives the highest return for Kawa Co's shareholders and also makes a good return for Lahla Co's shareholders. The impact on capital structure from this method is a higher percentage of equity and therefore scope to raise more finance through debt if required.

However, concerns that Lahla Co is unlisted and complications arising from this might make the cash payment method the preferred one for Kawa Co shareholders. The current cash offer is less than the value generated from the demerger and therefore unlikely to be accepted. Therefore, a cash offer to match the benefit from the demerger would need to be made. The initial cash offer and a higher revised cash offer would have a significant impact on Lahla Co's capital structure in terms of increased debt. Therefore, it is recommended that Lahla Co should consider equity finance through a partial listing. This would also enable Kawa Co's shareholders to trade their shares and thereby make the deal look better for them.

Report compiled by:

Date

(Note: Credit will be given for alternative and valid discursive comments)

Appendices:

Appendix 1: Estimate of Kawa Co cost of capital (Part (c)(i))

Kawa Co, market value of equity = $24.5\% \times (\$2.45 \times 2,000 \text{ million shares}) = \$1,200\text{m}$ approximately

Kawa Co, market value of debt = $\$400\text{m}$

Kawa Co, estimate of equity beta = $1.15 \times (\$1,200\text{m} + \$400\text{m} \times 0.8)/\$1,200\text{m} = 1.46$

Kawa Co, cost of equity = $3\% + 7.2\% \times 1.46 = 13.51\%$

Kawa Co, post tax cost of debt = $4.4\% \times 0.8 = 3.52\%$

Kawa Co, cost of capital = $(13.51\% \times \$1,200\text{m} + 3.52\% \times \$400\text{m})/(\$1,200\text{m} + \$400\text{m}) = 11.01\%$, say 11%

Appendix 2: Estimate of Kawa Co equity value if demerger is undertaken (Part (c)(i))

Current sales revenue attributable to Kawa Co = $20\% \times \$4,500\text{m} = \900m

Per year, sales revenue growth rate = 6%

Profit before interest and tax (PBIT) = 21%

Tax rate = 20%

Additional asset investment = $\$0.25/\1

Cost of capital (appendix 1) = 11%

Per year, free cash flow growth rate after first four years = 2.5%

Cash flows, years 1 to 4 (\$m)

Year	1	2	3	4
Sales revenue	954.0	1,011.2	1,071.9	1,136.2
PBIT	200.3	212.4	225.1	238.6
Tax	40.1	42.5	45.0	47.7
Additional asset investment	13.5	14.3	15.2	16.1
Free cash flows	146.7	155.6	164.9	174.8
Present value of free cash flows (11%)	132.2	126.3	120.6	115.1

Corporate value, years 1 to 4: $\$494.2\text{m}$

Corporate value, year 5 onwards: $(\$174.8\text{m} \times 1.025/(0.11 - 0.025)) \times 1.11^{-4} = \$1,388.5\text{m}$

Total corporate value: $\$1,882.7\text{m}$

Value attributable to equity: $75\% \times \$1,882.8\text{m} = \$1,412.0\text{m}$

Per share value = $\$1,412.0\text{m}/2,000 \text{ million shares} = \0.71 per share

Kawa Co original value = $\$1,200\text{m}/2,000 \text{ million shares} = \0.60 per share

Gain = $(\$0.71 - \$0.60)/\$0.60 = 18.3\%$, if Kawa Co gets demerged

Appendix 3: Sale of Kawa Co to Lahla Co (Part (c)(ii))

Lahla Co PE ratio = $90\% \times 15.61 = 14.05$

Lahla Co equity value = $14.05 \times \$171.0\text{m} = \$2,402.6\text{m}$

Kawa Co equity value = $\$1,200\text{m}$

Kawa Co estimate of PE ratio = $\$1,200\text{m}/\$117.1\text{m} = 10.25$

Profits after tax of combined company = $\$171.0\text{m} + \$117.1\text{m} + \$62\text{m} = \350.1m

Average PE ratio of combined company = $(14.05 + 10.25)/2 = 12.15$

Estimate of equity value of combined company = $\$350.1\text{m} \times 12.15 = \$4,253.7\text{m}$

Additional equity value created from combining the two companies = $\$4,253.7\text{m} - (\$1,200\text{m} + \$2,402.6\text{m}) = \651.1m

Cash offer

Chakula Co's shareholders will receive $\$0.66 \text{ per share}$ from sale of Kawa Co, or $\$0.66 \times 2,000 \text{ million shares} = \$1,320\text{m}$ in total

Kawa Co original value per share = $\$0.60$

Gain = $\$0.06/\$0.60 = 10\%$

Lahla Co's total shareholders' value is estimated at = $\$4,253.7\text{m} - \$1,320\text{m} = \$2,933.7\text{m}$, or $\$2,933.7\text{m}/1,200 \text{ million shares} = \2.44 share

Lahla Co estimate of original value = $\$2,402.6\text{m}/1,200 \text{ million} = \2 per share

Gain = $\$0.44/\$2 = 22\%$

Share-for-share offer

Additional shares issued by Lahla Co = $2,000 \text{ million}/3 = 667 \text{ million}$

Equity value of combined company = $\$4,253.7\text{m}$

Per share value = $\$4,253.7\text{m}/1,867 \text{ million shares} = \2.28

Gain to Kawa Co's shareholders

$\$2.28 - (\$0.60 \times 3) = \$0.48$

$\$0.48/\$1.80 = 26.7\%$

Gain to Lahla Co's shareholders from combining the company

$(\$2.28 - \$2)/\$2 = 14.0\%$

Lahla Co: Impact on capital structure from the two payment methods

Lahla Co, before acquisition

Market value of equity: \$2,402.6m (see above)

Market value of debt = $40/60 \times \$2,402.6\text{m} = \$1,601.7\text{m}$

Combined company, cash payment through debt borrowing

Market value of equity: \$2,933.7m or 46.9%

Market value of debt = $\$1,601.7\text{m} + \$1,320\text{m} + \$400\text{m}^* = \$3,321.7\text{m}$ or 53.1%

(Note: $\$2,933.7\text{m} + \$3,321.7\text{m} = \$6,255.4\text{m}$; $46.9\% = (\$2,933.7\text{m}/\$6,255.4\text{m}) \times 100\%$ and $53.1\% = (\$3,321.7\text{m}/\$6,255.4\text{m}) \times 100\%$)

Combined company, share-for-share exchange

Market value of equity: \$4,253.7m or 68.0%

Market value of debt: $\$1,601.7\text{m} + \$400\text{m}^* = \$2,001.7\text{m}$ or 32.0%

(Note: $\$4,253.7\text{m} + \$2,001.7\text{m} = \$6,255.4\text{m}$; $68.0\% = (\$4,253.7\text{m}/\$6,255.4\text{m}) \times 100\%$ and $32.0\% = (\$2,001.7\text{m}/\$6,255.4\text{m}) \times 100\%$)

* In the above cases, when the two companies are combined, it is assumed that Lahla Co will continue to service loan notes B or cancel them by paying them off through an equivalent borrowing.

2 Robson

(a) Project cash flows: All figures are in \$m

Year	0	1	2	3	4
Cash flows	(120.0)	20.9	20.6	28.7	104.6
Discount factors – 14% (w1)	1.000	0.877	0.769	0.675	0.592
Present values	(120.0)	18.3	15.8	19.4	61.9

Base case net present value = (\$4.6m)

Base case net present value is negative and on this basis should therefore be rejected.

Financing side effects: All figures are in \$m

Issue costs (w2)	(2.0)
Tax shield on subsidised loan (w3)	0.9
Tax shield on bank loan (w4)	2.0
Subsidy benefit (w5)	5.7
Total benefit of financing side effects	6.6

Recommendation

The adjusted present value of the project is \$2.0m and so the project should be accepted.

Workings:

Working 1 (w1): Ungeared cost of equity

$$\beta_a = \beta_e \times [MV_e / (MV_e + MV_d (1 - t))]$$

$$\beta_a = 1.418 \times [5 / (5 + 1(0.8))] = 1.222$$

$$K_{eu} = r_f + \beta_a (r_m - r_f) = 0.03 + (1.222 \times 0.09) = 14\%$$

Working 2 (w2): Issue costs

$$\$100\text{m} \times 0.02 = \$2,000,000$$

(Note: Issue costs are payable out of cash reserves, so the finance does not need to be grossed up)

Working 3 (w3): Tax shield on subsidised loan

$$\text{Annuity factor (9\%, 4 years)} = 3.240$$

$$\$40\text{m} \times 0.035 \times 0.20 \times 3.240 = \$907,200$$

(Note: The risk free rate would also be acceptable as a discount rate)

Working 4 (w4): Tax shield on bank loan

Annual repayment = $\$50\text{m}/3 \cdot 240 = \$15,432,098$

Year	1 \$ 000	2 \$ 000	3 \$ 000	4 \$ 000
Opening balance	50,000	39,068	27,152	14,164
Interest at 9%	4,500	3,516	2,444	1,275
Repayment	(15,432)	(15,432)	(15,432)	(15,432)
Closing balance	39,068	27,152	14,164	7
Tax relief on interest (20%)	900	703	489	255
Discount factor (9%)	0.917	0.842	0.772	0.708
Present value	825	592	378	181

Total present value = \$1,976,000

Working 5 (w5): Subsidy benefit

Subsidy benefit = $\$40\text{m} \times (0.09 - 0.035) \times 0.80 \times 3 \cdot 240 = \$5,702,400$

(b) Factors each capital provider may consider**External shareholders**

The chief executive's optimism regarding the rights issue may be misplaced. Robson Co's shareholders may question the need for another rights issue so soon after the last one. Nor may they have the funds available to take up their rights, particularly when there has been a series of fund raising exercises in the last six years. Whilst in theory the shareholders are able to sell their rights, this would mean accepting a dilution in their voting power, which may not be acceptable. Therefore it is possible a rights issue could fail. Even if Robson Co has the issue underwritten, failure of the rights issue would have an adverse impact on Robson Co's share price and the market's confidence in the board.

Shareholders may question the logic behind the new project and whether the forecast results can be delivered. They may need reassurance that lessons from the past have been learnt. The underwriting costs have been ignored in the financial appraisal even though these are likely to be significant and may prove fatal to the final outcome, particularly when the project's APV is quite marginal at \$2m.

The loan will mean that Robson Co's gearing once again exceeds the average and shareholders will require higher returns to compensate for the increase in financial risk. The shareholders may question whether the commitment to service and repay the new loans may mean that Robson Co will have difficulty paying an acceptable level of dividend.

Founding directors

The directors may also lack the funds required to take up their rights but their participation is likely to be critical to the success of the rights issue. Any reluctance to participate could send a signal to external investors that the risks involved in the new project are too high, particularly when there are information asymmetries about the project.

Tensions between the board and shareholders have increased to the extent that the shareholder action group only narrowly lost a recent vote to replace the CEO. If the directors do not take up all of their rights and the rights issue is still successful, the balance of power may alter in favour of the action group if there was ever a renewed effort to remove the board.

Subsidised loan provider

The subsidised loan programme provides capital for investment with the objective of boosting employment in a deprived part of the country. Since the funds ultimately originate from the taxpayer, the government is accountable for any funding decisions made. Robson Co's ability to service and ultimately repay the debt is therefore paramount. Robson Co's credit rating provides an assessment of the probability of default and the recent downgrade may cause concern. Even though Robson Co is unable to provide assets for security, the directors may still be faced with other covenants, for example, restrictions on dividends or further borrowing which may upset shareholders.

The subsidy means demand for such loans is likely to be high and the selection criteria difficult, so it is unlikely that the outcome is a foregone conclusion in the way Robson Co's CEO suggests. Based on the information provided, it is unclear whether the new project would meet those selection criteria. Although Robson Co's new project is to be located in an area targeted for regeneration, it remains the case that the objective of the move is to automate the production line. Whilst jobs may still be created in a deprived area, net job creation nationwide is still likely to be negative. Whether such a policy would be attractive to the government, or the taxpayer, remains to be seen.

(Note: Credit will be given for alternative and valid comments)

3 Gogarth

(a) Net receipt

$$\$37,400,000 - \$14,500,000 = \$22,900,000$$

Forward contract

$$\$22,900,000 / 0.2374 = \text{MR}96,461,668$$

Futures

Buy MR September futures

Basis

Assume that basis reduces to zero at contract maturity in a linear fashion.

$$\text{Predicted futures rate} = 0.2366 + [(0.2378 - 0.2366) \times 2/3] = 0.2374$$

$$\text{Alternatively, use spot rate, } 0.2358 + [(0.2378 - 0.2358) \times 4/5] = 0.2374$$

$$\text{Expected receipt} = \$22,900,000 / 0.2374 = \text{MR}96,461,668$$

$$\text{Number of contracts} = \text{MR}96,461,668 / \text{MR}500,000 = 192.9, \text{ say } 193$$

$$\text{Amount over-hedged} = (500,000 \times 193 \times \$0.2374) - \$22,900,000 = \$9,100$$

$$\text{Payment at forward rate} = \$9,100 / 0.2370 = \text{MR}38,397$$

Outcome

	MR
Futures (500,000 x 193)	96,500,000
Payment on forward market	(38,397)
	96,461,603

Options

Purchase MR September call options

$$\text{Receipt} = \$22,900,000 / 0.2368 = \text{MR}96,706,081$$

$$\text{Number of contracts} = \text{MR}96,706,081 / \text{MR}500,000 = 193.4 \text{ contracts, approximately } 193 \text{ contracts}$$

$$\text{Premium} = 193 \times \$0.0014 \times 500,000 = \$135,100$$

$$\text{Premium in MR, translated at spot rate} = \$135,100 / 0.2355 = \text{MR}573,673$$

$$\text{Amount under-hedged} = \$22,900,000 - (193 \times 500,000 \times \$0.2368) = \$48,800$$

$$\text{Translated at forward rate} = \$48,800 / 0.2374 = \text{MR}205,560$$

Outcome, assuming options are exercised

	MR
Options (500,000 x 193)	96,500,000
Receipt on forward market	205,560
Premium	(573,673)
	96,131,887

Recommendations

The forward contract gives a marginally higher receipt than the futures. Futures would be subject to basis risk, the risk that the difference between the futures price and spot rate does not decrease linearly towards the maturity of futures. This means that the receipt may be uncertain. Futures also require a margin payment, an initial payment of cash into a margin account operated by the futures exchange, with further payments if losses are made on contracts.

Options give a lower receipt, because of the need to pay a premium. Gogarth Co may consider options if it considers there is a chance that the dollar will be in a stronger position against the Malaysian ringgit than suggested by the forward rate, or if one or other transaction is likely to fall through.

Overall, Gogarth Co should choose the forward contract as it offers the marginally higher receipt and is not subject to basis risk.

Note: Other valid recommendations could be made.

(b) Advantages of exchange-traded options

The price and contract details of exchange-traded options are readily available and do not need to be negotiated by Gogarth Co, unlike over-the-counter options.

Gogarth Co can sell exchange-traded options at any time before expiry. It may not be possible to sell over-the-counter options, although they can be allowed to lapse if not required.

Exchange-traded options are less risky for Gogarth Co than over-the-counter options, due to their being subject to transparency and other regulations set by the exchange. The exchange also requires participants to set up an initial margin, so that counterparty risk should not be a problem, although margin requirements will impact Gogarth Co's cash flow.

Exchange-traded options are generally American style options, which means that they can be exercised at any time before their expiry date. This may be useful for Gogarth Co if transaction settlement dates are uncertain. Over-the-counter options tend to be European style options, which means that they can only be exercised on their expiry.

Disadvantages of exchange-traded options

The contract sizes and maturity date of exchange-traded options are fixed, whereas those of over-the-counter options can be negotiated to meet Gogarth Co's risk exposure. The standardisation of exchange-traded options also means that they are subject to hedge inefficiencies if the exact amount being hedged is not matched by the option contracts.

Over-the-counter options have a greater range of currencies available than exchange-traded options. This may be useful for Gogarth Co if it is looking to expand significantly abroad.

Exchange-traded options tend to be for shorter terms than those available for over-the-counter options. Longer-term options may be better for Gogarth Co depending on transaction settlement arrangements.

- (c) Economic risk is the longer-term risk that the present value of future cash flows may be increased or reduced by exchange rate movements. The treasury function will be involved in the development of longer-term responses, as the derivatives the treasury function will use for hedging of short-term exchange risk will not be appropriate.

Risk analysis

The treasury function needs to identify the cash flows which may be affected by exchange rate movements. These may not just include transactions with overseas customers and suppliers. Home market sales can also be affected if, for example, the currency of the country where a foreign competitor is based weakens against the ringgit and the competitor can then afford to charge cheaper ringgit prices.

The treasury function must also identify the factors affecting exchange rate movements in the longer term and assess what their impact is likely to be. This could include predicted movements, for example, changes in the economic cycle, and also the impact of sudden economic shocks. The treasury function will need to assess the impact of these exchange rate movements on Gogarth Co. This will include consideration of the other impacts which the factors affecting exchange rates will have, for example, a change in interest rate policy affecting demand for electrical equipment directly as well as influencing exchange rate levels.

Risk management

The treasury function will be particularly involved in determining funding policy in the context of the need to manage economic risk. One aspect of economic risk management is matching any assets held in a foreign country with a loan in that country's currency. The treasury function will determine the suitability of borrowing abroad and the best possible arrangement if foreign currency loans are required.

Economic risk can also be managed by diversifying customer, supplier and operational bases and changing pricing policy. The treasury function will be involved in assessing the possible impacts of policy changes. However, the decisions will be taken in the context of operational considerations, such as supplier management, and wider strategic considerations, such as scope of operations.

		Marks
1	Chakula	
(a)	Shareholders	2–3
	Other stakeholders	2–3
	Other comments	1–2
	Max	5
(b)	3 marks for discussing each of the two propositions	6
(c) (i)	(Appendices 1 and 2)	
	Kawa Co demerged	
	Kawa Co, value of equity	1
	Kawa Co, equity beta	1
	Kawa Co, cost of equity	1
	Kawa Co, cost of debt	1
	Kawa Co, cost of capital	1
	Sales revenue years 1 to 4	1
	PBIT years 1 to 4	1
	Taxation years 1 to 4	1
	Additional asset investment years 1 to 4	1
	Corporate value	2
	Value per share	1
		12
(ii)	(Appendix 3)	
	Lahla Co, current equity value	1
	Kawa Co, estimate of PE ratio	1
	Combined company, current equity value	1
	Estimate of additional value	1
	Cash offer: gain (both groups of shareholders)	2
	Share offer: gain (both groups of shareholders)	3
	Financing implications	3
		12
(iii)	Report	
	Evaluation of financial and other factors	7–8
	(Evaluation report could include, for example, financial returns from demerger and each form of consideration, exit strategies, concerns about becoming minority shareholders, but also concerns for majority shareholders on the impact minority shareholders may have, assumptions made and whether the value created from the share-for-share exchange is realistic)	
	Impact on the capital structure	3–4
	Max	11
Professional marks for part (c)		
	Report format	1
	Structure and presentation of the report	3
		4
	Total	50

		<i>Marks</i>
2	Robson	
(a)	Base case	1
	Ungeared cost of equity	2
	Issue costs	1
	Tax shield on subsidised loan	2
	Tax shield on bank loan	4
	Subsidy benefit	1
	Adjusted present value	1
	Recommendations	2
		14
(b)	Shareholders (e.g. fund availability, control, track record)	4–5
	Directors (e.g. signalling, asymmetries, action group)	3–4
	Subsidised loan provider (e.g. job creation, default risk, covenants)	3–4
	Max	11
	Total	25
3	Gogarth	
(a)	Netting of receipt and payment	1
	Forward contract	1
	Futures	
	Buy September futures/options and number of contracts	1
	Lock in rate	2
	Expected receipt	1
	Receipt from over/under hedge	1
	Options	
	Number of contracts	1
	Premium	2
	Receipt from over/under hedge	1
	Outcome	1
	Discussion	3–4
	Max	15
(b)	Advantages of exchange-traded options	2–3
	Disadvantages of exchange-traded options	2–3
	Max	5
(c)	Understanding of economic risk	1
	Up to 2 marks for each well-developed point	
	(Points could include identification of cash flows affected, identification of influences on exchange rate, role in implementing economic risk management, role in advising on economic risk management)	
	Max	5
	Total	25

Advanced Financial Management

Sample Exam – December 2021

Get to know your exam

These graphical representations are intended to give an indication of exam requirements and associated question content.

Please note that you will not be able to complete answers within these documents and in isolation they will not sufficiently prepare you for your exam.

We encourage you to visit the ACCA Practice Platform in order to attempt up to date practice exams within the computer-based exam environment.

Instruction screens

AFM September/December 2021 exam (21/22 syllabus)

Instructions (1 of 4)

The instructions displayed below are representative of those displayed in the live exam. Where there are differences between this past content exam and the live exam these are explained.

General Instructions

- In this exam, the instruction screens are not timed however in the live exam they will be available for a maximum of 10 minutes prior to the exam starting.
- In the live exam, the stated exam time will automatically start once the 10 minute period has passed (or earlier if you choose to start the exam within the 10 minute period).
- A copy of the instruction screens can be accessed at any time during the exam by selecting the Help button provided.

The Workspace

- Your exam consists of a number of questions. Each question is presented in a workspace. Each workspace will include:
 - Introductory information for the question.
 - Exhibits – these contain the question scenario content broken down into sections or sources of information you will need to answer the question requirement(s).
 - Requirements – these list the requirement(s) you are expected to answer in the workspace.
 - Response Options – you may be presented with one or more word processor and/or spreadsheet response options within which to construct your answer.
- When selected, the exhibits, requirements and response options will display in windows, which can be moved and resized as required.
- You can close each window individually or close all windows at once by selecting  **Close All** on the top toolbar. This minimises the open windows to the left-hand side of the screen. Note that any answers you have entered into the response options will be saved and you can re-open them and change your answers at any time.
- There is a splitter bar which you can move to see more or less of the left or right hand side of the screen as required.
- You can highlight or strikethrough text in the introductory information, PDF exhibits or requirements by selecting  **Highlight** or  **Strikethrough**.
- You can copy and paste between exhibits, requirements and response options by using the Ctrl-C (Copy) and Ctrl-V (Paste) shortcuts.

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Instruction screens (continued)

AFM September/December 2021 exam (21/22 syllabus)

Instructions (2 of 4)

Navigation

- The question number you are viewing is displayed in the top display bar. You can hide or restore this display by selecting .
- You can navigate between screens by selecting **Next** → or ← **Previous** or by clicking on a question number from the Navigator or Item Review screens.
- You can revisit questions and change your answers at any time during the exam.

Flag for Review

- If you wish to revisit/review a question later in the exam, click  **Flag for Review**.
- Click the button again to remove the flag.

Help

- Click the **Help** button provided to access:
 - A copy of these exam instructions.
 - Help and guidance on workspaces.
 - Formulae sheets/tax tables if your exam requires these. Note that the name of the Help button will indicate if formulae sheets/tax tables are available i.e.  **Help**,  **Help/Formulae Sheet** or  **Help/Tax Tables**

Calculator

- You have the option to use the on-screen standard or scientific calculators by selecting  **Calculator**.
- Note that in the live exam you are also permitted to use your own calculator providing it does not have the facility to store or display text.

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Instruction screens (continued)

AFM September/December 2021 exam (21/22 syllabus)

Instructions (3 of 4)

Workings/Scratch Pad

- You may use an on-screen Scratch Pad to make notes/workings by selecting  **Scratch Pad**.
- You can cut and copy text from the Scratch Pad and paste into your response options either by using the Ctrl-C (Copy), Ctrl-X (Cut) and Ctrl-V (Paste) shortcuts, or the on-screen buttons.
- The Scratch Pad retains all notes/workings entered for all workspaces and these are available for the duration of the exam. They will not be submitted for marking after your live exam.
- You will also be provided with paper for notes/workings for your live exam, should you prefer to use it. This will not be submitted for marking. It will be collected at the end of the exam and must not be removed from the exam room.
- **Important:**
 - The notes/workings entered onto the Scratch Pad or your workings paper during the live exam will not be marked.
 - If you want the marker to see any notes/workings for question requirements in the live exam you must show them within the relevant response options.

Symbol

- You can add a selection of currency symbols to your answers by selecting  **Symbol** on the top toolbar.

Navigator Screen

- Navigator can be accessed at any time during the exam by selecting  **Navigator**.
- This screen allows you to jump to any question number in the exam.
- It also allows you to see the status of each question and whether it has been viewed, attempted or flagged for review.

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Instruction screens (continued)

AFM September/December 2021 exam (21/22 syllabus)

Instructions (4 of 4)

Reviewing the Exam

- You can review this exam once you have attempted any, or all, of the questions.
- To do this:
 - Navigate to the **last question** in the exam.
 - Click **Next →**.
- This takes you to the Item Review screen.

Item Review Screen

- This screen gives you a final opportunity to see the flag and attempted status of each question and to review your answers before you exit the exam.
- You can select individual questions you wish to revisit, or quickly access groups of questions from the Item Review screen.
- During the item review period Navigator is not available however you can navigate to questions by selecting **Next →**, **← Previous** or **Review Screen**.
- When reviewing questions you can change your answers and click **Review Screen** to view any updated status on the Item Review screen.

Ending the Exam

- Once you have completed your item review and wish to finally end the exam click **End Exam** from the Item Review Screen.
- Once you end the exam, you cannot revisit any questions. Your answers will then be available for marking and review.

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Exam summary screen

AFM September/December 2021 exam (21/22 syllabus)

Exam Summary

Time allowed: This exam is not timed.

This exam is divided into two sections:

Section A

- One question worth 50 marks.
- 50 marks in total.

Section B

- Two questions, each worth 25 marks.
- 50 marks in total.

All questions are compulsory.

Select **Next** to start your exam.

← Previous Next →

Exam questions

Section A – summary screen

AFM September/December 2021 exam (21/22 syllabus)

Flag for Review

Section A

This section of the exam contains **one question**.

This question is worth **50 marks** and is compulsory.

This exam section is worth **50 marks** in total.

Important:

In your live exam, you should:

- Indicate which requirement each of your responses relate to so that this is clear for markers.
- Show all notes/workings that you want the marker to see within your responses. Remember, any notes/workings made on the Scratch Pad or on your workings paper will not be marked.

Select **Next** to continue.

Navigator Next →

Section A – question 1

AFM September/December 2021 exam (21/22 syllabus)

Symbol

Highlight

Strikethrough

Calculator

Scratch Pad

1 of 3

Close All

Flag for Review

Exhibits

1. Zhichi Co policy failures

2. New project discount rate

3. New project cash flows

4. Financing the new project

5. Future project funding

Requirements

Requirements (50 marks)

Response Options

Word Processor

Spreadsheet

The following exhibits, available on the left-hand side of the screen, provide information relevant to the question:

1. Zhichi Co policy failures

2. New project discount rate

3. New project cash flows

4. Financing the new project

5. Future project funding

This information should be used to answer the question requirements within your chosen response option(s).

Help/Formulae Sheet

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Requirements

Requirements (50 marks)	
(a) Discuss and justify the actions Zhichi Co should take to address the three financial strategy policy failures.	(8 marks)
(b) Prepare a report for the board of directors (BoD) of Zhichi Co which:	
(i) Estimates an appropriate discount rate to use to determine the net present value of the new project based on all-equity finance;	(6 marks)
(ii) Estimates the net present value of the new project, assuming that it is all-equity financed;	(9 marks)
(iii) Estimates the adjusted present value of the new project;	(7 marks)
(iv) Evaluates whether the new project should be undertaken, and: • discusses the assumptions made in the estimates above; • discusses whether the adjusted present value method would be more appropriate than the conventional net present value method to evaluate the new project.	(10 marks)
Professional marks will be awarded in part (b) for the format, structure and presentation of the report.	(4 marks)
(c) Compare and contrast whether Zhichi Co should raise future funding through new debt finance or through asset securitisation.	(6 marks)

Exhibit 1

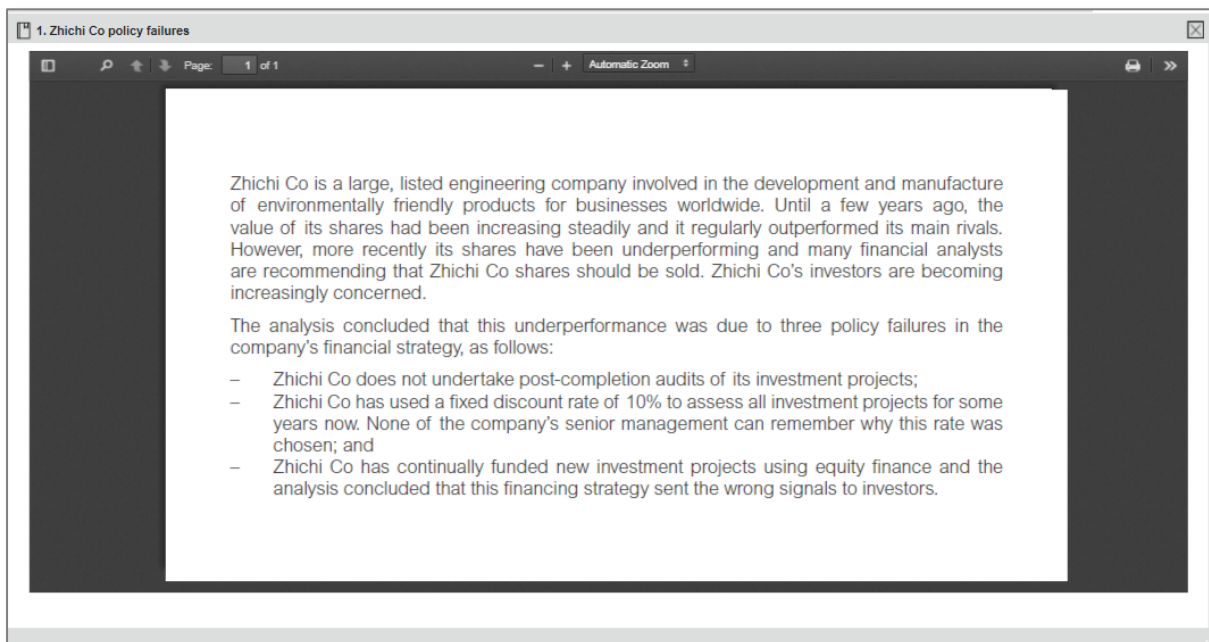


Exhibit 2

2. New project discount rate

Page: 1 of 1

Automatic Zoom

Zhichi Co is considering a new project to manufacture environmentally friendly motor scooters which are fully carbon neutral. This is a diversification into a new business area for Zhichi Co in which it has no previous experience. Zhichi Co's chief financial officer is of the opinion that Zhichi Co should determine an appropriate discount rate for the project based on an initial assumption that the project will be all-equity financed.

Liyu Co would be a competitor to Zhichi Co as it manufactures environmentally friendly motor scooters, as well as equipment for wind farms. Approximately 60% of Liyu Co's business is manufacturing motor scooters and the remaining 40% is manufacturing wind farm equipment.

Given below is the most recent financial information relating to Liyu Co.

Share capital (\$0.25 nominal value)	\$20,000,000
Reserves	\$27,436,000
Market value of equity	\$172,000,000
Market value of debt	\$48,260,000
Equity beta	1.20

Sanwenyu Co is involved in the manufacture of equipment for wind farms. It has estimated its cost of equity as 15.4%, and it is financed 20% by debt and 80% by equity in market value terms.

The estimated risk-free rate is 4.8% and the market risk premium is 8%. The corporation tax rate applicable to all the companies is 20%.

Exhibit 3

3. New project cash flows

Page: 1 of 1

Automatic Zoom

Zhichi Co expects the new project manufacturing environmentally friendly motor scooters to last for four years. The project will require an immediate expenditure of \$70m for plant and machinery. After the project ends in four years' time, it is anticipated that the project will be sold for \$42m, inclusive of any inflationary increase.

The following estimates of revenues and costs, relating directly to the project, have been made:

- In the first year, sales revenue is expected to be \$10m and then increase to \$40m in year two. In the final two years of the project, sales revenue will grow by 20% each year.
- Costs are estimated at 120% of sales revenue for the first year and 80% of sales revenue for the second year before reducing to 40% of sales revenue for each of the final two years.

Substantial initial working capital of \$10m will be required at the start of year one of the project. Subsequently, working capital of 15% of sales revenue for that year will be required at the start of years two to four. Any remaining working capital will be released at the end of the project.

Zhichi Co pays corporation tax of 20% every year. Tax is payable with a year's time delay and any tax losses from the project are set against the company's profits from other projects. Tax allowable depreciation is available on the expenditure on the plant and machinery for the project at an annual rate of 15% on a reducing balance basis. It is anticipated that the plant and machinery will have a realisable value of \$20m at the end of the project, and this realisable value is included in the project's estimated sale price of \$42m.

Exhibit 4

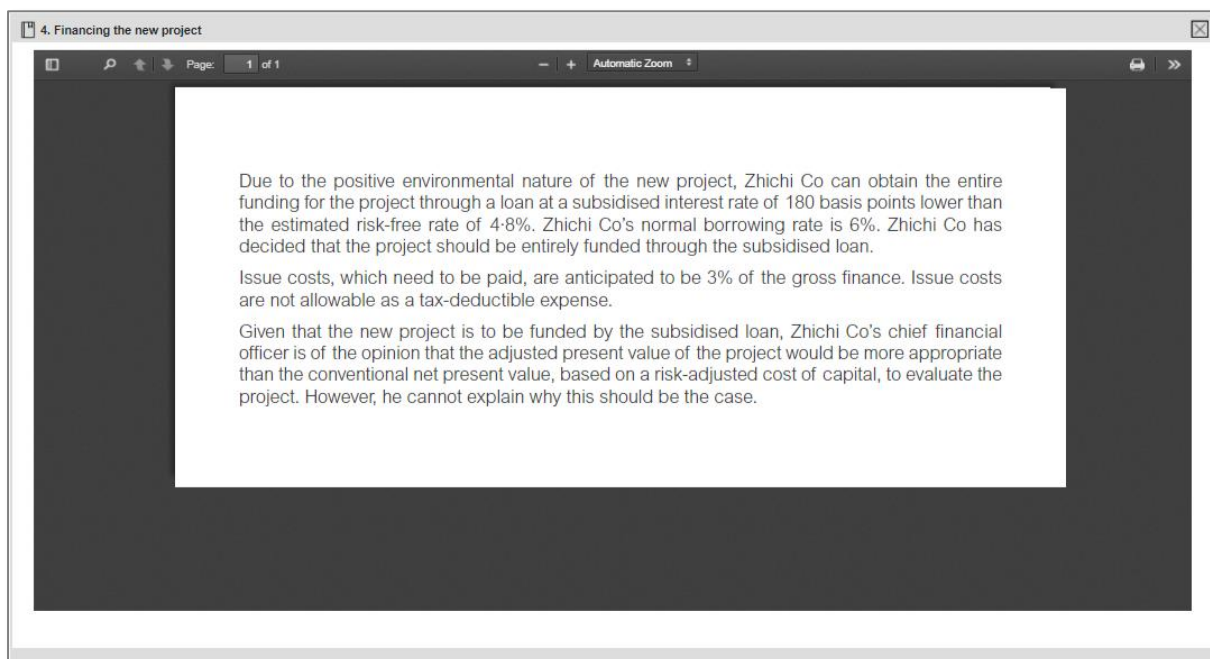
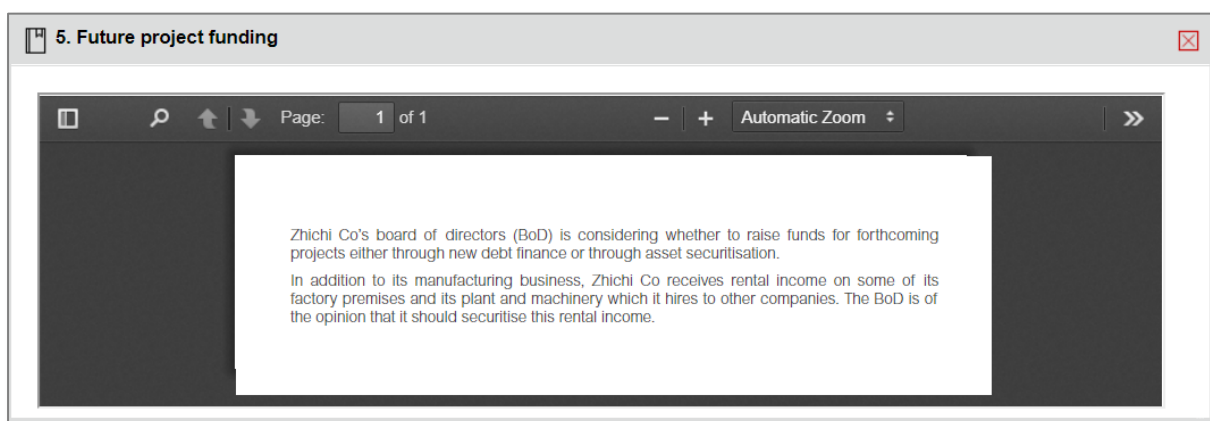


Exhibit 5



Section B – summary screen

AFM September/December 2021 exam (21/22 syllabus)

Flag for Review

Section B

This section of the exam contains **two questions**.
Each question is worth **25 marks** and is compulsory.
This exam section is worth **50 marks** in total.

Important:
In your live exam, you should:

- Indicate which requirement each of your responses relate to so that this is clear for markers.
- Show all notes/workings that you want the marker to see within your responses. Remember, any notes/workings made on the Scratch Pad or on your workings paper will not be marked.

Select **Next** to continue.

← Previous  Navigator Next →

Section B – question 1

AFM September/December 2021 exam (21/22 syllabus)

2 of 3

Symbol

Highlight

Strikethrough

Calculator

Scratch Pad

Close All

Flag for Review

Exhibits

1. Brandon Co

2. Transaction to be hedged

Requirements

Requirements (25 marks)

Response Options

Word Processor

Spreadsheet

The following exhibits, available on the left-hand side of the screen, provide information relevant to the question:

1. Brandon Co

2. Transaction to be hedged

This information should be used to answer the question requirements within your chosen response option(s).

Help/Formulae Sheet

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Requirements

Requirements (25 marks)

(a) Explain how the functional areas of a treasury department could add value to Brandon Co's restructuring plans and for each functional area, discuss the advantages of a centralised treasury department. (7 marks)

(b) Recommend a hedging strategy for the \$36m loan based on the hedging choices the treasury manager is considering, if the central bank base rate increases to 6.6%. Support your answer with appropriate calculations. (15 marks)

Note: Up to 4 marks are available for discussion.

(c) Briefly explain what is meant by smoothing and how this technique could be used to manage Brandon Co's interest rate risk. (3 marks)

Exhibit 1

1. Brandon Co

Page: 1 of 1 Automatic Zoom

Brandon Co is a holding company, operating a small chain of luxury department stores in city centre locations throughout the eurozone. Under the existing structure, the stores are grouped into a number of regional subsidiaries, each with their own head office and treasury function. The board plans to undertake a restructuring exercise in response to a significant decline in revenue over recent years. This will involve closing unprofitable stores within the eurozone whilst expanding into other more profitable locations worldwide. The proposal represents a significant expansion for the company and will be financed by a combination of debt and equity.

The restructuring is due to be discussed at the upcoming board meeting, including the possibility of centralising the treasury function.

Exhibit 2

2. Transaction to be hedged

Page: 1 of 1
100%

Assume today's date is 1 November. Brandon Co's treasury manager predicts a short-term loan of \$36m will be required next year on 31 January to fund an investment in non-current assets as part of the overseas expansion. The loan will be repaid on 31 May.

Brandon Co can borrow at the central bank base rate plus 40 basis points. The central bank base rate is currently 5.7%. Interest rates have been relatively volatile over the last few years and the finance director is concerned that the central bank could increase the base rate to 6.6% between now and 31 January in response to expectations about inflationary pressures within the economy. Recent media reports have also speculated on the possibility of a reduction in the base rate.

Hedging

Brandon Co's treasury manager would like to investigate smoothing as a technique to manage the company's future exposure to interest rate risk but for the purposes of this transaction is considering the following hedging possibilities with the objective of minimising exposure to interest rate risk:

- Forward rate agreements (FRAs)
- Interest rate futures
- Options on interest rate futures

The following data is available for each of the above:

FRAs

The following FRA rates are available:

3-4	5.82%
3-7	5.90%
4-8	5.99%

Three-month \$ futures, \$500,000 contract size
Prices are quoted in basis points at 100 – annual % yield.

December:	94.07
March:	93.95
June:	93.82

Options on three-month \$ futures, \$500,000 contract size, option premiums are in annual %.

Strike price	Calls			Puts		
	December	March	June	December	March	June
93-75	0.168	0.238	0.323	0.025	0.087	0.163

Assume futures and options contracts are settled at the end of each month. Basis is assumed to diminish to zero at contract maturity at a constant rate, based on monthly time intervals. It is also assumed that there is no basis risk and there are no margin requirements.

Section B – question 2

AFM September/December 2021 exam (21/22 syllabus)

Symbol

Highlight

Strikethrough

Calculator

Scratch Pad

Close All

Flag for Review

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Exhibits

1. Hanwood Shoes Co

2. Sale of children's shoes division

3. Impact and consequences of sale

4. Hanwood Shoes Co's SOFP

Requirements

Requirements (25 marks)

Response Options

Word Processor

Spreadsheet

The following exhibits, available on the left-hand side of the screen, provide information relevant to the question:

1. Hanwood Shoes Co

2. Sale of children's shoes division

3. Impact and consequences of sale

4. Hanwood Shoes Co's SOFP

This information should be used to answer the question requirements within your chosen response option(s).

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Requirements

Requirements (25 marks)

(a) Calculate the expected sales price of the children's shoes division and demonstrate its impact on Hanwood Shoes Co's statement of financial position and forecast earnings per share.

(13 marks)

(b) Discuss whether Hanwood Shoes Co's investors are likely to be satisfied with the proposed sale of the children's shoes division and its consequences for profits and funding.

(7 marks)

(c) Discuss the implications for future profits and liquidity of the alternative suggestion made by the non-executive director.

(5 marks)

Exhibit 1

1. Hanwood Shoes Co

Page: 1 of 1

Automatic Zoom

Hanwood Shoes Co started trading 40 years ago, manufacturing and selling children's shoes in shops. The shoes have been of higher quality than shoes produced by its competitors, although also more expensive. School shoe sales in particular have been a consistent large generator of cash, helping to fund the company's expansion.

About 15 years ago, Hanwood Shoes Co started manufacturing and selling adults' shoes. Hanwood Shoes Co's adults' shoes are currently sold at a higher profit margin than children's shoes. Adults' shoe sales now generate the majority of Hanwood Shoes Co's profits. The adults' shoes sold are a mixture of formal types with a long lifespan and some fashionable shoes. The inventory turnover period for adults' shoes is, on average, significantly higher than for children's shoes.

Hanwood Shoes Co is organised on the basis of two separate divisions for children's and adults' shoes. Its shops sell either children's shoes or adults' shoes, but not both. Hanwood Shoes Co also sells adult shoes to other retailers.

Five years ago, Hanwood Shoes Co restructured its cost base. It moved production of children's shoes to new facilities in its home country and outsourced the production of adults' shoes to foreign suppliers. However, Hanwood Shoes Co's board now predicts that profits from children's shoes will increase at a slower rate than adults' shoes. It expects that in the next few years, a greater share of the children's shoes market will be taken by companies with a lower cost base.

Hanwood Shoes Co's board is aware that it must convince investors that any major strategic changes it makes will result in an increase in earnings per share. Investors will also consider a change in earnings per share against any change in the business risk profile of the company.

Exhibit 2

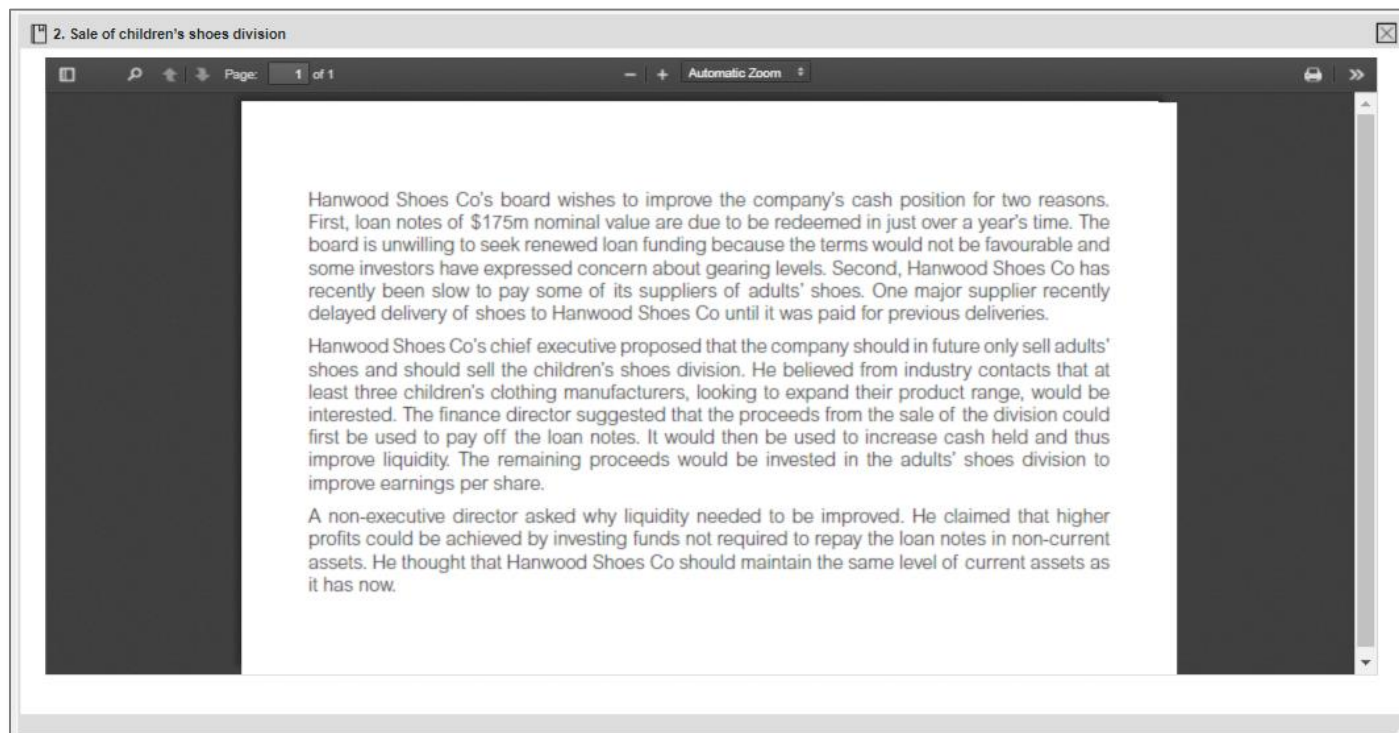


Exhibit 3

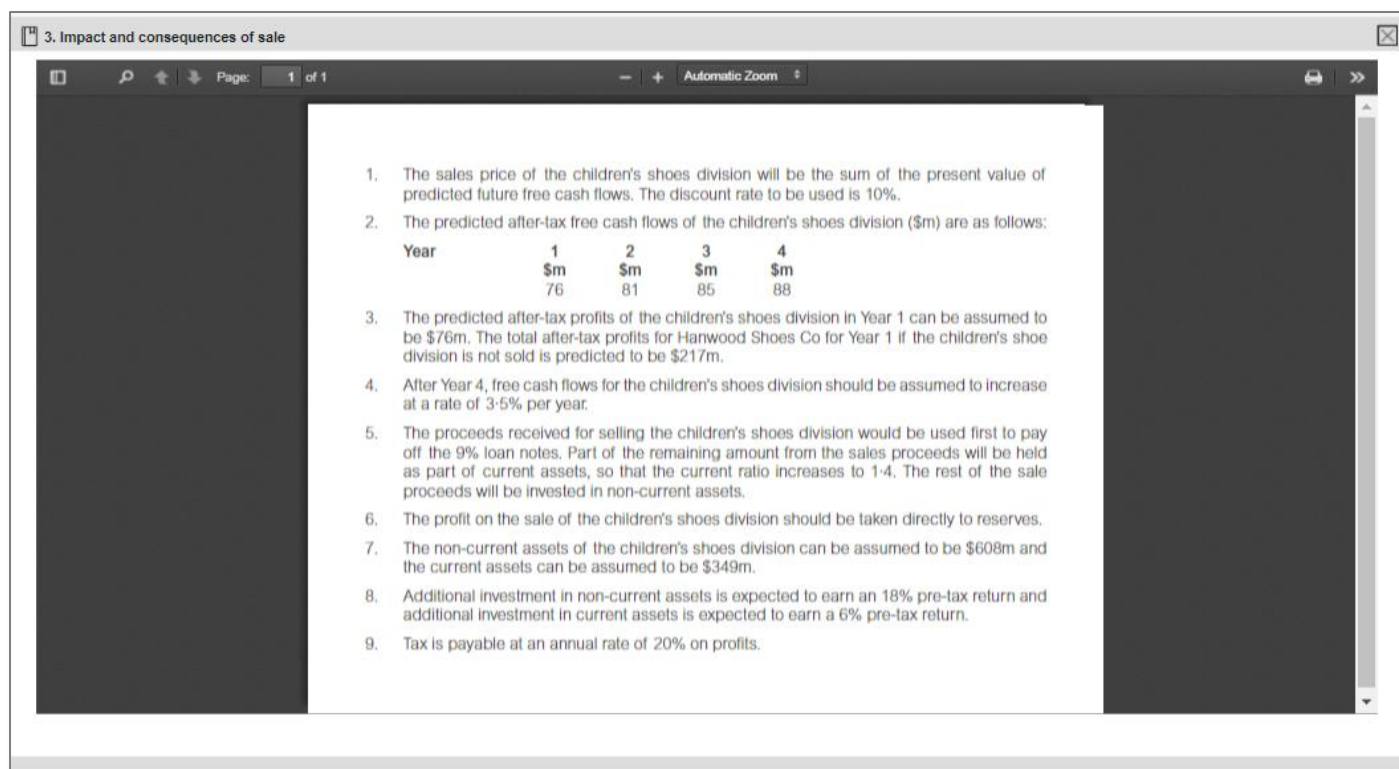


Exhibit 4

4. Hanwood Shoes Co's SOFP									
Edit Format									
100%									
A1									
	A	B	C	D	E	F	G	H	I
1	Hanwood Shoes Co's current statement of financial position								
2			\$m						
3	Assets								
4		Non-current assets	1,200						
5		Current assets	909						
6		Total assets	2,109						
7	Equity and liabilities								
8		Called-up share capital	50						
9		Reserves	737						
10		Total equity	787						
11		Non-current liabilities							
12		9% loan notes	175						
13		7% loan notes	145						
14		Bank loans	108						
15		Total non-current liabilities	428						
16		Current liabilities	894						
17		Total equity and liabilities	2,109						
18									
19									
20									
21									

Answers

1 (a) Post-completion audits

A post-completion audit is a part of monitoring and appraising capital investment projects. Its aims are to compare income, costs and timing of projects against the original budget and any subsequent changes to the budget as more information becomes known. It can be a useful learning tool to help Zhichi Co manage projects more effectively and efficiently in the future. This can lead to setting realistic budget targets, better control of costs and reduction of bottlenecks to help complete projects on time. This is especially so for similar projects. It can be an expensive process and its usefulness may be limited to projects which are unique. In order to maximise the effectiveness of post-completion audits, Zhichi Co would need to ensure that sufficient resources are provided to undertake the audit and the aims of the audit are clear and achievable.

Fixed discount rate

A fixed discount rate to appraise new investment projects, which Zhichi Co uses, can be ineffective when a decision is being made whether or not to undertake the project. This is because projects will have different risks attached to them and therefore the returns required from these projects would differ. This could result in low-risk projects being rejected which could have added to Zhichi Co's corporate value, and high-risk projects being accepted which could reduce corporate value. Zhichi Co should instead estimate an appropriate discount (or hurdle) rate which accounts for the risk of the project. In this way, the company can assess the value of projects more accurately and thereby add to its corporate value.

New project finance

Observations of how companies raise new finance for projects show that they typically prefer internal funds before accessing external markets. If new finance is raised through external markets, then companies prefer debt to equity. Debt finance may have advantages such as tax benefits and controlling the actions of managers. Investors have less information compared to managers and directors in a company, and use companies' actions on raising finance as a signal. Issuing debt finance can also be seen as a sign of confidence that the company can fulfil its interest payment commitments, and can therefore be considered to be stable and less risky by investors. Information asymmetry between investors and a company's managers sends signals that the company is only raising equity finance when share prices have peaked or shares are over-valued. This causes share prices to fall following announcements that a company is raising new equity finance. It is likely that Zhichi Co has experienced this. Therefore, Zhichi Co should finance through a long-term strategy of internal finance, followed by debt issues and then equity issues as a last resort.

(b) Report to the board of directors (BoD), Zhichi Co

Introduction

This report evaluates whether or not the new motor scooter project should be undertaken. It discusses the assumptions made in estimating the value of the project and whether the adjusted present value (APV) method, which is used to estimate the value of the project, is more appropriate than the conventional net present value (NPV) method.

Evaluation

Although the NPV based on an all-equity financed discount rate is negative \$0.9m (appendix 2), when the impact of the financing side effects is taken into account, the APV is positive \$4.9m (appendix 3). On that basis, the project should be accepted if the project is funded using the subsidised loan, but not necessarily without it.

However, both values are marginal and a small change in the variables (see below) could easily mean that the project is no longer viable. In addition to this, it should be noted that a large proportion of the present value of cash inflows from the project, \$51.5m (appendix 2), occur in the fourth year of the project. Projections further into the future tend to be more uncertain.

Zhichi Co may benefit from undertaking sensitivity and scenario analysis to assess the impact of changes in the input variables instead relying solely on the results of appendices 2 and 3.

Assumptions

The assumptions made in each of the three appendices are discussed in turn.

In appendix one, Liyu Co's and Sanwenyu Co's asset betas are calculated by degearing each company's equity beta to eliminate the company specific financial risk. The asset betas of both companies represent just the business risk element. It is assumed that Sanwenyu Co's asset beta represents the business risk of the wind farm business and Liyu Co's asset beta represents the business risks of both the wind farm and the environmentally friendly motor scooter businesses. From these it is assumed that the asset beta, representing a suitable proxy for the business risk of environmentally friendly motor scooters, can be computed and used to estimate the all-equity financed discount rate.

In appendix two, it is assumed that all input variables are known with certainty or reasonable accuracy. It is also assumed that these variables, and the factors which determine the variables, do not change in the future. Uncertainty increases as cash flows are predicted further into the future, and the majority of the positive cash flows for the new project occur in year 4.

In appendix three, it is assumed that the interest rates of the subsidised loan and the corporate tax rates remain unchanged for the period of the project. The normal borrowing rate of 6% is used to determine the present value of the financing side effects, although the risk-free rate could also be used and this will give a higher APV. The debt capacity of Zhichi Co could change as a result of undertaking the project.

In the computations, debt beta is assumed to be zero, although in practice, corporate debt is not free of default risk.

Adjusted present value or net present value

With NPV, future cash flows are discounted using Zhichi Co's average cost of capital (discount rate) since a positive NPV will ensure that the minimum return requirement of all Zhichi Co's investors is met. However, the discount rate often does not take into account (i) the changing business risk profile (since the project is a diversification) nor (ii) the changing financial risk profile (since the new project will be entirely financed by debt). With the new project, both these risks are changing and the APV method takes both changes into account.

Furthermore, the APV method will provide significantly more information about the sources of value and also about the different levels of risk applicable to different cash flows. When using the average cost of capital as the discount rate to generate the NPV, it is not possible to tell where the project's value is generated from, whether the value is from undertaking the project or from the changing capital structure. It also assumes that all cash flows have the same risk profile and should therefore be discounted at the same rate. The APV method considers the risk elements separately and considers the cash flow impact of each. It also assigns a suitable cost of capital which is relevant to each cash flow (for example, the ungeared cost of equity to base case NPV and the cost of debt to the financing side effects).

APV does not normally take into account costs of financial distress, possibility of tax exhaustion and agency costs related to financing using debt. However, in Zhichi Co's case, none of these is likely to be an issue because it has only used equity financing previously and therefore the impact of the above is likely to be minimal.

For these reasons, APV is the more appropriate method to use to evaluate Zhichi Co's new project.

Conclusion

After considering the assumptions made in the calculations and discussing why the APV is the more appropriate method, the recommendation is that the new project is undertaken because it generates a positive APV. However, sensitivity and scenario analysis should be undertaken because of the assumptions made and because the decision to accept is marginal.

Report compiled by:

Date

Appendices:

Appendix 1: (Part (b)(i)):

Liyu Co

$$\text{Asset beta} = 1.2 \times \$172\text{m} / (\$172\text{m} + \$48.26\text{m} \times 0.8) = 0.98$$

Sanwenyu Co

$$\text{Equity beta} = (15.4\% - 4.8\%) / 8\% = 1.325$$

$$\text{Asset beta} = 1.325 \times 0.8 / (0.8 + 0.2 \times 0.8) = 1.1 \text{ approx.}$$

[Assumption: asset beta of 1.1 reflects the business risk attributable to manufacturing wind farms and related equipment]

Asset beta attributable to business risk of manufacturing motor scooters

$$0.98 = 0.6 \times [\text{asset beta, motor scooters}] + 0.4 \times 1.1$$

$$\text{Asset beta} = (0.98 - 0.44) / 0.6 = 0.9$$

$$\text{Base case discount rate} = 4.8\% + 8\% \times 0.9 = 12\%$$

Appendix 2: (Part (b)(ii)): Motor scooter project, all-equity financed

Year	0	1	2	3	4	5
	\$m	\$m	\$m	\$m	\$m	\$m
Sales revenue		10.0	40.0	48.0	57.6	
Costs		(12.0)	(32.0)	(19.2)	(23.0)	
Cash flow before tax		(2.0)	8.0	28.8	34.6	
Taxation (w1)			2.5	0.2	(4.2)	(2.3)
Investment	(70.0)				42.0	
Working capital (w2)	(10.0)	4.0	(1.2)	(1.4)	8.6	
Cash flows	(80.0)	2.0	9.3	27.6	81.0	(2.3)
Discounted at 12% (appendix 1)	1	0.893	0.797	0.712	0.636	0.567
Present value of cash flows	(80.0)	1.8	7.4	19.7	51.5	(1.3)

All-equity financed net present value is approximately \$(0.9)m.

Working 1 (w1): Taxation

Year	TAD (Tax allowable depreciation) (\$m)	Balance (\$m)
1	Investment TAD (15%)	70.0 (10.5)
		59.5
2	TAD (15%)	(8.9)
		50.6
3	TAD (15%)	(7.6)
		43.0
4	Balancing allowance	(23.0)
		20.0

Year	1 \$m	2 \$m	3 \$m	4 \$m
Cash flow before tax	(2.0)	8.0	28.8	34.6
TAD	(10.5)	(8.9)	(7.6)	(23.0)
Taxable cash flows	(12.5)	(0.9)	21.2	11.6
Tax (20%)	(2.5)	(0.2)	4.2	2.3
Year payable	2	3	4	5

Working 2 (w2) Working capital

Year	0 \$m	1 \$m	2 \$m	3 \$m	4 \$m
Required		10	6.0	7.2	8.6
Invested/(released)	10	(4)	1.2	1.4	(8.6)

Appendix 3: (Part (b)(iii): Motor scooter project, adjusted present value**Issue costs**

$$3/97 \times \$80\text{m} = \$2.5\text{m}$$

Tax shield

$$\text{Annual interest payable} = \$80\text{m} \times 0.03 \times 0.2 = \$0.5\text{m}$$

$$\text{Present value of interest payable} = \$0.5\text{m} \times (4.212 - 0.943) = \$1.6\text{m}$$

Subsidised loan benefit and cost

$$\text{Present value of subsidised loan benefit} = \$80\text{m} \times 0.03 \times 3.465 = \$8.3\text{m}$$

$$\text{Present value of tax shield lost} = \$80\text{m} \times 0.03 \times 0.2 \times (4.212 - 0.943) = \$1.6\text{m}$$

[Note: Although 6% is used as the discount rate to calculate the present value of the tax shield and subsidised loan benefits, the risk-free rate of 4.8%, could be used as an alternative]

Adjusted present value

$$\$ (0.9)\text{m (from appendix 2)} + \$ (2.5)\text{m} + \$ 1.6\text{m} + \$ 8.3\text{m} + \$ (1.6)\text{m} = \$ 4.9\text{m}$$

- (c) There are many similarities between debt finance and asset securitisation, and Zhichi Co would face similar obligations which it would need to fulfil. Conventional debt finance would raise an initial borrowed amount, on which Zhichi Co will pay interest and make capital repayments as required. When using asset securitisation, the borrowing aspect for Zhichi Co is similar in terms of repayments of interest and capital. The main differences are the bases on which the borrowing takes place.

In order to mitigate or take account of risk, lenders of conventional debt would probably impose restrictive covenants on Zhichi Co and require a charge to be placed on specific assets or a pool of Zhichi Co's assets. It is likely that Zhichi Co would find it easier to raise conventional debt either through approaching specific lenders or issuing debt securities like bonds. It is also likely that raising conventional debt would be cheaper. This is because markets for conventional debt are more readily available.

Asset securitisation provides a means to capitalise future income flows into an asset, in this case Zhichi Co's rental income, and then use this asset as a means of borrowing funds. Converting the future rental income into an asset would likely be undertaken through the use of a special purpose vehicle (SPV). Zhichi Co would then manage the rent collection and use the rental income stream to pay the interest and capital over a specified time period.

It is likely that obtaining funding through asset securitisation can be expensive due to search costs, management costs, legal fees and continuing administrative costs. Additionally, it is unlikely that Zhichi Co would be able to borrow funds equivalent to the full value of the rental income, and if rental incomes fall, Zhichi Co would need to fund the interest and capital from alternative sources. This could open up Zhichi Co to lower credit ratings and increased risk of financial distress.

The advantages of asset securitisation are that Zhichi Co's debt capacity is maintained for future projects, and it provides a way for Zhichi Co of utilising income from non-core business areas.

Ultimately, the choice to Zhichi Co between debt finance and asset securitisation is that the latter probably provides greater flexibility and opportunity, but at a higher cost.

(Note: Credit will be given for alternative and valid comments and approaches, where relevant, for all parts of the question)

- 2 (a) There are a number of ways the functional areas of a treasury department could add value to Brandon Co's expansion plans.

Liquidity management

As Brandon Co expands, the increase in the volume of transactions across the group will require a greater emphasis on liquidity management, including the need to support Brandon Co's working capital requirement. A centralised treasury department could add value by netting off subsidiaries' debit and credit balances, which would reduce the number of transactions and hence transaction costs.

Currency management

The expansion outside the eurozone introduces foreign exchange risk into the group and the need to manage the impact of currency flows on Brandon Co's earnings and shareholder value. Effective currency management, including the use of derivatives, can have a significant impact on shareholder value for companies with overseas assets. A centralised treasury department will have the resources to employ experts with the knowledge required to manage these risks. It would also be possible to employ techniques across the group, such as matching income earned by one subsidiary with expenditure by another subsidiary in the same currency, reducing risk and the need for external hedging methods.

Funding

The treasury department will play a role in sourcing the appropriate debt instruments and matching maturities to the time horizon of the investment as well as managing the relationship with lenders. A centralised treasury department could add value by pooling funding requirements across the group to achieve better rates by borrowing in bulk.

Corporate finance

The corporate finance function would undertake the equity issue(s) to raise part of the funds required for the overseas expansion and be involved with the strategic decision-making such as formulating a dividend policy which meets the needs of shareholders. A centralised treasury department would have the appropriate level of oversight of Brandon Co's group activities to provide operational, strategic and corporate finance advice to enhance value. It can advise on short-term investment strategies, evaluate subsidiary performance and manage Brandon Co's financial structure to minimise the cost of capital.

(b) FRAs

FRA rate 5.90% (3–7) since the funds will be required in three months' time for a period of four months.

If central bank rate increases to 6.6%

	\$
Interest payment: $(6.6\% + 0.4\%) \times 4/12 \times \$36,000,000$	840,000
Receipt from bank: $(6.6\% - 5.9\%) \times 4/12 \times \$36,000,000$	(84,000)
Net borrowing cost	756,000
Effective annual interest rate: $756,000/36,000,000 \times 12/4$	6.3%

Futures

Hedge against an increase in interest rates, therefore go short in futures market. Use March contracts, as funds will be required on 31 January.

Number of contracts = $\$36,000,000/\$500,000 \times 4 \text{ months}/3 \text{ months} = 96 \text{ contracts}$

Basis

Spot price (1 November) – futures price = basis

$(100 - 5.70) - 93.95 = 0.35$

Unexpired basis on 31 January = $2/5 \times 0.35 = 0.14$

If central bank rate increases to 6.6%

	\$
Interest payment as above	840,000
Expected futures price: $100 - 6.6 - 0.14 = 93.26$	
Profit on the futures market: $(0.9395 - 0.9326) \times \$500,000 \times 3/12 \times 96$	(82,800)
Net borrowing cost	757,200
Effective annual interest rate: $757,200/36,000,000 \times 12/4$	6.31%

Options on interest rate futures

Buy March put options to hedge against an increase in interest rates. As above, 96 contracts are required.

$$\text{Premium} = 0.00087 \times \$500,000 \times 3/12 \times 96 = \$10,440$$

If central bank base rate increases to 6.6%

	Buy put
Exercise price	93.75
Expected futures price, as above	93.26
Exercise?	Yes
Gain in basis points	49
	\$
Interest payment as above	840,000
Profit on options $0.0049 \times \$500,000 \times 3/12 \times 96$	(58,800)
Premium	10,440
Net borrowing cost	<u>791,640</u>
Effective annual interest rate: $791,640/36,000,000 \times 12/4$	6.60%

Advice

The forward rate agreement and futures market provide broadly similar results. The outcome of the forward rate agreement hedge is marginally lower and may be the preferred choice. In theory, both methods provide a fixed interest payment and a certain cash outflow but futures contracts are subject to basis risk and margin requirements.

If the central bank rate increases to 6.6%, the option is the least attractive hedging method due to the expensive premium. Unlike forward rate agreements and futures contracts, however, options do not have to be exercised and allow Brandon Co to benefit from upside potential if the central bank base rate falls. Treasury staff would need to assess the likelihood and amount of a possible reduction in interest rates when assessing the viability of the option.

If the board is to achieve its objective of minimising risk exposure, the certain cash flow associated with the forward rate agreement will be more attractive than the variable outcome associated with the options.

(c) Smoothing

Smoothing is a relatively straightforward technique for managing interest rate risk. Under this approach, Brandon Co would make an assessment about possible interest rate movements and divide its debt between variable and fixed rate loans according to what it could afford.

If the central bank increases the base rate to 6.6%, only the variable rate loans will cost more. The fixed rate will be relatively cheaper and will reduce the overall financial burden compared to the scenario where all the loans are at a variable rate. If the base rate falls, Brandon Co benefits from a cheaper variable rate but will still have to service its fixed rate debt at the higher rate.

3 (a) Proceeds from sales of children's shoes division

Year	1	2	3	4
	\$m	\$m	\$m	\$m
Free cash flows	76	81	85	88
Discount factor 10%	0.909	0.826	0.751	0.683
Present value	<u>69</u>	<u>67</u>	<u>64</u>	<u>60</u>
Present value	260			

Growth rate year 5 onwards = 3.5%

Present value in year 5 onwards = $(\$88 \times 1.035 / (0.10 - 0.035)) \times 0.683 = \957m

Total present value = $\$260\text{m} + \$957\text{m} = \$1,217\text{m}$

Impact on statement of financial position (\$m)

Profit on sale = $\$1,217\text{m} - \$608\text{m} - \$349\text{m} = \260m

Current assets receipt from sale = $\$1,217\text{m}$

Current assets adjustment = $(\$894\text{m} \times 1.4) - (\$909\text{m} - \$349\text{m}) - \$1,217\text{m} = (\$525\text{m})$

Gain in current assets = $\$1,217\text{m} - \$525\text{m} = \$692\text{m}$

Alternative working

Gain in current assets = $(\$894\text{m} \times 1.4) - (\$909\text{m} - \$349\text{m}) = \692m

Non-current asset adjustment = $\$525\text{m} - \$175\text{m} = \$350\text{m}$

	Original \$m	Sales \$m	Adjustments \$m	Final \$m
Assets				
Non-current assets	1,200	(608)	350	942
		1,217		
Current assets	909	(349)	(525)	1,252
Total assets	<u>2,109</u>			<u>2,194</u>
Equity and liabilities				
Called-up share capital	50			50
Reserves	737	260		997
Total equity	<u>787</u>			<u>1,047</u>
Non-current liabilities				
9% loan notes	175		(175)	0
7% loan notes	145			145
Bank loans	108			108
Total non-current liabilities	<u>428</u>			<u>253</u>
Current liabilities	894			894
Total equity and liabilities	<u>2,109</u>			<u>2,194</u>

Impact on eps (\$m)

	Current forecast	Revised forecast
Predicted post tax profits	217	217
Less: Profits from children's shoes		(76)
Add: Interest saved, net of tax (\$175m x 9% x (1 - 0.2))		13
Add: Return on additional non-current assets (\$350m x 18% x (1 - 0.2))		50
Add: Return on additional current assets (\$692m x 6% x (1 - 0.2))		33
Adjusted profits	<u>217</u>	<u>237</u>
Number of shares	50m	50m
Adjusted eps	<u>\$4.34</u>	<u>\$4.74</u>

(b) Sale

Investors may query the need for the sale at this time. Although the board expects the children's shoes market to become more competitive in a few years' time, this has not yet happened. Investors may wonder if a better sales price could be obtained in a few years' time with a few more years' growth. Although some investors are concerned about gearing, a sale to reduce debt funding may seem to be a forced sale, and investors may ask whether alternative strategies could resolve the problems.

Investors may also wonder why Hanwood Shoes Co is not exploring the possibility of selling the production facilities (if sale proceeds equalled net book value, that would be enough to pay off the 9% loan notes). Hanwood Shoes Co could then use the same outsourcing business model for producing children's shoes which it does for adults' shoes.

Investors may also be concerned whether Hanwood Shoes Co is achieving best value from the sale if the company is going to be selling the division at \$1,217m. Assuming a higher growth rate (perhaps by taking the geometric average of growth over the next four years) would give a higher selling price. Hanwood Shoes Co could also be expected to ask for a premium on the sale, given that a number of companies looking to develop their sales of children's shoes may be interested. That said, the cash flows in the calculation are taken to infinity, which is optimistic.

Investors would also wonder about the decreased diversification that the sale would mean. Although adults' shoes achieve higher margins than children's shoes, they may need more investment to maintain their position, and the appeal of fashionable ranges may be uncertain. Children's school shoes have been a reliable cash generator. Ultimately also, the investors may feel that they are investing in a smaller company, given the decrease in the non-current base and range of products, and may want to invest elsewhere in a company offering better growth prospects.

Profits

The forecast increase in earnings per share is 9.2%, but investors will question whether the assumptions on which the forecast is based are optimistic. They will also take into account the risks of the business being less diversified as it will solely be selling adult shoes. However, part of the increase in profits is a fall in the commitment to paying interest, which investors may view positively if they are concerned about gearing.

Funding

The sale will help fund an improvement in the gearing of the company. The funds will not only be available to pay off the 9% loan notes immediately, they could also be used to pay off the bank loan and/or the other loan notes in time, or reduce trade payables as they may be being used as a source of short-term funding. However, investors might also wonder how future investment in adults' shoe shops would be funded, particularly if future sales did not meet current expectations. Current lenders

may have placed restrictions on Hanwood Shoes Co seeking new borrowing from other sources. If Hanwood Shoes Co was able to seek additional loan capital, it would have a smaller non-current asset base to offer as security, which may affect lenders' attitudes.

(c) Implications for profits

If the surplus after the loan repayment was solely used to fund non-current asset investment, this would, based on the assumptions about the return on non-current assets, increase earnings by \$33m ($(\$1,252\text{m} - \$909\text{m}) \times (18 - 6)\% \times (1 - 0.2)$), representing a total increase of \$53m or 24.4%. This would be more acceptable to the shareholders. However, it might not be possible to sustain the 18% return on non-current assets over a larger investment, certainly in the short term. Hanwood Shoes Co might also consider the impact of the total investment occurring over a longer period of time.

Implications for liquidity

However, the proposed policy would not result in the one-off improvement in liquidity. Hanwood Shoes Co is currently having difficulty paying suppliers, suggesting its current ratio is too low at present. Going forward, the increased volume of activity together with the longer turnover time of adult shoes compared with children's shoes would probably mean an increase in inventory, and decreased liquidity, making it more difficult to pay suppliers on time. Problems paying suppliers would become more significant as Hanwood Shoes Co would be dependent in future on suppliers for all its shoes, rather than having children's shoes made in the factory.

Hanwood Shoes Co might have to review its mix of working capital, particularly how much inventory it holds, in order to maintain sufficient liquidity. There might need to be a reduction in inventory holding times and a reduction in lines being stocked, particularly slow-moving lines.

		Max	Marks
1	(a) 2–3 marks per policy failure		8
	(For example, post-completion audits: compare to budget, useful for future budget development, can be expensive and needs sufficient resources, all projects can be unique and different so not universal. Fixed discount rate: need project specific discount rate to take account of project risk, otherwise low-risk projects can be rejected and high-risk projects might be accepted. Using equity finance. Not observed in practice. Instead pecking order of internal, then debt, then equity sources of finance. Due to information asymmetry, signalling and tax advantage of debt)		
	(b) (i) (Appendix 1)		
	Liyu Co, asset beta		1
	Sanwenyu Co, equity beta		2
	Sanwenyu Co, asset beta		1
	Asset beta attributable to business risk of motor scooter manufacturing		1
	Cost of equity/discount rate		1
			6
	(ii) (Appendix 2)		
	Sales revenue		1
	Costs		2
	Tax allowable depreciation		1
	Tax and timing		2
	Working capital invested/released: years 0 and 1		1
	Working capital invested/released: years 2–4		1
	Cash flows and all-equity financed NPV		1
			9
	(iii) (Appendix 3)		
	Issue costs		1
	Present value of tax shield		2
	Present value of loan subsidy benefit and cost		3
	Adjusted present value		1
			7
	(iv) Report		
	Evaluation		2–3
	Discussion of assumptions (For example, rival companies' asset betas represent business risk, all-equity financed project from asset beta, variable known with certainty and will not change, interest rates will not change, debt capacity is sufficient, cost of debt represents appropriate discount rate)		4–5
	Discussion of appropriateness of APV (For example, limitations of cost of capital re changes in business and financial risk, APV takes account of both risks, APV separates out areas where value is derived from, APV takes into account different levels of risk and required return, but APV does not consider financial distress, tax exhaustion or agency issues)		4–5
		Max	10
	Professional marks for part (b)		
	Report format		1
	Structure and presentation of the report		3
			4
	(c) Up to 2 marks per relevant point (Maximum 3 marks if only either debt finance or asset securitisation discussed) (For example, similarity of interest and capital repayment, conventional debt imposes restrictive covenants, security provided on assets, securitisation's security provided by pooling income, creating assets of this pool and creating different risk levels of securities, securitisation is expensive and not all income is pooled, but it gives more flexibility and does not impact debt capacity so it provides flexibility to the company)	Max	6
		Total	50

		Marks
2	(a) Functional areas (e.g. liquidity management, currency management, funding and corporate finance)	3–4
	Advantages of centralised treasury (e.g. netting off, expertise, pooling, strategic oversight and dividend policy)	3–4
	Max	7
	(b) FRA	2
	Go short on March futures	1
	Number of contracts	1
	Basis	1
	Impact of interest rate increase with futures	2
	Buy March puts	1
	Premium calculation	1
	Exercise	1
	Impact of interest rate increase on options	2
	Discussion	3–4
	Max	15
	(c) Smoothing explanation	1–2
	Smoothing application	1–2
	Max	3
	Total	25
3	(a) PV of free cash flows Years 1–4	2
	PV of free cash flows Year 5 onwards	2
	Revised statement of financial position	4
	Revised eps	5
		13
	(b) Up to 2 marks per relevant point	Max 7
	(relevant points can include need to sell, timing, amount, less diversification, alternative strategies, impact on eps impact on funding)	
	(c) Impact on profits	2–3
	Impact on liquidity	2–3
	Max	5
		25